

# **APPENDIX F**

## Placeholder Sheet for Supplemental Content

Application Number: 18743352

Document Date: 06/14/2024

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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

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### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

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### Documents

**TOTAL DOCUMENTS: 9**

| DOCUMENT                                                                                                   | PAGES | DESCRIPTION                  | SIZE (KB) |
|------------------------------------------------------------------------------------------------------------|-------|------------------------------|-----------|
| 2024-06-14 ADS 27843-2603.pdf                                                                              | 12    | Application Data Sheet       | 2263 KB   |
| 2024-06-14 Track One Request 27843-2603.pdf                                                                | 2     | Track One Request            | 125 KB    |
| Continuation Application 27843-2603-SPEC.docx                                                              | 142   | Specification                | 473 KB    |
| Warning: The automatic document description has been replaced. Bookmarks were found and have been removed. |       |                              |           |
| 2024-06-14 Auxillary Application 27843-2603.pdf                                                            | 142   | Auxiliary PDF of Application | 1460 KB   |

|                                                 |   |                                                    |         |
|-------------------------------------------------|---|----------------------------------------------------|---------|
| 2024-06-14 Drawings 27843-2603.pdf              | 2 | Drawings-only black and white line drawings        | 921 KB  |
| 2024-06-14 Executed Declarations 27843-2603.pdf | 7 | Oath or Declaration filed                          | 1107 KB |
| 2024-06-14 POA 27843-2603.pdf                   | 2 | Power of Attorney                                  | 384 KB  |
| 2024-06-14 IDS 27843-2603.pdf                   | 5 | Information Disclosure Statement (IDS) Form (SB08) | 1336 KB |
| 2024-06-14 IDS Cert 27843-2603.pdf              | 1 | Transmittal Letter                                 | 96 KB   |

## Digest

| DOCUMENT                                        | MESSAGE DIGEST(SHA-512)                                                                                                          |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 2024-06-14 ADS 27843-2603.pdf                   | 826C6DE15FB2DF38E41E7748262E13C584F5A2B8EFE50E3304D4FFE9C1CD42455032DCE490288B5030CAC6EEBD7B88F69D0F1307B59D23906995D4D3C12B4C06 |
| 2024-06-14 Track One Request 27843-2603.pdf     | B928C329C6EF2F6F73EB0610D77715313E8AE5FD159E7C3FFFD25F279FA8DB1223001C4E8E3D5605614CFE260238024FD6B19632F46FCCCEFF3B2F63F22809F9 |
| Continuation Application 27843-2603-SPEC.docx   | EC55F689E2DFDD5D9CEFC18686C54EFB7FFAE47D230D730E2E7F0E8CC46A0267FB78FEA64606ADEC78AD4A14667AE378621F6FF13532680D3ACF7B7B1FAEC844 |
| 2024-06-14 Auxillary Application 27843-2603.pdf | 5C1149B69387D61BC5A167B8559AF371051D133743B336470018BC4C0053865B647F3A6949156E0B705FE735C082DA070A3F1A42A8E7E21BEBADD3BC56071011 |
| 2024-06-14 Drawings 27843-2603.pdf              | 9AB96A2C5B3DDDE71A947908E8FF539F186FD59A46C0AEFD020CA67798899464039EF7CAFF18C34881D70972FB4A6AE65A9DCA4C909F27E0768325D895A6831E |

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|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-06-14 Executed<br>Declarations 27843-2603.pdf | 3D82108E6B8FBE2A54F4E039581A98B4946CAAEFCE08AB0AA<br>A9519FD2F10B5E7F25F783E15BB693D413D356107944DDA785<br>D4131861133142162765AA1994D86 |
| 2024-06-14 POA 27843-<br>2603.pdf                  | 89A3DA24917135E9795358BDAF298BA7286D52EAD560FA9221<br>46447CC48E19611C543D29CD122CC2E1685B7DB2153E6123F<br>F21E8BEF4524AFF37BB5EC3272602 |
| 2024-06-14 IDS 27843-2603.pdf                      | A1752416FD4662808A776258F77C118DD5982556172846AD6C<br>3D0EBB95BC59E4B6E50E9CECE357D5409B0A45363D3FEB14<br>B2CC398A0FF0717B8D321338A40D43 |
| 2024-06-14 IDS Cert 27843-<br>2603.pdf             | AC81C8033D21BDB59CFBA1CFA67786F9F25BF3729C69C9741<br>7AD84DDB7A94F23A260EB6D9869B901C84684724BD7DAD1D<br>1A48C2394A8795EA347078E24DD9EDD |

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

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**Claims**

1. A detector for identifying at least one material property  $m$  comprising  
  
at least one illumination source, wherein the illumination source is configured for generating at  
5 least one illumination pattern for illumination of the object, wherein the illumination pattern  
comprises at least one pattern selected from the group consisting of: at least one point  
pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at  
least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at  
least one pre-known feature; at least one regular pattern; at least one triangular pattern; at  
10 least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising  
convex uniform tilings; at least one line pattern comprising at least one line; at least one line  
pattern comprising at least two lines; and at least one line pattern comprising parallel or  
crossing lines;  
  
at least one sensor element comprising a matrix of optical sensors, the optical sensors each  
15 having a light-sensitive area, wherein the sensor element is configured for recording at least  
one reflection image of a light beam originating from at least one object; and  
  
at least one evaluation device configured for determining the material property by evaluation  
of at least one beam profile of the reflection image,  
  
wherein the evaluation device is configured for determining at least one material feature  $\phi_2 m$   
20 by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein  
the material dependent image filter is at least one filter selected from the group consisting of:  
a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-  
level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a  
grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter;

and a linear combination thereof;

wherein the evaluation device is configured for determining the material property  $m$  by evaluating the material feature  $\phi 2m$ .

2. The detector according claim 1, wherein the material dependent image filter is a spot shape  
5 filter.
3. The detector according to claim 1, wherein the material dependent image filter is Law's energy filter.
4. The detector according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a  
10 transparency, and a deviation from a Lambertian surface reflection.
5. The detector according to claim 1, wherein the evaluation device is configured for determining at least one distance feature  $\phi 1z$  by applying at least one distance dependent image filter  $\Phi 1$  to the reflection image, and  
wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  by  
15 evaluating the distance feature  $\phi 1z$ .
6. The detector according to claim 5, wherein the distance dependent image filter is a structured-light-filter.
7. The detector according to claim 1, wherein the sensor element comprises at least one CMOS sensor.
- 20 8. The detector according to claim 7, wherein the CMOS sensor comprises at least 100000 optical sensors.



9. The detector according to claim 8, wherein the CMOS sensor is sensitive in the range of 700 nm to 1000 nm.
10. The detector according to claim 1, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.
- 5 11. The detector according to claim 10, wherein the illumination source is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.
12. The detector according to claim 11, wherein the illumination source emits light having a wavelength of 700 nm to 1000 nm.
- 10 13. The detector according to claim 12, wherein the illumination source comprises at least one vertical cavity surface-emitting laser.
14. The detector according to claim 13, wherein the illumination source is integrated into a smartphone.
15. The detector according to claim 1, wherein the illumination source emits light having a  
15 wavelength of 700 nm to 1000 nm.
16. The detector according to claim 15, wherein the illumination source is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.
17. The detector according to claim 1, wherein the illumination source is integrated into a  
20 smartphone.
18. The detector according to claim 17, wherein the illumination source and the optical sensors is

arranged in a common plane.

19. The detector according to claim 18, wherein the illumination source emits light having a wavelength of 700 nm to 1000 nm.
20. The detector according to claim 1, wherein the object is a human.
- 5 21. The detector according to claim 1, wherein the detector is configured for identification and/or classification of the material.
22. The detector according to claim 20, wherein the detector is configured for determining and/or validating whether a surface to be examined is or comprises human skin.
23. The detector according to claim 1, wherein the detector is configured for detection of human  
10 skin.
24. The detector according to claim 23, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, and a deviation from a Lambertian surface reflection.
25. The detector according to claim 1, wherein the evaluation device is configured for performing  
15 at least one image analysis and/or image processing in order to identify a reflection feature in the reflection image.
26. The detector according to claim 25, wherein the evaluation device is adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature.
- 20 27. The detector according to claim 25, wherein the evaluation device is adapted to determine at least one reference feature in at least one reference image corresponding to the at least one

reflection feature.

28. The detector according to claim 25, wherein the image analysis and/or image processing comprises selection of at least one region of interest.

29. The detector according to claim 25, wherein the image analysis and/or image processing

5 comprises a background correction.

30. The detector according to claim 25, wherein the image analysis and/or image processing comprises applying a principle curvature-based region detector.

**Detector for identifying at least one material property****Abstract**

- 5 Described herein is a detector for identifying at least one material property  $m$ . The detector includes at least one sensor element including a matrix of optical sensors, the optical sensors each having a light-sensitive area. The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. The detector includes at least one evaluation device configured for determining the material property  $m$  by evaluation of
- 10 at least one beam profile of the reflection image and evaluation of a material feature  $\phi_2 m$ .

**Detector for identifying at least one material property**CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. Patent Application No. 18/425,144, filed January 29, 2024, which is a continuation application of U.S. Patent Application No. 17/439,492, filed September 15, 2021, which is a U.S. National Phase Application of International Patent Application No. PCT/EP20/56759, filed March 13, 2020, which claims priority to European Patent Application No. 19163250.4, filed March 15, 2019, the entire contents of which are hereby incorporated by reference herein.

DescriptionField of the invention

The invention relates to a detector, a detector system and a method for determining at least one material property of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

A large number of methods for material classification and identification are known from the prior art. For example, material identification can be used in machine vision applications, medical applications and security applications, for image classification purposes or in gesture recognition algorithms.

For example, US 2016/0206216 A1 describes a device, system and method for skin detection. The device comprises a thermal sensor input for obtaining thermal sensor data of a scene, a light sensor input for obtaining light sensor data of the scene, and an evaluation unit for analyzing the obtained thermal sensor data and the obtained light sensor data and for detecting skin areas within the scene based on said analysis. US 2016/155006 A1 describes a device and a corresponding method for skin detection. The device comprises an illumination unit configured to project a predetermined illumination pattern onto a scene, an imaging unit configured to acquire an image of the scene, and an evaluation unit configured to evaluate the acquired image by analyzing the imaged illumination pattern as reproduced in the image and to detect skin areas within the

image and distinguish them from non-skin areas within the image based on said analysis.

5 Material classification and identification is usually done in through beam mode, namely shining light through a sample and analyzing extinction. Further, material classification and identification can be done in reflective mode in controlled environments. However, in reflective mode usually wavelengths larger than 1000 nm are necessary in order to yield reliable results. Thus, despite the advantages of known methods for material identification, new concepts for reliable identification of materials are desired.

10 Specifically, identifying a material from its reflection beam profile would be highly desirable. However, most image filters used for beam profile analysis (BPA) yield features that depend on both, distance and material such that reliable identification and classification of the material is not possible. A typical example is the width of a beam profile reflected from a translucent material.

15 DE 198 46 619 A1 describes structured surface appearance quality determining equipment, which evaluates electrical measurement signal from photosensor array to derive structure code characterizing structure-dependent characteristic of measurement surface.

20 CN 108 363 482 A describes a method for controlling smart televisions via three-dimensional gestures on basis of binocular structured light. The method comprises the following steps of: synchronously acquiring images through a binocular camera; reconstructing a three-dimensional image according to an obtained left-right view;

25 carrying out preprocessing; recognizing a gesture action after three-dimensional gestures are segmented; and converting the recognized gesture action into an operation instruction for a smart television, and executing the operation instruction.

US 2018/033146 A1 describes systems and methods for determining a depth map and a reflectivity map from a structured light image. The depth map can be determined by capturing the structured light image and then using a triangulation method to determine a depth map based on the dots in the captured structured light image. The reflectivity map can be determined based on the depth map and based on performing additional analysis of the dots in the captured structured light image.

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#### Problem addressed by the invention

40 It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may identify at least one material property of an object, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

5 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

10 As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B,  
15 no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

20 Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

25 Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict  
30 the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction  
35 regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

40 In a first aspect of the present invention a detector for identifying at least one material property m is disclosed.

As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property

characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term “biological tissue” generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term “detection of biological tissue” refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from



human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

A detector for identifying at least one material property  $m$  comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter, or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed.

Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having  $m$  rows and  $n$  columns, with  $m$ ,  $n$ , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e.  $n > 1$ ,  $m > 1$ . Thus, as an example,  $n$  may be 2 to 16 or higher and  $m$  may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example,  $n$  and  $m$  may be selected such that  $0.3 \leq m/n \leq 3$ , such as by choosing  $m/n = 1:1$ ,  $4:3$ ,  $16:9$  or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing  $m=2$ ,  $n=2$  or  $m=3$ ,  $n=3$  or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g.  $\pm 20^\circ$  or less, preferably a tolerance of  $\pm 10^\circ$  or less, more preferably a tolerance of  $\pm 5^\circ$  or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g.  $\pm 20^\circ$  or less, preferably a tolerance of  $\pm 10^\circ$  or less, more preferably a tolerance of  $\pm 5^\circ$  or less. Thus, as an example, tolerances of less than  $20^\circ$ , specifically less than  $10^\circ$  or even less than  $5^\circ$ , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by

the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein,

5 the term "the optical sensors each having at least one light sensitive area" refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term "optical sensor" furthermore refers to a light-sensitive device configured for generating at least one output signal. In case the detector comprises a

10 plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area

15 optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

20 The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical

25 pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of

30 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™

35 GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the

40 optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one photoconductive sensor such as a PbS or

PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

5 The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or 15 may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or 20 partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

25 Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

30 The light-sensitive areas specifically may be oriented towards the object. As used herein, the term "is oriented towards the object" generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from  $0^\circ$ , such as an angle in the range of  $20^\circ$  to  $90^\circ$ , preferably  $80^\circ$  to  $90^\circ$  such as  $90^\circ$ . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the 35 optical axis. As used herein, the term "essentially perpendicular" refers to the condition of a perpendicular orientation, with a tolerance of e.g.  $\pm 20^\circ$  or less, preferably a tolerance of  $\pm 10^\circ$  or less, more preferably a tolerance of  $\pm 5^\circ$  or less. Similarly, the term "essentially parallel" refers to the condition of a parallel orientation, with a 40

tolerance of e.g.  $\pm 20^\circ$  or less, preferably a tolerance of  $\pm 10^\circ$  or less, more preferably a tolerance of  $\pm 5^\circ$  or less.

5 The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. As used herein, the term "reflection image" refers to an image determined by the sensor element comprising at least one reflection feature. As furthermore used herein, without limitation, the term "reflection image" specifically may relate to data recorded by using the sensor element, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element.

10 The reflection image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to "pixels", reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. As used herein, the term "reflection feature" refers to a feature in an image plane generated by

15 the object in response to illumination, for example with at least one illumination feature. As used herein, the term "determining at least one reflection feature" refers to imaging and/or recording at least one light beam generated by the object in response to illumination with a light beam, in particular with the at least one illumination feature. In particular, the sensor element may be configured for determining and/or imaging and/or

20 recording at least reflection image. The reflection image may comprise at least one reflection pattern comprising the at least one reflection feature.

The detector comprises at least one illumination source. The detector may comprise at least one illumination source configured for illuminating the object with at least one

25 illumination beam. The illumination source may be configured for projecting at least one illumination pattern comprising the at least one illumination feature on at least one surface of the object. As used herein, the term "at least one illumination source" refers to at least one arbitrary device configured for providing at least one illumination light beam, specifically the at least one illumination pattern, for illumination of the object. The

30 illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam.

35 The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least

40 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0  $\mu\text{m}$ . Specifically, the light in the part of the near infrared region where silicon photodiodes

are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source may be adapted to emit light at a single wavelength. In other

5       embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

10

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers,

15       hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs

20       and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at

25       least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

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The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of

35       illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams

40       may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or

- arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.
- 10 The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the illumination light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for
- 15 deflecting the illuminating light beam onto the optical axis. As an example, the illumination light beam, such as the laser light beam, and the optical axis may include an angle of less than  $10^\circ$ , preferably less than  $5^\circ$  or even less than  $2^\circ$ . Other embodiments, however, are feasible. Further, the illumination light beam may be on the optical axis or off the optical axis. As an example, the illumination light beam may be
- 20 parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.
- Specifically, the illumination source and the optical sensors may be arranged in a
- 25 common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.
- The illumination source may be configured for generating at least one illumination
- 30 pattern for illumination of the object. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern;
- 35 at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may
- 40 comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object. As used herein, the term "illumination feature"

refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. For example, the illumination pattern may comprise at least one line pattern. For example, the illumination pattern may comprise at least one stripe pattern. For example, the illumination pattern may comprise at least one checkerboard pattern. For example, the illumination pattern may comprise at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. For example, the illumination source may be adapted to generate and/or to project a cloud of points. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. The illumination source may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, for generating and projecting the illumination pattern, the illumination source may comprise at least one laser source and at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one point pattern. As further used herein, the term “projecting at least one illumination pattern” refers to providing the at least one illumination pattern for illuminating the at least one object. The projected illumination pattern may be scarce, as only a single illumination feature, such as a single point, may be present. For increasing reliability, the illumination pattern may comprise several illumination features such as several points. If the pattern is scarce, a single image can be used for detection of biological tissue and face identification simultaneously.

For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.



As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam, an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a cream, an oil, a powder, a carpet, a juice, a suspension, a paint, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These material properties may be used to distinguish skin from other materials, specifically by analyzing the back scattering profile.

The least one light beam may propagate from the object towards the detector. The light beam may originate from the object or may originate from an illumination source, such

- as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively to the at least one illumination source, the detector may use radiation already present in the scene such as from at least one ambient light source.
- 10 The sensor element may be configured for recording a beam profile of at least one reflection feature of the reflection image. The evaluation device may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element. For example, in

case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

For example, the illumination source may be adapted to generate and/or to project a  
 5 cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by  
 10 the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction.  
 15 The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

The evaluation device may be configured for determining the material property  $m$  by  
 20 evaluating the beam profile of the reflection image. As used herein, the term “beam profile of the reflection image” refers to at least one intensity distribution of at least one of the reflection features of the reflection image, such as of a light spot on the sensor element, as a function of the pixel. The beam profile of the reflection image, also denoted reflection beam profile, may be selected from the group consisting of a  
 25 trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. As used herein, the term “evaluating the beam profile” refers to applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. As used herein, the term “image” refers to a two-dimensional  
 30 function,  $f(x,y)$ , wherein brightness and/or color values are given for any  $x,y$ -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the  
 35 beam profile. Specifically, the image filter  $\Phi$  maps an image  $f$ , or a region of interest in the image, onto a real number,  $\Phi(f(x,y)) = \varphi$ , wherein  $\varphi$  denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may  
 40 be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device is configured for determining at least one distance feature  $\varphi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image. As used

herein, the term “distance” refers to the distance of the object, specifically between the object and the detector. As used herein, the term “distance dependent” image filter refers to an image having a distance dependent output. The output of the distance dependent image filter is denoted herein “distance feature  $\phi_{1z}$ ” or “distance dependent feature  $\phi_{1z}$ ”. The distance feature may be or may comprise at least one information about the distance of the object such as at least one measure for the distance of the object, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter  $\Phi_{1other}$  may correlate to one or more of the distance dependent image filters  $\Phi_z$  by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$ . The similarity of two image filters  $\Phi_i$  and  $\Phi_j$  may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j))}{\sigma_i \sigma_j},$$

wherein  $\mu$  and  $\sigma$  are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature  $\phi_{1z}$  by evaluation of the combined signal Q. The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor

signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby

5 generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or

10 comparison result. As used herein, the term “combined signal Q” refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In particular, the combined signal may be a quotient signal. The combined signal Q may be determined by using various means. As an example, a

15 software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

20 The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the

25 **distance feature**  $\phi_{1z}$  for determining the **distance feature**  $\phi_{1z}$ . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

30 wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and  $E(x, y; z_o)$  denotes the beam profile given at the object distance  $z_o$ . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross

35 section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance  $L(z_o)$  and beam shape  $S(x, y; z_o)$ . Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition,

40 using the combined signal allows determination of the distance  $z_o$  independent from the

object size. Thus, the combined signal allows determination of the distance  $z_0$  independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature  $\phi_{1z}$  may be a function of the combined signal  $Q$ ,  $\phi_{1z} = \phi_{1z}(Q)$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $Q$ . Further, as an example, the object distance  $z_0$  may be a function of the distance dependent feature  $\phi_{1z}$ ,  $z_0 = z_0(\phi_{1z})$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $\phi_{1z}$ . Thus, the object distance  $z_0$  may be a function of the combined signal  $Q$ ,  $z_0 = z_0(Q)$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $Q$ .

The light-sensitive areas of at least two optical sensors may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal  $Q$ .

The evaluation device may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole

area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

5 The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, 10 the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as 15 determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

20 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

25 The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be 30 used as the physical basis of the method.

The evaluation device specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor 35 signals. As an example, Q may simply be determined as  $Q = s_1/s_2$  or  $Q = s_2/s_1$ , with  $s_1$  denoting the first sensor signal and  $s_2$  denoting the second sensor signal. Additionally or alternatively, Q may be determined as  $Q = a \cdot s_1 / b \cdot s_2$  or  $Q = b \cdot s_2 / a \cdot s_1$ , with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as  $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$ , with a, b, c and 40 d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as  $Q = s_1 / (s_1 + s_2)$ . Other combined or quotient signals are feasible.

Typically, the combined signal  $Q$  is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient  $Q=s_1/s_2$  is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the setup described above, both the first signal  $s_1$  and the second signal  $s_2$  decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal  $s_1$  decreases more rapidly than the second signal  $s_2$ , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the combined signal  $Q$  may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the incident light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

With respect to further details and embodiments of evaluation of the combined signal  $Q$  reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device may be configured for determining from the image the **distance feature**  $\phi_{1z}$  of the object by optimizing at least one blurring function  $f_a$ . The determined distance feature  $\phi_{1z}$  may correspond directly to the longitudinal coordinate of the object. The **distance feature**  $\phi_{1z}$  may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. As used herein, the term “blurring function  $f_a$ ”, also referred to as blur kernel or point spread function, refers to a response function of the detector to the illumination from the object. Specifically, the blurring function models the blur of a defocused object.



The at least one blurring function  $f_a$  may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

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The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image  $i_b$ . The evaluation device may be configured for reconstructing the distance feature  $\phi_{1z}$  from the blurred image  $i_b$  and the blurring function  $f_a$ . The distance feature  $\phi_{1z}$  may be determined by minimizing a difference between the blurred image  $i_b$  and the convolution of the blurring function  $f_a$  with at least one further image  $i'_b$ ,

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$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters  $\sigma$  of the blurring function.  $\sigma(z)$  is a set of distance dependent blurring parameters. The further image may be blurred or sharp. As used herein, the term “sharp” or “sharp image” refers to a blurred image having a maximum contrast. The at least one further image may be generated from the blurred image  $i_b$  by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the **distance feature**  $\phi_{1z}$ .

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The evaluation device may be configured for determining at least one combined distance information  $z$  considering the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter. The combined distance information  $z$  may be a real function depending on the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter. The combined distance information  $z$  may be a rational or irrational polynomial of the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield advantageous distance measurement results with enhanced precision.

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The evaluation device may be configured for determining at least one combined distance information using at least one recursive filter. The recursive filter may be at least one Kalman filter or at least one Extended Kalman filter (EKF). The combined distance information  $z$  may be obtained using a real function  $z=f(z_{DPR}, z_{DFD})$ , wherein  $z_{DPR}$  is the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and  $z_{DFD}$  is the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter, such as the arithmetic or geometric mean, a polynomial, preferably a polynomial up to the eighth order in  $z_{DPR}$  and  $z_{DFD}$ . The function  $f$  may be or may be based on a look-up table of prerecorded values. For example, the evaluation device may comprise at least one data storage device configured for storing the prerecorded values and/or one or more look-up tables. The function  $f$  may be based on a look-up table in combination with an

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- interpolation scheme for interpolation between the values in the look-up table. The interpolation scheme may be a linear interpolation, a splines interpolation, or the like. The  $z_{DFD}$  and  $z_{DPR}$  may be used as input variables within the recursive filter, in combination with a model or model function such as the function  $f$ , concerning the relationship between  $z$ ,  $z_{DFD}$ , and  $z_{DPR}$ . The model or model function may comprise statistical assumptions and/or statistical models concerning the distances  $z$ ,  $z_{DFD}$ , and  $z_{DPR}$  such as a distribution, such as a Gaussian distribution of a measured distance  $z$ ,  $z_{DFD}$ , and/or  $z_{DPR}$  around a real distance  $z_{real}$ .
- 10 The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters. The further parameters may comprise further information from the sensor element, for example the CMOS sensor, such as information about quality and/or noise of recorded data and/or information about overexposure and/or information about underexposure or the like. The
- 15 detector may comprise at least one further sensor configured for determining the further sensor data. The recursive filter may be configured for determining the combined distance information considering the further sensor data. The further sensor may be at least one sensor selected from the group consisting of: a temperature sensor, an illumination sensor such as a control sensor for determining illumination information, an
- 20 inertial measurement unit; a gyroscope. The model and/or the Kalman filter may comprise further input parameters such as the further sensor data, e.g. temperature and/or detector motion from a gyroscope and/or information from an inertial measurement unit, and/or information from an illumination sensor) and/or further parameters such as the quality/noise of recorded data, overexposure, underexposure, or
- 25 the like. The further sensor data may be provided by the sensor element, in particular by at least one CMOS sensor, and/or by further image analysis. The model and/or the Kalman filter may comprise the at least one material feature  $\phi_{2m}$  as input variable.
- For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image
- 30 filter. For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one
- 35 first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for selecting at least one image determined by the first sensor element or the
- 40 second sensor element as reflection image and for selecting at least one image determined by the other one of the first sensor element or the second sensor element as reference image. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of

recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature  $\phi_{1z}$  determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval  $\pm \varepsilon$ .

The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term “essentially identical” refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teach-in of the epipolar geometry of the specific camera.

The evaluation device may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. As used herein,

the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term “reference feature” refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval  $\pm \varepsilon$  or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine the displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to  $\phi_{1z} \pm \varepsilon$ .

The evaluation device may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image

- position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image
- 5 position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.
- 10 The evaluation device may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. As used herein, the term “displacement” refers to a difference between a position in the
- 15 reference image to a position in the reflection image. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one
- 20 data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference
- 25 feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least
- 30 one storage device of the evaluation device.

Additionally or alternatively, the evaluation device may be configured for performing the following steps:

- 35 - Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- 40 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

- Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as  $\varepsilon$ -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or reference feature in shorter distance and/or  $\varepsilon$ -weighted distance to the reference feature and/or reflection feature.
- 5 The evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature  $\phi_{2m}$ ” or “material dependent feature  $\phi_{2m}$ ”. The material feature may be or may comprise at least one information about the at least one material property of the object.

- The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter  $\Phi_{2other}$  may correlate to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$ .

- The material dependent image filter may be at least one arbitrary filter  $\Phi$  that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis  $H_0$  is rejected and an alternative hypothesis  $H_1$  is accepted.
- 40 The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of  $N_B$  Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

- wherein each of the  $N_B$  Gaussian radial basis functions is defined by a center  $(x_{lk}, y_{lk})$ , a prefactor,  $a_{lk}$ , and an exponential factor  $\alpha = 1/\epsilon$ . The exponential factor is identical for all Gaussian functions in all images. The center-positions,  $x_{lk}, y_{lk}$ , are identical for all images  $f_k: (x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$ . Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for  $f_k(x, y)$  in combination with the following parameter table:

| Image Index | Material classifier, Material Index | Distance z | Parameters                              |
|-------------|-------------------------------------|------------|-----------------------------------------|
| $k = 0$     | Skin, $m = 0$                       | 0.4 m      | $(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$ |
| $k = 1$     | Skin, $m = 0$                       | 0.6 m      | $(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$ |
| $k = 2$     | Fabric, $m = 1$                     | 0.6 m      | $(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$ |
| $\vdots$    |                                     | $\vdots$   |                                         |
| $k = N$     | Material $J$ , $m = J - 1$          |            | $(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$ |

- The values for  $x, y$ , are integers corresponding to pixels with  $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$ . The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for  $f_k$  in combination with a parameter set to obtain a continuous description of  $f_k$ . **The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for  $x, y$ , in  $f_k(x, y)$ .** For example, for pixel (6,9), the value  $f_k(6,9)$  may be computed.

- Subsequently, for each image  $f_k$ , the feature value  $\varphi_k$  corresponding to the filter  $\Phi$  may be calculated,  $\Phi(f_k(x, y), z_k) = \varphi_k$ , **wherein  $z_k$  is a distance value corresponding to the image  $f_k$  from the predefined data set.** This yields a dataset with corresponding generated feature values  $\varphi_k$ . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by  $H_0: \mu_1 = \mu_2 = \dots = \mu_J$ , wherein  $\mu_m$  is the expectation value of each material-group corresponding to the feature values  $\varphi_k$ . Index  $m$  denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish

between at least two material classifiers. The alternative hypothesis may be given by  $H_1$ :  $\exists m, m' : \mu_m \neq \mu_{m'}$ . As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical.

As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the  $J$  materials, i.e. in total  $J$  mean values,

10  $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$ , for  $m \in [0, 1, \dots, J-1]$ , wherein  $N_m$  gives the number of feature values for each of the  $J$  materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all  $N$  feature values  $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$ . The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left( \sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

15 The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left( \frac{d_1}{2}, \frac{d_2}{2} \right)$ , where  $d_1 = N - J, d_2 = J - 1$ ,
- $F(x) = 1 - CDF(x)$
- 20 ○  $p = F(mssb/mssw)$

Herein,  $I_x$  is the regularized incomplete Beta-Function,  $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$ , with the Euler Beta-Function  $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$  and  $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$  being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if  $p \leq 0.075$ , preferably  $p \leq 0.05$ , more preferably  $p \leq 0.025$ , and most preferably  $p \leq 0.01$ . For example, in case the pre-defined level of significance is  $\alpha = 0.075$ , the image filter may pass the hypothesis testing if the p-value is smaller than  $\alpha = 0.075$ . In this case the Null-hypothesis  $H_0$  can be rejected and the alternative hypothesis  $H_1$  can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image  $f$  may



be given by a function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$ , wherein the background of the image  $f$  may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The  
 5 luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where  $f$  is the spot image. The distance of the spot is denoted by  $z$ , where  $z$  may be  
 10 obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by  $n \in \mathbb{R}^3$  and can be obtained as the normal of the surface spanned by at least three measured points. The vector  $d_{ray} \in \mathbb{R}^3$  is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-  
 15 photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system,  $d_{ray}$ , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output  
 20 dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein  $0 < \alpha, \beta < 1$  are weights for the spot height  $h$ , and  $H$  denotes the Heavyside  
 25 function, i.e.  $H(x) = 1 : x \geq 0$ ,  $H(x) = 0 : x < 0$ . The spot height  $h$  may be determined by

$$h = \int_{B_r} f(x) dx,$$

where  $B_r$  is an inner circle of a spot with radius  $r$ .

For example, the material dependent image filter may be a squared norm gradient. This  
 30 material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein  $\mu$  is the mean value given by  $\mu = \int (f(x)) dx$ .

5

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

10

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein  $\mathcal{F}$  is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance  $z$ , as described in the formula above.

15

The distance  $z$  may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern  $N$  can be described in an empirical way by

20

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where  $f_0$  is an image of a despeckled spot.  $N(X)$  is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of  $f$ , i.e.  $f_0 \approx \mathcal{F}(f)$ , wherein  $\mathcal{F}$  is a smoothness operator like a Gaussian filter or median filter. Thus, an

25

approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein  $\text{Var}$  denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix  $M_{f,\rho}(g_1g_2) = [p_{g_1g_2}]$ , whereas  $p_{g_1g_2}$  is the occurrence rate of the grey combination  $(g_1,g_2)=[f(x_1,y_1),f(x_2,y_2)]$ , and the relation  $\rho$  defines the distance between  $(x_1,y_1)$  and  $(x_2,y_2)$ , which is  $\rho$

5  $(x,y)=(x+a,y+b)$  with a and b selected from 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

10 For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

15 For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

20 For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

25

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector  $L_5=[1,4,6,4,1]$  and  $E_5=[-1,-2,0,-2,-1]$  and the matrices  $L_5(E_5)^T$  and  $E_5(L_5)^T$ .

The image  $f_k$  is convoluted with these matrices:

30 
$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i,y+j) L_5(E_5)^T$$

and

$$f_{k,E5L5}^*(x, y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x + i, y + j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L5E5}^*(x, y)}{\max(f_{k,L5E5}^*(x, y))} dx dy,$$

$$F = \int \frac{f_{k,E5L5}^*(x, y)}{\max(f_{k,E5L5}^*(x, y))} dx dy,$$

- 5 Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area  $\Omega 1$ , may be an  
 10 area wherein the function  $f$  is larger than  $\alpha$  times the maximum of  $f$ . A second area  $\Omega 2$ , may be an area wherein the function  $f$  is smaller than  $\alpha$  times the maximum of  $f$ , but larger than a threshold value  $\varepsilon$  times the maximum of  $f$ . Preferably  $\alpha$  may be 0.5 and  $\varepsilon$  may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example,  $\Omega 1$  may comprise speckles or  
 15 unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega 1} 1}{\int_{\Omega 2} 1},$$

wherein  $\Omega 1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$  and  $\Omega 2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$ .

The material property  $m$  and/or the longitudinal coordinate  $z$  may be determined by using a predetermined relationship between  $\phi_{1z}$ ,  $\phi_{2m}$  and  $z$ ,  $m$ . The evaluation device  
 20 may be configured for determining the material property  $m$  and/or the longitudinal coordinate  $z$  by evaluating the features  $\phi_{2m}$ ,  $\phi_{1z}$ . For example, the evaluation device may be configured for using at least one predetermined relationship between the distance feature  $\phi_{1z}$  and the longitudinal coordinate of the object for determining the longitudinal coordinate  $z$  of the object. The evaluation device may be configured for using at least  
 25 one predetermined relationship between the material feature  $\phi_{2m}$  and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship,  
 30 such as a lookup list or a lookup table. For example, the material property may be determined by evaluating  $\phi_{2m}$  subsequently after determining of the longitudinal coordinate  $z$  such that the information about the longitudinal coordinate  $z$  can be considered for evaluating of  $\phi_{2m}$ . Specifically, the material property  $m$  and/or the longitudinal coordinate  $z$  may be determined by a function  $z(\phi_{1z}, \phi_{2m})$  and/or by a  
 35 function  $m(\phi_{1z}, \phi_{2m})$ . The function may be predefined and/or predetermined. For example, the function may be a linear function.

In the ideal case, an image filter would yield features that are only dependent on distance or material properties. However, image filters used in beam profile analysis may yield features that depend on distance and material properties, such as translucency. At least one of the material dependent image filter or the distance dependent image filter may be a function of the feature  $\phi_{1z}$  or  $\phi_{2m}$  of the at least one other image filter. At least one of the material dependent image filter or the distance dependent image filter may be a function of the at least one other image filter. The evaluation device may be configured for determining whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the feature  $\phi_{1z}$  or  $\phi_{1m}$  of the other image filter and/or whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the at least one other image filter. Specifically, the evaluation device may be configured for determining a correlation coefficient of the material dependent image filter and the distance dependent image filter. In case the correlation coefficient of the material dependent image filter with the distance dependent image filter is close to 1 or -1, the distance may be projected out, by projecting the material feature on the principal axis with the lowest variance. As an example, the material feature may be projected onto an axis orthogonal to the correlating main component. In other words, the material feature may be projected onto the second main component. This may be done using a principal component analysis as known to the person skilled in the art. The evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image. In particular, the evaluation device may be configured for simultaneously determining features that are strongly correlated with distance and weakly correlated with material properties and features that are weakly correlated with distance and strongly correlated with material properties. Alternatively, the evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image. The evaluation device may be configured for determining at least one of  $z$  and/or  $m$  by applying at least one further filter depending on at least one of  $\phi_{1z}$  and  $\phi_{2m}$  to the reflection image.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. The above-mentioned operations, including analysis of the beam profile, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays

(FPGAs) which are configured for performing the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

- 5 The detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

The detector may comprise at least one optical element selected from the group  
10 consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device  
15 may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element,  
20 preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a  
25 measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin  
30 refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-  
35 selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

40

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical

setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate  $l$  is a coordinate along the optical axis and wherein  $d$  is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a  $z$ -axis and in which a distance from the  $z$ -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the  $z$ -axis may be considered a longitudinal direction, and a coordinate along the  $z$ -axis may be considered a longitudinal coordinate  $l$ . Any direction perpendicular to the  $z$ -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using at least one distance dependent filter using the depth-from-photon-ratio technique and/or the depth-from-defocus technique as outlined above. The detector may be configured for determining a position of the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at

least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

5 In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may be determined from which a center signal arises. This  
10 may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal  
15 sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other  
20 embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by  
25 the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further  
30 detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate  $x$ ,  $y$  of the object. The  
35 evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The use of a matrix of optical sensors provides a plurality of advantages and benefits.  
40 Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal



position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629

- 5 A1. Additionally or alternatively, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

10 The present invention discloses in a further aspect, a detector for identifying at least one material property  $m$  comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- 15 - at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\varphi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least
- 20 one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
- 25 wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a
- 30 further material dependent image filter  $\Phi_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being
- 35 one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-
- 40

occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,  
 wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$   
 5 and the material feature  $\phi_{2m}$ .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

- 10 The present invention discloses in a further aspect, a detector for identifying at least one material property  $m$  comprising
- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
  - 15 - at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
  - 20 wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance,
  - 25 wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

40 With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

In a further aspect of the present invention, a detector system is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or

more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0  $\mu$ m. Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured for enabling at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured for varying the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one

detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

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In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

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In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term “scenery” generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term “scan” generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

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Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term “dot” refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be

- illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as
- 5 exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.
- 10 For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or
- 15 alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0  $\mu\text{m}$ . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred.
- 20 Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment,
- 25 particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or
- 30 more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or
- 35 the handleability of the scanning system by the user.

- Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at
- 40 least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning

system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the

mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3D model by connecting adjacent points with planar surfaces. The 3D model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the



edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

- 5 As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.
- 10 As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.
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The transfer device can be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. This feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

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For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector.

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Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

5 The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the  
10 beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited,  
15 for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can  
20 be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

25 The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the  
30 near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm  
35 to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross  
40 section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region

and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
  - b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
  - b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
  - b3) determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p, is smaller or equal than a pre-defined level of significance,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- 5       a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
  - b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating
    - 10       comprises:
      - b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear
       - 15       combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
      - 20       b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image,
      - b3) determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .
- 25   The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise
- 30   using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in

35   further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps

40   of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera.

Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device configured for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements configured for generating electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device configured for performing 3D photography. The camera generally may be configured for acquiring a single image, such as a single 3D image, or may be configured for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera configured for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be configured for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate

system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device configured for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

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The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

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The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

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The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP<sup>TM</sup>), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

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The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

The human-machine interface may comprise a plurality of beacon devices which are configured for being at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by

any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

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The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device configured for generating the at least one light beam to be  
10 detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements configured for reflecting a light beam generated by a separate illumination source. The at least one beacon device may permanently or  
15 temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

20 Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-  
25 machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and  
30 integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as  
35 in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an  
40 example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be configured in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source configured for generating at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector configured for reflecting light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command configured for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts

of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is configured for gathering information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be configured for providing information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is configured for tracking a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the

control element is directly or indirectly attachable to or integratable into the object to be tracked.

5 The tracking system may be configured for initiating one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be configured for initiating at least one action in accordance with at least one actual position of the object. As an  
10 example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

15 As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system  
20 might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

25 In a further aspect of the present invention, an inertial measurement unit for use in an electronic device disclosed. The inertial measurement unit is adapted to receive data determined by at least one detector according to the present invention. The inertial measurement unit further is adapted to receive data determined by at least one further  
30 sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one  
35 further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

40 The inertial measurement unit may comprise the detector according to the present invention and/or may be connected to the detector, via at least one data connection. The evaluation device and/or at least one processing device of the inertial measurement unit may be configured for determining the at least one combined distance information, in particular using the at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data

and/or further parameters such as further sensor data from the further sensor of the inertial measurement unit. With respect to definitions and embodiments of the inertial measurement unit reference is made to the description of the detector.

- 5 Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for identifying at least one material property  $m$  comprising

- 10 - at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
- 15 wherein the evaluation device is configured for determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further
- 20 distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
- 25 wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast
- 30 filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the
- 35 standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being
- 40 one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

- 5 Embodiment 2: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance.

Embodiment 3: The detector according to the preceding embodiment, wherein  $p \leq 0.075$ , preferably  $p \leq 0.05$ , more preferably  $p \leq 0.025$ , most preferably  $p \leq 0.01$ .

- 15 Embodiment 4: The detector according to any one of the preceding embodiments, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

- 20 Embodiment 5: The detector according to any one of the preceding embodiments, wherein the further distance dependent image filter  $\Phi_{1other}$  correlates to one or more of the distance dependent image filters  $\Phi_z$  by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$ .

- 25 Embodiment 6: The detector according to any one of the preceding embodiments, wherein the further material dependent image filter  $\Phi_{2other}$  correlates to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$ .

- 30 Embodiment 7: The detector according to any one of the preceding embodiments, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by using a predetermined relationship between  $\phi_{1z}$ ,  $\phi_{2m}$  and  $z$ ,  $m$ .

- 35 Embodiment 8: The detector according to any one of the preceding embodiments, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by a function  $z(\phi_{1z}, \phi_{2m})$  and/or  $m(\phi_{1z}, \phi_{2m})$ .

- 40 Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

Embodiment 10: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the feature  $\phi_{1z}$  or  $\phi_{1m}$  of the other image filter or whether at least

one of  $\Phi_1$  or  $\Phi_2$  is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

- 5 Embodiment 11: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of  $z$  and/or  $m$  by applying at least one further filter depending on at least one of  $\phi_{1z}$  and/or  $\phi_{2m}$  to the reflection image.
- 10 Embodiment 12: The detector according to any one of the preceding embodiments, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal  $Q$  from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal  $Q$  by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the
- 15 sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal  $Q$  and the **distance feature**  $\phi_{1z}$  for determining the distance feature  $\phi_{1z}$ .

- Embodiment 13: The detector according to the preceding embodiment, wherein the
- 20 evaluation device is configured for deriving the combined signal  $Q$  by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y, z_o) dx dy}{\iint_{A_2} E(x, y, z_o) dx dy}$$

- wherein  $x$  and  $y$  are transversal coordinates,  $A_1$  and  $A_2$  are different areas of the beam profile and  $E(x, y, z_o)$  denotes the beam profile given at the object distance  $z_o$ , wherein each of the sensor signals comprises at least one information of at least one area of the
- 25 beam profile of the light beam propagating from the object to the detector.

- Embodiment 14: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a
- 30 second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the
- 35 second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of
- 40 photons in the second area of the beam profile.



Embodiment 16: The detector according to any one of the preceding embodiments, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is  
 5 configured for determining the distance feature  $\phi_{1z}$  by optimizing at least one blurring function  $f_a$ , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 17: The detector according to the preceding embodiment, wherein the  
 10 reflection image is a blurred image  $i_b$ , wherein the evaluation device is configured for reconstructing the distance feature  $\phi_{1z}$  from the blurred image  $i_b$  and the blurring function  $f_a$ , wherein the distance feature  $\phi_{1z}$  is determined by minimizing a difference between the blurred image  $i_b$  and the convolution of the blurring function  $f_a$  with at least one further image  $i'_b$ ,

15 
$$\min\|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$
  
 by varying the parameters  $\sigma$  of the blurring function.

Embodiment 18: The detector according to any one of the two preceding embodiments, wherein the at least one blurring function  $f_a$  is a function or composite function  
 20 composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 19: The detector according to any one of the preceding embodiments,  
 25 wherein the sensor element comprises at least one CMOS sensor.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of  
 30 the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one  
 35 hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 21: The detector according to the preceding embodiment, wherein the  
 40 illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of

optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one  
 5 image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 23: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the  
 10 reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature  $\phi_{1z}$  and an error interval  $\pm \varepsilon$ , wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for  
 15 matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal  
 20 coordinate and the displacement.

Embodiment 24: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is  
 25 adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature  $\phi_{1z}$  and to determine an extent of the displacement region along the epipolar line corresponding to the error interval  $\pm \varepsilon$ .

30 Embodiment 25: A detector for identifying at least one material property  $m$  comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating  
 35 from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\varphi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least  
 40 one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination

thereof by  $|\rho_{\phi_{1\text{other}}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

Embodiment 26: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

Embodiment 27: The detector according to any one of embodiments 25 and 26, wherein  $p \leq 0.075$ , preferably  $p \leq 0.05$ , more preferably  $p \leq 0.025$ , most preferably  $p \leq 0.01$ .

Embodiment 28: The detector according to any one of embodiments 25 to 27, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 29: The detector according to any one of embodiments 25 to 28, wherein the further distance dependent image filter  $\Phi_{1\text{other}}$  correlates to one or more of the distance dependent image filters  $\Phi_z$  by  $|\rho_{\phi_{1\text{other}}, \Phi_z}| \geq 0.60$ , preferably by  $|\rho_{\phi_{1\text{other}}, \Phi_z}| \geq 0.80$ .

Embodiment 30: The detector according to any one of embodiments 25 to 29, wherein the further material dependent image filter  $\Phi_{2\text{other}}$  correlates to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho_{\phi_{2\text{other}}, \Phi_m}| \geq 0.60$ , preferably by  $|\rho_{\phi_{2\text{other}}, \Phi_m}| \geq 0.80$ .

Embodiment 31: The detector according to any one of embodiments 25 to 30, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2\text{other}}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation,

- the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.
- 5      Embodiment 32: The detector according to any one of embodiments 25 to 31, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by using a predetermined relationship between  $\phi_{1z}$ ,  $\phi_{2m}$  and  $z$ ,  $m$ .
- 15      Embodiment 33: The detector according to any one of embodiments 25 to 32, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by a function  $z(\phi_{1z}, \phi_{2m})$  and/or  $m(\phi_{1z}, \phi_{2m})$ .
- 20      Embodiment 34: The detector according to any one of embodiments 25 to 33, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
- 25      Embodiment 35: The detector according to any one of embodiments 25 to 34, wherein the evaluation device is configured for determining whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the feature  $\phi_{1z}$  or  $\phi_{1m}$  of the other image filter or whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
- 30      Embodiment 36: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of  $z$  and/or  $m$  by applying at least one further filter depending on at least one of  $\phi_{1z}$  and  $\phi_{2m}$  to the reflection image.
- 35      Embodiment 37: The detector according to any one of embodiments 25 to 36, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal  $Q$  from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal  $Q$  by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal  $Q$  and the distance feature  $\phi_{1z}$  for determining the distance feature  $\phi_{1z}$ .
- 40      Embodiment 38: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal  $Q$  by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y, z_o) dx dy}{\iint_{A_2} E(x, y, z_o) dx dy}$$

wherein  $x$  and  $y$  are transversal coordinates,  $A_1$  and  $A_2$  are different areas of the beam profile and  $E(x, y, z_o)$  denotes the beam profile given at the object distance  $z_o$ , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

Embodiment 39: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 40: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 41: The detector according to any one of embodiments 25 to 40, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature  $\phi_{1z}$  by optimizing at least one blurring function  $f_a$ , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 42: The detector according to the preceding embodiment, wherein the reflection image is a blurred image  $i_b$ , wherein the evaluation device is configured for reconstructing the distance feature  $\phi_{1z}$  from the blurred image  $i_b$  and the blurring function  $f_a$ , wherein the distance feature  $\phi_{1z}$  is determined by minimizing a difference between the blurred image  $i_b$  and the convolution of the blurring function  $f_a$  with at least one further image  $i'_b$ ,

by varying the parameters  $\sigma$  of the blurring function.

$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

Embodiment 43: The detector according to any one of embodiments 25 to 42, wherein the at least one blurring function  $f_a$  is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox

function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

5 Embodiment 44: The detector according to any one of embodiments 25 to 43, wherein the sensor element comprises at least one CMOS sensor.

10 Embodiment 45: The detector according to any one of embodiments 25 to 44, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one  
15 hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

20 Embodiment 46: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

25 Embodiment 47: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor  
30 element as reference image.

35 Embodiment 48: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature  $\phi_{1z}$  and an error interval  $\pm \varepsilon$ , wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the  
40 displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 49: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature  $\phi_{1z}$  and to determine an extent of the displacement region along the epipolar line corresponding to the error interval  $\pm \varepsilon$ .

Embodiment 50: A detector for identifying at least one material property  $m$  comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

Embodiment 51: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance.

Embodiment 52: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

Embodiment 53: The detector according to any one of embodiments 50 and 52, wherein  $p \leq 0.075$ , preferably  $p \leq 0.05$ , more preferably  $p \leq 0.025$ , most preferably  $p \leq 0.01$ .

5 Embodiment 54: The detector according to any one of embodiments 50 to 53, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

10 Embodiment 55: The detector according to any one of embodiments 50 to 54, wherein the further distance dependent image filter  $\Phi_{1\text{other}}$  correlates to one or more of the distance dependent image filters  $\Phi_z$  by  $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$ .

15 Embodiment 56: The detector according to any one of embodiments 50 to 55, wherein the further material dependent image filter  $\Phi_{2\text{other}}$  correlates to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$ .

20 Embodiment 57: The detector according to any one of embodiments 50 to 56, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof, or a  
25 further material dependent image filter  $\Phi_{2\text{other}}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof  
30 by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

35 Embodiment 58: The detector according to any one of embodiments 50 to 57, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by using a predetermined relationship between  $\phi_{1z}$ ,  $\phi_{2m}$  and  $z$ ,  $m$ .

40 Embodiment 59: The detector according to any one of embodiments 50 to 58, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by a function  $z(\phi_{1z}, \phi_{2m})$  and/or  $m(\phi_{1z}, \phi_{2m})$ .



Embodiment 60: The detector according to any one of embodiments 50 to 59, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

- 5 Embodiment 61: The detector according to any one of embodiments 50 to 60, wherein the evaluation device is configured for determining whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the feature  $\phi_{1z}$  or  $\phi_{1m}$  of the other image filter or whether at least one of  $\Phi_1$  or  $\Phi_2$  is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent  
10 image filter sequentially or recursively to the reflection image.

- Embodiment 62: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of  $z$  and/or  $m$  by applying at least one further filter depending on at least one of  $\phi_{1z}$  and  $\phi_{2m}$  to the reflection image.

- 15 Embodiment 63: The detector according to any one of embodiments 50 to 62, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal  $Q$  from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal  $Q$  by one or more of dividing the sensor  
20 signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal  $Q$  and the distance feature  $\phi_{1z}$  for determining the distance feature  $\phi_{1z}$ .

- 25 Embodiment 64: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal  $Q$  by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y, z_o) dx dy}{\iint_{A_2} E(x, y, z_o) dx dy}$$

- wherein  $x$  and  $y$  are transversal coordinates,  $A_1$  and  $A_2$  are different areas of the beam profile and  $E(x, y, z_o)$  denotes the beam profile given at the object distance  $z_o$ , wherein  
30 each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

- Embodiment 65: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a  
35 first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- Embodiment 66: The detector according to the preceding embodiment, wherein the  
40 evaluation device is configured for determining the first area of the beam profile and the

second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 67: The detector according to any one of embodiments 50 to 66, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature  $\phi_{1z}$  by optimizing at least one blurring function  $f_a$ , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 68: The detector according to the preceding embodiment, wherein the reflection image is a blurred image  $i_b$ , wherein the evaluation device is configured for reconstructing the distance feature  $\phi_{1z}$  from the blurred image  $i_b$  and the blurring function  $f_a$ , wherein the distance feature  $\phi_{1z}$  is determined by minimizing a difference between the blurred image  $i_b$  and the convolution of the blurring function  $f_a$  with at least one further image  $i'_b$ ,

$$\min\|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters  $\sigma$  of the blurring function.

Embodiment 69: The detector according to any one of embodiments 50 to 68, wherein the at least one blurring function  $f_a$  is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 70: The detector according to any one of embodiments 50 to 69, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 71: The detector according to any one of embodiments 50 to 70, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 72: The detector according to the embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

5 Embodiment 73: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one  
10 image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 74: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the  
15 reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature  $\phi_{1z}$  and an error interval  $\pm \varepsilon$ , wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for  
20 matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal  
25 coordinate and the displacement.

Embodiment 75: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is  
30 adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature  $\phi_{1z}$  and to determine an extent of the displacement region along the epipolar line corresponding to the error interval  $\pm \varepsilon$ .

35 Embodiment 76: A detector system, the detector system comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the  
40 object, holdable by the object and integratable into the object.

Embodiment 77: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein

the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 78: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 79: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 80: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of embodiments 1 to 24, 25 to 49, or 50 to 75, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 81: A camera for imaging at least one object, the camera comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75.

Embodiment 82: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

Embodiment 83: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 1 to 24, the method comprising the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

5 b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

10 b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

15 b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

35 b3) determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

40 Embodiment 84: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 25 to 49, the method comprising the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

- 5        b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
- 10        b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p, is smaller or equal than a pre-defined level of significance,
- 15        b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .
- 20

Embodiment 85: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 50 to 75, the method comprising the following steps:

25

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- 30        b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
- 35        b1) determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
- 40        b2) determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image,

b3) determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $\phi_{1z}$  and the material feature  $\phi_{2m}$ .

Embodiment 86: A use of the detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

#### Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a detector according to the present invention;

Figure 2 shows the statistics of the resulting material features of the spot shape material dependent image filter applied to the data set; and

Figure 3 shows a beam profile image.

#### Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for identifying at least one material property  $m$  of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible,

such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

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The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e.  $n > 1$ ,  $m > 1$ . Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that  $0.3 \leq m/n \leq 3$ , such as by choosing  $m/n = 1:1$ ,  $4:3$ ,  $16:9$  or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing  $m=2$ ,  $n=2$  or  $m=3$ ,  $n=3$  or the like.

40



The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

The detector 110 may further comprise an illumination source 124. As an example, the illumination source 124 may be configured for generating an illumination light beam 126 for illuminating the object 112. The detector 110 may be configured such that the illumination light beam 126 propagates from the detector 110 towards the object 112 along an optical axis 128 of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 128.

The illumination source 124 may comprise at least one light source. The illumination source 124 may comprise a plurality of light sources. The illumination source 124 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 124 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0  $\mu\text{m}$ . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source 124 may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers,

semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 124 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points, for example the illumination source 124 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 124 may be integrated into a housing of the detector 110.

The illumination light beam 126 may propagate from the object 112 towards the detector 110. The detector 110 may be used in active and/or passive illumination scenarios. For example, the at least one illumination source 124 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. Additionally or alternatively to the at least one illumination source 124, the detector 110 may use radiation already present in the scene such as from at least one ambient light source.

The illumination source 124 may illuminate the at least one object 112 with at least one illumination pattern. The illumination pattern may comprise a plurality of points as image features. These points are illustrated as light beam 126 emerging from the illumination source 124.

The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

Each optical sensor 120 may be designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam 130 propagating from the object 112 to the detector 110. The sensor element 116 may be configured for determining at least one reflection image of at least one reflection feature generated by the reflection light beam 130 on the optical sensors 120. The reflection image may be data recorded by using the sensor element 116, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element 116. The reflection image itself may comprise pixels, the pixels of the image

correlating to pixels of the matrix 118 of the sensor element 116. The matrix 118 may comprise the reflection image 126. For example, in case of illumination with a point pattern, the reflection image may comprise points as reflection features. These points may result from reflection light beams 130 originating from the at least one object 112.

5 The sensor element 116 may be configured for recording a beam profile of at least one reflection feature of the reflection image. The detector 110 comprises at least one evaluation device 132. The evaluation device 132 may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at  
10 least one light spot, provided by the sensor element 116. The evaluation device 132 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least  
15 one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness  
20 channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point  
25 operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a  
30 differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter;  
35 applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by  
40 recognizing an object within an image generated by the sensor element 116. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

The evaluation device 132 may be configured for determining the material property  $m$  by evaluating the beam profile of the reflection image. The beam profile of the reflection image may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

5 The evaluation device 132 may be configured for applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. Specifically, the image filter  $\Phi$  maps an image  $f$ , or a region of interest in the image, onto a real number,  $\Phi(f(x,y)) = \varphi$ , wherein  $\varphi$  denotes a feature, in particular a distance feature in case of  
 10 distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

15 The evaluation device 132 is configured for determining at least one distance feature  $\varphi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image. The distance feature may be or may comprise at least one information about the distance of the object 112 such as at least one measure for the distance of the object 112, a  
 20 distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear  
 25 combination thereof by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter  $\Phi_{1other}$  may correlate to one or more of the distance dependent image filters  $\Phi_z$  by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$ . The similarity of two image filters  $\Phi_i$  and  $\Phi_j$  may be assessed through the correlation of their  
 30 features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov((\Phi_i(f(x,y)) - \mu_i)(\Phi_j(f(x,y)) - \mu_j))}{\sigma_i \sigma_j},$$

wherein  $\mu$  and  $\sigma$  are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images  
 35 may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen  
 40 as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded

in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature  $\phi_{1z}$  by evaluation of the combined signal Q. The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device 132, as an example, may comprise at least one divider 134, wherein the divider 134 is configured for deriving the quotient signal. The divider 134 may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device 132 may be configured for using at least one predetermined relationship between the combined signal Q and the **distance feature**  $\phi_{1z}$  for determining the **distance feature**  $\phi_{1z}$ . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and  $E(x, y, z_o)$  denotes the beam profile given at the object distance  $z_o$ . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance  $L(z_o)$  and beam shape  $S(x, y; z_o)$ . Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance  $z_o$  independent from the object size. Thus, the combined signal allows determination of the distance  $z_o$  independent from the material properties and/or reflective properties and/or scattering properties of the object 112 and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an

example, the distance dependent feature  $\phi_{1z}$  may be a function of the combined signal  $Q$ ,  $\phi_{1z} = \phi_{1z}(Q)$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $Q$ . Further, as an example, the object distance  $z_0$  may be a function of the distance dependent feature  $\phi_{1z}$ ,  $z_0 = z_0(\phi_{1z})$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $\phi_{1z}$ . Thus, the object distance  $z_0$  may be a function of the combined signal  $Q$ ,  $z_0 = z_0(Q)$ , whereas the function may be a linear, quadratic, or higher order polynomial in  $Q$ .

The light-sensitive areas 122 of at least two optical sensors 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

The evaluation device 132 may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 132 may be adapted for determining an area integral of the beam profile. The evaluation device 132 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 132 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device

may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device 132 specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as  $Q = s_1/s_2$  or  $Q = s_2/s_1$ , with  $s_1$  denoting the first sensor signal and  $s_2$  denoting the second sensor signal. Additionally or alternatively, Q may be determined as  $Q = a \cdot s_1 / b \cdot s_2$  or  $Q = b \cdot s_2 / a \cdot s_1$ , with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as  $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$ , with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as  $Q = s_1 / (s_1 + s_2)$ . Other combined or quotient signals are feasible.

With respect to further details and embodiments of evaluation of the combined signal Q and determination of the longitudinal coordinate z reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device 132 may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device 132 may be configured for determining from the image the **distance feature**  $\phi_{1z}$  of the object by optimizing at least one blurring function  $f_a$ . The determined distance feature  $\phi_{1z}$  may correspond directly to the longitudinal coordinate of the object. The **distance feature**  $\phi_{1z}$  may be determined by using at least one convolution-based algorithm such as a depth-

from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function  $f_a$  may be a function or composite function composed from

5 at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one

10 blurring function. The reflection image may be a blurred image  $i_b$ . The evaluation device may be configured for reconstructing the distance feature  $\phi_{1z}$  from the blurred image  $i_b$  and the blurring function  $f_a$ . The distance feature  $\phi_{1z}$  may be determined by minimizing a difference between the blurred image  $i_b$  and the convolution of the blurring function  $f_a$  with at least one further image  $i'_b$ ,

$$15 \quad \min \| (i'_b * f_a(\sigma(\phi_{1z})) - i_b) \|,$$

by varying the parameters  $\sigma$  of the blurring function.  $\sigma(z)$  is a set of distance dependent blurring parameters. The further image may be blurred or sharp. The at least one further image may be generated from the blurred image  $i_b$  by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the

20 **distance feature**  $\phi_{1z}$ .

The evaluation device 132 may be configured for determining at least one combined distance information  $z$  considering the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and the **distance feature**  $\phi_{1z}$  determined by applying the

25 depth-from-defocus filter. The combined distance information  $z$  may be a real function depending on the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter. The combined distance information  $z$  may be a rational or irrational polynomial of the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-photon-ratio filter and

30 the **distance feature**  $\phi_{1z}$  determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield to advantageous distance measurement results with enhanced precision.

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For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. The detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element and at least one

40 second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in



particular at least one second reflection feature. The evaluation device 132 may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 132 may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature  $\phi_{1z}$  determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval  $\pm \varepsilon$ .

The evaluation device 132 may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 132 may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device 132 may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The evaluation device 132 may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device 132 may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teach-in of the epipolar geometry of the specific camera.

The evaluation device 132 may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the

reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The

5 displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device 132 may be adapted to

10 determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval  $\pm \varepsilon$  or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different

15 directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device 132 may determine the displacement region around the image position

20 of the reflection feature. The evaluation device 132 may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to  $\phi_{1z} \pm \varepsilon$ .

The evaluation device 132 may be configured for matching the selected feature of the

25 reflection pattern with at least one feature of the reference pattern within the displacement region. The evaluation device 132 may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling

30 algorithm. The evaluation device 132 may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device 132 may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation

35 device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device 132 may determine a displacement region around the image position of each reflection feature. The evaluation device 132 may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement

40 region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device 132 may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to

the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

5 The evaluation device 132 may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device 132 may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-  
10 empiric relationship and an analytically derived relationship. The evaluation device 132 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device 132 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the  
15 matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 132 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature  
20 the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 132.

Additionally or alternatively, the evaluation device 132 may be configured for performing the following steps:

- 25 - Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a  
30 direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within  
35 the assigned displacement region along the assigned epipolar line.

The evaluation device 132 is configured for determining at least one material feature  $\phi_{2m}$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image. The material feature may be or may comprise at least one information about the at least  
40 one material property of the object 112.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-

occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter  $\Phi_{2\text{other}}$  which correlates to one or more of the

5 luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$  with  $\Phi_m$  being one of the luminance filter, the spot shape filter, the

10 squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter  $\Phi_{2\text{other}}$  may correlate to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$ , preferably by  $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$ .

15 With respect to the description of exemplary material dependent image filter, reference is made to the description of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter such as a Gaussian filter or median filter, the grey-level-occurrence-based contrast filter, the grey-level-occurrence-

20 based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, and the threshold area filter given above.

The material dependent image filter may be at least one arbitrary filter  $\Phi$  that passes a

25 hypothesis testing. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. The beam profile image may be given by a sum of  $N_B$  Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the  $N_B$  Gaussian radial basis functions is defined by a center  $(x_{lk}, y_{lk})$ , a prefactor,  $a_{lk}$ , and an exponential factor  $\alpha = 1/\epsilon$ . The exponential factor is identical for

35 all Gaussian functions in all beam profile images. The center-positions,  $x_{lk}$ ,  $y_{lk}$ , are identical for all images  $f_k: (x_0, x_1, \dots, x_{N_B}), (y_0, y_1, \dots, y_{N_B})$ . The beam profile images may be generated by using the above formula for  $f_k(x, y)$  in combination with the following parameter table:

| Image Index | Material classifier, Material Index | Distance z | Parameters                            |
|-------------|-------------------------------------|------------|---------------------------------------|
| $k = 0$     | Skin, $m = 0$                       | 0.4 m      | $(a_{00}, a_{10}, \dots, a_{N_B-10})$ |

|          |                            |          |                                         |
|----------|----------------------------|----------|-----------------------------------------|
| $k = 1$  | Skin, $m = 0$              | 0.6 m    | $(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$ |
| $k = 2$  | Fabric, $m = 1$            | 0.6 m    | $(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$ |
| $\vdots$ |                            | $\vdots$ |                                         |
| $k = N$  | Material $J$ , $m = J - 1$ |          | $(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$ |

The values for  $x, y$ , are integers corresponding to pixels with  $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$ . The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for  $f_k$  in combination with a parameter set to

- 5 obtain a continuous description of  $f_k$ . **The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for  $x, y$ , in  $f_k(x, y)$ .** For example, for pixel (6,9), the value  $f_k(6,9)$  may be computed. Figure 3 shows an embodiment of a beam profile image defined by an exemplary data set.

- 10 An exemplary data set is presented in the following. The following parameter tables list the center positions,  $x_{lk} = x_l, y_{lk} = y_l$ , of all Gaussian functions  $g_{lk}$  for all images  $f_k$ :

|       |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|
| $l$   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| $x_l$ | 14 | 11 | 10 | 19 | 15 | 20 | 12 | 17 |
| $y_l$ | 17 | 11 | 17 | 19 | 20 | 20 | 12 | 20 |
| $l$   | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| $x_l$ | 13 | 18 | 10 | 14 | 19 | 15 | 16 | 12 |
| $y_l$ | 17 | 19 | 8  | 18 | 14 | 11 | 22 | 17 |
| $l$   | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| $x_l$ | 25 | 13 | 18 | 8  | 9  | 10 | 14 | 19 |
| $y_l$ | 15 | 20 | 20 | 15 | 14 | 13 | 24 | 24 |
| $l$   | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| $x_l$ | 10 | 9  | 14 | 12 | 19 | 20 | 17 | 15 |
| $y_l$ | 24 | 16 | 15 | 11 | 13 | 19 | 13 | 14 |
| $l$   | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 |
| $x_l$ | 16 | 13 | 12 | 20 | 18 | 17 | 21 | 13 |
| $y_l$ | 19 | 10 | 22 | 14 | 10 | 18 | 15 | 23 |
| $l$   | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| $x_l$ | 8  | 10 | 8  | 14 | 19 | 11 | 15 | 14 |
| $y_l$ | 12 | 14 | 18 | 21 | 23 | 15 | 16 | 8  |
| $l$   | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 |
| $x_l$ | 16 | 20 | 11 | 17 | 16 | 15 | 21 | 13 |
| $y_l$ | 13 | 16 | 16 | 8  | 16 | 13 | 21 | 13 |
| $l$   | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 |
| $x_l$ | 22 | 18 | 23 | 14 | 19 | 11 | 15 | 16 |

$y_l$  | 23 15 12 22 18 10 23 10

The following parameter tables list the material classifiers, in particular for a white Teflon target (denoted control), a fabric, a dark skin (denoted as dark\_skin), a pale skin (denoted pale\_skin), and a highly translucent skin (denoted trans), and the distance  $z$  for each image  $f_k$ , referenced by the image index  $k$ :

| $k$ | material classifier | Distance $z$ | $k$ | material classifier | Distance $z$ | $k$ | material classifier | Distance $z$ |
|-----|---------------------|--------------|-----|---------------------|--------------|-----|---------------------|--------------|
| 0   | control             | 0.4          | 40  | dark_skin           | 0.4          | 80  | control             | 0.6          |
| 1   | control             | 0.4          | 41  | dark_skin           | 0.4          | 81  | control             | 0.6          |
| 2   | control             | 0.4          | 42  | dark_skin           | 0.4          | 82  | control             | 0.6          |
| 3   | control             | 0.4          | 43  | dark_skin           | 0.4          | 83  | control             | 0.6          |
| 4   | control             | 0.4          | 44  | dark_skin           | 0.4          | 84  | control             | 0.6          |
| 5   | control             | 0.4          | 45  | dark_skin           | 0.4          | 85  | control             | 0.6          |
| 6   | control             | 0.4          | 46  | dark_skin           | 0.4          | 86  | control             | 0.6          |
| 7   | control             | 0.4          | 47  | dark_skin           | 0.4          | 87  | control             | 0.6          |
| 8   | control             | 0.4          | 48  | pale_skin           | 0.4          | 88  | control             | 0.6          |
| 9   | control             | 0.4          | 49  | pale_skin           | 0.4          | 89  | control             | 0.6          |
| 10  | control             | 0.4          | 50  | pale_skin           | 0.4          | 90  | control             | 0.6          |
| 11  | control             | 0.4          | 51  | pale_skin           | 0.4          | 91  | control             | 0.6          |
| 12  | control             | 0.4          | 52  | pale_skin           | 0.4          | 92  | control             | 0.6          |
| 13  | control             | 0.4          | 53  | pale_skin           | 0.4          | 93  | control             | 0.6          |
| 14  | control             | 0.4          | 54  | pale_skin           | 0.4          | 94  | control             | 0.6          |
| 15  | control             | 0.4          | 55  | pale_skin           | 0.4          | 95  | control             | 0.6          |
| 16  | fabric              | 0.4          | 56  | pale_skin           | 0.4          | 96  | fabric              | 0.6          |
| 17  | fabric              | 0.4          | 57  | pale_skin           | 0.4          | 97  | fabric              | 0.6          |
| 18  | fabric              | 0.4          | 58  | pale_skin           | 0.4          | 98  | fabric              | 0.6          |
| 19  | fabric              | 0.4          | 59  | pale_skin           | 0.4          | 99  | fabric              | 0.6          |
| 20  | fabric              | 0.4          | 60  | pale_skin           | 0.4          | 100 | fabric              | 0.6          |
| 21  | fabric              | 0.4          | 61  | pale_skin           | 0.4          | 101 | fabric              | 0.6          |
| 22  | fabric              | 0.4          | 62  | pale_skin           | 0.4          | 102 | fabric              | 0.6          |
| 23  | fabric              | 0.4          | 63  | pale_skin           | 0.4          | 103 | fabric              | 0.6          |
| 24  | fabric              | 0.4          | 64  | trans               | 0.4          | 104 | fabric              | 0.6          |
| 25  | fabric              | 0.4          | 65  | trans               | 0.4          | 105 | fabric              | 0.6          |
| 26  | fabric              | 0.4          | 66  | trans               | 0.4          | 106 | fabric              | 0.6          |
| 27  | fabric              | 0.4          | 67  | trans               | 0.4          | 107 | fabric              | 0.6          |
| 28  | fabric              | 0.4          | 68  | trans               | 0.4          | 108 | fabric              | 0.6          |
| 29  | fabric              | 0.4          | 69  | trans               | 0.4          | 109 | fabric              | 0.6          |
| 30  | fabric              | 0.4          | 70  | trans               | 0.4          | 110 | fabric              | 0.6          |
| 31  | fabric              | 0.4          | 71  | trans               | 0.4          | 111 | fabric              | 0.6          |
| 32  | dark_skin           | 0.4          | 72  | trans               | 0.4          | 112 | dark_skin           | 0.6          |
| 33  | dark_skin           | 0.4          | 73  | trans               | 0.4          | 113 | dark_skin           | 0.6          |
| 34  | dark_skin           | 0.4          | 74  | trans               | 0.4          | 114 | dark_skin           | 0.6          |
| 35  | dark_skin           | 0.4          | 75  | trans               | 0.4          | 115 | dark_skin           | 0.6          |
| 36  | dark_skin           | 0.4          | 76  | trans               | 0.4          | 116 | dark_skin           | 0.6          |
| 37  | dark_skin           | 0.4          | 77  | trans               | 0.4          | 117 | dark_skin           | 0.6          |
| 38  | dark_skin           | 0.4          | 78  | trans               | 0.4          | 118 | dark_skin           | 0.6          |
| 39  | dark_skin           | 0.4          | 79  | trans               | 0.4          | 119 | dark_skin           | 0.6          |

| $k$ | material classifier | Distance $z$ | $k$ | material classifier | Distance $z$ | $k$ | material classifier | Distance $z$ |
|-----|---------------------|--------------|-----|---------------------|--------------|-----|---------------------|--------------|
| 120 | dark_skin           | 0.6          | 160 | control             | 0.8          | 200 | dark_skin           | 0.8          |
| 121 | dark_skin           | 0.6          | 161 | control             | 0.8          | 201 | dark_skin           | 0.8          |
| 122 | dark_skin           | 0.6          | 162 | control             | 0.8          | 202 | dark_skin           | 0.8          |
| 123 | dark_skin           | 0.6          | 163 | control             | 0.8          | 203 | dark_skin           | 0.8          |
| 124 | dark_skin           | 0.6          | 164 | control             | 0.8          | 204 | dark_skin           | 0.8          |
| 125 | dark_skin           | 0.6          | 165 | control             | 0.8          | 205 | dark_skin           | 0.8          |
| 126 | dark_skin           | 0.6          | 166 | control             | 0.8          | 206 | dark_skin           | 0.8          |
| 127 | dark_skin           | 0.6          | 167 | control             | 0.8          | 207 | dark_skin           | 0.8          |
| 128 | pale_skin           | 0.6          | 168 | control             | 0.8          | 208 | pale_skin           | 0.8          |
| 129 | pale_skin           | 0.6          | 169 | control             | 0.8          | 209 | pale_skin           | 0.8          |
| 130 | pale_skin           | 0.6          | 170 | control             | 0.8          | 210 | pale_skin           | 0.8          |
| 131 | pale_skin           | 0.6          | 171 | control             | 0.8          | 211 | pale_skin           | 0.8          |
| 132 | pale_skin           | 0.6          | 172 | control             | 0.8          | 212 | pale_skin           | 0.8          |
| 133 | pale_skin           | 0.6          | 173 | control             | 0.8          | 213 | pale_skin           | 0.8          |
| 134 | pale_skin           | 0.6          | 174 | control             | 0.8          | 214 | pale_skin           | 0.8          |
| 135 | pale_skin           | 0.6          | 175 | control             | 0.8          | 215 | pale_skin           | 0.8          |
| 136 | pale_skin           | 0.6          | 176 | fabric              | 0.8          | 216 | pale_skin           | 0.8          |
| 137 | pale_skin           | 0.6          | 177 | fabric              | 0.8          | 217 | pale_skin           | 0.8          |
| 138 | pale_skin           | 0.6          | 178 | fabric              | 0.8          | 218 | pale_skin           | 0.8          |
| 139 | pale_skin           | 0.6          | 179 | fabric              | 0.8          | 219 | pale_skin           | 0.8          |
| 140 | pale_skin           | 0.6          | 180 | fabric              | 0.8          | 220 | pale_skin           | 0.8          |
| 141 | pale_skin           | 0.6          | 181 | fabric              | 0.8          | 221 | pale_skin           | 0.8          |
| 142 | pale_skin           | 0.6          | 182 | fabric              | 0.8          | 222 | pale_skin           | 0.8          |
| 143 | pale_skin           | 0.6          | 183 | fabric              | 0.8          | 223 | pale_skin           | 0.8          |
| 144 | trans               | 0.6          | 184 | fabric              | 0.8          | 224 | trans               | 0.8          |
| 145 | trans               | 0.6          | 185 | fabric              | 0.8          | 225 | trans               | 0.8          |
| 146 | trans               | 0.6          | 186 | fabric              | 0.8          | 226 | trans               | 0.8          |
| 147 | trans               | 0.6          | 187 | fabric              | 0.8          | 227 | trans               | 0.8          |
| 148 | trans               | 0.6          | 188 | fabric              | 0.8          | 228 | trans               | 0.8          |
| 149 | trans               | 0.6          | 189 | fabric              | 0.8          | 229 | trans               | 0.8          |
| 150 | trans               | 0.6          | 190 | fabric              | 0.8          | 230 | trans               | 0.8          |
| 151 | trans               | 0.6          | 191 | fabric              | 0.8          | 231 | trans               | 0.8          |
| 152 | trans               | 0.6          | 192 | dark_skin           | 0.8          | 232 | trans               | 0.8          |
| 153 | trans               | 0.6          | 193 | dark_skin           | 0.8          | 233 | trans               | 0.8          |
| 154 | trans               | 0.6          | 194 | dark_skin           | 0.8          | 234 | trans               | 0.8          |
| 155 | trans               | 0.6          | 195 | dark_skin           | 0.8          | 235 | trans               | 0.8          |
| 156 | trans               | 0.6          | 196 | dark_skin           | 0.8          | 236 | trans               | 0.8          |
| 157 | trans               | 0.6          | 197 | dark_skin           | 0.8          | 237 | trans               | 0.8          |
| 158 | trans               | 0.6          | 198 | dark_skin           | 0.8          | 238 | trans               | 0.8          |
| 159 | trans               | 0.6          | 199 | dark_skin           | 0.8          | 239 | trans               | 0.8          |

The following parameter tables list the prefactors  $a_{lk}$  for the Gaussian functions  $g_{lk}$  in each image  $f_k$  referenced by the image index  $k$  and the Gaussian index  $l$ :

| $k$ | $a_{0k}$ | $a_{1k}$ | $a_{2k}$ | $a_{3k}$ | $a_{4k}$ | $a_{5k}$ | $a_{6k}$ | $a_{7k}$ |
|-----|----------|----------|----------|----------|----------|----------|----------|----------|
| 0   | 291.94   | -7.20    | -14.72   | -22.75   | 32.57    | 16.92    | -45.50   | -23.31   |
| 1   | 300.91   | -7.58    | -15.22   | -14.41   | 32.18    | 15.96    | -44.41   | -23.92   |
| 2   | 297.04   | -6.42    | -13.94   | -10.30   | 31.19    | 16.44    | -43.68   | -22.27   |
| 3   | 289.98   | -7.67    | -14.20   | -13.60   | 31.68    | 17.16    | -44.04   | -21.41   |
| 4   | 295.94   | -7.72    | -12.93   | -16.81   | 32.77    | 15.92    | -45.15   | -22.30   |
| 5   | 295.37   | -6.87    | -13.30   | -15.51   | 33.10    | 16.70    | -45.18   | -22.29   |
| 6   | 298.03   | -6.79    | -14.89   | -12.68   | 32.43    | 14.46    | -44.47   | -22.71   |

|    |        |       |        |        |       |       |        |        |
|----|--------|-------|--------|--------|-------|-------|--------|--------|
| 7  | 296.95 | -7.09 | -14.98 | -13.54 | 31.56 | 15.69 | -45.09 | -20.03 |
| 8  | 306.69 | -6.91 | -14.94 | -14.27 | 33.28 | 16.02 | -43.93 | -20.21 |
| 9  | 291.40 | -7.73 | -13.91 | -14.95 | 30.51 | 16.18 | -45.86 | -21.61 |
| 10 | 301.21 | -7.21 | -14.05 | -17.46 | 32.02 | 17.26 | -45.02 | -22.89 |
| 11 | 309.78 | -6.93 | -13.47 | -19.23 | 33.01 | 14.94 | -43.61 | -20.86 |
| 12 | 298.62 | -7.09 | -14.75 | -18.43 | 32.59 | 18.97 | -44.11 | -22.73 |
| 13 | 296.09 | -7.18 | -14.25 | -15.83 | 31.94 | 14.62 | -43.72 | -19.77 |
| 14 | 301.13 | -7.24 | -14.44 | -19.11 | 33.54 | 17.01 | -44.08 | -21.39 |
| 15 | 304.28 | -7.08 | -14.10 | -17.26 | 32.54 | 16.81 | -44.40 | -21.14 |
| 16 | 246.71 | -3.77 | -13.62 | 1.43   | 19.38 | -2.03 | -52.50 | -19.62 |
| 17 | 258.85 | -3.11 | -12.88 | -3.91  | 19.02 | -1.05 | -49.64 | -21.50 |
| 18 | 263.06 | -2.91 | -13.72 | 1.65   | 19.03 | -1.26 | -51.02 | -21.58 |
| 19 | 251.12 | -4.15 | -14.47 | -6.19  | 19.00 | -1.26 | -50.65 | -22.40 |
| 20 | 250.05 | -2.91 | -13.84 | 0.30   | 19.48 | -1.79 | -49.89 | -20.92 |
| 21 | 249.35 | -2.80 | -13.64 | -0.25  | 18.90 | 0.42  | -51.29 | -19.21 |
| 22 | 260.34 | -2.57 | -14.30 | 1.83   | 17.72 | -1.18 | -51.27 | -24.07 |
| 23 | 246.64 | -3.68 | -13.92 | -2.29  | 19.79 | -0.20 | -50.33 | -23.09 |
| 24 | 245.58 | -2.91 | -15.25 | -6.59  | 18.34 | -1.75 | -50.89 | -24.38 |
| 25 | 249.09 | -3.09 | -14.13 | 3.85   | 19.58 | -1.58 | -50.05 | -22.87 |
| 26 | 255.98 | -2.91 | -13.67 | 6.77   | 18.09 | -2.15 | -48.40 | -21.74 |
| 27 | 249.87 | -3.43 | -13.89 | 1.80   | 17.14 | -2.32 | -50.95 | -25.65 |
| 28 | 247.42 | -3.34 | -13.32 | 2.32   | 17.69 | -1.29 | -50.40 | -23.82 |
| 29 | 253.00 | -3.32 | -12.60 | 4.98   | 17.09 | -1.16 | -49.54 | -25.89 |
| 30 | 254.40 | -3.44 | -14.69 | 1.64   | 20.18 | -1.15 | -49.66 | -22.73 |
| 31 | 256.40 | -3.75 | -13.62 | -8.47  | 19.57 | -1.08 | -52.04 | -21.62 |
| 32 | 24.07  | -2.06 | 2.98   | -22.09 | 14.88 | 8.52  | 3.28   | -4.27  |
| 33 | 24.00  | -1.30 | 3.15   | -22.28 | 14.65 | 9.22  | 3.94   | -4.75  |
| 34 | 20.76  | -1.69 | 3.24   | -22.34 | 14.81 | 8.95  | 3.76   | -4.69  |
| 35 | 22.23  | -2.51 | 3.10   | -22.85 | 14.01 | 9.71  | 3.29   | -4.61  |
| 36 | 20.33  | -2.23 | 3.04   | -21.40 | 15.61 | 8.99  | 3.00   | -4.12  |
| 37 | 21.12  | -2.09 | 3.25   | -26.46 | 14.68 | 8.75  | 4.67   | -5.05  |
| 38 | 21.89  | -1.92 | 3.05   | -22.32 | 14.65 | 8.25  | 3.77   | -4.15  |
| 39 | 14.81  | -2.07 | 2.82   | -20.92 | 14.96 | 9.05  | 3.73   | -5.40  |

| k  | a8k     | a9k     | a10k | a11k   | a12k   | a13k  | a14k  | a15k  |
|----|---------|---------|------|--------|--------|-------|-------|-------|
| 0  | -95.90  | -116.79 | 0.49 | -92.61 | -76.01 | 36.70 | 0.46  | 78.40 |
| 1  | -99.07  | -117.11 | 0.57 | -89.09 | -76.31 | 37.12 | -0.60 | 86.28 |
| 2  | -92.44  | -115.78 | 0.59 | -93.89 | -76.95 | 35.15 | 0.42  | 83.74 |
| 3  | -96.50  | -115.93 | 0.49 | -92.28 | -75.29 | 36.48 | -0.17 | 70.45 |
| 4  | -100.83 | -120.58 | 0.57 | -95.39 | -73.94 | 37.36 | -0.26 | 79.80 |
| 5  | -94.89  | -117.93 | 0.59 | -93.45 | -75.67 | 36.88 | 0.22  | 74.29 |
| 6  | -94.27  | -119.97 | 0.51 | -91.14 | -74.61 | 35.76 | -0.72 | 73.46 |
| 7  | -94.78  | -114.15 | 0.55 | -91.40 | -71.49 | 36.55 | -0.61 | 73.29 |
| 8  | -97.53  | -112.61 | 0.54 | -91.30 | -70.94 | 36.62 | 0.39  | 75.30 |
| 9  | -97.85  | -108.91 | 0.57 | -94.07 | -72.52 | 36.34 | -0.32 | 76.34 |
| 10 | -97.23  | -115.13 | 0.64 | -95.38 | -74.62 | 36.22 | -0.33 | 74.11 |
| 11 | -96.18  | -115.91 | 0.45 | -93.13 | -77.72 | 37.20 | -0.07 | 81.82 |
| 12 | -100.79 | -113.16 | 0.52 | -90.33 | -72.38 | 36.46 | -1.02 | 74.17 |
| 13 | -98.70  | -124.58 | 0.53 | -93.48 | -74.52 | 36.90 | -0.78 | 74.38 |
| 14 | -93.07  | -116.00 | 0.52 | -92.98 | -75.28 | 36.84 | -0.77 | 76.49 |
| 15 | -106.83 | -117.44 | 0.52 | -94.22 | -75.28 | 36.42 | -0.51 | 79.82 |
| 16 | -88.58  | -87.41  | 0.63 | -70.05 | -37.06 | 27.50 | 3.58  | 84.26 |
| 17 | -87.71  | -82.32  | 0.54 | -68.18 | -40.98 | 28.10 | 3.20  | 88.34 |
| 18 | -76.19  | -80.81  | 0.53 | -66.95 | -35.36 | 27.23 | 3.61  | 87.52 |



|    |        |        |       |        |        |       |      |       |
|----|--------|--------|-------|--------|--------|-------|------|-------|
| 19 | -80.42 | -84.95 | 0.55  | -67.91 | -38.39 | 27.70 | 3.39 | 90.71 |
| 20 | -87.87 | -76.07 | 0.74  | -66.04 | -39.03 | 27.55 | 2.84 | 83.83 |
| 21 | -93.60 | -80.04 | 0.45  | -69.48 | -38.83 | 27.01 | 3.71 | 83.32 |
| 22 | -88.04 | -78.17 | 0.62  | -72.00 | -39.30 | 27.24 | 3.08 | 89.23 |
| 23 | -88.92 | -76.16 | 0.62  | -66.79 | -39.81 | 27.64 | 3.01 | 84.46 |
| 24 | -90.47 | -77.18 | 0.58  | -66.41 | -35.84 | 27.85 | 3.13 | 94.66 |
| 25 | -89.84 | -83.96 | 0.70  | -64.36 | -36.65 | 26.40 | 3.38 | 86.08 |
| 26 | -81.82 | -72.08 | 0.59  | -66.39 | -38.13 | 27.45 | 3.47 | 83.27 |
| 27 | -80.57 | -82.11 | 0.65  | -66.61 | -35.78 | 28.23 | 3.99 | 86.32 |
| 28 | -93.40 | -81.19 | 0.66  | -68.45 | -39.71 | 26.68 | 3.19 | 88.66 |
| 29 | -92.92 | -79.14 | 0.68  | -64.90 | -38.61 | 28.11 | 3.79 | 83.05 |
| 30 | -91.82 | -79.09 | 0.57  | -66.39 | -37.74 | 27.24 | 4.20 | 82.26 |
| 31 | -90.60 | -82.20 | 0.51  | -67.61 | -35.58 | 28.10 | 3.02 | 89.40 |
| 32 | -0.54  | -16.54 | -0.31 | -3.67  | -33.80 | 4.91  | 1.87 | 11.78 |
| 33 | -0.17  | -16.06 | -0.31 | -2.39  | -32.09 | 4.41  | 1.86 | 11.66 |
| 34 | -3.04  | -15.03 | -0.34 | -5.36  | -31.98 | 4.17  | 1.76 | 11.91 |
| 35 | -5.65  | -15.79 | -0.35 | -2.71  | -32.21 | 4.77  | 1.90 | 10.71 |
| 36 | -6.54  | -15.97 | -0.29 | -3.29  | -33.09 | 4.26  | 1.69 | 14.99 |
| 37 | -1.22  | -16.20 | -0.29 | -2.79  | -32.90 | 4.42  | 1.93 | 9.92  |
| 38 | -3.85  | -13.17 | -0.22 | -2.98  | -33.98 | 4.68  | 1.84 | 11.96 |
| 39 | -5.99  | -12.87 | -0.34 | -2.53  | -34.59 | 3.97  | 1.80 | 11.77 |

| k  | a16k  | a17k  | a18k  | a19k    | a20k  | a21k  | a22k   | a23k |
|----|-------|-------|-------|---------|-------|-------|--------|------|
| 0  | 3.86  | 18.38 | 66.68 | 70.89   | 46.30 | 51.83 | 63.61  | 6.33 |
| 1  | -1.41 | 18.18 | 66.65 | -116.27 | 47.93 | 53.00 | -19.71 | 6.14 |
| 2  | 5.49  | 17.64 | 67.40 | 86.57   | 47.99 | 51.08 | -20.20 | 6.06 |
| 3  | 1.41  | 18.29 | 68.83 | -73.86  | 49.40 | 52.06 | -9.49  | 6.14 |
| 4  | 0.58  | 17.86 | 70.11 | 73.41   | 47.45 | 52.04 | 19.20  | 6.06 |
| 5  | -0.96 | 18.19 | 71.29 | 12.08   | 48.40 | 52.10 | 36.63  | 6.18 |
| 6  | -0.98 | 18.97 | 69.33 | 107.17  | 48.12 | 51.68 | -30.34 | 6.21 |
| 7  | 1.44  | 17.76 | 67.10 | -119.11 | 46.64 | 51.64 | -19.85 | 6.11 |
| 8  | 2.83  | 18.10 | 66.62 | 42.19   | 47.28 | 52.29 | -24.88 | 6.17 |
| 9  | -0.87 | 17.50 | 68.71 | -129.91 | 45.89 | 51.63 | -43.44 | 6.23 |
| 10 | 1.24  | 18.26 | 66.88 | -12.32  | 47.49 | 52.45 | 45.87  | 6.14 |
| 11 | 0.39  | 18.50 | 68.03 | -103.43 | 47.42 | 52.40 | -25.17 | 6.01 |
| 12 | 1.60  | 18.13 | 67.42 | -159.30 | 48.41 | 52.43 | -8.06  | 6.08 |
| 13 | 2.92  | 17.75 | 68.36 | 139.40  | 45.89 | 52.74 | -16.56 | 6.19 |
| 14 | 1.69  | 17.80 | 68.48 | -16.76  | 46.43 | 52.86 | -15.34 | 6.20 |
| 15 | 0.66  | 17.51 | 68.30 | -16.84  | 46.61 | 52.23 | -12.07 | 6.00 |
| 16 | 1.53  | 10.12 | 45.53 | -105.48 | 51.47 | 53.43 | 29.32  | 3.91 |
| 17 | -0.83 | 10.67 | 47.22 | -65.54  | 49.74 | 55.32 | -3.14  | 3.90 |
| 18 | -0.05 | 11.56 | 46.07 | -38.72  | 50.65 | 54.17 | -71.99 | 3.87 |
| 19 | 1.11  | 10.54 | 47.56 | -22.99  | 50.63 | 54.47 | 19.05  | 3.81 |
| 20 | -2.77 | 10.45 | 46.29 | 102.12  | 50.39 | 54.05 | -21.67 | 3.88 |
| 21 | 1.51  | 11.03 | 46.20 | -61.41  | 52.28 | 55.89 | 51.76  | 4.01 |
| 22 | -2.08 | 11.82 | 45.96 | 121.36  | 49.18 | 54.28 | -44.05 | 3.89 |
| 23 | -0.13 | 10.89 | 46.21 | 34.94   | 52.82 | 54.93 | 26.48  | 3.82 |
| 24 | -3.52 | 10.89 | 49.08 | 97.01   | 50.10 | 55.70 | 4.07   | 3.65 |
| 25 | 2.07  | 9.39  | 46.08 | -53.59  | 51.65 | 53.65 | 4.94   | 3.97 |
| 26 | 0.44  | 11.04 | 47.23 | -188.70 | 49.01 | 55.21 | 30.06  | 4.08 |
| 27 | -4.62 | 9.33  | 47.59 | 141.42  | 51.09 | 54.20 | -1.25  | 3.95 |
| 28 | -1.78 | 10.70 | 45.70 | -47.29  | 49.84 | 54.81 | 40.79  | 3.71 |
| 29 | -2.45 | 10.65 | 47.01 | -32.51  | 51.18 | 53.56 | 11.77  | 3.93 |
| 30 | 0.03  | 10.49 | 43.45 | -149.80 | 50.93 | 53.95 | 14.36  | 3.83 |

|    |       |       |       |        |       |       |        |      |
|----|-------|-------|-------|--------|-------|-------|--------|------|
| 31 | 3.37  | 10.57 | 46.51 | 20.87  | 49.67 | 54.60 | 22.79  | 3.96 |
| 32 | 0.86  | 7.55  | 19.63 | -6.16  | 7.11  | 9.03  | -20.02 | 1.08 |
| 33 | -1.54 | 7.76  | 20.22 | 38.99  | 7.26  | 8.99  | -10.41 | 1.12 |
| 34 | -0.19 | 7.43  | 18.78 | 21.86  | 7.75  | 8.86  | -16.09 | 1.15 |
| 35 | -2.22 | 7.79  | 19.15 | 4.23   | 7.44  | 8.75  | 19.94  | 1.06 |
| 36 | -1.11 | 7.83  | 20.15 | 20.92  | 6.25  | 8.80  | -31.48 | 1.06 |
| 37 | -0.31 | 7.90  | 19.64 | 8.33   | 7.22  | 9.07  | 18.83  | 1.12 |
| 38 | -1.59 | 7.27  | 19.37 | -18.56 | 6.82  | 8.51  | 21.93  | 1.10 |
| 39 | -1.18 | 7.41  | 19.11 | 85.48  | 7.72  | 8.69  | -17.48 | 1.01 |

| k  | a24k  | a25k  | a26k  | a27k  | a28k  | a29k   | a30k   | a31k   |
|----|-------|-------|-------|-------|-------|--------|--------|--------|
| 0  | -2.45 | 11.03 | 80.28 | 37.44 | 26.08 | -32.86 | -12.63 | 179.34 |
| 1  | 4.01  | 11.52 | 78.78 | 37.44 | 24.32 | -34.91 | -12.16 | 180.51 |
| 2  | -3.35 | 10.90 | 81.50 | 37.26 | 24.80 | -33.69 | -10.77 | 180.89 |
| 3  | 1.73  | 10.84 | 80.38 | 37.51 | 25.47 | -36.57 | -15.88 | 183.66 |
| 4  | 1.27  | 9.42  | 81.14 | 38.14 | 25.13 | -35.66 | -13.87 | 180.51 |
| 5  | -1.05 | 10.89 | 78.62 | 35.99 | 24.44 | -34.31 | -10.65 | 181.52 |
| 6  | -0.46 | 10.30 | 81.92 | 35.89 | 24.74 | -34.63 | -15.73 | 180.07 |
| 7  | 2.40  | 10.57 | 78.99 | 37.01 | 23.94 | -34.87 | -12.35 | 183.67 |
| 8  | 1.32  | 10.56 | 80.87 | 37.45 | 24.70 | -35.58 | -16.41 | 180.05 |
| 9  | 1.05  | 9.87  | 79.26 | 37.63 | 24.52 | -33.58 | -12.25 | 178.55 |
| 10 | -0.78 | 9.63  | 81.05 | 38.38 | 25.20 | -33.14 | -13.41 | 176.95 |
| 11 | 3.17  | 10.63 | 81.55 | 37.98 | 24.25 | -32.39 | -8.97  | 178.12 |
| 12 | -0.12 | 10.57 | 81.11 | 38.18 | 25.06 | -32.51 | -14.47 | 180.98 |
| 13 | -3.92 | 11.23 | 81.59 | 37.07 | 24.28 | -35.06 | -13.10 | 179.20 |
| 14 | -1.14 | 11.04 | 81.63 | 36.75 | 25.25 | -34.70 | -11.65 | 180.54 |
| 15 | 1.46  | 11.05 | 80.55 | 36.57 | 23.89 | -33.64 | -12.62 | 181.91 |
| 16 | -5.70 | 11.70 | 54.00 | 34.44 | 12.44 | -14.39 | -12.01 | 171.54 |
| 17 | -0.70 | 12.09 | 55.21 | 34.71 | 9.92  | -17.13 | -11.70 | 168.02 |
| 18 | -3.57 | 11.69 | 53.52 | 34.81 | 11.95 | -13.55 | -12.37 | 172.64 |
| 19 | -3.32 | 12.04 | 54.84 | 35.03 | 11.31 | -16.56 | -8.97  | 169.62 |
| 20 | 2.59  | 12.21 | 55.48 | 35.95 | 11.23 | -16.53 | -9.56  | 169.59 |
| 21 | 2.54  | 12.19 | 54.25 | 35.20 | 10.76 | -16.90 | -12.28 | 169.58 |
| 22 | -3.44 | 11.69 | 55.10 | 35.89 | 12.10 | -18.59 | -8.83  | 171.87 |
| 23 | 0.56  | 12.26 | 53.50 | 35.39 | 10.43 | -17.85 | -6.90  | 169.62 |
| 24 | 1.18  | 11.85 | 54.90 | 34.61 | 11.57 | -14.31 | -8.59  | 170.29 |
| 25 | 4.59  | 13.16 | 55.42 | 35.65 | 11.43 | -20.08 | -11.70 | 170.01 |
| 26 | -0.12 | 11.06 | 54.09 | 34.94 | 11.37 | -15.16 | -13.76 | 172.27 |
| 27 | 1.10  | 11.81 | 55.45 | 34.85 | 10.99 | -17.00 | -9.50  | 170.91 |
| 28 | 2.78  | 11.51 | 56.06 | 34.44 | 10.57 | -13.57 | -8.79  | 172.45 |
| 29 | -1.28 | 11.48 | 55.08 | 35.02 | 10.89 | -15.74 | -9.63  | 169.75 |
| 30 | 2.04  | 12.16 | 55.11 | 35.67 | 10.86 | -17.71 | -6.55  | 170.86 |
| 31 | 0.32  | 12.82 | 53.12 | 34.92 | 10.54 | -18.42 | -8.44  | 169.18 |
| 32 | 0.20  | 0.74  | 50.97 | -0.31 | 16.61 | -0.13  | -0.95  | 4.08   |
| 33 | -2.15 | 1.02  | 49.63 | -0.32 | 16.22 | 0.23   | 0.22   | 1.95   |
| 34 | 1.20  | 0.57  | 50.53 | -0.69 | 16.60 | 0.05   | -0.14  | 2.81   |
| 35 | 2.07  | 0.84  | 50.16 | -0.49 | 16.35 | 0.75   | -0.83  | 2.72   |
| 36 | 0.80  | 0.74  | 49.51 | 0.39  | 16.42 | -0.53  | -1.24  | 3.37   |
| 37 | -1.45 | 1.05  | 52.96 | -0.54 | 16.22 | 0.77   | -0.23  | 3.04   |
| 38 | 2.48  | 0.88  | 49.48 | 0.67  | 16.29 | -0.46  | -1.51  | 3.83   |
| 39 | -0.81 | 0.65  | 50.20 | -0.45 | 16.38 | -0.34  | -0.14  | 3.97   |

| k  | a32k   | a33k   | a34k  | a35k  | a36k | a37k   | a38k   | a39k  |
|----|--------|--------|-------|-------|------|--------|--------|-------|
| 0  | -19.43 | -15.41 | -0.54 | 35.39 | 4.01 | 146.63 | -31.68 | -3.15 |
| 1  | -18.88 | -15.15 | -0.37 | 34.72 | 4.18 | 148.07 | -31.76 | -3.56 |
| 2  | -16.02 | -15.31 | -0.38 | 33.72 | 4.18 | 145.90 | -32.03 | -3.18 |
| 3  | -17.80 | -15.75 | -0.39 | 36.39 | 4.08 | 145.63 | -31.54 | -3.47 |
| 4  | -15.90 | -15.69 | -0.30 | 34.77 | 4.13 | 149.16 | -31.20 | -3.47 |
| 5  | -18.98 | -15.68 | -0.58 | 33.97 | 4.12 | 147.36 | -31.71 | -3.33 |
| 6  | -15.90 | -15.37 | -0.49 | 34.31 | 4.29 | 146.96 | -30.78 | -3.23 |
| 7  | -14.81 | -15.73 | -0.31 | 36.06 | 4.14 | 150.82 | -32.18 | -3.67 |
| 8  | -15.48 | -15.58 | -0.58 | 36.41 | 4.13 | 146.32 | -31.10 | -3.39 |
| 9  | -16.55 | -15.89 | -0.37 | 32.95 | 3.99 | 141.73 | -30.59 | -3.45 |
| 10 | -18.58 | -15.48 | -0.48 | 33.49 | 4.40 | 148.23 | -30.74 | -3.45 |
| 11 | -15.63 | -15.39 | -0.38 | 35.93 | 4.06 | 144.15 | -30.84 | -3.40 |
| 12 | -17.34 | -15.78 | -0.58 | 35.76 | 4.10 | 140.54 | -32.14 | -3.58 |
| 13 | -16.01 | -15.44 | -0.51 | 35.03 | 4.19 | 142.21 | -31.70 | -3.44 |
| 14 | -19.00 | -15.39 | -0.63 | 35.03 | 4.06 | 146.35 | -31.59 | -3.67 |
| 15 | -16.05 | -15.61 | -0.51 | 35.06 | 4.04 | 144.24 | -31.00 | -3.59 |
| 16 | -1.73  | -11.69 | 0.83  | 17.68 | 4.06 | 115.00 | -27.19 | -5.03 |
| 17 | -4.29  | -12.41 | 1.18  | 16.91 | 3.80 | 117.37 | -26.45 | -5.33 |
| 18 | -2.73  | -12.25 | 1.21  | 17.51 | 3.83 | 119.84 | -28.87 | -5.24 |
| 19 | -1.43  | -12.78 | 1.09  | 16.03 | 3.87 | 118.57 | -27.48 | -5.28 |
| 20 | -1.08  | -12.82 | 1.18  | 16.30 | 3.94 | 118.99 | -26.64 | -4.93 |
| 21 | -3.12  | -12.39 | 0.98  | 17.09 | 3.77 | 116.73 | -27.35 | -5.06 |
| 22 | -2.46  | -12.29 | 1.12  | 15.90 | 3.83 | 119.51 | -27.39 | -4.75 |
| 23 | -1.88  | -12.16 | 1.06  | 16.70 | 3.81 | 120.08 | -27.74 | -5.30 |
| 24 | -3.35  | -12.24 | 1.09  | 14.93 | 3.90 | 117.65 | -27.21 | -5.13 |
| 25 | -1.85  | -11.74 | 1.20  | 18.98 | 3.86 | 118.21 | -27.50 | -5.19 |
| 26 | -2.53  | -12.60 | 1.34  | 18.04 | 3.83 | 118.75 | -26.64 | -5.50 |
| 27 | -4.14  | -11.95 | 1.01  | 18.05 | 3.80 | 120.99 | -27.10 | -4.89 |
| 28 | -2.49  | -12.69 | 1.15  | 18.06 | 3.69 | 117.75 | -26.73 | -5.11 |
| 29 | -2.52  | -12.69 | 1.30  | 15.46 | 4.07 | 116.11 | -27.32 | -5.01 |
| 30 | -1.51  | -12.10 | 1.15  | 17.59 | 3.63 | 123.35 | -27.39 | -5.32 |
| 31 | -4.63  | -12.59 | 1.03  | 17.39 | 3.88 | 119.73 | -27.34 | -5.17 |
| 32 | -5.22  | 1.64   | 2.40  | 15.00 | 2.34 | 28.20  | -5.04  | -5.62 |
| 33 | -5.96  | 1.34   | 2.52  | 14.98 | 2.36 | 28.53  | -4.99  | -5.55 |
| 34 | -5.47  | 1.64   | 2.42  | 14.89 | 2.39 | 29.51  | -4.77  | -5.28 |
| 35 | -4.58  | 1.75   | 2.52  | 14.33 | 2.30 | 31.32  | -5.18  | -5.29 |
| 36 | -6.27  | 1.70   | 2.46  | 14.36 | 2.42 | 31.69  | -5.05  | -5.67 |
| 37 | -5.85  | 1.50   | 2.37  | 14.83 | 2.33 | 29.47  | -5.19  | -5.62 |
| 38 | -4.86  | 1.37   | 2.45  | 14.20 | 2.32 | 30.31  | -5.13  | -5.42 |
| 39 | -6.24  | 1.66   | 2.42  | 14.65 | 2.30 | 31.65  | -5.26  | -5.40 |

| k | a40k   | a41k    | a42k   | a43k   | a44k   | a45k   | a46k    | a47k  |
|---|--------|---------|--------|--------|--------|--------|---------|-------|
| 0 | -10.74 | -128.40 | -12.06 | -27.68 | -47.57 | 129.47 | -144.26 | 5.82  |
| 1 | -10.79 | -126.25 | -6.60  | -27.37 | 30.05  | 129.30 | -151.57 | 0.18  |
| 2 | -10.74 | -126.91 | 22.40  | -24.92 | 8.00   | 127.04 | -148.75 | 4.11  |
| 3 | -10.77 | -125.83 | -0.47  | -27.97 | -22.77 | 127.09 | -144.81 | -5.16 |
| 4 | -10.78 | -123.99 | 31.65  | -28.65 | 24.12  | 128.49 | -154.50 | 13.78 |
| 5 | -10.58 | -125.15 | -1.09  | -27.15 | -24.40 | 127.57 | -149.54 | 5.15  |
| 6 | -10.85 | -124.37 | -9.75  | -27.32 | -4.55  | 126.41 | -145.43 | -4.92 |

|    |        |         |        |        |        |        |         |        |
|----|--------|---------|--------|--------|--------|--------|---------|--------|
| 7  | -10.65 | -127.00 | 0.46   | -27.02 | -13.23 | 129.75 | -144.09 | -13.95 |
| 8  | -11.02 | -126.29 | -13.90 | -26.51 | -7.75  | 127.78 | -152.92 | 8.36   |
| 9  | -10.68 | -124.07 | -7.82  | -27.67 | -23.63 | 127.87 | -150.02 | 8.26   |
| 10 | -10.83 | -124.06 | 47.68  | -28.79 | 7.39   | 124.57 | -149.34 | -3.62  |
| 11 | -10.80 | -128.19 | -2.30  | -25.63 | -6.57  | 129.60 | -150.58 | -9.25  |
| 12 | -10.87 | -124.73 | -54.90 | -24.31 | 41.58  | 134.82 | -151.79 | -4.73  |
| 13 | -10.82 | -127.07 | -0.98  | -28.09 | 7.94   | 130.29 | -148.24 | -10.15 |
| 14 | -11.01 | -125.00 | -2.61  | -26.15 | -9.59  | 129.00 | -148.98 | -19.14 |
| 15 | -10.69 | -125.55 | 6.16   | -27.16 | 5.81   | 130.83 | -152.90 | -9.64  |
| 16 | -11.43 | -133.31 | 27.27  | -18.30 | -26.33 | 148.91 | -121.02 | 7.67   |
| 17 | -11.74 | -134.46 | -52.34 | -16.89 | -23.99 | 151.23 | -120.04 | -19.73 |
| 18 | -11.45 | -130.80 | 18.52  | -16.85 | -48.77 | 150.98 | -122.92 | 10.85  |
| 19 | -11.27 | -133.08 | -29.76 | -17.58 | 3.69   | 149.84 | -121.37 | -4.93  |
| 20 | -11.27 | -134.65 | 8.68   | -16.64 | 34.24  | 149.00 | -118.06 | -9.55  |
| 21 | -11.71 | -137.87 | 30.52  | -14.52 | 0.44   | 149.28 | -119.23 | 7.09   |
| 22 | -11.44 | -133.43 | 27.53  | -14.99 | 8.31   | 149.32 | -122.39 | -19.43 |
| 23 | -11.34 | -135.85 | -13.46 | -14.85 | 0.51   | 149.46 | -118.80 | 12.65  |
| 24 | -11.46 | -134.89 | -32.30 | -18.64 | 3.14   | 145.98 | -119.97 | 5.91   |
| 25 | -10.92 | -135.14 | -16.97 | -16.50 | 9.98   | 148.08 | -126.66 | 0.91   |
| 26 | -11.38 | -132.17 | 38.54  | -18.23 | 23.72  | 150.56 | -114.45 | 1.06   |
| 27 | -11.44 | -134.82 | -43.20 | -14.82 | -21.37 | 150.11 | -123.74 | -7.24  |
| 28 | -11.64 | -136.08 | -8.20  | -17.05 | 0.87   | 153.36 | -125.50 | 18.53  |
| 29 | -11.25 | -132.85 | -28.20 | -17.77 | -16.83 | 146.70 | -118.61 | -2.81  |
| 30 | -11.21 | -133.55 | -22.12 | -16.42 | -9.17  | 150.89 | -123.39 | 7.29   |
| 31 | -11.50 | -135.59 | 16.35  | -17.71 | -39.44 | 149.92 | -117.21 | -17.87 |
| 32 | -1.75  | -17.34  | -4.56  | -15.38 | -4.53  | 27.52  | -4.76   | -15.17 |
| 33 | -1.66  | -18.40  | -1.44  | -15.69 | -11.54 | 27.41  | -2.83   | -8.66  |
| 34 | -1.72  | -17.15  | 3.10   | -14.93 | -13.85 | 27.41  | 0.29    | 18.03  |
| 35 | -1.60  | -18.29  | 10.53  | -16.01 | 10.69  | 27.36  | -0.67   | -6.10  |
| 36 | -1.70  | -17.23  | -4.85  | -14.95 | 7.00   | 25.49  | -2.44   | -2.75  |
| 37 | -1.65  | -17.09  | -18.55 | -15.00 | 3.78   | 27.40  | 1.75    | 3.17   |
| 38 | -1.68  | -17.34  | 10.25  | -15.93 | -2.93  | 27.08  | -9.32   | 12.10  |
| 39 | -1.66  | -16.95  | 6.80   | -16.81 | -3.07  | 27.33  | -0.14   | -2.79  |

| k  | a48k  | a49k  | a50k   | a51k  | a52k   | a53k    | a54k  | a55k  |
|----|-------|-------|--------|-------|--------|---------|-------|-------|
| 0  | 37.14 | 79.32 | -51.17 | -1.76 | 107.54 | -136.45 | -0.70 | 49.45 |
| 1  | 39.73 | 78.51 | -53.02 | -3.67 | 107.24 | -137.40 | -0.92 | 48.53 |
| 2  | 38.32 | 79.55 | -53.33 | 6.10  | 109.08 | -137.60 | -0.44 | 47.48 |
| 3  | 43.04 | 78.76 | -49.78 | -5.20 | 109.63 | -137.79 | -0.41 | 49.72 |
| 4  | 41.97 | 78.51 | -51.25 | -3.82 | 108.30 | -134.67 | -0.72 | 47.28 |
| 5  | 43.91 | 79.20 | -53.25 | 5.02  | 104.46 | -134.65 | -0.72 | 50.25 |
| 6  | 41.71 | 79.71 | -51.24 | 9.31  | 106.65 | -135.02 | -0.97 | 48.74 |
| 7  | 44.52 | 80.29 | -48.51 | -2.75 | 107.04 | -140.33 | 0.13  | 49.50 |
| 8  | 39.20 | 80.04 | -52.02 | -1.21 | 109.14 | -135.03 | -0.87 | 48.70 |
| 9  | 42.96 | 79.10 | -48.95 | 0.10  | 107.27 | -135.84 | -1.15 | 51.11 |
| 10 | 42.98 | 80.11 | -51.37 | 2.86  | 107.63 | -136.54 | -1.02 | 49.77 |
| 11 | 41.67 | 78.52 | -54.87 | 9.25  | 107.78 | -137.56 | -0.36 | 49.93 |
| 12 | 41.54 | 78.66 | -49.72 | 12.58 | 108.48 | -138.05 | -1.51 | 50.82 |
| 13 | 45.53 | 79.40 | -53.00 | -0.90 | 106.29 | -138.06 | -1.22 | 48.46 |
| 14 | 41.05 | 78.05 | -50.05 | 5.44  | 108.17 | -134.77 | -1.00 | 49.65 |
| 15 | 40.71 | 80.07 | -56.73 | -3.43 | 106.52 | -136.43 | -0.06 | 50.09 |
| 16 | 37.90 | 83.28 | -64.08 | -0.55 | 90.49  | -120.94 | 4.79  | 58.73 |

|    |       |       |        |        |       |         |       |       |
|----|-------|-------|--------|--------|-------|---------|-------|-------|
| 17 | 39.35 | 84.15 | -61.34 | 14.09  | 91.22 | -115.20 | 4.42  | 58.38 |
| 18 | 39.25 | 83.87 | -64.28 | -2.73  | 90.15 | -119.81 | 5.17  | 59.14 |
| 19 | 44.51 | 83.85 | -64.77 | -5.83  | 91.64 | -120.52 | 4.96  | 59.80 |
| 20 | 39.23 | 84.11 | -64.08 | -0.47  | 90.61 | -119.79 | 4.41  | 58.04 |
| 21 | 33.64 | 84.97 | -66.24 | 2.00   | 90.84 | -119.05 | 4.36  | 55.65 |
| 22 | 40.09 | 84.70 | -68.59 | 8.97   | 89.54 | -122.30 | 4.21  | 58.89 |
| 23 | 39.94 | 84.58 | -64.95 | 9.90   | 90.45 | -118.55 | 4.20  | 58.47 |
| 24 | 41.44 | 83.21 | -61.49 | 21.96  | 86.40 | -121.01 | 5.88  | 59.02 |
| 25 | 43.13 | 84.06 | -71.09 | 5.92   | 91.68 | -121.74 | 5.43  | 58.39 |
| 26 | 40.08 | 84.02 | -62.57 | -16.45 | 90.01 | -124.94 | 5.37  | 59.22 |
| 27 | 42.77 | 83.17 | -60.93 | -3.80  | 90.21 | -119.34 | 4.16  | 59.00 |
| 28 | 38.19 | 83.14 | -68.75 | 0.93   | 89.45 | -118.83 | 4.65  | 58.62 |
| 29 | 40.28 | 84.08 | -66.06 | 2.04   | 90.85 | -121.14 | 4.95  | 58.58 |
| 30 | 38.60 | 83.57 | -63.85 | 4.02   | 90.34 | -121.73 | 5.02  | 59.52 |
| 31 | 38.72 | 83.98 | -62.23 | -8.14  | 90.87 | -118.15 | 4.47  | 57.21 |
| 32 | 11.57 | 21.55 | -10.03 | -6.33  | 68.62 | -1.53   | -1.26 | 5.29  |
| 33 | 12.52 | 21.60 | -9.90  | 5.14   | 67.28 | -2.03   | -1.17 | 6.06  |
| 34 | 10.52 | 21.27 | -9.92  | 7.48   | 62.26 | -1.26   | -1.10 | 6.49  |
| 35 | 9.72  | 21.00 | -9.55  | 4.65   | 67.09 | -2.90   | -1.42 | 5.32  |
| 36 | 10.18 | 21.30 | -10.46 | 0.53   | 68.50 | -1.87   | -1.28 | 5.86  |
| 37 | 11.41 | 21.45 | -10.21 | -5.58  | 69.19 | -2.94   | -0.93 | 6.38  |
| 38 | 10.20 | 21.55 | -10.48 | -0.02  | 72.00 | -2.62   | -1.45 | 5.82  |
| 39 | 10.76 | 21.90 | -9.25  | 2.27   | 70.02 | -3.28   | -1.69 | 5.49  |

| k  | a56k   | a57k   | a58k  | a59k  | a60k   | a61k  | a62k    | a63k    |
|----|--------|--------|-------|-------|--------|-------|---------|---------|
| 0  | -12.86 | 129.34 | -0.04 | 14.92 | 138.37 | -1.55 | 89.62   | 30.73   |
| 1  | -5.85  | 128.69 | 0.01  | 15.19 | 137.24 | -1.35 | -3.02   | -80.51  |
| 2  | -4.35  | 128.38 | 0.04  | 15.14 | 134.51 | -1.28 | 41.45   | -64.49  |
| 3  | -12.07 | 129.28 | 0.00  | 15.99 | 138.44 | -1.79 | 43.17   | -32.41  |
| 4  | -29.61 | 130.64 | 0.05  | 15.92 | 138.34 | -1.07 | -61.32  | -48.06  |
| 5  | -8.28  | 128.31 | 0.04  | 16.59 | 134.68 | -1.42 | 22.30   | 2.66    |
| 6  | 19.35  | 130.30 | 0.02  | 13.87 | 136.49 | -1.67 | -84.92  | 0.79    |
| 7  | 24.36  | 127.83 | 0.04  | 15.94 | 138.86 | -1.90 | 0.65    | -3.65   |
| 8  | -14.01 | 128.40 | 0.03  | 15.30 | 137.88 | -1.44 | -4.38   | -31.18  |
| 9  | 0.91   | 129.12 | 0.02  | 14.12 | 135.10 | -1.55 | 69.96   | 4.18    |
| 10 | 23.68  | 129.32 | 0.07  | 16.18 | 140.35 | -1.81 | -95.92  | -40.96  |
| 11 | -17.89 | 127.87 | -0.02 | 14.63 | 137.22 | -1.64 | -53.97  | -21.93  |
| 12 | -16.16 | 127.54 | 0.03  | 15.68 | 139.52 | -1.33 | -2.36   | -18.42  |
| 13 | -9.14  | 128.00 | 0.04  | 15.45 | 137.55 | -1.47 | -154.65 | -37.01  |
| 14 | 6.32   | 128.51 | 0.01  | 15.55 | 135.27 | -1.61 | -75.40  | 17.78   |
| 15 | -56.04 | 128.95 | -0.01 | 15.81 | 140.72 | -1.50 | 15.30   | -0.58   |
| 16 | 11.07  | 103.86 | 0.32  | 12.40 | 106.49 | -3.15 | -17.66  | -6.34   |
| 17 | 1.20   | 103.57 | 0.30  | 11.80 | 105.25 | -3.35 | 158.30  | -11.96  |
| 18 | 15.81  | 104.02 | 0.31  | 12.73 | 105.76 | -3.14 | 27.25   | 38.46   |
| 19 | -38.30 | 105.03 | 0.26  | 13.24 | 105.45 | -3.08 | 55.65   | -53.71  |
| 20 | -11.59 | 103.10 | 0.28  | 12.71 | 104.03 | -3.53 | -163.96 | 49.02   |
| 21 | -12.52 | 102.37 | 0.36  | 13.19 | 102.12 | -2.92 | -173.39 | -5.65   |
| 22 | -9.62  | 103.25 | 0.33  | 12.39 | 108.93 | -2.99 | 52.22   | -161.59 |
| 23 | 25.92  | 103.48 | 0.29  | 11.74 | 106.99 | -2.55 | 105.11  | -85.85  |
| 24 | -5.50  | 103.73 | 0.35  | 12.51 | 104.40 | -3.33 | 69.39   | -5.04   |
| 25 | -10.31 | 105.43 | 0.32  | 12.82 | 105.99 | -3.71 | -26.84  | -118.93 |
| 26 | 18.71  | 106.00 | 0.30  | 10.96 | 106.89 | -3.08 | -32.36  | -84.52  |

|    |        |        |      |       |        |       |        |        |
|----|--------|--------|------|-------|--------|-------|--------|--------|
| 27 | 20.64  | 104.18 | 0.31 | 10.98 | 105.86 | -2.84 | -60.72 | -27.19 |
| 28 | -15.68 | 105.50 | 0.30 | 12.70 | 106.94 | -2.97 | 88.69  | 23.52  |
| 29 | -13.79 | 104.57 | 0.35 | 11.66 | 106.15 | -3.25 | 26.48  | -34.70 |
| 30 | 0.58   | 103.49 | 0.33 | 12.68 | 105.58 | -2.79 | -41.33 | 6.57   |
| 31 | -41.18 | 104.89 | 0.34 | 12.31 | 105.97 | -3.75 | 29.53  | -6.35  |
| 32 | 1.27   | 44.57  | 0.12 | 13.53 | 42.01  | 2.37  | 17.83  | 3.10   |
| 33 | 6.12   | 45.81  | 0.10 | 13.35 | 42.85  | 2.35  | -40.89 | -6.28  |
| 34 | -2.92  | 44.37  | 0.10 | 13.49 | 42.40  | 2.50  | 118.60 | -8.15  |
| 35 | -5.72  | 45.65  | 0.09 | 13.59 | 42.95  | 2.35  | -24.50 | 42.41  |
| 36 | 8.16   | 46.19  | 0.13 | 13.25 | 42.19  | 2.58  | -23.66 | -14.21 |
| 37 | -2.52  | 45.21  | 0.10 | 14.30 | 42.49  | 2.21  | 10.46  | 8.05   |
| 38 | 1.03   | 44.35  | 0.08 | 13.11 | 42.30  | 2.30  | -19.72 | -34.53 |
| 39 | 7.84   | 44.14  | 0.07 | 12.77 | 41.49  | 2.14  | -87.32 | -9.49  |

| k  | a0k   | a1k   | a2k  | a3k    | a4k   | a5k   | a6k  | a7k   |
|----|-------|-------|------|--------|-------|-------|------|-------|
| 40 | 19.90 | -2.13 | 3.40 | -23.58 | 14.75 | 9.48  | 3.86 | -5.28 |
| 41 | 18.83 | -2.16 | 2.85 | -22.88 | 14.32 | 8.82  | 4.18 | -4.88 |
| 42 | 20.70 | -1.65 | 3.27 | -20.35 | 15.25 | 9.24  | 3.58 | -4.35 |
| 43 | 20.25 | -1.77 | 3.20 | -22.22 | 15.27 | 9.16  | 3.48 | -3.80 |
| 44 | 24.14 | -2.06 | 3.32 | -21.09 | 15.40 | 8.40  | 4.18 | -4.42 |
| 45 | 18.27 | -1.69 | 3.27 | -23.09 | 14.59 | 8.75  | 3.83 | -4.73 |
| 46 | 25.79 | -2.16 | 2.67 | -23.50 | 13.93 | 9.50  | 3.72 | -4.28 |
| 47 | 20.38 | -2.29 | 3.08 | -22.68 | 15.06 | 8.65  | 2.52 | -5.19 |
| 48 | 50.37 | -1.77 | 3.59 | -24.60 | 18.28 | 10.50 | 5.85 | -7.96 |
| 49 | 49.85 | -2.31 | 3.47 | -28.05 | 18.13 | 10.17 | 7.16 | -7.75 |
| 50 | 41.67 | -2.30 | 3.79 | -26.10 | 18.67 | 10.55 | 6.25 | -8.25 |
| 51 | 42.09 | -2.00 | 3.53 | -25.53 | 18.63 | 10.87 | 5.39 | -5.86 |
| 52 | 50.69 | -2.14 | 3.14 | -26.33 | 19.35 | 9.35  | 6.52 | -7.00 |
| 53 | 42.49 | -1.43 | 3.63 | -27.22 | 19.41 | 9.91  | 6.84 | -7.48 |
| 54 | 53.71 | -2.44 | 3.50 | -31.61 | 18.14 | 11.45 | 6.06 | -9.48 |
| 55 | 46.99 | -2.19 | 3.48 | -29.49 | 18.31 | 10.81 | 6.44 | -6.38 |
| 56 | 51.56 | -2.37 | 2.97 | -27.93 | 19.01 | 11.39 | 5.33 | -6.22 |
| 57 | 46.94 | -2.41 | 3.00 | -30.06 | 17.38 | 9.09  | 6.34 | -7.10 |
| 58 | 41.31 | -2.75 | 3.65 | -25.85 | 19.08 | 9.83  | 6.07 | -6.96 |
| 59 | 44.49 | -2.22 | 3.61 | -32.55 | 18.31 | 10.01 | 6.66 | -7.11 |
| 60 | 45.40 | -2.42 | 3.22 | -27.27 | 18.00 | 11.33 | 6.80 | -7.85 |
| 61 | 48.69 | -2.05 | 3.04 | -27.02 | 19.11 | 10.47 | 6.21 | -8.31 |
| 62 | 51.91 | -2.10 | 3.23 | -24.02 | 18.61 | 10.14 | 6.24 | -8.18 |
| 63 | 49.12 | -2.03 | 3.79 | -26.05 | 19.92 | 9.39  | 6.41 | -7.48 |
| 64 | 14.82 | -1.26 | 2.54 | -14.92 | 10.38 | 5.70  | 4.36 | -2.13 |
| 65 | 18.85 | -1.68 | 2.35 | -15.32 | 10.60 | 5.67  | 4.25 | -2.41 |
| 66 | 12.30 | -1.35 | 2.98 | -13.25 | 10.11 | 6.36  | 5.11 | -2.88 |
| 67 | 10.43 | -0.99 | 2.47 | -13.28 | 10.38 | 6.20  | 5.13 | -2.15 |
| 68 | 10.66 | -1.07 | 2.81 | -16.65 | 10.22 | 5.83  | 3.98 | -2.43 |
| 69 | 14.27 | -1.20 | 2.24 | -13.79 | 9.68  | 4.91  | 3.57 | -2.61 |
| 70 | 14.63 | -0.95 | 2.44 | -13.66 | 10.38 | 4.72  | 4.73 | -2.55 |
| 71 | 13.43 | -1.33 | 2.57 | -16.19 | 9.82  | 5.72  | 4.41 | -2.84 |
| 72 | 12.29 | -1.25 | 2.24 | -13.42 | 9.93  | 5.64  | 4.34 | -2.66 |
| 73 | 13.65 | -1.66 | 2.41 | -16.32 | 10.52 | 5.40  | 4.92 | -2.50 |
| 74 | 13.37 | -1.27 | 2.47 | -15.49 | 9.98  | 5.94  | 3.65 | -1.62 |
| 75 | 14.10 | -1.64 | 2.34 | -14.76 | 10.05 | 5.40  | 4.82 | -2.77 |
| 76 | 16.53 | -1.24 | 1.83 | -18.07 | 9.51  | 6.44  | 4.40 | -2.85 |

|    |       |       |      |        |       |      |      |       |
|----|-------|-------|------|--------|-------|------|------|-------|
| 77 | 15.73 | -1.15 | 2.18 | -13.81 | 9.36  | 5.62 | 4.61 | -2.36 |
| 78 | 17.20 | -1.32 | 2.68 | -13.10 | 9.52  | 5.70 | 4.78 | -2.81 |
| 79 | 18.10 | -1.03 | 2.61 | -14.46 | 10.11 | 5.73 | 4.83 | -2.49 |

| k  | a8k    | a9k    | a10k  | a11k  | a12k   | a13k | a14k | a15k  |
|----|--------|--------|-------|-------|--------|------|------|-------|
| 40 | -0.48  | -14.13 | -0.26 | -4.85 | -32.91 | 4.45 | 1.62 | 14.53 |
| 41 | -2.55  | -14.69 | -0.26 | -2.59 | -32.83 | 4.48 | 1.64 | 10.87 |
| 42 | -2.32  | -13.82 | -0.25 | -3.21 | -32.11 | 4.71 | 1.92 | 12.41 |
| 43 | 0.25   | -13.86 | -0.32 | -0.74 | -32.76 | 4.46 | 2.18 | 10.08 |
| 44 | -2.98  | -14.88 | -0.29 | -0.15 | -33.19 | 4.33 | 1.78 | 12.83 |
| 45 | 0.26   | -14.82 | -0.34 | -3.73 | -35.01 | 4.39 | 1.93 | 11.13 |
| 46 | -0.55  | -13.52 | -0.27 | -2.21 | -32.26 | 4.23 | 1.58 | 13.57 |
| 47 | -2.58  | -14.57 | -0.27 | -1.27 | -32.66 | 4.42 | 1.55 | 11.73 |
| 48 | -9.49  | -17.26 | -0.40 | -7.16 | -36.29 | 5.87 | 2.34 | 25.11 |
| 49 | -15.26 | -17.26 | -0.39 | -9.71 | -37.64 | 6.04 | 2.21 | 24.19 |
| 50 | -17.74 | -16.05 | -0.38 | -8.14 | -37.86 | 5.44 | 2.03 | 27.35 |
| 51 | -7.51  | -17.69 | -0.44 | -7.78 | -36.26 | 6.10 | 2.27 | 26.16 |
| 52 | -16.09 | -16.80 | -0.45 | -7.39 | -35.79 | 6.14 | 2.29 | 22.75 |
| 53 | -21.20 | -18.29 | -0.46 | -6.47 | -37.56 | 6.01 | 2.73 | 22.73 |
| 54 | -14.84 | -18.42 | -0.34 | -6.94 | -35.71 | 6.25 | 2.47 | 23.41 |
| 55 | -16.75 | -17.24 | -0.39 | -7.01 | -38.93 | 6.20 | 2.67 | 26.48 |
| 56 | -8.39  | -22.20 | -0.43 | -6.67 | -40.21 | 5.83 | 2.38 | 22.94 |
| 57 | -14.66 | -19.71 | -0.36 | -8.63 | -40.12 | 6.05 | 2.57 | 23.51 |
| 58 | -16.77 | -15.89 | -0.40 | -8.32 | -38.01 | 6.03 | 2.53 | 23.62 |
| 59 | -16.13 | -20.60 | -0.48 | -7.39 | -36.88 | 6.09 | 2.39 | 23.28 |
| 60 | -14.21 | -16.48 | -0.41 | -8.28 | -39.00 | 5.52 | 2.26 | 23.17 |
| 61 | -14.30 | -22.10 | -0.43 | -8.26 | -37.30 | 6.59 | 2.03 | 22.66 |
| 62 | -17.72 | -18.12 | -0.41 | -7.89 | -37.81 | 6.13 | 2.04 | 25.89 |
| 63 | -17.68 | -18.26 | -0.39 | -6.30 | -36.11 | 5.40 | 2.04 | 21.67 |
| 64 | 2.73   | -5.49  | -0.33 | 1.00  | -13.27 | 1.08 | 2.61 | 11.70 |
| 65 | -4.20  | -5.79  | -0.38 | -1.97 | -13.62 | 1.34 | 2.28 | 11.01 |
| 66 | -5.39  | -6.55  | -0.36 | 0.72  | -12.79 | 0.98 | 2.48 | 13.80 |
| 67 | -4.50  | -5.76  | -0.33 | -1.87 | -13.93 | 0.96 | 2.58 | 11.78 |
| 68 | -4.01  | -5.60  | -0.37 | -1.22 | -14.53 | 1.28 | 2.37 | 12.48 |
| 69 | -6.29  | -5.07  | -0.33 | 0.39  | -14.47 | 1.50 | 2.31 | 13.61 |
| 70 | -6.11  | -5.86  | -0.40 | -0.83 | -14.26 | 1.23 | 2.63 | 10.62 |
| 71 | -8.03  | -5.00  | -0.32 | -0.78 | -12.45 | 1.14 | 2.62 | 12.27 |
| 72 | -6.81  | -6.73  | -0.38 | 0.34  | -14.42 | 1.52 | 2.49 | 12.46 |
| 73 | -7.12  | -6.95  | -0.33 | 0.12  | -13.53 | 1.04 | 2.45 | 10.32 |
| 74 | -6.57  | -3.28  | -0.31 | 0.22  | -14.17 | 1.09 | 2.41 | 13.68 |
| 75 | -6.00  | -3.61  | -0.33 | -0.67 | -13.40 | 1.23 | 2.67 | 12.09 |
| 76 | -5.23  | -7.88  | -0.38 | -0.19 | -14.11 | 1.19 | 2.52 | 13.16 |
| 77 | -7.27  | -3.96  | -0.32 | 1.76  | -14.15 | 1.23 | 2.79 | 11.77 |
| 78 | -3.39  | -7.77  | -0.34 | -0.85 | -13.00 | 1.32 | 2.61 | 10.48 |
| 79 | -4.11  | -6.10  | -0.36 | -1.01 | -13.69 | 1.30 | 2.55 | 11.87 |

| k  | a16k  | a17k | a18k  | a19k   | a20k | a21k | a22k  | a23k |
|----|-------|------|-------|--------|------|------|-------|------|
| 40 | -1.34 | 7.31 | 20.08 | -43.84 | 7.06 | 8.83 | 3.34  | 1.08 |
| 41 | -1.10 | 7.79 | 19.71 | 8.41   | 7.32 | 8.92 | 15.86 | 1.04 |
| 42 | 1.31  | 8.00 | 19.19 | -20.28 | 6.83 | 8.82 | 24.92 | 1.08 |

|    |       |       |       |        |       |       |        |      |
|----|-------|-------|-------|--------|-------|-------|--------|------|
| 43 | 1.45  | 7.98  | 19.87 | 46.25  | 6.78  | 8.45  | -2.96  | 1.00 |
| 44 | 0.33  | 7.34  | 19.54 | -38.90 | 6.92  | 8.67  | -12.94 | 1.11 |
| 45 | 1.75  | 7.10  | 20.14 | -21.37 | 6.82  | 8.73  | -27.46 | 1.13 |
| 46 | 0.61  | 8.00  | 20.03 | -28.34 | 7.09  | 8.93  | 31.83  | 1.12 |
| 47 | -0.64 | 7.87  | 19.53 | -46.57 | 7.31  | 8.89  | -30.39 | 1.07 |
| 48 | -0.15 | 11.06 | 25.54 | -2.80  | 12.05 | 11.55 | -13.03 | 1.27 |
| 49 | -0.12 | 11.11 | 25.52 | 8.12   | 12.20 | 12.65 | -6.52  | 1.32 |
| 50 | -1.32 | 10.83 | 24.11 | 37.99  | 12.63 | 11.38 | 32.37  | 1.35 |
| 51 | 0.13  | 10.49 | 25.74 | 8.31   | 12.13 | 11.86 | -5.37  | 1.28 |
| 52 | -1.83 | 11.17 | 26.01 | 33.77  | 12.42 | 11.00 | 34.57  | 1.29 |
| 53 | -1.54 | 11.10 | 25.30 | 42.41  | 11.53 | 11.16 | 50.78  | 1.36 |
| 54 | 0.86  | 10.54 | 23.84 | -10.95 | 12.27 | 11.29 | 7.27   | 1.27 |
| 55 | 0.11  | 11.02 | 25.47 | -32.01 | 11.40 | 11.13 | 0.00   | 1.32 |
| 56 | 1.25  | 10.62 | 26.27 | 11.39  | 11.63 | 11.39 | 0.65   | 1.32 |
| 57 | -1.16 | 10.55 | 25.32 | 76.10  | 12.37 | 11.06 | 17.89  | 1.41 |
| 58 | -0.85 | 11.14 | 24.88 | 25.65  | 11.81 | 11.47 | -34.03 | 1.31 |
| 59 | -1.57 | 10.72 | 25.21 | -19.84 | 11.67 | 12.12 | -12.65 | 1.17 |
| 60 | 0.05  | 10.51 | 24.90 | -78.84 | 12.63 | 12.01 | -6.74  | 1.37 |
| 61 | -1.47 | 10.82 | 25.26 | 46.37  | 12.04 | 11.15 | -18.96 | 1.39 |
| 62 | 1.03  | 10.49 | 25.79 | 18.87  | 12.11 | 11.18 | -7.43  | 1.35 |
| 63 | 0.25  | 10.96 | 26.25 | 59.58  | 11.55 | 11.69 | -2.09  | 1.42 |
| 64 | -0.42 | 5.86  | 11.16 | 66.99  | 8.70  | 6.34  | 31.58  | 1.06 |
| 65 | 1.35  | 5.34  | 11.18 | 14.04  | 8.27  | 6.43  | 59.34  | 1.01 |
| 66 | 0.05  | 5.54  | 10.70 | -52.21 | 8.29  | 6.36  | 6.84   | 1.05 |
| 67 | 0.45  | 5.43  | 11.31 | 18.29  | 7.77  | 6.55  | 8.15   | 1.01 |
| 68 | -0.61 | 6.04  | 9.77  | 28.80  | 7.99  | 6.69  | 3.61   | 1.10 |
| 69 | -0.84 | 5.57  | 10.58 | 42.30  | 8.11  | 6.44  | -41.16 | 1.11 |
| 70 | 0.34  | 5.48  | 10.22 | 22.47  | 8.03  | 6.57  | -21.22 | 1.11 |
| 71 | -0.40 | 6.13  | 10.13 | -19.36 | 8.25  | 6.63  | -9.41  | 1.03 |
| 72 | -0.74 | 5.96  | 10.71 | 15.47  | 7.99  | 6.60  | -11.79 | 1.10 |
| 73 | 0.30  | 5.77  | 10.78 | -35.45 | 8.02  | 7.23  | -7.79  | 1.12 |
| 74 | 0.31  | 6.04  | 9.85  | -2.67  | 7.43  | 6.23  | -16.70 | 0.99 |
| 75 | 0.07  | 5.67  | 11.17 | -30.70 | 7.82  | 6.03  | -41.32 | 1.03 |
| 76 | 0.91  | 6.31  | 10.37 | 65.63  | 7.65  | 6.38  | -19.29 | 1.11 |
| 77 | -0.55 | 5.90  | 10.02 | 41.19  | 7.65  | 6.85  | 26.33  | 1.08 |
| 78 | 0.38  | 5.93  | 10.95 | -33.73 | 8.60  | 6.71  | 15.83  | 1.07 |
| 79 | 0.08  | 5.82  | 10.18 | 55.66  | 8.03  | 6.07  | 15.49  | 1.06 |

| k  | a24k  | a25k  | a26k  | a27k  | a28k  | a29k  | a30k  | a31k |
|----|-------|-------|-------|-------|-------|-------|-------|------|
| 40 | -1.53 | 1.02  | 52.10 | -0.48 | 16.66 | 0.51  | -0.48 | 3.17 |
| 41 | 1.12  | 0.85  | 50.58 | -0.66 | 16.94 | -0.20 | -1.60 | 1.47 |
| 42 | -0.14 | 0.54  | 49.36 | -0.28 | 16.62 | -0.54 | -0.84 | 2.34 |
| 43 | 1.89  | -0.17 | 50.32 | -0.77 | 16.71 | 0.11  | -0.84 | 2.40 |
| 44 | 2.72  | 0.93  | 51.66 | 0.03  | 16.64 | -0.52 | -0.93 | 3.31 |
| 45 | 0.90  | 0.89  | 50.79 | -0.49 | 16.85 | 0.29  | -0.30 | 2.02 |
| 46 | 0.92  | 1.06  | 49.05 | -0.55 | 15.95 | -0.36 | -0.75 | 2.36 |
| 47 | -1.10 | 0.50  | 50.20 | -0.34 | 16.34 | -0.29 | -0.96 | 2.86 |
| 48 | 1.43  | 2.74  | 66.38 | -2.26 | 21.67 | 0.95  | -5.11 | 7.11 |
| 49 | 0.16  | 3.22  | 63.74 | -1.00 | 21.20 | 1.23  | -4.04 | 5.34 |
| 50 | 1.56  | 2.91  | 65.05 | -1.22 | 20.65 | 1.57  | -3.43 | 4.36 |
| 51 | -0.08 | 2.84  | 63.90 | -2.64 | 21.54 | -0.16 | -4.06 | 5.01 |
| 52 | -1.22 | 3.14  | 64.73 | -2.15 | 20.73 | 1.26  | -3.17 | 6.76 |



|    |       |      |       |       |       |       |       |       |
|----|-------|------|-------|-------|-------|-------|-------|-------|
| 53 | 2.12  | 3.35 | 63.77 | -2.31 | 20.64 | 1.90  | -2.79 | 4.15  |
| 54 | -0.09 | 3.01 | 65.03 | -2.19 | 21.60 | 0.00  | -3.75 | 3.41  |
| 55 | 2.71  | 3.52 | 62.85 | -2.18 | 21.41 | 1.02  | -3.95 | 4.22  |
| 56 | -1.83 | 3.25 | 64.90 | -2.23 | 20.87 | 1.17  | -5.24 | 6.48  |
| 57 | -1.32 | 3.01 | 65.44 | -1.67 | 20.99 | 0.86  | -4.19 | 4.77  |
| 58 | -1.32 | 3.42 | 67.01 | -1.82 | 20.89 | 0.54  | -2.96 | 6.01  |
| 59 | -1.20 | 3.13 | 67.59 | -2.06 | 21.62 | 0.29  | -4.53 | 7.54  |
| 60 | 3.64  | 2.66 | 65.10 | -1.51 | 20.53 | 2.64  | -2.72 | 4.43  |
| 61 | -2.11 | 2.97 | 65.31 | -2.06 | 21.06 | 1.06  | -5.37 | 5.61  |
| 62 | 1.09  | 2.90 | 65.18 | -2.28 | 21.55 | 1.93  | -3.25 | 4.31  |
| 63 | 3.76  | 3.61 | 62.63 | -3.22 | 19.95 | 0.38  | -4.80 | 6.77  |
| 64 | 2.01  | 3.49 | 35.08 | -1.73 | 7.78  | -0.78 | 0.64  | -7.62 |
| 65 | 0.65  | 3.75 | 37.57 | -1.59 | 8.16  | 0.25  | 1.16  | -6.40 |
| 66 | -0.59 | 3.12 | 35.67 | -1.59 | 8.70  | -0.65 | 0.44  | -7.70 |
| 67 | 0.84  | 3.74 | 36.42 | -1.44 | 8.46  | 0.59  | 0.32  | -7.58 |
| 68 | 1.82  | 3.81 | 36.20 | -0.96 | 8.04  | 0.50  | 0.78  | -6.98 |
| 69 | -3.57 | 3.82 | 34.70 | -1.26 | 8.63  | 0.05  | 0.45  | -6.48 |
| 70 | -0.26 | 3.34 | 35.18 | -1.92 | 8.23  | -0.19 | 0.54  | -6.45 |
| 71 | 0.42  | 3.34 | 36.22 | -1.78 | 7.93  | 0.71  | 1.11  | -6.50 |
| 72 | -2.08 | 3.40 | 36.98 | -1.86 | 8.30  | -0.10 | -0.07 | -6.00 |
| 73 | 0.88  | 3.39 | 36.43 | -1.80 | 8.82  | 1.05  | 2.09  | -6.28 |
| 74 | 0.13  | 3.49 | 37.23 | -1.68 | 7.89  | 0.03  | 0.84  | -6.37 |
| 75 | 0.72  | 3.63 | 34.17 | -1.43 | 8.40  | -0.15 | 1.45  | -6.18 |
| 76 | 1.16  | 3.70 | 37.28 | -1.47 | 8.24  | 0.62  | 0.36  | -6.41 |
| 77 | -1.66 | 3.54 | 36.23 | -1.32 | 8.74  | -0.46 | 1.47  | -6.34 |
| 78 | 0.57  | 3.54 | 37.16 | -1.52 | 8.39  | 0.23  | 0.68  | -5.44 |
| 79 | 1.19  | 3.07 | 36.90 | -2.02 | 8.87  | 0.19  | 0.45  | -6.90 |

| k  | a32k  | a33k | a34k | a35k  | a36k | a37k  | a38k  | a39k  |
|----|-------|------|------|-------|------|-------|-------|-------|
| 40 | -5.34 | 1.56 | 2.37 | 14.21 | 2.38 | 29.25 | -5.24 | -4.95 |
| 41 | -5.33 | 1.96 | 2.51 | 14.89 | 2.30 | 29.91 | -4.95 | -5.64 |
| 42 | -4.23 | 1.47 | 2.50 | 14.89 | 2.39 | 30.13 | -5.49 | -5.67 |
| 43 | -5.61 | 1.39 | 2.40 | 14.13 | 2.39 | 30.04 | -5.16 | -5.43 |
| 44 | -5.26 | 1.42 | 2.47 | 14.70 | 2.47 | 29.51 | -5.46 | -5.65 |
| 45 | -4.85 | 1.38 | 2.49 | 14.53 | 2.35 | 31.15 | -5.49 | -5.35 |
| 46 | -5.17 | 1.42 | 2.66 | 14.21 | 2.40 | 31.82 | -5.15 | -5.79 |
| 47 | -5.68 | 1.57 | 2.53 | 15.07 | 2.40 | 29.24 | -5.09 | -5.72 |
| 48 | -6.00 | 2.58 | 2.50 | 14.63 | 3.20 | 38.48 | -3.62 | -5.61 |
| 49 | -6.67 | 2.35 | 2.72 | 14.89 | 3.23 | 37.63 | -3.69 | -6.09 |
| 50 | -5.81 | 2.30 | 2.52 | 15.66 | 3.28 | 38.95 | -4.26 | -6.18 |
| 51 | -5.13 | 2.44 | 2.67 | 15.54 | 3.18 | 36.41 | -4.09 | -5.84 |
| 52 | -4.42 | 1.99 | 2.54 | 15.71 | 3.13 | 39.60 | -4.01 | -6.18 |
| 53 | -5.04 | 2.06 | 2.66 | 17.16 | 3.12 | 36.76 | -4.33 | -6.06 |
| 54 | -6.10 | 2.53 | 2.50 | 16.41 | 3.27 | 37.65 | -4.36 | -6.06 |
| 55 | -5.57 | 2.26 | 2.60 | 16.96 | 3.16 | 40.02 | -3.85 | -6.00 |
| 56 | -4.10 | 1.92 | 2.64 | 14.69 | 3.17 | 39.48 | -4.48 | -6.07 |
| 57 | -4.38 | 2.29 | 2.61 | 15.62 | 3.22 | 38.06 | -3.97 | -6.03 |
| 58 | -5.50 | 2.09 | 2.58 | 16.36 | 3.27 | 40.47 | -3.28 | -5.79 |
| 59 | -4.45 | 2.46 | 2.78 | 16.60 | 3.15 | 39.11 | -3.78 | -6.16 |
| 60 | -3.34 | 2.51 | 2.52 | 15.15 | 3.23 | 38.56 | -4.02 | -5.90 |
| 61 | -5.04 | 2.20 | 2.58 | 16.64 | 3.16 | 38.05 | -4.02 | -5.69 |
| 62 | -4.07 | 2.51 | 2.61 | 16.88 | 3.27 | 38.29 | -3.92 | -6.25 |

|    |       |      |      |       |      |       |       |       |
|----|-------|------|------|-------|------|-------|-------|-------|
| 63 | -5.47 | 2.46 | 2.56 | 17.19 | 3.20 | 36.88 | -3.81 | -5.98 |
| 64 | -3.26 | 1.93 | 2.58 | 6.64  | 1.79 | 15.55 | -0.68 | -6.62 |
| 65 | -1.96 | 1.80 | 2.75 | 6.53  | 1.80 | 14.47 | -1.32 | -6.52 |
| 66 | -2.93 | 1.95 | 2.58 | 6.37  | 1.80 | 13.31 | -1.55 | -6.16 |
| 67 | -2.63 | 1.77 | 2.78 | 6.77  | 1.79 | 14.86 | -0.93 | -6.19 |
| 68 | -3.09 | 2.07 | 2.62 | 6.63  | 1.85 | 13.31 | -0.92 | -6.44 |
| 69 | -2.86 | 1.95 | 2.66 | 6.79  | 1.74 | 13.74 | -1.00 | -6.61 |
| 70 | -1.71 | 1.92 | 2.68 | 5.20  | 1.78 | 15.31 | -1.48 | -6.46 |
| 71 | -3.02 | 1.93 | 2.72 | 6.27  | 1.83 | 13.72 | -1.27 | -6.62 |
| 72 | -2.50 | 2.08 | 2.62 | 6.55  | 1.75 | 12.96 | -1.31 | -6.45 |
| 73 | -2.41 | 2.11 | 2.70 | 6.32  | 1.82 | 15.33 | -1.06 | -6.63 |
| 74 | -2.92 | 2.10 | 2.73 | 6.78  | 1.77 | 15.18 | -1.17 | -6.35 |
| 75 | -2.50 | 1.93 | 2.62 | 6.95  | 1.85 | 15.54 | -1.22 | -6.55 |
| 76 | -2.98 | 2.03 | 2.66 | 6.13  | 1.77 | 14.15 | -1.40 | -6.31 |
| 77 | -2.17 | 1.67 | 2.73 | 6.52  | 1.86 | 12.99 | -0.97 | -6.39 |
| 78 | -3.26 | 1.85 | 2.79 | 5.93  | 1.78 | 15.29 | -1.24 | -6.51 |
| 79 | -2.63 | 2.13 | 2.65 | 6.26  | 1.82 | 14.10 | -1.09 | -6.49 |

| k  | a40k  | a41k   | a42k   | a43k   | a44k   | a45k  | a46k   | a47k   |
|----|-------|--------|--------|--------|--------|-------|--------|--------|
| 40 | -1.63 | -17.33 | -15.14 | -16.17 | -29.42 | 28.83 | -1.30  | -12.03 |
| 41 | -1.62 | -16.73 | 15.47  | -15.91 | -10.45 | 27.24 | -4.18  | -15.77 |
| 42 | -1.76 | -17.69 | -4.99  | -16.03 | 12.60  | 28.26 | -1.18  | 19.16  |
| 43 | -1.73 | -16.97 | 12.09  | -15.62 | -11.85 | 26.94 | -0.37  | 0.46   |
| 44 | -1.61 | -17.12 | -5.19  | -15.90 | 2.02   | 26.26 | -0.46  | -5.74  |
| 45 | -1.70 | -17.11 | 2.87   | -15.47 | -1.41  | 27.27 | 2.06   | 2.50   |
| 46 | -1.71 | -18.19 | -5.68  | -15.33 | -1.19  | 27.62 | 1.22   | 2.53   |
| 47 | -1.69 | -17.28 | -11.69 | -15.44 | -22.75 | 27.70 | -1.62  | 1.81   |
| 48 | -2.35 | -23.98 | -5.23  | -19.65 | 22.91  | 35.28 | -12.08 | -13.87 |
| 49 | -2.43 | -27.18 | 5.93   | -18.98 | -27.08 | 38.11 | -20.01 | -4.01  |
| 50 | -2.31 | -25.96 | -1.21  | -17.47 | -3.80  | 36.94 | -16.15 | 0.74   |
| 51 | -2.43 | -25.54 | -5.39  | -18.91 | 2.69   | 37.51 | -17.32 | -10.40 |
| 52 | -2.34 | -25.63 | -0.54  | -16.74 | -13.52 | 38.11 | -12.87 | 1.63   |
| 53 | -2.28 | -26.61 | 18.39  | -19.47 | 38.54  | 36.67 | -15.14 | -8.79  |
| 54 | -2.24 | -25.23 | 0.96   | -18.66 | -4.01  | 37.97 | -12.54 | 3.16   |
| 55 | -2.46 | -26.20 | 3.85   | -20.41 | -20.02 | 38.86 | -9.05  | -6.67  |
| 56 | -2.20 | -26.94 | -5.01  | -19.81 | 20.93  | 36.49 | -12.64 | 1.44   |
| 57 | -2.24 | -26.94 | -1.68  | -18.68 | -1.71  | 36.78 | -9.21  | 13.52  |
| 58 | -2.27 | -24.52 | 0.95   | -18.90 | -12.42 | 39.65 | -21.43 | -15.86 |
| 59 | -2.33 | -25.67 | -10.70 | -19.74 | 1.12   | 39.45 | -18.24 | 7.89   |
| 60 | -2.30 | -26.36 | 11.17  | -18.81 | -2.69  | 39.28 | -17.71 | -4.19  |
| 61 | -2.29 | -24.66 | 3.78   | -19.38 | -19.97 | 38.21 | -17.79 | -5.47  |
| 62 | -2.42 | -24.80 | 11.78  | -19.14 | 3.75   | 38.29 | -17.86 | -0.73  |
| 63 | -2.44 | -24.51 | -21.78 | -18.50 | 23.88  | 36.49 | -15.04 | -0.89  |
| 64 | -1.83 | -13.40 | -11.14 | -13.47 | 6.68   | 20.28 | -11.60 | 8.74   |
| 65 | -1.62 | -15.70 | 1.68   | -12.90 | 6.44   | 19.59 | -9.48  | -7.91  |
| 66 | -1.80 | -13.18 | -15.05 | -13.18 | 1.99   | 18.97 | -9.93  | -3.74  |
| 67 | -1.63 | -13.75 | -7.33  | -13.37 | -23.16 | 18.82 | -6.33  | -1.89  |
| 68 | -1.76 | -13.83 | 8.46   | -12.67 | 5.10   | 20.29 | -9.76  | 5.80   |
| 69 | -1.74 | -13.90 | 1.52   | -12.82 | -22.21 | 18.64 | -11.50 | -0.64  |
| 70 | -1.78 | -14.92 | 1.24   | -12.89 | 4.97   | 19.29 | -7.71  | -4.83  |
| 71 | -1.90 | -13.09 | -9.94  | -13.40 | -18.68 | 19.39 | -5.61  | 6.31   |
| 72 | -1.81 | -14.30 | -9.16  | -13.84 | 13.20  | 18.53 | -6.02  | -1.96  |

|    |       |        |       |        |       |       |        |       |
|----|-------|--------|-------|--------|-------|-------|--------|-------|
| 73 | -1.76 | -14.66 | -2.27 | -12.31 | -4.24 | 18.90 | -6.41  | 14.15 |
| 74 | -1.79 | -13.88 | -6.26 | -12.21 | -0.23 | 19.15 | -10.59 | -0.15 |
| 75 | -1.64 | -13.15 | 2.92  | -13.45 | -9.89 | 19.10 | -10.12 | -5.30 |
| 76 | -1.77 | -13.07 | 4.78  | -13.52 | -1.98 | 19.49 | -8.17  | -0.11 |
| 77 | -1.67 | -13.32 | 0.73  | -13.84 | 5.22  | 20.52 | -11.56 | -5.77 |
| 78 | -1.77 | -13.99 | -4.65 | -12.55 | -8.03 | 18.97 | -7.84  | 0.31  |
| 79 | -1.70 | -13.62 | 3.32  | -13.01 | 5.30  | 20.11 | -7.49  | 7.78  |

| k  | a48k  | a49k  | a50k   | a51k   | a52k  | a53k  | a54k  | a55k |
|----|-------|-------|--------|--------|-------|-------|-------|------|
| 40 | 10.47 | 21.27 | -10.21 | 0.10   | 68.25 | -3.70 | -1.40 | 6.15 |
| 41 | 10.05 | 21.29 | -10.73 | -5.80  | 66.81 | -3.54 | -1.12 | 5.55 |
| 42 | 9.65  | 21.44 | -10.41 | 3.16   | 64.95 | -2.16 | -1.26 | 5.51 |
| 43 | 10.09 | 20.98 | -9.71  | -1.86  | 66.84 | -1.27 | -1.51 | 5.68 |
| 44 | 10.76 | 21.46 | -10.07 | -3.57  | 67.32 | -1.94 | -1.25 | 6.11 |
| 45 | 12.36 | 21.09 | -11.19 | -6.00  | 66.15 | -1.00 | -1.45 | 5.81 |
| 46 | 9.57  | 21.06 | -10.42 | -7.61  | 64.58 | -2.32 | -1.27 | 4.69 |
| 47 | 10.48 | 21.27 | -8.46  | -5.23  | 67.39 | -2.78 | -1.50 | 5.91 |
| 48 | 19.85 | 26.07 | -17.78 | 5.77   | 76.70 | -5.97 | -1.33 | 7.34 |
| 49 | 17.76 | 26.10 | -18.03 | -11.44 | 73.40 | -5.48 | -1.67 | 5.96 |
| 50 | 19.37 | 25.92 | -17.34 | -2.21  | 75.97 | -4.95 | -1.61 | 6.77 |
| 51 | 19.60 | 26.19 | -15.93 | 1.24   | 77.29 | -6.90 | -0.94 | 6.74 |
| 52 | 20.65 | 25.47 | -17.51 | -0.40  | 74.81 | -5.15 | -1.53 | 6.92 |
| 53 | 20.51 | 24.90 | -17.03 | 6.78   | 78.03 | -4.57 | -1.38 | 7.08 |
| 54 | 19.00 | 25.89 | -16.67 | -8.22  | 78.44 | -7.80 | -0.90 | 5.93 |
| 55 | 19.61 | 26.07 | -16.10 | -4.31  | 75.29 | -5.28 | -0.84 | 7.33 |
| 56 | 17.19 | 25.63 | -17.16 | -4.83  | 76.12 | -6.28 | -1.30 | 7.15 |
| 57 | 16.12 | 26.62 | -18.85 | -10.03 | 76.11 | -7.80 | -1.48 | 6.26 |
| 58 | 22.83 | 25.80 | -18.76 | 0.78   | 79.84 | -6.48 | -1.62 | 6.51 |
| 59 | 18.77 | 25.66 | -19.39 | -3.45  | 75.39 | -5.64 | -1.89 | 5.89 |
| 60 | 18.65 | 26.05 | -17.83 | 9.61   | 75.07 | -4.95 | -1.50 | 7.25 |
| 61 | 20.10 | 25.50 | -17.72 | 5.46   | 79.24 | -5.64 | -0.75 | 6.04 |
| 62 | 18.06 | 26.18 | -16.77 | -0.60  | 75.60 | -5.70 | -1.29 | 5.48 |
| 63 | 20.15 | 26.14 | -18.30 | 0.52   | 77.40 | -5.65 | -1.63 | 6.15 |
| 64 | 4.17  | 11.78 | -11.14 | -6.19  | 47.76 | 4.84  | -0.10 | 1.22 |
| 65 | 4.95  | 11.91 | -11.29 | -7.40  | 49.57 | 5.36  | -0.13 | 0.55 |
| 66 | 5.23  | 11.69 | -11.24 | 2.25   | 47.65 | 3.93  | -0.44 | 1.66 |
| 67 | 4.34  | 11.68 | -12.65 | -2.83  | 44.20 | 6.00  | -0.18 | 1.83 |
| 68 | 4.77  | 11.73 | -10.81 | -7.30  | 48.16 | 4.80  | 0.07  | 1.50 |
| 69 | 5.63  | 12.02 | -11.26 | -5.67  | 46.34 | 4.37  | -0.32 | 1.27 |
| 70 | 6.80  | 11.44 | -12.79 | 4.52   | 46.72 | 4.09  | -0.01 | 1.03 |
| 71 | 4.40  | 11.61 | -10.90 | 3.28   | 46.91 | 5.31  | -0.15 | 1.20 |
| 72 | 5.63  | 11.86 | -9.84  | -2.76  | 47.55 | 4.03  | -0.09 | 1.07 |
| 73 | 5.57  | 11.65 | -11.73 | 4.00   | 48.09 | 5.29  | -0.14 | 1.55 |
| 74 | 6.56  | 12.02 | -11.78 | 5.11   | 47.54 | 4.43  | -0.34 | 1.50 |
| 75 | 4.76  | 11.61 | -12.34 | -1.54  | 46.82 | 4.11  | -0.51 | 1.09 |
| 76 | 3.83  | 11.41 | -11.62 | 5.85   | 45.35 | 5.55  | -0.03 | 1.42 |
| 77 | 5.23  | 11.93 | -10.66 | -2.98  | 46.92 | 4.34  | -0.39 | 1.13 |
| 78 | 6.95  | 12.23 | -11.66 | 2.36   | 44.43 | 5.72  | -0.37 | 1.19 |
| 79 | 5.55  | 11.92 | -10.93 | 2.02   | 45.36 | 5.01  | -0.28 | 1.38 |

| k  | a56k   | a57k  | a58k | a59k  | a60k  | a61k | a62k    | a63k   |
|----|--------|-------|------|-------|-------|------|---------|--------|
| 40 | -13.21 | 46.06 | 0.12 | 12.76 | 40.97 | 2.21 | -5.81   | 30.19  |
| 41 | 5.60   | 43.87 | 0.12 | 12.55 | 40.42 | 2.20 | -5.66   | 10.59  |
| 42 | 1.30   | 43.78 | 0.08 | 13.33 | 42.36 | 2.34 | 104.95  | 11.44  |
| 43 | -7.10  | 45.74 | 0.12 | 12.70 | 41.80 | 2.17 | 53.45   | 1.42   |
| 44 | -15.01 | 43.41 | 0.08 | 13.65 | 40.16 | 2.34 | 7.38    | 7.04   |
| 45 | -4.50  | 44.35 | 0.10 | 13.38 | 42.10 | 2.24 | -60.08  | 29.55  |
| 46 | -6.08  | 43.93 | 0.11 | 12.88 | 41.53 | 2.30 | 43.67   | -37.28 |
| 47 | -1.42  | 43.77 | 0.10 | 12.83 | 42.02 | 2.03 | -33.22  | -1.93  |
| 48 | 7.40   | 55.67 | 0.34 | 15.31 | 49.80 | 3.58 | -32.69  | -7.42  |
| 49 | 0.03   | 55.98 | 0.35 | 15.00 | 48.94 | 3.07 | -12.52  | -12.64 |
| 50 | 13.40  | 56.33 | 0.38 | 16.14 | 50.79 | 2.89 | 79.93   | -57.05 |
| 51 | -14.93 | 53.37 | 0.35 | 15.22 | 51.09 | 2.85 | 70.33   | 27.06  |
| 52 | 2.81   | 54.09 | 0.35 | 15.14 | 49.42 | 3.15 | 3.52    | -55.33 |
| 53 | -6.01  | 56.32 | 0.35 | 14.90 | 50.48 | 2.85 | -83.24  | -53.13 |
| 54 | -22.63 | 56.03 | 0.32 | 15.27 | 49.90 | 2.96 | 131.84  | -18.47 |
| 55 | -12.38 | 57.33 | 0.34 | 14.90 | 48.67 | 2.74 | 61.03   | 8.29   |
| 56 | 15.92  | 55.65 | 0.35 | 16.30 | 50.40 | 2.95 | -128.30 | -29.08 |
| 57 | 5.86   | 54.69 | 0.34 | 14.83 | 50.68 | 3.04 | 60.23   | 14.95  |
| 58 | 6.32   | 56.35 | 0.37 | 16.09 | 50.08 | 3.27 | -61.65  | -31.02 |
| 59 | -4.18  | 54.81 | 0.39 | 15.12 | 49.89 | 2.74 | 48.95   | -25.32 |
| 60 | -1.82  | 55.12 | 0.32 | 16.16 | 49.81 | 2.53 | 23.54   | 38.64  |
| 61 | -6.60  | 56.15 | 0.31 | 14.82 | 50.61 | 3.48 | -15.81  | -62.06 |
| 62 | 13.52  | 55.60 | 0.36 | 14.65 | 50.81 | 2.75 | -55.36  | 2.84   |
| 63 | 6.26   | 55.11 | 0.35 | 16.22 | 50.09 | 2.78 | -38.96  | -9.46  |
| 64 | -6.41  | 21.29 | 0.06 | 13.13 | 25.53 | 2.19 | 3.75    | 28.78  |
| 65 | 5.40   | 20.54 | 0.03 | 12.43 | 26.21 | 2.26 | -53.85  | -56.62 |
| 66 | -1.36  | 21.26 | 0.08 | 12.95 | 24.86 | 1.82 | 19.76   | -12.86 |
| 67 | 2.78   | 19.53 | 0.06 | 13.77 | 25.62 | 1.69 | -28.92  | 2.03   |
| 68 | -2.41  | 19.89 | 0.07 | 12.87 | 24.40 | 2.26 | 4.00    | 8.27   |
| 69 | -0.42  | 20.76 | 0.08 | 13.75 | 25.35 | 2.40 | -7.41   | -16.89 |
| 70 | 2.68   | 21.11 | 0.06 | 12.80 | 26.02 | 1.63 | -34.09  | -1.22  |
| 71 | -2.07  | 20.29 | 0.06 | 13.23 | 24.64 | 2.14 | -0.73   | 27.99  |
| 72 | 13.77  | 19.43 | 0.03 | 12.95 | 25.40 | 2.05 | -41.90  | 41.49  |
| 73 | 6.38   | 21.39 | 0.07 | 12.67 | 25.45 | 2.27 | 64.46   | 10.51  |
| 74 | 14.28  | 20.99 | 0.05 | 12.33 | 26.63 | 1.89 | 72.91   | 18.93  |
| 75 | 11.50  | 20.47 | 0.07 | 13.27 | 25.94 | 1.95 | 95.35   | 4.58   |
| 76 | -2.82  | 20.91 | 0.06 | 12.71 | 25.71 | 2.04 | 58.62   | 12.36  |
| 77 | -11.93 | 21.90 | 0.02 | 13.81 | 25.86 | 2.21 | -10.84  | 12.70  |
| 78 | -11.93 | 20.49 | 0.07 | 13.10 | 25.82 | 2.10 | 20.80   | 15.62  |
| 79 | -4.90  | 21.84 | 0.05 | 13.09 | 25.78 | 1.86 | -24.02  | 21.73  |

| k  | a0k    | a1k    | a2k    | a3k   | a4k   | a5k  | a6k   | a7k   |
|----|--------|--------|--------|-------|-------|------|-------|-------|
| 80 | 237.62 | -19.42 | -15.00 | 22.16 | 76.44 | 7.36 | 55.13 | 20.11 |
| 81 | 245.81 | -19.54 | -15.13 | 18.05 | 77.32 | 7.05 | 55.20 | 19.51 |
| 82 | 239.03 | -19.32 | -14.10 | 14.18 | 76.56 | 7.65 | 54.42 | 21.17 |
| 83 | 239.85 | -19.51 | -14.78 | 16.79 | 75.94 | 7.74 | 53.67 | 19.67 |
| 84 | 236.54 | -19.33 | -14.34 | 17.57 | 76.25 | 5.88 | 54.22 | 20.28 |
| 85 | 233.79 | -19.10 | -14.73 | 18.82 | 77.02 | 7.74 | 55.42 | 19.18 |
| 86 | 244.29 | -19.38 | -14.24 | 21.06 | 76.84 | 7.05 | 53.39 | 21.57 |
| 87 | 245.46 | -19.23 | -14.63 | 19.52 | 76.89 | 6.20 | 55.06 | 19.29 |
| 88 | 237.27 | -19.52 | -15.00 | 11.51 | 75.62 | 6.84 | 54.76 | 18.98 |

|     |        |        |        |        |       |       |       |       |
|-----|--------|--------|--------|--------|-------|-------|-------|-------|
| 89  | 246.30 | -19.38 | -14.77 | 19.82  | 76.40 | 7.87  | 53.51 | 19.31 |
| 90  | 239.99 | -19.66 | -14.51 | 23.01  | 78.38 | 7.62  | 54.41 | 20.58 |
| 91  | 246.68 | -19.32 | -14.48 | 19.30  | 76.38 | 6.57  | 54.02 | 19.59 |
| 92  | 244.90 | -18.74 | -14.76 | 19.33  | 75.50 | 8.02  | 53.86 | 19.84 |
| 93  | 240.25 | -19.46 | -14.08 | 15.51  | 77.25 | 6.68  | 54.41 | 20.96 |
| 94  | 236.25 | -19.15 | -14.29 | 15.40  | 77.92 | 8.29  | 54.72 | 20.92 |
| 95  | 234.53 | -19.51 | -14.79 | 15.95  | 76.96 | 7.71  | 53.95 | 19.60 |
| 96  | 225.80 | -14.96 | -12.69 | 32.62  | 79.10 | -8.15 | 47.15 | 29.30 |
| 97  | 224.72 | -15.06 | -12.31 | 38.11  | 78.67 | -8.34 | 47.15 | 28.12 |
| 98  | 219.05 | -15.08 | -12.67 | 33.06  | 78.99 | -8.10 | 46.83 | 29.35 |
| 99  | 225.28 | -15.07 | -12.13 | 34.86  | 78.98 | -7.64 | 46.64 | 29.02 |
| 100 | 219.18 | -15.13 | -12.51 | 32.24  | 79.59 | -8.61 | 46.55 | 28.06 |
| 101 | 223.01 | -14.98 | -12.59 | 32.56  | 78.79 | -7.85 | 47.48 | 29.74 |
| 102 | 226.92 | -15.06 | -12.82 | 35.50  | 78.13 | -8.46 | 47.53 | 29.39 |
| 103 | 218.47 | -15.15 | -12.70 | 31.58  | 77.22 | -8.28 | 47.09 | 28.69 |
| 104 | 217.40 | -15.08 | -12.47 | 31.83  | 77.80 | -8.14 | 47.10 | 29.56 |
| 105 | 211.59 | -15.13 | -12.39 | 34.55  | 78.57 | -8.92 | 46.36 | 27.44 |
| 106 | 218.37 | -14.92 | -13.15 | 31.41  | 78.23 | -8.59 | 47.33 | 28.58 |
| 107 | 213.20 | -15.28 | -12.69 | 32.20  | 78.63 | -8.15 | 47.25 | 28.56 |
| 108 | 211.17 | -15.16 | -12.35 | 33.61  | 78.13 | -8.64 | 46.45 | 29.12 |
| 109 | 225.61 | -15.22 | -12.14 | 30.36  | 76.95 | -7.72 | 47.83 | 29.32 |
| 110 | 225.64 | -15.32 | -12.32 | 30.59  | 79.54 | -8.44 | 47.85 | 29.54 |
| 111 | 224.91 | -15.05 | -12.61 | 33.73  | 78.88 | -8.69 | 47.23 | 29.07 |
| 112 | -89.51 | -0.54  | 3.33   | -33.39 | 1.42  | 11.15 | 2.84  | -3.79 |
| 113 | -86.30 | -0.64  | 3.48   | -32.33 | 1.63  | 11.26 | 2.88  | -3.91 |
| 114 | -80.17 | -0.41  | 3.36   | -33.65 | 1.40  | 11.45 | 2.87  | -2.39 |
| 115 | -86.94 | -0.54  | 3.66   | -33.07 | 1.89  | 10.91 | 3.66  | -2.75 |
| 116 | -80.07 | -0.67  | 3.73   | -32.76 | 1.05  | 11.32 | 2.95  | -2.93 |
| 117 | -87.45 | -0.91  | 3.55   | -31.94 | 1.72  | 11.40 | 3.03  | -3.38 |
| 118 | -80.27 | -0.56  | 3.49   | -31.59 | 1.38  | 11.19 | 3.53  | -3.49 |
| 119 | -90.36 | -0.70  | 3.58   | -32.58 | 1.11  | 10.70 | 3.54  | -3.50 |

| k  | a8k     | a9k     | a10k  | a11k   | a12k    | a13k  | a14k   | a15k  |
|----|---------|---------|-------|--------|---------|-------|--------|-------|
| 80 | -186.45 | -104.53 | -2.49 | -80.09 | -134.45 | 10.24 | -14.72 | 95.88 |
| 81 | -192.97 | -107.51 | -2.51 | -80.49 | -133.61 | 10.69 | -14.16 | 94.57 |
| 82 | -187.71 | -108.95 | -2.46 | -79.35 | -133.97 | 10.30 | -14.36 | 97.20 |
| 83 | -196.02 | -105.38 | -2.44 | -78.86 | -135.96 | 9.68  | -14.04 | 92.95 |
| 84 | -194.61 | -105.27 | -2.45 | -79.94 | -133.74 | 10.20 | -14.61 | 95.05 |
| 85 | -198.68 | -106.99 | -2.42 | -81.39 | -132.72 | 10.63 | -14.47 | 94.83 |
| 86 | -194.46 | -107.19 | -2.42 | -82.50 | -134.72 | 10.43 | -14.56 | 95.38 |
| 87 | -200.13 | -106.43 | -2.49 | -78.22 | -134.53 | 10.86 | -14.80 | 93.82 |
| 88 | -193.98 | -106.70 | -2.47 | -81.14 | -133.25 | 10.47 | -14.52 | 93.65 |
| 89 | -196.03 | -107.53 | -2.45 | -80.79 | -133.78 | 9.95  | -14.23 | 95.79 |
| 90 | -197.87 | -106.99 | -2.47 | -79.13 | -135.12 | 10.49 | -14.67 | 92.70 |
| 91 | -196.90 | -110.92 | -2.43 | -82.52 | -133.55 | 10.59 | -14.29 | 95.36 |
| 92 | -197.47 | -106.85 | -2.47 | -82.24 | -133.42 | 10.28 | -14.49 | 95.94 |
| 93 | -195.18 | -104.82 | -2.46 | -82.04 | -134.55 | 10.47 | -14.71 | 94.09 |
| 94 | -195.83 | -106.77 | -2.42 | -79.90 | -131.30 | 10.06 | -14.41 | 92.75 |
| 95 | -195.81 | -104.92 | -2.43 | -82.97 | -133.36 | 10.49 | -14.20 | 92.97 |
| 96 | -181.47 | -115.06 | -2.02 | -70.47 | -131.47 | 11.38 | -17.41 | 86.99 |
| 97 | -188.01 | -114.36 | -2.00 | -72.63 | -131.68 | 10.99 | -17.24 | 89.31 |
| 98 | -180.37 | -115.61 | -1.98 | -70.70 | -131.42 | 10.72 | -17.68 | 87.15 |

|     |         |         |       |        |         |       |        |        |
|-----|---------|---------|-------|--------|---------|-------|--------|--------|
| 99  | -180.37 | -114.48 | -2.02 | -67.98 | -131.22 | 10.86 | -17.79 | 89.21  |
| 100 | -182.21 | -111.90 | -2.00 | -70.02 | -132.86 | 10.41 | -17.62 | 92.41  |
| 101 | -178.91 | -117.06 | -2.03 | -71.92 | -129.11 | 11.31 | -17.42 | 90.89  |
| 102 | -181.48 | -113.66 | -2.01 | -69.24 | -130.05 | 11.09 | -17.57 | 94.75  |
| 103 | -180.28 | -117.23 | -2.03 | -71.54 | -128.29 | 10.24 | -17.24 | 92.37  |
| 104 | -185.23 | -115.73 | -2.01 | -69.82 | -129.67 | 11.10 | -17.39 | 87.03  |
| 105 | -182.95 | -111.77 | -2.03 | -71.93 | -130.92 | 11.05 | -17.50 | 90.66  |
| 106 | -180.42 | -115.91 | -2.02 | -71.93 | -132.11 | 10.58 | -17.60 | 89.10  |
| 107 | -184.49 | -115.82 | -2.04 | -70.93 | -133.27 | 10.63 | -17.39 | 92.69  |
| 108 | -182.04 | -114.89 | -2.02 | -73.68 | -130.74 | 10.27 | -17.30 | 88.48  |
| 109 | -187.39 | -116.94 | -2.00 | -70.45 | -133.71 | 10.58 | -17.37 | 88.44  |
| 110 | -183.45 | -115.03 | -2.00 | -72.48 | -131.80 | 10.55 | -17.51 | 87.44  |
| 111 | -178.97 | -116.76 | -2.03 | -70.39 | -131.76 | 11.18 | -17.83 | 90.02  |
| 112 | 53.55   | 15.02   | -0.01 | 26.67  | -5.88   | -9.21 | 0.67   | -27.37 |
| 113 | 55.15   | 13.52   | -0.06 | 23.88  | -7.50   | -9.22 | 0.32   | -26.84 |
| 114 | 55.34   | 13.76   | -0.05 | 25.76  | -6.52   | -9.00 | 0.45   | -23.41 |
| 115 | 56.13   | 12.27   | -0.04 | 26.72  | -7.43   | -8.65 | 0.23   | -25.40 |
| 116 | 58.20   | 12.54   | 0.00  | 27.33  | -8.21   | -9.11 | 0.39   | -25.47 |
| 117 | 53.40   | 11.55   | -0.04 | 24.89  | -6.52   | -9.32 | 0.47   | -26.03 |
| 118 | 57.18   | 15.39   | -0.04 | 26.82  | -6.54   | -9.12 | 0.17   | -26.77 |
| 119 | 53.75   | 13.14   | -0.04 | 25.75  | -5.81   | -9.26 | 0.44   | -26.64 |

| k   | a16k  | a17k  | a18k  | a19k   | a20k  | a21k   | a22k   | a23k |
|-----|-------|-------|-------|--------|-------|--------|--------|------|
| 80  | 1.25  | 31.48 | 36.45 | 5.03   | -4.19 | -13.70 | 34.65  | 3.63 |
| 81  | 1.58  | 31.20 | 35.32 | 3.71   | -4.06 | -13.75 | -25.86 | 3.60 |
| 82  | 0.39  | 32.65 | 36.13 | -66.31 | -4.02 | -13.44 | 2.89   | 3.60 |
| 83  | 1.13  | 32.53 | 35.72 | 9.35   | -4.30 | -13.69 | -12.59 | 3.59 |
| 84  | 1.32  | 32.05 | 35.43 | 77.97  | -4.23 | -13.75 | -11.80 | 3.64 |
| 85  | 0.40  | 32.67 | 35.83 | 10.64  | -4.27 | -13.54 | -13.35 | 3.59 |
| 86  | 1.24  | 31.25 | 36.79 | -0.20  | -4.24 | -13.69 | -7.93  | 3.61 |
| 87  | 0.63  | 32.68 | 36.03 | -9.71  | -3.66 | -13.65 | 28.19  | 3.65 |
| 88  | 1.29  | 31.34 | 35.88 | 45.80  | -4.03 | -13.27 | -1.63  | 3.60 |
| 89  | -1.15 | 32.21 | 35.54 | 38.92  | -4.00 | -13.62 | -23.70 | 3.68 |
| 90  | -0.43 | 31.99 | 35.51 | -10.68 | -4.34 | -13.83 | -29.95 | 3.70 |
| 91  | 0.06  | 31.39 | 37.08 | 16.14  | -3.99 | -13.78 | -12.77 | 3.64 |
| 92  | -2.56 | 32.15 | 36.26 | -18.55 | -3.82 | -13.57 | -2.25  | 3.67 |
| 93  | -1.69 | 31.85 | 35.62 | -46.41 | -3.96 | -13.64 | -21.34 | 3.64 |
| 94  | 0.62  | 32.69 | 35.98 | -64.99 | -4.18 | -13.56 | -7.76  | 3.68 |
| 95  | 2.32  | 32.37 | 36.68 | 2.72   | -3.94 | -13.79 | -10.48 | 3.62 |
| 96  | 1.91  | 27.17 | 35.46 | 31.42  | -6.83 | -14.35 | -4.68  | 3.04 |
| 97  | -0.84 | 28.04 | 36.09 | 66.63  | -6.64 | -14.40 | -26.19 | 3.11 |
| 98  | -0.22 | 28.28 | 35.61 | -27.73 | -6.88 | -14.78 | -1.46  | 3.08 |
| 99  | 0.40  | 28.54 | 36.22 | -57.05 | -6.45 | -14.68 | 10.02  | 3.12 |
| 100 | -1.97 | 28.51 | 35.25 | -1.24  | -6.95 | -14.54 | -11.66 | 3.11 |
| 101 | -0.27 | 28.64 | 35.64 | -32.40 | -6.88 | -14.72 | -4.98  | 3.09 |
| 102 | -0.47 | 27.32 | 35.97 | 27.27  | -6.83 | -14.57 | 10.59  | 3.08 |
| 103 | 0.44  | 28.14 | 35.71 | -30.29 | -7.15 | -14.77 | -9.52  | 3.05 |
| 104 | 1.54  | 28.04 | 35.70 | 17.22  | -7.02 | -14.61 | -8.51  | 3.08 |
| 105 | 1.01  | 27.15 | 36.02 | 15.69  | -6.67 | -14.79 | -14.83 | 3.10 |
| 106 | -0.91 | 27.98 | 35.25 | 12.21  | -6.68 | -14.71 | -3.81  | 3.04 |
| 107 | 0.73  | 26.90 | 35.44 | -45.54 | -7.21 | -14.30 | -8.87  | 3.12 |
| 108 | 0.21  | 28.51 | 35.12 | -4.38  | -6.82 | -14.75 | -32.65 | 3.15 |

|     |       |       |       |        |       |        |       |      |
|-----|-------|-------|-------|--------|-------|--------|-------|------|
| 109 | 1.95  | 28.09 | 35.19 | -12.43 | -7.11 | -14.39 | 17.18 | 3.04 |
| 110 | -0.93 | 28.33 | 36.14 | -55.64 | -6.91 | -14.73 | 9.19  | 3.13 |
| 111 | 1.20  | 28.73 | 35.63 | -23.48 | -6.74 | -14.51 | 0.11  | 3.05 |
| 112 | -0.02 | -4.26 | 4.39  | -21.84 | -1.95 | -1.90  | 4.25  | 0.50 |
| 113 | -0.19 | -5.10 | 4.80  | -4.81  | -1.96 | -1.52  | 0.62  | 0.54 |
| 114 | -0.55 | -5.07 | 4.62  | 20.67  | -2.08 | -1.80  | 1.84  | 0.56 |
| 115 | -0.12 | -4.51 | 4.51  | 41.62  | -1.85 | -1.93  | 2.57  | 0.56 |
| 116 | -0.88 | -3.89 | 4.43  | -8.71  | -1.84 | -1.74  | -0.13 | 0.55 |
| 117 | -0.21 | -4.95 | 4.50  | -8.92  | -1.94 | -1.54  | -3.25 | 0.54 |
| 118 | -0.11 | -4.62 | 4.46  | 15.90  | -1.87 | -1.91  | 1.65  | 0.54 |
| 119 | -0.29 | -4.89 | 4.58  | -6.43  | -1.68 | -1.77  | -5.02 | 0.52 |

| k   | a24k  | a25k  | a26k   | a27k   | a28k  | a29k   | a30k   | a31k   |
|-----|-------|-------|--------|--------|-------|--------|--------|--------|
| 80  | -0.72 | 13.41 | 156.23 | -22.05 | 54.69 | -21.12 | -36.70 | -16.35 |
| 81  | 1.23  | 12.76 | 154.12 | -21.44 | 53.34 | -21.60 | -37.59 | -15.05 |
| 82  | 1.64  | 13.48 | 158.87 | -21.61 | 53.88 | -20.64 | -36.38 | -19.94 |
| 83  | -0.66 | 13.19 | 153.77 | -22.15 | 54.43 | -20.25 | -34.46 | -17.11 |
| 84  | 0.56  | 13.61 | 156.37 | -21.57 | 54.48 | -19.82 | -35.80 | -16.25 |
| 85  | -0.14 | 13.52 | 158.07 | -21.30 | 54.41 | -20.91 | -34.52 | -12.68 |
| 86  | 0.41  | 13.54 | 156.75 | -21.79 | 53.12 | -20.37 | -36.33 | -13.44 |
| 87  | -1.00 | 12.92 | 156.99 | -21.66 | 53.95 | -19.99 | -35.01 | -12.89 |
| 88  | -0.16 | 14.20 | 155.55 | -21.40 | 54.41 | -20.89 | -34.07 | -15.57 |
| 89  | -1.85 | 12.45 | 156.90 | -21.26 | 53.80 | -21.84 | -36.75 | -17.41 |
| 90  | 0.51  | 13.56 | 157.68 | -21.81 | 54.28 | -20.43 | -34.55 | -16.46 |
| 91  | 2.46  | 13.10 | 155.86 | -22.04 | 54.75 | -20.86 | -34.88 | -13.51 |
| 92  | -1.96 | 13.68 | 155.14 | -22.09 | 53.98 | -20.80 | -37.23 | -20.04 |
| 93  | 4.18  | 13.40 | 158.36 | -21.82 | 54.19 | -20.46 | -36.36 | -16.43 |
| 94  | 3.18  | 13.43 | 156.65 | -22.12 | 53.60 | -20.05 | -35.90 | -16.26 |
| 95  | -1.71 | 13.55 | 155.07 | -22.08 | 53.90 | -20.33 | -36.81 | -16.48 |
| 96  | -1.87 | 11.55 | 132.79 | -19.54 | 51.83 | -3.17  | -27.20 | 2.49   |
| 97  | 1.90  | 11.74 | 132.36 | -19.78 | 51.36 | -2.55  | -26.79 | -1.40  |
| 98  | 2.83  | 11.98 | 133.42 | -19.47 | 52.29 | -2.64  | -26.69 | 2.37   |
| 99  | -2.49 | 11.82 | 130.84 | -19.85 | 52.62 | -2.99  | -27.51 | 0.39   |
| 100 | -2.51 | 12.23 | 132.00 | -19.92 | 52.38 | -2.79  | -27.31 | 1.30   |
| 101 | -0.33 | 11.25 | 135.64 | -19.17 | 52.55 | -3.21  | -28.44 | 0.79   |
| 102 | -0.66 | 11.59 | 132.50 | -19.63 | 51.89 | -2.65  | -28.62 | 0.35   |
| 103 | 4.76  | 11.46 | 133.48 | -19.67 | 51.71 | -3.21  | -25.00 | 0.87   |
| 104 | 3.28  | 11.84 | 132.09 | -19.71 | 52.27 | -3.36  | -27.45 | 2.13   |
| 105 | 0.21  | 11.47 | 129.46 | -19.72 | 51.54 | -2.97  | -27.22 | -2.11  |
| 106 | 1.65  | 12.36 | 134.76 | -19.93 | 52.29 | -2.91  | -28.74 | 3.66   |
| 107 | 5.53  | 11.96 | 132.76 | -19.63 | 52.03 | -3.21  | -29.41 | -1.15  |
| 108 | -3.77 | 11.16 | 131.28 | -19.55 | 51.37 | -2.87  | -27.01 | 1.87   |
| 109 | -3.12 | 11.83 | 134.62 | -19.65 | 52.18 | -2.48  | -26.46 | 0.83   |
| 110 | 2.85  | 12.08 | 133.21 | -19.44 | 52.47 | -3.09  | -27.62 | -1.18  |
| 111 | 0.07  | 11.77 | 131.28 | -19.90 | 52.21 | -2.69  | -27.81 | 2.83   |
| 112 | 2.49  | -2.66 | 30.15  | -1.56  | 4.27  | -0.80  | -2.30  | -49.05 |
| 113 | -0.21 | -3.10 | 31.04  | -1.60  | 4.31  | -0.43  | -2.86  | -49.25 |
| 114 | 1.94  | -2.82 | 31.03  | -1.86  | 4.12  | -1.00  | -2.97  | -50.39 |
| 115 | 1.08  | -2.63 | 31.21  | -1.55  | 3.77  | -0.67  | -3.06  | -50.02 |
| 116 | -0.40 | -2.46 | 31.05  | -1.70  | 4.43  | -0.82  | -2.46  | -50.67 |
| 117 | -0.06 | -2.74 | 31.50  | -1.52  | 4.41  | -0.77  | -2.77  | -50.13 |
| 118 | -0.80 | -2.66 | 31.91  | -1.62  | 3.94  | -0.64  | -2.97  | -49.29 |

119 | -0.25 | -2.75 | 32.83 | -1.54 | 3.88 | -0.49 | -2.99 | -49.55

| k   | a32k    | a33k | a34k  | a35k  | a36k  | a37k   | a38k  | a39k  |
|-----|---------|------|-------|-------|-------|--------|-------|-------|
| 80  | -95.69  | 0.82 | -1.10 | 46.86 | -0.54 | 130.43 | -0.31 | -6.35 |
| 81  | -95.93  | 0.92 | -0.98 | 46.06 | -0.58 | 127.11 | 0.13  | -6.43 |
| 82  | -96.16  | 0.86 | -1.01 | 45.27 | -0.54 | 127.80 | -0.38 | -6.37 |
| 83  | -95.92  | 0.73 | -0.94 | 45.00 | -0.54 | 128.49 | 0.19  | -6.34 |
| 84  | -95.27  | 0.81 | -1.04 | 44.91 | -0.64 | 126.65 | -0.16 | -6.39 |
| 85  | -92.81  | 0.66 | -0.86 | 47.10 | -0.53 | 130.24 | 0.06  | -6.42 |
| 86  | -96.29  | 0.72 | -0.91 | 45.81 | -0.46 | 128.86 | -0.40 | -6.39 |
| 87  | -96.00  | 1.01 | -0.93 | 46.00 | -0.58 | 128.57 | 0.25  | -6.31 |
| 88  | -94.99  | 0.62 | -0.94 | 45.13 | -0.55 | 128.39 | 0.16  | -6.34 |
| 89  | -96.60  | 0.78 | -0.89 | 45.47 | -0.52 | 126.84 | -0.11 | -6.40 |
| 90  | -94.71  | 1.23 | -1.19 | 45.58 | -0.62 | 128.54 | -0.77 | -6.34 |
| 91  | -94.84  | 0.65 | -1.13 | 45.76 | -0.67 | 128.77 | -0.04 | -6.34 |
| 92  | -98.94  | 0.90 | -1.10 | 46.93 | -0.58 | 126.04 | 0.20  | -6.35 |
| 93  | -95.55  | 0.79 | -1.08 | 45.13 | -0.51 | 126.68 | 0.11  | -6.38 |
| 94  | -95.94  | 0.89 | -0.91 | 46.23 | -0.61 | 127.94 | -0.21 | -6.32 |
| 95  | -95.34  | 1.31 | -1.05 | 44.16 | -0.61 | 129.03 | -0.25 | -6.34 |
| 96  | -107.90 | 0.76 | 0.00  | 43.98 | -0.72 | 132.70 | 4.65  | -5.94 |
| 97  | -107.59 | 0.99 | -0.20 | 44.15 | -0.85 | 132.48 | 4.73  | -5.90 |
| 98  | -107.87 | 0.85 | -0.02 | 44.18 | -0.77 | 134.56 | 4.47  | -5.93 |
| 99  | -107.13 | 0.84 | -0.11 | 44.50 | -0.88 | 132.73 | 5.06  | -5.89 |
| 100 | -107.14 | 0.81 | -0.29 | 43.22 | -0.80 | 132.40 | 4.81  | -5.89 |
| 101 | -107.45 | 1.18 | -0.09 | 44.08 | -0.83 | 129.23 | 4.52  | -5.89 |
| 102 | -107.72 | 0.94 | -0.04 | 43.68 | -0.76 | 128.11 | 4.76  | -5.93 |
| 103 | -108.87 | 0.85 | -0.20 | 44.01 | -0.84 | 131.58 | 4.83  | -5.90 |
| 104 | -106.70 | 0.96 | -0.25 | 42.66 | -0.82 | 129.83 | 5.01  | -5.99 |
| 105 | -106.40 | 0.98 | -0.08 | 43.16 | -0.85 | 130.13 | 4.95  | -5.86 |
| 106 | -106.63 | 1.09 | -0.06 | 44.91 | -0.82 | 129.33 | 4.92  | -5.93 |
| 107 | -107.55 | 1.00 | -0.13 | 43.78 | -0.79 | 131.18 | 4.41  | -6.02 |
| 108 | -105.69 | 1.12 | -0.27 | 44.38 | -0.87 | 131.80 | 5.34  | -5.86 |
| 109 | -105.87 | 1.07 | 0.08  | 43.11 | -0.79 | 133.55 | 4.85  | -5.86 |
| 110 | -107.71 | 0.90 | -0.20 | 43.99 | -0.87 | 133.70 | 4.92  | -6.02 |
| 111 | -106.95 | 0.78 | -0.10 | 44.61 | -0.76 | 130.77 | 4.70  | -5.93 |
| 112 | -4.44   | 1.70 | 1.70  | 0.99  | -0.98 | -0.86  | 2.56  | -2.05 |
| 113 | -4.14   | 1.83 | 1.66  | 1.66  | -0.97 | -1.69  | 2.43  | -2.09 |
| 114 | -4.10   | 1.81 | 1.72  | 0.63  | -0.94 | -2.98  | 2.44  | -1.99 |
| 115 | -4.14   | 1.83 | 1.69  | 0.96  | -0.99 | -3.20  | 2.68  | -2.10 |
| 116 | -4.28   | 1.80 | 1.62  | 0.80  | -0.98 | -3.96  | 2.41  | -2.02 |
| 117 | -3.84   | 1.93 | 1.78  | 0.41  | -1.00 | -4.36  | 2.44  | -2.03 |
| 118 | -4.50   | 1.70 | 1.76  | 0.90  | -1.00 | -2.13  | 2.54  | -2.05 |
| 119 | -4.48   | 1.88 | 1.66  | 0.14  | -0.95 | -2.51  | 2.49  | -2.04 |

| k  | a40k | a41k  | a42k  | a43k   | a44k   | a45k   | a46k   | a47k  |
|----|------|-------|-------|--------|--------|--------|--------|-------|
| 80 | 3.40 | 19.84 | 3.52  | -78.11 | -37.18 | -14.13 | -71.01 | 6.01  |
| 81 | 3.35 | 20.61 | -0.60 | -78.24 | -9.91  | -13.29 | -75.61 | -1.14 |
| 82 | 3.41 | 20.64 | 0.82  | -76.73 | 19.98  | -13.42 | -70.88 | -6.79 |
| 83 | 3.46 | 20.20 | -0.67 | -78.09 | -14.48 | -13.16 | -78.82 | 0.37  |
| 84 | 3.44 | 21.50 | 0.15  | -78.53 | -27.77 | -12.11 | -74.33 | -4.33 |



|     |      |       |       |        |        |        |        |        |
|-----|------|-------|-------|--------|--------|--------|--------|--------|
| 85  | 3.42 | 20.07 | -1.55 | -77.34 | -15.72 | -13.50 | -71.87 | -5.76  |
| 86  | 3.47 | 19.94 | -3.51 | -80.03 | 9.79   | -13.72 | -76.54 | -2.25  |
| 87  | 3.49 | 20.55 | 6.58  | -77.18 | 14.38  | -13.78 | -73.69 | -8.28  |
| 88  | 3.47 | 20.61 | 10.20 | -77.82 | -7.27  | -12.83 | -76.26 | -5.58  |
| 89  | 3.41 | 20.55 | 2.07  | -76.65 | 5.56   | -12.96 | -75.26 | -0.05  |
| 90  | 3.43 | 20.12 | -1.73 | -76.33 | -3.90  | -13.35 | -72.26 | 5.49   |
| 91  | 3.44 | 20.60 | 0.77  | -77.71 | -11.89 | -13.52 | -68.06 | -2.05  |
| 92  | 3.46 | 20.20 | 3.82  | -77.63 | -0.28  | -12.97 | -79.25 | -10.64 |
| 93  | 3.43 | 20.64 | 1.29  | -77.80 | -18.28 | -12.80 | -72.32 | 0.38   |
| 94  | 3.44 | 20.07 | 0.60  | -76.68 | 15.97  | -13.62 | -76.04 | 6.19   |
| 95  | 3.41 | 20.74 | -1.12 | -77.81 | 2.76   | -13.11 | -72.34 | -2.04  |
| 96  | 3.65 | 25.16 | 2.76  | -72.70 | -2.57  | -15.00 | -62.60 | -2.69  |
| 97  | 3.65 | 25.10 | 3.51  | -72.06 | -1.31  | -16.34 | -66.97 | 5.31   |
| 98  | 3.66 | 25.35 | 0.27  | -73.24 | 8.88   | -15.21 | -61.74 | 2.10   |
| 99  | 3.62 | 25.43 | -5.10 | -73.18 | -11.54 | -15.52 | -62.32 | 1.04   |
| 100 | 3.61 | 25.28 | 0.40  | -72.53 | -1.10  | -15.47 | -70.69 | -4.99  |
| 101 | 3.62 | 24.49 | -3.88 | -73.96 | -0.97  | -15.72 | -68.40 | 2.03   |
| 102 | 3.58 | 25.52 | 2.92  | -73.28 | 4.49   | -15.85 | -61.30 | -5.42  |
| 103 | 3.58 | 25.33 | 1.26  | -73.93 | -13.86 | -15.93 | -64.63 | -4.88  |
| 104 | 3.65 | 25.41 | 3.75  | -73.70 | 4.01   | -15.94 | -63.18 | 5.70   |
| 105 | 3.65 | 25.65 | -0.10 | -71.82 | -4.99  | -15.83 | -71.39 | -2.33  |
| 106 | 3.63 | 24.95 | 1.65  | -72.28 | -7.68  | -15.71 | -61.19 | 0.91   |
| 107 | 3.64 | 25.07 | 0.78  | -73.83 | -7.90  | -15.93 | -61.00 | -4.62  |
| 108 | 3.65 | 24.83 | -0.31 | -74.01 | 5.44   | -15.62 | -62.16 | 0.71   |
| 109 | 3.62 | 24.79 | 1.58  | -72.87 | -16.26 | -15.41 | -54.83 | -3.32  |
| 110 | 3.57 | 24.99 | 4.36  | -73.64 | 3.01   | -16.03 | -57.92 | 2.51   |
| 111 | 3.63 | 24.88 | 2.53  | -72.63 | -2.20  | -15.87 | -69.05 | 2.72   |
| 112 | 0.31 | 4.79  | -0.41 | -1.00  | 4.16   | -7.16  | 62.77  | -1.29  |
| 113 | 0.36 | 4.94  | -1.85 | -0.77  | -0.50  | -6.81  | 58.70  | 2.31   |
| 114 | 0.36 | 5.23  | 1.31  | -1.09  | -3.95  | -6.47  | 60.91  | -2.13  |
| 115 | 0.37 | 5.08  | -0.23 | -1.37  | -5.48  | -6.74  | 63.68  | 2.69   |
| 116 | 0.37 | 4.81  | -0.94 | -1.59  | -1.18  | -7.42  | 64.90  | 2.51   |
| 117 | 0.32 | 5.26  | -0.40 | -1.36  | 7.83   | -6.88  | 69.96  | 1.83   |
| 118 | 0.34 | 4.88  | -1.21 | -1.55  | -8.41  | -6.50  | 60.24  | 2.64   |
| 119 | 0.30 | 5.04  | -0.69 | -1.16  | -8.28  | -6.90  | 62.01  | -0.35  |

| k  | a48k  | a49k   | a50k   | a51k  | a52k   | a53k   | a54k  | a55k   |
|----|-------|--------|--------|-------|--------|--------|-------|--------|
| 80 | 35.18 | -18.92 | -15.82 | -4.04 | 123.00 | -19.42 | -1.04 | -51.71 |
| 81 | 34.09 | -18.91 | -16.07 | 6.61  | 118.59 | -17.87 | -1.15 | -52.53 |
| 82 | 36.45 | -18.80 | -17.25 | 2.41  | 120.53 | -15.35 | -0.15 | -51.71 |
| 83 | 35.43 | -19.22 | -15.75 | -0.26 | 121.11 | -14.13 | -1.33 | -51.60 |
| 84 | 35.61 | -19.31 | -15.80 | 3.78  | 122.22 | -13.50 | -0.46 | -51.12 |
| 85 | 35.84 | -19.29 | -15.60 | -3.07 | 113.52 | -19.69 | -1.38 | -51.49 |
| 86 | 34.84 | -19.11 | -15.75 | 6.88  | 119.39 | -19.36 | -1.24 | -51.76 |
| 87 | 31.00 | -18.93 | -16.37 | 2.25  | 118.69 | -14.80 | -0.99 | -52.10 |
| 88 | 34.90 | -19.56 | -16.97 | -5.46 | 121.17 | -18.96 | -1.06 | -52.00 |
| 89 | 34.10 | -19.04 | -15.91 | 1.89  | 118.31 | -15.26 | -0.99 | -51.50 |
| 90 | 34.94 | -18.97 | -16.30 | 5.87  | 119.97 | -19.67 | -1.19 | -52.26 |
| 91 | 29.87 | -19.33 | -16.96 | 4.69  | 119.99 | -15.82 | -1.76 | -51.64 |
| 92 | 34.26 | -18.90 | -16.35 | -0.85 | 119.09 | -13.64 | -1.06 | -51.17 |
| 93 | 35.63 | -18.87 | -16.09 | -2.07 | 123.06 | -16.23 | -0.80 | -51.82 |
| 94 | 37.07 | -19.29 | -16.14 | 11.96 | 121.62 | -15.06 | -1.21 | -51.56 |

|     |       |        |        |        |        |        |       |        |
|-----|-------|--------|--------|--------|--------|--------|-------|--------|
| 95  | 36.78 | -19.43 | -16.06 | -2.11  | 119.46 | -17.76 | -1.23 | -51.70 |
| 96  | 21.23 | -22.94 | -15.56 | -2.98  | 85.63  | -20.85 | 2.54  | -46.27 |
| 97  | 24.91 | -22.30 | -14.76 | 2.60   | 81.48  | -21.58 | 2.96  | -44.93 |
| 98  | 21.70 | -22.33 | -14.50 | -3.51  | 88.57  | -21.05 | 1.94  | -46.62 |
| 99  | 22.56 | -21.99 | -14.26 | -10.16 | 85.86  | -20.91 | 2.38  | -45.87 |
| 100 | 22.17 | -22.94 | -14.66 | 8.34   | 86.53  | -21.30 | 2.37  | -46.47 |
| 101 | 21.08 | -23.03 | -14.09 | -0.27  | 87.85  | -22.21 | 2.51  | -46.73 |
| 102 | 19.67 | -22.16 | -14.80 | 3.10   | 85.87  | -21.61 | 2.82  | -47.10 |
| 103 | 22.49 | -22.60 | -14.67 | 3.54   | 87.94  | -21.15 | 2.48  | -46.57 |
| 104 | 22.32 | -22.83 | -15.55 | -3.03  | 89.65  | -19.85 | 2.31  | -46.25 |
| 105 | 22.19 | -22.25 | -14.09 | 5.15   | 84.82  | -18.96 | 2.62  | -47.25 |
| 106 | 21.33 | -22.63 | -15.21 | -6.48  | 86.04  | -20.04 | 2.40  | -47.84 |
| 107 | 21.61 | -22.56 | -13.73 | 2.23   | 87.92  | -19.20 | 2.10  | -46.74 |
| 108 | 19.32 | -22.17 | -15.69 | -1.33  | 86.07  | -18.90 | 2.44  | -46.14 |
| 109 | 21.27 | -22.66 | -13.80 | 0.11   | 86.80  | -20.63 | 2.40  | -46.11 |
| 110 | 21.06 | -22.97 | -14.68 | -2.90  | 86.30  | -22.66 | 2.28  | -46.75 |
| 111 | 22.19 | -22.73 | -15.73 | -5.40  | 87.87  | -19.19 | 2.68  | -46.05 |
| 112 | 2.40  | -6.68  | 6.28   | -2.72  | 70.14  | 32.73  | -4.02 | -3.00  |
| 113 | 1.66  | -6.52  | 6.68   | -1.88  | 71.24  | 32.05  | -4.00 | -1.76  |
| 114 | 2.74  | -6.41  | 6.80   | -2.46  | 72.66  | 32.14  | -4.04 | -3.13  |
| 115 | 2.97  | -6.78  | 6.95   | -1.73  | 71.59  | 31.97  | -3.94 | -3.26  |
| 116 | 1.87  | -6.57  | 6.62   | -2.56  | 71.55  | 32.38  | -3.96 | -3.04  |
| 117 | 2.38  | -6.59  | 7.29   | 0.80   | 73.77  | 32.78  | -4.12 | -2.99  |
| 118 | 2.23  | -6.52  | 7.40   | 4.06   | 73.25  | 32.26  | -4.00 | -2.40  |
| 119 | 2.04  | -6.41  | 6.41   | -1.61  | 69.82  | 31.86  | -3.80 | -3.61  |

| k   | a56k   | a57k   | a58k  | a59k  | a60k  | a61k  | a62k   | a63k    |
|-----|--------|--------|-------|-------|-------|-------|--------|---------|
| 80  | 7.22   | 128.52 | -1.88 | 43.51 | 39.31 | 15.41 | -14.23 | 23.79   |
| 81  | 16.34  | 128.59 | -1.91 | 43.36 | 39.03 | 15.48 | -28.20 | 5.78    |
| 82  | -6.02  | 129.83 | -1.88 | 43.69 | 37.71 | 15.66 | -12.79 | -13.12  |
| 83  | 0.26   | 129.91 | -1.89 | 43.46 | 38.12 | 15.62 | 39.21  | 11.40   |
| 84  | -13.98 | 128.38 | -1.87 | 43.41 | 37.06 | 15.59 | 13.00  | -4.79   |
| 85  | -20.46 | 130.06 | -1.87 | 43.02 | 38.18 | 15.67 | -23.98 | -72.21  |
| 86  | -2.59  | 127.98 | -1.90 | 43.07 | 37.21 | 15.58 | -30.51 | 7.52    |
| 87  | 11.87  | 128.01 | -1.88 | 43.97 | 37.46 | 15.56 | -27.53 | 39.33   |
| 88  | -8.42  | 128.67 | -1.91 | 43.50 | 40.10 | 15.60 | -39.33 | -13.06  |
| 89  | -19.59 | 130.39 | -1.86 | 43.48 | 38.13 | 15.66 | -15.58 | 38.94   |
| 90  | 4.00   | 129.94 | -1.85 | 43.36 | 37.51 | 15.62 | -16.66 | 20.91   |
| 91  | -15.39 | 128.60 | -1.89 | 43.80 | 38.56 | 15.60 | -8.93  | 13.06   |
| 92  | 13.67  | 129.15 | -1.87 | 42.93 | 37.77 | 15.50 | -15.23 | 5.92    |
| 93  | 3.75   | 129.33 | -1.92 | 43.30 | 37.36 | 15.57 | 3.32   | -17.65  |
| 94  | -15.76 | 130.12 | -1.87 | 43.29 | 37.20 | 15.39 | -11.95 | -15.64  |
| 95  | -20.12 | 129.29 | -1.85 | 43.60 | 40.08 | 15.90 | -60.06 | -10.83  |
| 96  | -17.10 | 114.33 | -2.02 | 39.62 | 9.75  | 12.73 | 52.83  | -29.46  |
| 97  | 0.00   | 113.68 | -2.03 | 39.64 | 9.45  | 12.67 | -31.55 | 15.89   |
| 98  | -5.74  | 115.79 | -2.01 | 39.91 | 10.43 | 13.02 | 45.48  | -21.84  |
| 99  | 1.37   | 115.20 | -2.04 | 40.55 | 7.81  | 12.62 | 7.80   | -103.58 |
| 100 | 4.74   | 115.96 | -2.03 | 40.13 | 11.11 | 12.59 | 40.56  | -60.33  |
| 101 | 9.00   | 115.12 | -2.03 | 40.05 | 9.51  | 12.81 | 8.19   | 18.79   |
| 102 | 15.33  | 113.84 | -2.02 | 39.69 | 9.57  | 12.73 | 29.45  | 79.46   |
| 103 | 9.99   | 114.81 | -2.04 | 40.46 | 9.24  | 12.55 | -24.76 | -12.46  |
| 104 | 2.29   | 114.70 | -2.04 | 40.01 | 10.31 | 12.96 | 8.96   | 24.34   |

|     |        |        |       |       |       |       |        |        |
|-----|--------|--------|-------|-------|-------|-------|--------|--------|
| 105 | -3.84  | 114.69 | -2.03 | 39.86 | 9.44  | 12.77 | -20.44 | -33.68 |
| 106 | -16.27 | 115.65 | -2.04 | 40.05 | 9.97  | 12.69 | 31.92  | -4.97  |
| 107 | -8.85  | 114.18 | -2.02 | 40.49 | 9.01  | 12.48 | 6.07   | -18.59 |
| 108 | 13.66  | 113.88 | -2.07 | 39.23 | 11.45 | 12.60 | 38.66  | -56.12 |
| 109 | -8.74  | 114.98 | -2.03 | 39.73 | 10.32 | 12.72 | -4.35  | -32.95 |
| 110 | 7.85   | 116.66 | -2.01 | 40.22 | 9.71  | 12.66 | -0.79  | -36.94 |
| 111 | 3.91   | 115.17 | -1.99 | 39.78 | 9.93  | 12.55 | 9.97   | 15.23  |
| 112 | 1.14   | 13.18  | -0.14 | 2.79  | 24.72 | 0.07  | -16.74 | 8.49   |
| 113 | 4.25   | 13.58  | -0.14 | 2.10  | 24.18 | 0.17  | 17.27  | 13.28  |
| 114 | -3.59  | 13.49  | -0.16 | 2.96  | 24.03 | 0.26  | 6.52   | 41.70  |
| 115 | 5.11   | 14.28  | -0.14 | 2.68  | 24.04 | 0.36  | 10.37  | 38.29  |
| 116 | 6.99   | 13.92  | -0.14 | 2.68  | 24.65 | 0.11  | 10.23  | -3.54  |
| 117 | 5.35   | 14.04  | -0.16 | 2.83  | 24.69 | 0.32  | -1.11  | 7.77   |
| 118 | -14.75 | 13.75  | -0.12 | 2.85  | 24.55 | 0.22  | -16.30 | 0.78   |
| 119 | 7.24   | 13.73  | -0.13 | 2.73  | 24.81 | 0.13  | -1.81  | 10.69  |

| k   | a0k    | a1k   | a2k   | a3k    | a4k  | a5k   | a6k  | a7k   |
|-----|--------|-------|-------|--------|------|-------|------|-------|
| 120 | -86.29 | -0.63 | 3.88  | -32.02 | 1.84 | 11.04 | 2.67 | -3.79 |
| 121 | -85.57 | -0.73 | 3.64  | -32.07 | 1.65 | 11.44 | 2.57 | -2.83 |
| 122 | -80.08 | -0.26 | 3.98  | -32.67 | 1.94 | 10.85 | 2.31 | -3.57 |
| 123 | -89.15 | -0.56 | 3.69  | -32.93 | 1.34 | 11.29 | 2.70 | -3.13 |
| 124 | -93.09 | -0.87 | 3.69  | -33.69 | 1.66 | 11.05 | 2.62 | -2.62 |
| 125 | -79.10 | -0.35 | 3.46  | -34.24 | 1.35 | 11.14 | 3.71 | -2.73 |
| 126 | -79.96 | -0.48 | 3.30  | -34.24 | 1.28 | 11.20 | 2.45 | -3.40 |
| 127 | -88.22 | -0.65 | 3.53  | -36.03 | 1.47 | 11.34 | 3.62 | -3.03 |
| 128 | -64.37 | -4.04 | 1.18  | -12.23 | 6.38 | 4.37  | 5.12 | -0.76 |
| 129 | -69.64 | -4.14 | 1.10  | -13.64 | 6.02 | 4.69  | 5.44 | -0.43 |
| 130 | -63.61 | -4.12 | 0.95  | -10.49 | 6.13 | 4.01  | 4.92 | -0.32 |
| 131 | -66.53 | -4.06 | 0.94  | -13.49 | 5.92 | 4.79  | 5.35 | -0.99 |
| 132 | -70.96 | -4.00 | 1.10  | -13.01 | 6.39 | 3.96  | 5.16 | -0.30 |
| 133 | -67.75 | -4.00 | 1.23  | -15.14 | 6.04 | 3.95  | 5.44 | -0.43 |
| 134 | -64.48 | -4.20 | 0.82  | -13.91 | 5.82 | 4.06  | 5.23 | -0.18 |
| 135 | -61.97 | -4.07 | 0.68  | -13.86 | 5.92 | 4.81  | 5.69 | -0.76 |
| 136 | -68.31 | -4.09 | 0.82  | -13.67 | 5.52 | 4.21  | 5.11 | -0.62 |
| 137 | -57.85 | -4.15 | 1.00  | -14.37 | 5.78 | 4.21  | 5.63 | -1.43 |
| 138 | -68.14 | -4.06 | 0.87  | -13.57 | 5.80 | 4.44  | 5.36 | -0.51 |
| 139 | -71.48 | -4.31 | 1.21  | -16.35 | 6.10 | 4.03  | 5.38 | -0.61 |
| 140 | -65.76 | -4.29 | 0.82  | -14.07 | 6.50 | 4.40  | 5.54 | -0.17 |
| 141 | -61.81 | -4.12 | 1.10  | -12.99 | 5.91 | 4.76  | 4.79 | -0.75 |
| 142 | -66.55 | -4.48 | 1.08  | -12.19 | 5.67 | 4.64  | 5.21 | -0.94 |
| 143 | -63.27 | -4.03 | 1.02  | -14.10 | 5.86 | 4.08  | 6.23 | -1.19 |
| 144 | -38.06 | -4.80 | -0.40 | -7.99  | 5.14 | 1.07  | 4.09 | -2.19 |
| 145 | -34.73 | -5.10 | -0.37 | -9.37  | 4.70 | 0.55  | 3.89 | -1.54 |
| 146 | -36.22 | -4.93 | -0.66 | -8.77  | 4.53 | 0.87  | 3.70 | -1.89 |
| 147 | -38.97 | -4.91 | -0.26 | -8.19  | 4.97 | 0.85  | 4.47 | -1.42 |
| 148 | -39.77 | -4.98 | -0.42 | -9.31  | 5.15 | 0.76  | 4.33 | -2.09 |
| 149 | -40.83 | -4.93 | -0.29 | -7.86  | 4.61 | 0.53  | 4.00 | -1.78 |
| 150 | -42.55 | -4.97 | -0.37 | -8.84  | 4.88 | 1.17  | 4.23 | -1.88 |
| 151 | -40.43 | -4.83 | -0.39 | -10.02 | 4.55 | 0.64  | 3.83 | -2.11 |
| 152 | -35.57 | -4.98 | -0.28 | -9.08  | 4.79 | 1.04  | 4.22 | -1.39 |
| 153 | -36.15 | -4.86 | -0.47 | -10.73 | 4.58 | 1.24  | 3.96 | -1.78 |
| 154 | -40.29 | -4.90 | -0.50 | -9.89  | 4.86 | 1.22  | 3.86 | -2.27 |

|     |        |       |       |       |      |      |      |       |
|-----|--------|-------|-------|-------|------|------|------|-------|
| 155 | -36.78 | -5.15 | -0.39 | -9.03 | 5.19 | 0.90 | 4.12 | -1.66 |
| 156 | -39.53 | -5.04 | -0.58 | -9.80 | 5.07 | 0.85 | 4.06 | -1.30 |
| 157 | -42.15 | -4.89 | -0.60 | -9.48 | 4.90 | 0.85 | 3.62 | -2.15 |
| 158 | -38.22 | -5.10 | -0.56 | -7.54 | 4.48 | 0.71 | 4.17 | -1.92 |
| 159 | -40.67 | -5.02 | -0.56 | -9.52 | 4.98 | 0.99 | 3.72 | -1.58 |

| k   | a8k   | a9k   | a10k  | a11k  | a12k   | a13k  | a14k  | a15k   |
|-----|-------|-------|-------|-------|--------|-------|-------|--------|
| 120 | 53.48 | 10.31 | -0.06 | 26.05 | -6.81  | -9.45 | 0.35  | -27.38 |
| 121 | 58.44 | 14.49 | -0.03 | 26.18 | -6.15  | -9.19 | 0.45  | -24.02 |
| 122 | 55.08 | 11.57 | -0.02 | 23.35 | -9.94  | -9.11 | 0.39  | -25.86 |
| 123 | 56.58 | 12.31 | -0.04 | 26.82 | -7.44  | -9.01 | 0.26  | -25.55 |
| 124 | 54.35 | 10.39 | -0.05 | 24.98 | -8.46  | -9.02 | 0.44  | -25.96 |
| 125 | 53.81 | 15.02 | -0.04 | 27.99 | -7.80  | -9.58 | 0.44  | -25.95 |
| 126 | 53.27 | 14.24 | -0.04 | 27.91 | -7.71  | -9.26 | 0.47  | -25.63 |
| 127 | 58.03 | 12.68 | -0.01 | 25.01 | -6.40  | -8.74 | 0.40  | -28.05 |
| 128 | 39.98 | -1.05 | -0.33 | 21.96 | -22.04 | -4.80 | -0.41 | -15.66 |
| 129 | 41.22 | -2.03 | -0.36 | 22.19 | -21.99 | -5.06 | -0.56 | -15.90 |
| 130 | 40.95 | -2.24 | -0.34 | 21.97 | -22.07 | -5.14 | -0.57 | -19.52 |
| 131 | 40.26 | -0.35 | -0.34 | 20.92 | -20.71 | -5.18 | -0.39 | -18.20 |
| 132 | 41.86 | -2.16 | -0.37 | 22.34 | -21.90 | -4.85 | -0.56 | -16.02 |
| 133 | 43.88 | -2.36 | -0.34 | 21.20 | -21.58 | -5.34 | -0.51 | -16.43 |
| 134 | 36.87 | -2.70 | -0.33 | 20.90 | -21.32 | -5.10 | -0.61 | -19.40 |
| 135 | 39.83 | -2.87 | -0.36 | 18.90 | -21.75 | -4.79 | -0.38 | -17.56 |
| 136 | 45.23 | -0.60 | -0.35 | 22.43 | -21.35 | -5.04 | -0.36 | -17.73 |
| 137 | 41.16 | -0.89 | -0.36 | 23.49 | -20.47 | -5.32 | -0.67 | -14.10 |
| 138 | 38.19 | -1.59 | -0.35 | 21.84 | -23.38 | -5.16 | -0.48 | -17.71 |
| 139 | 41.58 | -2.39 | -0.39 | 21.34 | -24.13 | -4.83 | -0.53 | -17.91 |
| 140 | 42.51 | -2.42 | -0.38 | 23.72 | -21.10 | -5.44 | -0.54 | -15.10 |
| 141 | 40.30 | 0.10  | -0.36 | 22.77 | -22.23 | -5.06 | -0.67 | -17.92 |
| 142 | 41.11 | 0.05  | -0.35 | 22.22 | -20.58 | -4.96 | -0.59 | -18.54 |
| 143 | 42.21 | -2.62 | -0.33 | 21.58 | -22.83 | -5.20 | -0.53 | -18.43 |
| 144 | 23.17 | 3.07  | -0.39 | 11.72 | -4.83  | -2.20 | 0.03  | -8.73  |
| 145 | 20.93 | 1.49  | -0.37 | 13.64 | -5.71  | -2.24 | -0.03 | -7.18  |
| 146 | 23.18 | 3.13  | -0.37 | 12.07 | -6.08  | -2.29 | -0.04 | -9.87  |
| 147 | 24.10 | 2.17  | -0.37 | 12.71 | -6.07  | -1.98 | 0.18  | -8.85  |
| 148 | 21.38 | 3.16  | -0.34 | 12.79 | -5.25  | -2.21 | -0.02 | -7.81  |
| 149 | 23.10 | 3.56  | -0.37 | 13.95 | -6.65  | -2.46 | -0.06 | -9.00  |
| 150 | 19.33 | 3.00  | -0.38 | 12.67 | -5.73  | -2.00 | -0.07 | -7.75  |
| 151 | 25.90 | 2.23  | -0.36 | 11.71 | -6.53  | -2.38 | 0.09  | -8.28  |
| 152 | 24.52 | 2.63  | -0.37 | 14.16 | -6.56  | -2.29 | 0.02  | -8.56  |
| 153 | 24.20 | 3.68  | -0.34 | 12.92 | -5.60  | -2.00 | -0.12 | -8.80  |
| 154 | 20.10 | 2.21  | -0.36 | 13.40 | -5.76  | -2.46 | 0.13  | -8.69  |
| 155 | 23.10 | 1.82  | -0.36 | 12.73 | -5.94  | -2.06 | 0.12  | -8.08  |
| 156 | 24.71 | 2.42  | -0.37 | 12.42 | -5.34  | -2.50 | -0.12 | -9.17  |
| 157 | 23.11 | 3.73  | -0.39 | 12.30 | -6.50  | -2.19 | -0.03 | -9.57  |
| 158 | 24.10 | 3.93  | -0.35 | 14.40 | -6.18  | -2.17 | 0.00  | -8.30  |
| 159 | 22.82 | 4.16  | -0.34 | 13.30 | -6.05  | -2.27 | 0.10  | -7.50  |

| k   | a16k | a17k  | a18k | a19k   | a20k  | a21k  | a22k | a23k |
|-----|------|-------|------|--------|-------|-------|------|------|
| 120 | 0.31 | -5.03 | 4.62 | -32.56 | -1.82 | -1.70 | 4.33 | 0.53 |

|     |       |       |      |        |       |       |       |      |
|-----|-------|-------|------|--------|-------|-------|-------|------|
| 121 | 0.72  | -4.01 | 4.85 | 1.24   | -1.88 | -1.74 | -1.33 | 0.54 |
| 122 | 0.45  | -4.10 | 4.84 | -2.36  | -1.67 | -1.69 | -2.20 | 0.55 |
| 123 | 0.11  | -4.61 | 4.79 | 14.98  | -1.97 | -1.76 | -1.86 | 0.53 |
| 124 | -0.04 | -4.58 | 4.63 | 0.19   | -1.97 | -1.92 | 1.82  | 0.55 |
| 125 | 0.56  | -4.69 | 4.83 | -20.06 | -1.93 | -1.77 | 3.49  | 0.53 |
| 126 | -0.20 | -4.71 | 4.49 | 4.85   | -1.85 | -1.65 | -2.92 | 0.54 |
| 127 | -0.06 | -4.54 | 4.62 | -14.70 | -1.93 | -1.83 | -4.25 | 0.55 |
| 128 | -0.59 | -1.46 | 4.32 | -19.70 | 1.52  | 0.78  | 1.79  | 0.51 |
| 129 | -0.28 | -1.64 | 4.09 | -34.57 | 1.15  | 0.54  | 5.08  | 0.51 |
| 130 | 0.18  | -1.25 | 4.46 | 22.73  | 1.33  | 0.70  | 7.53  | 0.48 |
| 131 | 1.03  | -1.55 | 4.43 | 8.79   | 1.40  | 0.59  | 5.14  | 0.47 |
| 132 | -0.07 | -1.65 | 4.37 | -34.22 | 1.36  | 0.55  | -0.52 | 0.48 |
| 133 | 0.09  | -1.60 | 4.25 | 6.16   | 1.45  | 0.59  | 7.45  | 0.50 |
| 134 | 0.59  | -1.67 | 4.41 | -25.72 | 1.17  | 0.73  | 6.02  | 0.48 |
| 135 | -1.51 | -1.42 | 4.83 | -0.65  | 1.22  | 0.56  | 7.84  | 0.49 |
| 136 | -0.64 | -1.43 | 4.64 | -25.29 | 1.16  | 0.77  | 2.70  | 0.47 |
| 137 | 0.38  | -1.91 | 4.26 | 29.50  | 1.25  | 0.50  | 3.59  | 0.47 |
| 138 | -0.51 | -1.60 | 4.68 | -16.14 | 1.52  | 0.61  | -4.79 | 0.49 |
| 139 | -0.07 | -1.17 | 4.53 | 42.81  | 1.29  | 0.66  | 6.75  | 0.50 |
| 140 | -0.70 | -1.19 | 4.06 | 6.12   | 1.09  | 0.81  | -3.05 | 0.55 |
| 141 | -0.06 | -1.58 | 4.40 | 6.81   | 1.20  | 0.47  | 3.59  | 0.52 |
| 142 | -0.30 | -1.72 | 3.98 | -8.17  | 1.33  | 0.76  | 4.02  | 0.50 |
| 143 | -0.08 | -1.32 | 4.27 | -7.54  | 1.36  | 0.36  | -7.33 | 0.50 |
| 144 | 0.45  | 0.82  | 3.10 | 11.79  | 3.12  | 2.48  | -2.30 | 0.22 |
| 145 | -0.01 | 0.44  | 2.79 | -4.24  | 3.08  | 2.36  | -0.35 | 0.25 |
| 146 | 0.37  | 0.43  | 2.61 | -14.16 | 3.16  | 2.41  | 0.10  | 0.20 |
| 147 | 0.50  | 0.65  | 2.30 | 16.23  | 3.18  | 2.47  | 13.38 | 0.22 |
| 148 | -0.08 | 0.27  | 2.63 | 15.55  | 3.20  | 2.50  | 6.03  | 0.22 |
| 149 | 0.77  | 0.39  | 2.89 | 11.11  | 3.10  | 2.45  | 1.68  | 0.23 |
| 150 | -0.07 | 0.70  | 2.73 | -7.13  | 3.16  | 2.45  | 6.97  | 0.23 |
| 151 | -0.14 | 0.72  | 2.70 | 16.50  | 3.19  | 2.47  | 7.78  | 0.23 |
| 152 | -0.05 | 0.58  | 3.11 | 5.90   | 3.31  | 2.60  | 5.82  | 0.21 |
| 153 | -0.41 | 0.77  | 2.48 | -3.59  | 3.23  | 2.64  | 1.59  | 0.21 |
| 154 | -0.34 | 0.61  | 2.63 | 13.77  | 3.13  | 2.70  | 5.05  | 0.23 |
| 155 | -0.58 | 0.21  | 2.43 | -4.03  | 3.19  | 2.42  | 0.11  | 0.21 |
| 156 | -0.25 | 0.71  | 2.85 | -12.11 | 3.07  | 2.63  | 6.59  | 0.24 |
| 157 | 0.54  | 0.53  | 2.69 | 2.39   | 3.20  | 2.50  | -3.10 | 0.21 |
| 158 | 0.28  | 0.84  | 2.99 | -14.40 | 3.08  | 2.52  | -6.42 | 0.19 |
| 159 | 0.26  | 0.25  | 2.33 | 8.40   | 3.20  | 2.36  | 5.99  | 0.25 |

| k   | a24k  | a25k  | a26k  | a27k  | a28k | a29k  | a30k  | a31k   |
|-----|-------|-------|-------|-------|------|-------|-------|--------|
| 120 | -3.15 | -2.74 | 32.68 | -1.28 | 4.17 | -0.54 | -2.97 | -49.03 |
| 121 | 0.37  | -2.71 | 32.30 | -1.60 | 4.71 | -0.70 | -2.92 | -49.08 |
| 122 | -1.57 | -2.50 | 33.88 | -1.53 | 4.70 | -0.59 | -2.22 | -49.68 |
| 123 | -2.35 | -2.72 | 30.56 | -1.41 | 4.83 | -0.77 | -2.27 | -49.43 |
| 124 | -1.34 | -2.41 | 29.44 | -1.53 | 4.20 | -0.64 | -3.11 | -49.42 |
| 125 | -0.38 | -2.90 | 32.38 | -1.44 | 3.98 | -0.92 | -3.34 | -49.54 |
| 126 | -0.34 | -2.87 | 30.15 | -1.49 | 4.13 | -0.79 | -3.50 | -50.50 |
| 127 | -1.26 | -2.59 | 31.97 | -1.33 | 3.99 | -0.36 | -2.67 | -50.02 |
| 128 | 1.14  | -0.88 | 32.09 | -0.37 | 9.19 | -2.91 | -3.97 | -36.02 |
| 129 | 0.01  | -0.90 | 29.34 | -1.01 | 8.94 | -2.71 | -3.49 | -35.29 |
| 130 | -0.35 | -0.83 | 32.11 | -0.90 | 9.10 | -3.05 | -4.29 | -34.94 |

|     |       |       |       |       |      |       |       |        |
|-----|-------|-------|-------|-------|------|-------|-------|--------|
| 131 | -1.21 | -1.12 | 32.67 | -0.71 | 9.25 | -3.00 | -3.36 | -36.80 |
| 132 | -0.17 | -1.05 | 30.63 | -0.91 | 8.77 | -2.89 | -4.80 | -36.50 |
| 133 | 1.87  | -0.79 | 31.14 | -0.52 | 9.94 | -2.94 | -3.83 | -36.11 |
| 134 | -0.47 | -1.01 | 30.51 | -0.75 | 9.13 | -3.33 | -3.84 | -35.71 |
| 135 | 0.30  | -0.80 | 30.54 | -1.22 | 9.54 | -2.83 | -3.54 | -35.94 |
| 136 | -1.37 | -0.96 | 32.28 | -0.65 | 9.08 | -2.91 | -3.37 | -36.61 |
| 137 | 2.06  | -1.05 | 29.71 | -0.50 | 9.28 | -2.99 | -3.87 | -35.73 |
| 138 | -0.62 | -0.65 | 28.44 | -0.93 | 9.22 | -2.73 | -3.47 | -36.62 |
| 139 | -1.69 | -0.79 | 32.28 | -0.62 | 9.13 | -2.77 | -3.37 | -36.46 |
| 140 | -1.42 | -0.48 | 31.05 | -0.86 | 9.25 | -3.21 | -3.49 | -35.39 |
| 141 | 2.17  | -0.73 | 31.65 | -0.97 | 9.05 | -3.07 | -4.55 | -35.42 |
| 142 | -1.91 | -0.93 | 29.79 | -0.57 | 8.76 | -2.92 | -3.90 | -35.86 |
| 143 | 0.66  | -0.81 | 33.30 | -0.84 | 9.12 | -3.15 | -4.24 | -37.51 |
| 144 | 1.11  | 0.42  | 18.05 | 0.99  | 2.98 | -0.40 | 1.34  | -22.51 |
| 145 | 0.18  | 0.19  | 18.46 | 0.91  | 2.43 | -0.95 | 1.18  | -21.18 |
| 146 | -0.04 | 0.32  | 17.97 | 0.82  | 2.36 | -0.36 | 0.97  | -22.37 |
| 147 | -0.54 | 0.58  | 18.34 | 1.07  | 2.36 | -0.56 | 0.77  | -22.28 |
| 148 | 0.51  | 0.19  | 18.43 | 0.91  | 2.45 | -0.37 | 0.93  | -22.38 |
| 149 | 0.27  | 0.47  | 18.05 | 0.97  | 2.15 | -0.67 | 1.19  | -22.81 |
| 150 | 0.84  | 0.36  | 17.25 | 1.26  | 2.25 | -0.60 | 1.62  | -22.12 |
| 151 | 1.21  | 0.45  | 18.18 | 1.26  | 2.77 | -0.41 | 1.26  | -22.47 |
| 152 | -1.01 | 0.39  | 18.00 | 1.08  | 2.27 | -0.42 | 1.60  | -23.23 |
| 153 | -0.96 | 0.54  | 17.76 | 1.28  | 2.68 | -0.65 | 1.19  | -22.35 |
| 154 | -1.06 | 0.43  | 18.31 | 0.95  | 1.97 | -0.58 | 1.72  | -22.32 |
| 155 | -1.09 | 0.56  | 16.71 | 0.46  | 2.04 | -0.37 | 1.65  | -21.98 |
| 156 | -1.60 | 0.60  | 18.48 | 0.96  | 2.26 | -0.62 | 1.03  | -22.74 |
| 157 | 1.04  | 0.44  | 16.36 | 0.86  | 2.27 | -0.58 | 1.05  | -23.32 |
| 158 | -0.46 | 0.44  | 16.89 | 0.97  | 2.06 | -0.66 | 1.13  | -21.35 |
| 159 | 0.16  | 0.47  | 18.03 | 1.13  | 2.51 | -0.59 | 1.34  | -22.34 |

| k   | a32k  | a33k  | a34k | a35k | a36k  | a37k  | a38k  | a39k  |
|-----|-------|-------|------|------|-------|-------|-------|-------|
| 120 | -4.42 | 1.84  | 1.76 | 1.01 | -0.98 | -2.13 | 2.45  | -2.01 |
| 121 | -4.37 | 2.00  | 1.74 | 0.71 | -0.97 | -1.82 | 2.59  | -2.07 |
| 122 | -3.96 | 1.87  | 1.66 | 0.88 | -0.98 | -1.29 | 2.49  | -2.04 |
| 123 | -3.97 | 1.99  | 1.80 | 0.12 | -0.92 | -1.94 | 2.43  | -2.10 |
| 124 | -4.29 | 1.75  | 1.75 | 0.79 | -0.97 | -3.66 | 2.39  | -2.04 |
| 125 | -4.25 | 1.90  | 1.82 | 1.10 | -0.98 | -2.15 | 2.39  | -2.09 |
| 126 | -4.33 | 1.92  | 1.73 | 1.23 | -0.96 | -3.61 | 2.43  | -2.04 |
| 127 | -4.28 | 1.86  | 1.88 | 1.40 | -1.02 | -3.55 | 2.51  | -1.98 |
| 128 | -7.95 | 0.11  | 1.68 | 8.51 | -0.87 | 10.98 | -2.80 | -2.96 |
| 129 | -8.25 | 0.12  | 1.63 | 9.09 | -0.88 | 8.94  | -2.75 | -3.10 |
| 130 | -8.36 | 0.16  | 1.71 | 8.72 | -0.86 | 9.22  | -2.60 | -3.06 |
| 131 | -8.59 | 0.00  | 1.59 | 7.62 | -0.86 | 11.42 | -2.57 | -2.99 |
| 132 | -8.02 | 0.21  | 1.68 | 8.66 | -0.87 | 9.90  | -2.69 | -3.05 |
| 133 | -8.27 | 0.18  | 1.74 | 8.45 | -0.92 | 9.04  | -2.93 | -3.06 |
| 134 | -8.28 | -0.01 | 1.69 | 8.82 | -0.93 | 9.85  | -2.73 | -3.04 |
| 135 | -8.91 | 0.14  | 1.71 | 8.75 | -0.83 | 9.35  | -2.74 | -3.04 |
| 136 | -8.32 | 0.14  | 1.68 | 8.33 | -0.91 | 9.73  | -2.51 | -3.02 |
| 137 | -8.23 | 0.27  | 1.76 | 8.53 | -0.86 | 10.43 | -2.72 | -2.95 |
| 138 | -8.15 | 0.11  | 1.60 | 8.97 | -0.93 | 10.95 | -2.78 | -2.98 |
| 139 | -8.33 | 0.06  | 1.70 | 8.68 | -0.89 | 10.59 | -3.01 | -3.00 |
| 140 | -7.64 | 0.16  | 1.68 | 8.39 | -0.82 | 9.51  | -2.84 | -3.02 |

|     |       |       |      |      |       |       |       |       |
|-----|-------|-------|------|------|-------|-------|-------|-------|
| 141 | -9.02 | 0.15  | 1.58 | 8.08 | -0.92 | 10.09 | -2.75 | -3.09 |
| 142 | -8.73 | 0.14  | 1.68 | 8.98 | -0.89 | 10.21 | -2.67 | -3.04 |
| 143 | -7.95 | 0.22  | 1.63 | 9.15 | -0.98 | 9.25  | -3.00 | -3.00 |
| 144 | -2.45 | -1.17 | 0.94 | 4.61 | -0.40 | 1.78  | -3.48 | -2.51 |
| 145 | -2.39 | -1.16 | 1.04 | 4.32 | -0.39 | 2.64  | -3.46 | -2.56 |
| 146 | -1.88 | -1.16 | 0.98 | 4.39 | -0.41 | 0.79  | -3.46 | -2.50 |
| 147 | -2.14 | -1.10 | 1.02 | 4.42 | -0.42 | 2.73  | -3.50 | -2.53 |
| 148 | -2.58 | -1.01 | 0.97 | 4.80 | -0.36 | 2.52  | -3.48 | -2.52 |
| 149 | -2.46 | -1.06 | 0.97 | 4.32 | -0.38 | 2.31  | -3.48 | -2.54 |
| 150 | -2.92 | -1.09 | 1.00 | 4.66 | -0.42 | 2.15  | -3.46 | -2.52 |
| 151 | -2.89 | -1.10 | 0.90 | 4.61 | -0.37 | 2.57  | -3.54 | -2.54 |
| 152 | -2.62 | -1.10 | 0.98 | 4.88 | -0.37 | 2.24  | -3.62 | -2.56 |
| 153 | -3.08 | -1.12 | 1.01 | 4.37 | -0.35 | 3.16  | -3.55 | -2.56 |
| 154 | -2.80 | -1.10 | 0.94 | 4.54 | -0.41 | 2.04  | -3.56 | -2.52 |
| 155 | -2.13 | -1.11 | 0.95 | 4.54 | -0.41 | 2.28  | -3.54 | -2.48 |
| 156 | -2.77 | -1.04 | 0.95 | 4.39 | -0.36 | 1.86  | -3.63 | -2.55 |
| 157 | -3.33 | -1.13 | 1.01 | 3.85 | -0.38 | 3.13  | -3.50 | -2.49 |
| 158 | -2.72 | -0.98 | 0.94 | 4.57 | -0.36 | 2.95  | -3.54 | -2.53 |
| 159 | -2.54 | -1.18 | 1.01 | 4.96 | -0.35 | 1.73  | -3.52 | -2.48 |

| k   | a40k  | a41k  | a42k  | a43k  | a44k  | a45k  | a46k  | a47k   |
|-----|-------|-------|-------|-------|-------|-------|-------|--------|
| 120 | 0.33  | 4.92  | -0.31 | -1.59 | 0.68  | -6.93 | 56.35 | 1.36   |
| 121 | 0.30  | 4.77  | 0.00  | -1.50 | 4.89  | -6.96 | 59.88 | -3.65  |
| 122 | 0.34  | 4.82  | 4.31  | -1.16 | -7.78 | -6.96 | 67.78 | -1.90  |
| 123 | 0.31  | 4.96  | -0.12 | -2.05 | -5.38 | -7.09 | 66.01 | 0.16   |
| 124 | 0.36  | 4.79  | 3.53  | -0.49 | 6.42  | -6.72 | 66.01 | 2.43   |
| 125 | 0.35  | 5.13  | -0.44 | -1.41 | -1.40 | -6.90 | 63.79 | 3.73   |
| 126 | 0.33  | 4.75  | -1.99 | -1.23 | -9.02 | -6.86 | 58.33 | 3.46   |
| 127 | 0.36  | 4.73  | 0.36  | -1.53 | 4.56  | -7.29 | 61.10 | 3.11   |
| 128 | -0.07 | -1.38 | 2.12  | -8.54 | -9.27 | -1.78 | 54.45 | -0.97  |
| 129 | -0.13 | -1.34 | 2.14  | -8.46 | 0.16  | -2.53 | 47.17 | -2.51  |
| 130 | -0.06 | -1.74 | -2.76 | -8.60 | 2.08  | -2.47 | 57.05 | -0.39  |
| 131 | -0.15 | -1.22 | 0.09  | -8.08 | 5.34  | -2.74 | 64.38 | -0.47  |
| 132 | -0.10 | -2.01 | -3.16 | -8.12 | -3.11 | -2.73 | 54.72 | 3.38   |
| 133 | -0.13 | -1.45 | 2.02  | -8.04 | 2.46  | -2.24 | 68.54 | 1.40   |
| 134 | -0.11 | -1.36 | -2.75 | -7.68 | 0.87  | -2.20 | 62.77 | 0.46   |
| 135 | -0.10 | -1.66 | 0.80  | -7.99 | -6.50 | -2.85 | 58.11 | -3.03  |
| 136 | -0.10 | -1.39 | -0.66 | -8.05 | -0.64 | -2.58 | 60.67 | -1.20  |
| 137 | -0.08 | -2.04 | -0.70 | -8.62 | 5.92  | -2.03 | 63.08 | 2.07   |
| 138 | -0.11 | -1.44 | -2.66 | -8.23 | -3.00 | -2.64 | 52.05 | -12.58 |
| 139 | -0.09 | -1.34 | 1.73  | -7.92 | 1.25  | -2.09 | 54.21 | -4.23  |
| 140 | -0.13 | -1.72 | -1.14 | -8.65 | 3.68  | -2.46 | 60.90 | -3.56  |
| 141 | -0.14 | -1.29 | -1.05 | -8.42 | -4.40 | -1.80 | 55.03 | -1.73  |
| 142 | -0.10 | -1.33 | -4.62 | -9.38 | -5.19 | -2.68 | 56.49 | -0.74  |
| 143 | -0.09 | -1.35 | -1.42 | -9.07 | 3.70  | -2.50 | 61.63 | 0.05   |
| 144 | -0.44 | -6.62 | -0.08 | -8.43 | 3.03  | 2.65  | 34.50 | 2.06   |
| 145 | -0.47 | -6.28 | 1.57  | -9.29 | -6.12 | 2.33  | 29.93 | -1.97  |
| 146 | -0.44 | -6.15 | -0.35 | -8.90 | -6.12 | 2.76  | 31.77 | -0.39  |
| 147 | -0.47 | -6.03 | 1.02  | -8.50 | -3.98 | 2.55  | 27.37 | 6.41   |
| 148 | -0.47 | -6.47 | 0.58  | -9.16 | -4.25 | 2.53  | 29.28 | -2.16  |
| 149 | -0.47 | -6.41 | 0.58  | -8.59 | -7.69 | 2.66  | 32.78 | -4.26  |
| 150 | -0.44 | -6.63 | 0.33  | -8.96 | 5.06  | 2.56  | 29.31 | 0.25   |

|     |       |       |       |       |        |      |       |       |
|-----|-------|-------|-------|-------|--------|------|-------|-------|
| 151 | -0.45 | -6.32 | 4.26  | -8.83 | -3.55  | 2.68 | 35.36 | -1.39 |
| 152 | -0.44 | -6.65 | 0.05  | -8.71 | -11.51 | 2.71 | 30.30 | 3.43  |
| 153 | -0.44 | -6.47 | -0.37 | -8.79 | 7.18   | 2.80 | 30.82 | -6.53 |
| 154 | -0.44 | -6.12 | 0.06  | -8.76 | 3.86   | 2.95 | 28.19 | -2.98 |
| 155 | -0.44 | -6.12 | 1.80  | -9.11 | -9.21  | 2.68 | 30.34 | 6.30  |
| 156 | -0.46 | -6.68 | -0.77 | -8.69 | 1.26   | 2.63 | 30.94 | -0.36 |
| 157 | -0.47 | -6.39 | -0.08 | -8.15 | 7.13   | 2.80 | 28.23 | 1.12  |
| 158 | -0.45 | -6.39 | -0.75 | -8.70 | 8.69   | 2.90 | 31.24 | 1.50  |
| 159 | -0.46 | -6.57 | -1.82 | -8.71 | -3.29  | 2.75 | 28.59 | -2.87 |

| k   | a48k  | a49k  | a50k | a51k  | a52k  | a53k  | a54k  | a55k  |
|-----|-------|-------|------|-------|-------|-------|-------|-------|
| 120 | 2.78  | -6.70 | 6.71 | -4.07 | 75.91 | 32.50 | -4.02 | -2.35 |
| 121 | 1.75  | -6.51 | 6.99 | 1.30  | 76.01 | 32.64 | -3.86 | -1.98 |
| 122 | 2.33  | -6.45 | 7.21 | -2.74 | 73.44 | 32.65 | -3.94 | -3.11 |
| 123 | 2.80  | -6.68 | 7.26 | -2.89 | 72.83 | 33.33 | -3.87 | -2.74 |
| 124 | 3.47  | -6.47 | 6.87 | 1.60  | 66.53 | 32.72 | -3.73 | -1.95 |
| 125 | 1.90  | -6.45 | 7.49 | 0.13  | 71.55 | 33.59 | -3.90 | -2.40 |
| 126 | 2.54  | -6.66 | 7.75 | -4.69 | 70.82 | 32.24 | -3.83 | -3.09 |
| 127 | 1.80  | -6.47 | 6.30 | -0.37 | 75.11 | 32.34 | -3.76 | -2.51 |
| 128 | 5.08  | -2.11 | 4.54 | -3.75 | 56.90 | 24.33 | -1.11 | -3.41 |
| 129 | 6.93  | -2.21 | 4.69 | -1.04 | 53.20 | 22.82 | -0.86 | -3.26 |
| 130 | 5.71  | -2.02 | 4.77 | -2.04 | 59.19 | 22.72 | -1.09 | -3.87 |
| 131 | 6.18  | -2.16 | 5.19 | -1.55 | 56.71 | 22.38 | -0.87 | -3.38 |
| 132 | 4.92  | -2.36 | 4.70 | -3.26 | 53.43 | 21.56 | -0.71 | -3.77 |
| 133 | 5.37  | -2.04 | 5.37 | 4.33  | 55.12 | 23.02 | -1.00 | -3.41 |
| 134 | 4.82  | -2.12 | 4.95 | 2.78  | 57.33 | 21.93 | -1.02 | -4.04 |
| 135 | 5.17  | -2.10 | 4.70 | 2.48  | 52.38 | 21.10 | -0.78 | -3.61 |
| 136 | 5.59  | -2.43 | 5.28 | -2.07 | 55.91 | 21.57 | -1.07 | -3.74 |
| 137 | 6.13  | -2.32 | 4.42 | -3.71 | 55.69 | 22.77 | -0.81 | -3.72 |
| 138 | 4.85  | -2.26 | 4.81 | 6.46  | 55.44 | 22.67 | -0.71 | -3.47 |
| 139 | 4.82  | -2.22 | 5.09 | 2.24  | 54.98 | 22.90 | -0.79 | -3.88 |
| 140 | 6.04  | -2.27 | 4.38 | -0.44 | 59.54 | 22.51 | -0.85 | -3.18 |
| 141 | 6.36  | -2.39 | 5.49 | -0.20 | 54.86 | 21.78 | -0.90 | -3.71 |
| 142 | 5.02  | -2.24 | 5.11 | 2.02  | 57.42 | 21.87 | -0.84 | -3.51 |
| 143 | 5.52  | -2.22 | 4.91 | 1.56  | 58.08 | 23.02 | -1.00 | -3.42 |
| 144 | -0.21 | 1.06  | 1.30 | -0.44 | 38.34 | 16.26 | 0.17  | -2.55 |
| 145 | -0.70 | 1.07  | 0.82 | 0.28  | 39.73 | 16.77 | 0.10  | -2.79 |
| 146 | -0.67 | 0.88  | 0.94 | 0.56  | 41.10 | 17.62 | 0.23  | -2.75 |
| 147 | -0.68 | 1.10  | 1.41 | 0.32  | 40.17 | 16.63 | 0.19  | -2.42 |
| 148 | -0.41 | 1.17  | 1.07 | 0.73  | 37.75 | 15.98 | 0.12  | -2.83 |
| 149 | -0.55 | 1.15  | 1.39 | 3.93  | 39.29 | 16.75 | 0.30  | -3.12 |
| 150 | -1.68 | 1.29  | 1.09 | 2.30  | 41.93 | 16.56 | 0.33  | -2.48 |
| 151 | -0.46 | 1.00  | 1.37 | -0.76 | 41.86 | 16.89 | 0.23  | -3.06 |
| 152 | -1.16 | 0.71  | 0.72 | -3.59 | 36.96 | 15.92 | 0.09  | -2.85 |
| 153 | -0.73 | 1.16  | 1.43 | -2.06 | 37.10 | 16.71 | 0.20  | -2.38 |
| 154 | -0.77 | 1.19  | 1.48 | 0.25  | 39.82 | 16.00 | 0.06  | -1.96 |
| 155 | -0.23 | 1.11  | 0.91 | 0.10  | 39.59 | 16.68 | 0.02  | -2.58 |
| 156 | -0.49 | 1.09  | 1.59 | 1.28  | 37.71 | 16.57 | 0.07  | -2.28 |
| 157 | -0.44 | 1.01  | 0.76 | 0.78  | 41.09 | 16.19 | 0.21  | -2.43 |
| 158 | -0.30 | 1.03  | 1.11 | -2.55 | 38.76 | 15.61 | 0.29  | -2.24 |
| 159 | -1.59 | 1.10  | 1.12 | -0.03 | 39.03 | 16.28 | 0.16  | -2.11 |



| k   | a56k   | a57k  | a58k  | a59k | a60k  | a61k  | a62k   | a63k   |
|-----|--------|-------|-------|------|-------|-------|--------|--------|
| 120 | -3.49  | 13.40 | -0.17 | 2.77 | 24.03 | 0.22  | -1.92  | 8.36   |
| 121 | -1.56  | 13.53 | -0.15 | 2.91 | 24.00 | 0.19  | -0.13  | 9.30   |
| 122 | -0.33  | 13.08 | -0.14 | 2.69 | 23.29 | -0.11 | -10.57 | 21.22  |
| 123 | 4.41   | 13.72 | -0.15 | 2.98 | 24.76 | 0.23  | -14.55 | 8.52   |
| 124 | -5.43  | 13.28 | -0.15 | 2.53 | 23.69 | 0.36  | 3.17   | 30.92  |
| 125 | -2.68  | 13.25 | -0.13 | 2.61 | 24.53 | 0.23  | 43.45  | 1.59   |
| 126 | 7.50   | 15.48 | -0.16 | 2.78 | 24.04 | 0.28  | -3.80  | 25.95  |
| 127 | 4.32   | 14.32 | -0.14 | 2.77 | 23.76 | 0.04  | 2.42   | -25.97 |
| 128 | -13.42 | 27.45 | -0.24 | 7.34 | 17.51 | 2.47  | 0.94   | 9.49   |
| 129 | 10.58  | 26.06 | -0.24 | 7.23 | 17.53 | 2.31  | 14.43  | 46.44  |
| 130 | 4.21   | 24.93 | -0.25 | 7.19 | 17.44 | 2.41  | -21.84 | -14.67 |
| 131 | 1.14   | 27.25 | -0.25 | 7.89 | 17.62 | 2.36  | -6.28  | -1.76  |
| 132 | -1.71  | 25.57 | -0.28 | 7.46 | 17.35 | 2.48  | 0.91   | 32.36  |
| 133 | 12.79  | 24.80 | -0.28 | 7.83 | 17.59 | 2.47  | -0.89  | -14.20 |
| 134 | 3.23   | 24.22 | -0.24 | 7.25 | 17.33 | 2.29  | 7.74   | 36.11  |
| 135 | 5.29   | 24.94 | -0.25 | 7.41 | 17.48 | 2.41  | 0.58   | 13.47  |
| 136 | -3.92  | 26.67 | -0.25 | 7.53 | 17.72 | 2.42  | -8.16  | 22.44  |
| 137 | -3.58  | 26.44 | -0.26 | 7.61 | 16.62 | 2.49  | -1.47  | -16.57 |
| 138 | 4.11   | 25.58 | -0.24 | 7.62 | 17.52 | 2.36  | -16.44 | -25.70 |
| 139 | -1.61  | 25.23 | -0.24 | 7.50 | 17.71 | 2.41  | -13.21 | 25.78  |
| 140 | -2.54  | 27.55 | -0.25 | 7.87 | 18.15 | 2.40  | 1.36   | 18.40  |
| 141 | -12.06 | 26.71 | -0.24 | 7.31 | 17.41 | 2.28  | -12.71 | 9.79   |
| 142 | 1.45   | 25.83 | -0.25 | 7.40 | 17.14 | 2.38  | -6.48  | -4.14  |
| 143 | 3.09   | 26.96 | -0.25 | 7.34 | 17.09 | 2.50  | -5.15  | 17.91  |
| 144 | -4.85  | 8.77  | -0.03 | 7.05 | 9.88  | 2.53  | 14.66  | 9.71   |
| 145 | -3.68  | 8.85  | -0.04 | 7.36 | 9.83  | 2.65  | 13.40  | 23.05  |
| 146 | -2.12  | 8.74  | -0.03 | 7.62 | 10.26 | 2.57  | -4.56  | -13.00 |
| 147 | 0.77   | 9.60  | -0.04 | 7.20 | 10.29 | 2.55  | -1.43  | 5.03   |
| 148 | 1.14   | 9.42  | -0.03 | 7.40 | 10.21 | 2.61  | -10.34 | 1.62   |
| 149 | 1.93   | 8.85  | -0.04 | 7.68 | 9.93  | 2.31  | -10.78 | 11.24  |
| 150 | -1.34  | 10.06 | -0.04 | 7.44 | 10.26 | 2.50  | 2.13   | 6.41   |
| 151 | 3.92   | 9.28  | -0.03 | 7.41 | 10.52 | 2.47  | -23.89 | 21.08  |
| 152 | 1.39   | 9.20  | -0.02 | 7.35 | 10.53 | 2.59  | 16.81  | 5.07   |
| 153 | -8.77  | 8.14  | -0.02 | 7.68 | 10.60 | 2.67  | 2.72   | -9.55  |
| 154 | -0.43  | 9.26  | -0.04 | 7.60 | 10.54 | 2.54  | -2.97  | 14.19  |
| 155 | 0.07   | 9.38  | -0.03 | 7.27 | 10.46 | 2.65  | 2.90   | 14.79  |
| 156 | -0.81  | 9.37  | -0.05 | 7.54 | 10.21 | 2.54  | 10.72  | 0.67   |
| 157 | -1.46  | 9.47  | -0.05 | 7.26 | 10.23 | 2.60  | -7.04  | -39.70 |
| 158 | -5.91  | 8.46  | -0.04 | 7.27 | 9.78  | 2.38  | 0.29   | 9.16   |
| 159 | -3.15  | 8.99  | -0.05 | 7.41 | 10.58 | 2.70  | -10.17 | 2.93   |

| k   | a0k     | a1k  | a2k  | a3k    | a4k  | a5k   | a6k  | a7k   |
|-----|---------|------|------|--------|------|-------|------|-------|
| 160 | -183.25 | 2.14 | 5.62 | -25.74 | 3.13 | 10.32 | 3.56 | 11.78 |
| 161 | -200.77 | 1.99 | 5.47 | -27.03 | 3.73 | 10.43 | 3.14 | 12.40 |
| 162 | -189.77 | 1.88 | 5.81 | -25.08 | 3.29 | 10.28 | 2.50 | 12.23 |
| 163 | -180.47 | 1.92 | 5.11 | -24.66 | 3.56 | 9.96  | 3.94 | 11.75 |
| 164 | -190.80 | 2.23 | 5.54 | -24.51 | 1.97 | 10.54 | 4.33 | 11.66 |
| 165 | -202.82 | 2.32 | 5.41 | -25.31 | 2.83 | 10.24 | 3.98 | 11.72 |
| 166 | -191.96 | 2.42 | 5.77 | -26.22 | 4.19 | 10.76 | 2.92 | 12.35 |

|     |         |      |      |        |       |       |      |       |
|-----|---------|------|------|--------|-------|-------|------|-------|
| 167 | -202.76 | 2.17 | 6.14 | -25.94 | 2.82  | 10.63 | 3.41 | 11.44 |
| 168 | -200.75 | 1.98 | 4.80 | -24.85 | 2.05  | 10.13 | 4.60 | 11.72 |
| 169 | -203.33 | 2.07 | 4.43 | -25.07 | 3.21  | 10.43 | 4.22 | 11.70 |
| 170 | -210.89 | 2.59 | 5.25 | -26.95 | 2.38  | 10.55 | 4.53 | 11.48 |
| 171 | -182.98 | 1.78 | 5.53 | -26.40 | 2.18  | 10.46 | 3.14 | 12.03 |
| 172 | -198.69 | 1.99 | 5.10 | -24.36 | 2.57  | 10.33 | 3.05 | 11.84 |
| 173 | -188.40 | 1.90 | 5.59 | -24.79 | 3.64  | 10.55 | 3.01 | 11.43 |
| 174 | -214.04 | 2.14 | 5.54 | -24.92 | 2.11  | 10.22 | 2.96 | 11.82 |
| 175 | -184.63 | 2.21 | 5.32 | -25.17 | 2.94  | 10.15 | 4.37 | 12.25 |
| 176 | -200.57 | 2.15 | 5.04 | -22.81 | 1.93  | 9.60  | 4.05 | 11.62 |
| 177 | -192.89 | 1.82 | 5.99 | -23.73 | 1.29  | 9.49  | 3.28 | 11.57 |
| 178 | -203.63 | 1.64 | 5.83 | -20.94 | -0.29 | 9.36  | 3.67 | 12.18 |
| 179 | -188.18 | 2.06 | 5.25 | -23.68 | 1.11  | 9.95  | 5.50 | 11.80 |
| 180 | -209.99 | 1.94 | 5.75 | -22.03 | 1.03  | 9.56  | 4.07 | 11.49 |
| 181 | -186.37 | 1.67 | 5.88 | -21.16 | 0.82  | 9.68  | 4.08 | 11.39 |
| 182 | -179.41 | 2.00 | 4.79 | -22.47 | 1.65  | 9.46  | 3.93 | 11.91 |
| 183 | -205.64 | 1.85 | 6.10 | -21.08 | 0.69  | 9.89  | 2.51 | 12.14 |
| 184 | -191.97 | 1.98 | 5.27 | -22.05 | 0.35  | 9.36  | 3.29 | 11.41 |
| 185 | -189.61 | 1.88 | 5.68 | -22.09 | 1.18  | 9.34  | 3.93 | 11.81 |
| 186 | -194.95 | 2.16 | 5.47 | -23.46 | 2.22  | 9.99  | 4.03 | 11.41 |
| 187 | -199.69 | 1.98 | 5.83 | -21.48 | 0.05  | 9.70  | 3.21 | 11.41 |
| 188 | -186.98 | 1.60 | 5.00 | -23.36 | 1.17  | 9.80  | 2.65 | 11.23 |
| 189 | -194.99 | 1.63 | 5.53 | -23.27 | 1.00  | 9.60  | 4.56 | 12.00 |
| 190 | -195.44 | 1.23 | 5.91 | -21.99 | 1.57  | 9.26  | 3.87 | 12.39 |
| 191 | -197.60 | 1.64 | 5.88 | -21.36 | 0.88  | 9.76  | 2.72 | 10.91 |
| 192 | -80.99  | 1.13 | 5.78 | -41.96 | -7.61 | 14.12 | 4.56 | -2.34 |
| 193 | -89.68  | 0.98 | 5.35 | -40.67 | -7.33 | 14.32 | 4.86 | -3.09 |
| 194 | -87.61  | 1.04 | 5.68 | -40.81 | -7.47 | 14.54 | 5.72 | -3.11 |
| 195 | -75.78  | 0.82 | 5.89 | -41.56 | -7.69 | 14.38 | 4.84 | -3.28 |
| 196 | -81.87  | 1.14 | 5.62 | -39.90 | -7.61 | 14.36 | 5.82 | -3.01 |
| 197 | -86.38  | 1.03 | 5.52 | -40.84 | -7.57 | 14.22 | 4.74 | -2.69 |
| 198 | -90.67  | 1.07 | 5.62 | -40.01 | -7.85 | 14.09 | 4.38 | -2.72 |
| 199 | -85.90  | 0.90 | 5.82 | -39.91 | -7.87 | 14.58 | 4.61 | -3.11 |

| k   | a8k    | a9k   | a10k  | a11k  | a12k   | a13k   | a14k  | a15k   |
|-----|--------|-------|-------|-------|--------|--------|-------|--------|
| 160 | 93.69  | 50.35 | -0.06 | 43.76 | -7.96  | -11.45 | -1.65 | -39.83 |
| 161 | 94.19  | 51.22 | 0.05  | 41.97 | -8.07  | -11.77 | -1.83 | -35.89 |
| 162 | 96.08  | 50.24 | -0.01 | 44.07 | -8.94  | -11.94 | -1.74 | -33.20 |
| 163 | 91.01  | 50.82 | -0.02 | 41.62 | -9.97  | -11.61 | -1.79 | -35.37 |
| 164 | 94.91  | 51.64 | -0.03 | 44.52 | -10.47 | -11.14 | -1.66 | -38.97 |
| 165 | 99.80  | 49.04 | -0.02 | 49.32 | -10.67 | -11.51 | -1.78 | -35.32 |
| 166 | 100.32 | 49.55 | 0.01  | 48.96 | -8.32  | -11.16 | -1.87 | -35.89 |
| 167 | 91.74  | 50.26 | -0.03 | 47.65 | -8.62  | -11.54 | -1.66 | -34.51 |
| 168 | 99.23  | 48.64 | -0.05 | 42.04 | -8.21  | -11.68 | -1.67 | -34.09 |
| 169 | 91.16  | 51.29 | -0.01 | 48.34 | -11.59 | -11.22 | -1.69 | -38.04 |
| 170 | 84.86  | 50.02 | 0.02  | 47.47 | -8.47  | -12.16 | -1.65 | -39.32 |
| 171 | 104.41 | 49.99 | -0.04 | 46.22 | -7.03  | -11.70 | -1.61 | -36.51 |
| 172 | 95.70  | 51.92 | 0.01  | 43.17 | -11.26 | -11.33 | -1.77 | -42.20 |
| 173 | 99.92  | 49.58 | 0.00  | 46.70 | -8.98  | -12.03 | -1.72 | -36.09 |
| 174 | 92.96  | 50.20 | -0.05 | 47.19 | -10.20 | -11.95 | -1.65 | -39.79 |
| 175 | 87.99  | 50.84 | -0.06 | 47.82 | -10.64 | -11.26 | -1.78 | -40.49 |
| 176 | 92.57  | 45.83 | -0.03 | 46.46 | -3.54  | -10.29 | -1.66 | -40.48 |

|     |        |       |       |       |       |        |       |        |
|-----|--------|-------|-------|-------|-------|--------|-------|--------|
| 177 | 103.07 | 45.08 | -0.08 | 46.56 | -6.48 | -10.76 | -1.59 | -34.80 |
| 178 | 89.56  | 45.20 | -0.04 | 45.63 | -3.64 | -10.46 | -1.56 | -43.17 |
| 179 | 100.68 | 43.71 | -0.03 | 49.09 | -2.93 | -10.14 | -1.61 | -40.29 |
| 180 | 105.61 | 44.15 | -0.06 | 48.19 | -3.47 | -9.72  | -1.61 | -40.95 |
| 181 | 91.34  | 45.93 | -0.02 | 50.35 | -3.56 | -10.55 | -1.62 | -40.77 |
| 182 | 87.76  | 44.55 | 0.00  | 46.54 | -7.49 | -10.17 | -1.51 | -43.01 |
| 183 | 97.67  | 44.55 | -0.04 | 49.44 | -4.48 | -10.41 | -1.59 | -36.79 |
| 184 | 93.30  | 44.01 | -0.02 | 47.30 | -2.87 | -10.32 | -1.78 | -37.03 |
| 185 | 101.12 | 47.15 | 0.00  | 45.58 | -6.35 | -9.75  | -1.68 | -42.95 |
| 186 | 92.30  | 43.85 | -0.03 | 49.68 | -5.53 | -10.36 | -1.73 | -39.98 |
| 187 | 90.84  | 43.94 | -0.02 | 44.30 | -1.86 | -10.72 | -1.55 | -37.11 |
| 188 | 101.37 | 42.91 | -0.04 | 48.77 | -4.67 | -9.64  | -1.62 | -45.48 |
| 189 | 95.02  | 44.73 | -0.04 | 45.22 | -2.35 | -10.64 | -1.65 | -39.73 |
| 190 | 93.66  | 43.89 | -0.01 | 45.99 | -6.29 | -9.88  | -1.58 | -39.34 |
| 191 | 103.16 | 44.83 | -0.01 | 47.51 | -5.19 | -10.19 | -1.53 | -36.73 |
| 192 | 59.69  | 57.64 | -0.04 | 16.47 | 24.54 | -9.22  | 2.64  | -37.59 |
| 193 | 69.07  | 58.35 | -0.09 | 18.14 | 26.55 | -9.27  | 2.42  | -38.58 |
| 194 | 64.30  | 59.42 | -0.09 | 19.73 | 26.26 | -9.01  | 2.47  | -37.31 |
| 195 | 64.31  | 57.75 | -0.08 | 18.64 | 26.38 | -9.27  | 2.49  | -39.18 |
| 196 | 66.35  | 57.50 | -0.09 | 17.61 | 26.14 | -9.63  | 2.47  | -34.66 |
| 197 | 70.49  | 57.92 | -0.09 | 19.08 | 25.79 | -9.25  | 2.47  | -36.50 |
| 198 | 67.45  | 59.20 | -0.07 | 19.74 | 26.43 | -9.27  | 2.38  | -38.96 |
| 199 | 69.13  | 57.68 | -0.11 | 19.25 | 26.73 | -9.21  | 2.41  | -38.46 |

| k   | a16k  | a17k  | a18k   | a19k   | a20k  | a21k  | a22k   | a23k  |
|-----|-------|-------|--------|--------|-------|-------|--------|-------|
| 160 | 0.05  | -6.82 | -24.67 | -24.74 | -2.61 | -2.86 | -2.71  | -1.60 |
| 161 | 0.00  | -6.42 | -24.61 | -59.26 | -2.55 | -2.53 | 6.00   | -1.60 |
| 162 | -0.30 | -6.70 | -24.98 | 8.36   | -2.47 | -2.78 | 20.37  | -1.59 |
| 163 | -0.11 | -5.77 | -24.59 | 50.59  | -2.62 | -2.67 | 6.97   | -1.61 |
| 164 | 0.19  | -6.04 | -24.78 | -12.92 | -2.65 | -2.69 | -11.54 | -1.60 |
| 165 | -0.02 | -5.73 | -24.78 | -25.13 | -2.64 | -2.84 | 8.52   | -1.63 |
| 166 | -0.06 | -7.15 | -24.72 | 10.36  | -2.90 | -2.60 | 11.33  | -1.60 |
| 167 | 0.05  | -5.02 | -24.62 | -18.12 | -2.65 | -2.70 | -9.13  | -1.60 |
| 168 | -0.18 | -5.74 | -24.65 | -43.31 | -2.61 | -2.74 | 2.60   | -1.63 |
| 169 | 0.05  | -3.92 | -24.69 | -0.11  | -2.62 | -2.81 | -0.57  | -1.59 |
| 170 | -0.22 | -6.42 | -24.74 | 31.59  | -2.54 | -2.42 | 4.93   | -1.60 |
| 171 | 0.38  | -6.29 | -24.79 | 17.32  | -2.47 | -2.80 | 3.16   | -1.58 |
| 172 | -0.20 | -4.47 | -24.71 | -76.32 | -2.72 | -2.81 | 14.78  | -1.58 |
| 173 | -0.02 | -3.41 | -24.55 | 12.62  | -2.59 | -2.77 | 13.40  | -1.61 |
| 174 | -0.19 | -6.36 | -24.80 | -30.09 | -2.58 | -2.56 | 11.87  | -1.60 |
| 175 | 0.01  | -5.58 | -24.80 | -0.33  | -2.57 | -2.98 | 8.06   | -1.59 |
| 176 | -0.21 | -7.35 | -22.87 | -2.52  | -2.79 | -3.04 | -3.04  | -1.43 |
| 177 | 0.42  | -6.96 | -22.77 | -14.25 | -2.87 | -2.93 | -3.93  | -1.45 |
| 178 | -0.02 | -6.32 | -22.67 | 6.85   | -2.92 | -3.19 | -7.99  | -1.45 |
| 179 | -0.09 | -6.74 | -22.78 | -7.19  | -2.85 | -2.83 | 8.51   | -1.45 |
| 180 | 0.17  | -6.39 | -22.66 | 56.84  | -2.89 | -2.75 | 1.92   | -1.46 |
| 181 | 0.19  | -5.99 | -22.80 | 8.81   | -2.85 | -2.82 | -0.75  | -1.45 |
| 182 | 0.14  | -5.99 | -22.76 | 24.31  | -2.86 | -2.90 | -4.31  | -1.46 |
| 183 | -0.08 | -7.56 | -22.53 | 13.39  | -2.79 | -2.74 | 16.60  | -1.45 |
| 184 | -0.11 | -6.77 | -22.95 | 19.90  | -3.09 | -2.74 | 1.85   | -1.47 |
| 185 | 0.08  | -6.63 | -22.79 | 28.47  | -2.86 | -3.04 | -8.57  | -1.44 |
| 186 | -0.04 | -6.29 | -22.94 | 53.97  | -2.89 | -3.03 | -0.93  | -1.44 |

|     |       |       |        |        |       |       |       |       |
|-----|-------|-------|--------|--------|-------|-------|-------|-------|
| 187 | -0.14 | -6.33 | -22.59 | 99.70  | -3.14 | -2.75 | -4.33 | -1.43 |
| 188 | 0.00  | -7.06 | -22.71 | -0.85  | -2.92 | -2.89 | 0.54  | -1.45 |
| 189 | -0.07 | -8.13 | -22.59 | -18.23 | -2.91 | -3.01 | -8.11 | -1.44 |
| 190 | -0.17 | -7.08 | -22.39 | 23.48  | -2.85 | -2.87 | -3.17 | -1.46 |
| 191 | -0.06 | -7.27 | -22.67 | 41.89  | -2.92 | -2.76 | -3.93 | -1.46 |
| 192 | -0.08 | -4.09 | -16.51 | -9.54  | -3.59 | -2.93 | -0.63 | -1.16 |
| 193 | -0.14 | -3.11 | -16.34 | 18.48  | -3.60 | -3.00 | 1.15  | -1.14 |
| 194 | 0.04  | -3.51 | -16.64 | 17.91  | -3.51 | -2.91 | 1.10  | -1.13 |
| 195 | -0.17 | -3.94 | -16.52 | 15.91  | -3.51 | -2.89 | -1.44 | -1.13 |
| 196 | 0.08  | -3.88 | -16.58 | -20.06 | -3.47 | -2.90 | 5.56  | -1.13 |
| 197 | 0.09  | -3.67 | -16.49 | -16.57 | -3.57 | -2.88 | 3.05  | -1.13 |
| 198 | 0.05  | -3.94 | -16.42 | -16.93 | -3.42 | -2.97 | 1.59  | -1.11 |
| 199 | 0.02  | -4.02 | -16.51 | -25.71 | -3.60 | -2.90 | 2.24  | -1.14 |

| k   | a24k  | a25k  | a26k  | a27k  | a28k   | a29k | a30k | a31k   |
|-----|-------|-------|-------|-------|--------|------|------|--------|
| 160 | -1.72 | -4.57 | -3.31 | -5.64 | 1.91   | 1.82 | 6.79 | -74.96 |
| 161 | 2.72  | -4.07 | 0.55  | -5.64 | 1.76   | 1.89 | 6.87 | -73.97 |
| 162 | -4.32 | -4.19 | -2.77 | -5.46 | 2.01   | 1.92 | 7.05 | -73.97 |
| 163 | -1.39 | -4.01 | -0.95 | -5.48 | 2.46   | 1.69 | 6.74 | -74.71 |
| 164 | -0.61 | -4.33 | -1.97 | -5.78 | 2.93   | 1.84 | 6.85 | -75.67 |
| 165 | 2.85  | -4.36 | -3.46 | -5.41 | 2.81   | 1.98 | 6.35 | -75.37 |
| 166 | -1.58 | -3.98 | -1.06 | -5.35 | 1.94   | 1.90 | 6.59 | -74.94 |
| 167 | -0.38 | -5.08 | -3.13 | -5.36 | 2.77   | 1.81 | 6.39 | -75.00 |
| 168 | 2.47  | -3.87 | -0.88 | -5.81 | 2.49   | 1.71 | 7.18 | -75.15 |
| 169 | -2.51 | -4.45 | -6.66 | -4.91 | 1.83   | 2.03 | 6.90 | -76.07 |
| 170 | 6.57  | -3.95 | -4.60 | -5.75 | 1.82   | 2.01 | 6.96 | -73.68 |
| 171 | 2.24  | -3.92 | -1.15 | -5.47 | 2.71   | 1.61 | 6.63 | -73.51 |
| 172 | 3.40  | -4.54 | -2.52 | -5.54 | 2.40   | 2.00 | 6.80 | -73.48 |
| 173 | -0.75 | -4.31 | 2.15  | -5.78 | 2.35   | 1.56 | 6.88 | -74.65 |
| 174 | 3.78  | -4.76 | -3.69 | -5.35 | 1.90   | 1.99 | 7.05 | -75.07 |
| 175 | 2.70  | -4.13 | -3.26 | -5.61 | 1.83   | 1.74 | 6.62 | -75.22 |
| 176 | -3.40 | -4.70 | -1.81 | -4.85 | 1.49   | 1.10 | 5.28 | -69.91 |
| 177 | 3.89  | -4.52 | -2.20 | -5.42 | 1.47   | 1.05 | 5.44 | -68.29 |
| 178 | -0.89 | -4.57 | -0.54 | -5.06 | 1.72   | 1.23 | 5.32 | -68.69 |
| 179 | 0.34  | -4.88 | -0.94 | -5.10 | 1.60   | 1.22 | 5.19 | -67.29 |
| 180 | 2.91  | -4.43 | -2.20 | -5.38 | 1.18   | 1.28 | 5.03 | -68.04 |
| 181 | 0.20  | -4.51 | 0.35  | -5.67 | 1.11   | 1.13 | 5.20 | -69.65 |
| 182 | -5.24 | -4.31 | -1.03 | -5.08 | 1.14   | 1.21 | 4.73 | -67.72 |
| 183 | 2.46  | -4.05 | 0.58  | -5.67 | 0.91   | 1.03 | 5.01 | -68.14 |
| 184 | -1.50 | -4.78 | -1.48 | -5.34 | 1.04   | 1.23 | 5.58 | -68.25 |
| 185 | 0.24  | -5.14 | 0.73  | -4.87 | 1.21   | 1.05 | 5.17 | -68.14 |
| 186 | -0.78 | -4.30 | -4.96 | -5.25 | 1.03   | 1.33 | 5.06 | -68.46 |
| 187 | 1.72  | -5.38 | 0.77  | -4.80 | 1.35   | 0.99 | 5.47 | -67.65 |
| 188 | -0.43 | -5.19 | -0.03 | -5.43 | 1.03   | 1.19 | 5.13 | -71.14 |
| 189 | 4.51  | -4.64 | -3.26 | -4.93 | 0.78   | 0.95 | 5.52 | -68.34 |
| 190 | 0.38  | -4.65 | 0.48  | -4.86 | 1.19   | 1.08 | 5.19 | -69.56 |
| 191 | -3.32 | -4.37 | 4.83  | -5.59 | 1.22   | 1.00 | 5.24 | -71.35 |
| 192 | -1.10 | -6.19 | 10.44 | -5.27 | -10.64 | 1.78 | 8.50 | -46.49 |
| 193 | -0.22 | -5.99 | 10.70 | -5.10 | -10.32 | 1.84 | 8.95 | -46.44 |
| 194 | 1.19  | -6.00 | 10.41 | -5.23 | -9.90  | 1.72 | 8.65 | -45.75 |
| 195 | -0.99 | -6.04 | 11.34 | -5.36 | -10.16 | 1.78 | 8.59 | -44.99 |
| 196 | -1.78 | -5.87 | 9.88  | -5.37 | -10.11 | 1.82 | 8.86 | -46.13 |

|     |       |       |       |       |        |      |      |        |
|-----|-------|-------|-------|-------|--------|------|------|--------|
| 197 | 0.12  | -6.17 | 11.67 | -5.32 | -10.56 | 1.82 | 8.33 | -46.32 |
| 198 | 0.29  | -6.03 | 10.23 | -5.05 | -10.16 | 1.82 | 8.42 | -44.92 |
| 199 | -2.02 | -5.95 | 9.82  | -5.22 | -9.75  | 1.57 | 8.69 | -45.94 |

| k   | a32k  | a33k | a34k | a35k  | a36k  | a37k   | a38k  | a39k  |
|-----|-------|------|------|-------|-------|--------|-------|-------|
| 160 | -7.10 | 3.84 | 2.29 | 7.25  | -1.41 | -42.89 | -2.47 | -2.31 |
| 161 | -7.16 | 4.00 | 2.40 | 6.79  | -1.29 | -44.10 | -2.43 | -2.37 |
| 162 | -6.88 | 3.82 | 2.18 | 6.67  | -1.35 | -44.13 | -2.49 | -2.35 |
| 163 | -7.18 | 3.48 | 2.31 | 6.99  | -1.38 | -42.27 | -2.53 | -2.32 |
| 164 | -6.67 | 3.67 | 2.34 | 6.50  | -1.34 | -42.33 | -2.46 | -2.34 |
| 165 | -8.06 | 3.80 | 2.15 | 6.43  | -1.41 | -42.56 | -2.48 | -2.36 |
| 166 | -6.97 | 3.84 | 2.07 | 6.30  | -1.37 | -41.59 | -2.35 | -2.30 |
| 167 | -7.50 | 3.63 | 2.27 | 8.27  | -1.29 | -43.63 | -2.40 | -2.33 |
| 168 | -7.23 | 3.84 | 2.14 | 7.07  | -1.36 | -42.42 | -2.44 | -2.32 |
| 169 | -6.82 | 3.72 | 2.20 | 6.12  | -1.31 | -43.38 | -2.53 | -2.33 |
| 170 | -6.81 | 3.71 | 2.21 | 7.09  | -1.38 | -41.57 | -2.45 | -2.37 |
| 171 | -6.97 | 3.74 | 2.16 | 7.13  | -1.35 | -42.14 | -2.49 | -2.32 |
| 172 | -6.01 | 3.77 | 2.13 | 7.48  | -1.41 | -42.50 | -2.42 | -2.30 |
| 173 | -6.58 | 3.62 | 2.20 | 6.64  | -1.30 | -41.40 | -2.46 | -2.35 |
| 174 | -6.09 | 3.80 | 2.19 | 6.27  | -1.46 | -42.84 | -2.53 | -2.33 |
| 175 | -7.31 | 3.70 | 2.41 | 6.68  | -1.30 | -41.21 | -2.56 | -2.35 |
| 176 | -6.83 | 3.47 | 2.53 | 4.38  | -1.11 | -39.26 | -2.01 | -2.17 |
| 177 | -7.14 | 3.45 | 2.16 | 4.46  | -1.08 | -38.15 | -2.01 | -2.17 |
| 178 | -6.84 | 3.45 | 2.44 | 4.82  | -1.20 | -37.87 | -1.96 | -2.14 |
| 179 | -7.17 | 3.28 | 2.32 | 4.76  | -1.15 | -38.48 | -2.04 | -2.17 |
| 180 | -6.73 | 3.38 | 2.39 | 3.85  | -1.07 | -36.51 | -1.99 | -2.13 |
| 181 | -6.45 | 3.47 | 2.29 | 5.05  | -0.93 | -37.26 | -1.92 | -2.14 |
| 182 | -6.41 | 3.52 | 2.16 | 4.22  | -1.01 | -40.58 | -2.04 | -2.16 |
| 183 | -6.37 | 3.34 | 2.26 | 4.47  | -1.11 | -38.40 | -1.98 | -2.14 |
| 184 | -6.07 | 3.42 | 2.22 | 5.10  | -1.07 | -38.98 | -1.95 | -2.18 |
| 185 | -6.69 | 3.57 | 2.23 | 4.02  | -1.03 | -37.01 | -2.01 | -2.19 |
| 186 | -6.10 | 3.26 | 2.21 | 4.04  | -1.15 | -37.83 | -1.92 | -2.15 |
| 187 | -6.41 | 3.36 | 2.25 | 4.21  | -1.12 | -41.20 | -2.00 | -2.14 |
| 188 | -6.48 | 3.30 | 2.49 | 4.27  | -1.03 | -39.02 | -1.98 | -2.14 |
| 189 | -6.73 | 3.32 | 2.34 | 4.37  | -1.12 | -37.91 | -1.96 | -2.18 |
| 190 | -6.61 | 3.66 | 2.16 | 4.43  | -1.14 | -37.43 | -1.86 | -2.15 |
| 191 | -7.03 | 3.45 | 2.56 | 4.75  | -1.13 | -38.65 | -1.94 | -2.17 |
| 192 | 14.34 | 3.23 | 0.52 | -6.89 | -0.44 | -54.93 | -0.40 | -0.38 |
| 193 | 15.28 | 3.27 | 0.54 | -6.69 | -0.47 | -54.20 | -0.34 | -0.37 |
| 194 | 15.25 | 3.25 | 0.65 | -6.55 | -0.50 | -55.28 | -0.39 | -0.36 |
| 195 | 15.22 | 3.25 | 0.61 | -6.99 | -0.47 | -54.89 | -0.36 | -0.38 |
| 196 | 15.23 | 3.32 | 0.62 | -6.76 | -0.44 | -55.82 | -0.36 | -0.31 |
| 197 | 14.98 | 3.30 | 0.73 | -7.01 | -0.45 | -54.51 | -0.36 | -0.36 |
| 198 | 14.34 | 3.25 | 0.68 | -7.27 | -0.47 | -54.67 | -0.36 | -0.37 |
| 199 | 14.44 | 3.17 | 0.51 | -6.82 | -0.48 | -54.34 | -0.37 | -0.36 |

| k   | a40k | a41k | a42k  | a43k  | a44k  | a45k  | a46k   | a47k  |
|-----|------|------|-------|-------|-------|-------|--------|-------|
| 160 | 0.37 | 6.16 | -4.89 | -3.95 | -0.18 | -5.80 | 216.68 | -4.24 |
| 161 | 0.44 | 6.30 | -5.02 | -1.35 | 9.89  | -6.94 | 215.70 | 0.94  |
| 162 | 0.47 | 5.80 | -2.58 | -2.61 | 8.27  | -5.37 | 202.79 | -3.53 |

|     |      |      |       |       |       |       |        |       |
|-----|------|------|-------|-------|-------|-------|--------|-------|
| 163 | 0.41 | 6.05 | 2.06  | -4.03 | 5.60  | -6.26 | 205.63 | -4.57 |
| 164 | 0.40 | 5.86 | 4.56  | -3.05 | -1.66 | -6.13 | 201.34 | -2.44 |
| 165 | 0.43 | 6.14 | -4.27 | -1.20 | -3.83 | -6.40 | 219.57 | -2.51 |
| 166 | 0.35 | 6.13 | -2.92 | -2.09 | 8.70  | -6.09 | 207.33 | 1.18  |
| 167 | 0.39 | 5.87 | -2.87 | -4.76 | 8.99  | -7.35 | 200.76 | 5.54  |
| 168 | 0.38 | 6.11 | 2.58  | -1.78 | -3.50 | -5.84 | 212.19 | -5.32 |
| 169 | 0.41 | 5.65 | -0.03 | -1.87 | 9.47  | -6.14 | 212.81 | 2.41  |
| 170 | 0.44 | 5.74 | -2.44 | -1.68 | 6.05  | -6.48 | 202.50 | 3.86  |
| 171 | 0.45 | 6.48 | -1.42 | -1.59 | 4.23  | -6.51 | 218.85 | -1.59 |
| 172 | 0.38 | 6.75 | 2.44  | -0.51 | 1.69  | -6.58 | 204.05 | -0.83 |
| 173 | 0.46 | 5.72 | -0.38 | -0.25 | 5.15  | -5.50 | 191.42 | 0.98  |
| 174 | 0.45 | 6.33 | 3.98  | -2.61 | -5.38 | -5.59 | 205.05 | -3.88 |
| 175 | 0.40 | 5.97 | 6.85  | -2.28 | 7.55  | -6.99 | 200.65 | -0.70 |
| 176 | 0.45 | 7.06 | -0.46 | -1.72 | 1.09  | -6.70 | 193.26 | 3.23  |
| 177 | 0.44 | 6.59 | 0.91  | -1.08 | -0.45 | -7.42 | 197.52 | 2.04  |
| 178 | 0.46 | 7.05 | -2.54 | -0.60 | 1.25  | -6.31 | 187.34 | 0.15  |
| 179 | 0.47 | 6.80 | -3.35 | -0.78 | 2.85  | -7.77 | 196.66 | 2.32  |
| 180 | 0.45 | 6.54 | -0.56 | -2.66 | 5.28  | -7.34 | 204.22 | -2.50 |
| 181 | 0.49 | 6.72 | 2.50  | -0.06 | 5.80  | -6.91 | 198.68 | -4.05 |
| 182 | 0.51 | 7.05 | -4.15 | -2.54 | -1.36 | -7.48 | 201.35 | 0.50  |
| 183 | 0.45 | 7.01 | -1.19 | -0.48 | 3.29  | -8.19 | 205.06 | -2.27 |
| 184 | 0.50 | 6.51 | 1.42  | -1.52 | 2.04  | -7.31 | 184.98 | 5.86  |
| 185 | 0.46 | 6.88 | 0.90  | -0.33 | 1.45  | -7.19 | 200.08 | -1.33 |
| 186 | 0.41 | 6.38 | -0.92 | 0.12  | -2.64 | -8.02 | 193.03 | 4.59  |
| 187 | 0.48 | 7.11 | -1.69 | -1.28 | -2.29 | -7.25 | 195.79 | -4.40 |
| 188 | 0.44 | 6.54 | 3.58  | -0.63 | 5.67  | -6.94 | 198.26 | -5.30 |
| 189 | 0.42 | 6.39 | -4.58 | -1.29 | 7.96  | -7.29 | 198.65 | -3.19 |
| 190 | 0.46 | 6.64 | -4.68 | -1.11 | -4.70 | -7.46 | 196.34 | 3.04  |
| 191 | 0.45 | 6.81 | 2.51  | -2.38 | 0.31  | -7.21 | 195.56 | 1.44  |
| 192 | 0.51 | 7.88 | -0.86 | 5.86  | -1.23 | -9.89 | 31.77  | -1.15 |
| 193 | 0.49 | 7.70 | 0.24  | 5.63  | -4.70 | -9.82 | 32.51  | -0.43 |
| 194 | 0.54 | 7.71 | -0.61 | 5.51  | -5.12 | -9.75 | 32.83  | 1.99  |
| 195 | 0.49 | 7.83 | 0.62  | 5.83  | 2.58  | -9.67 | 33.03  | 0.79  |
| 196 | 0.52 | 7.78 | -1.62 | 5.24  | -1.38 | -9.49 | 37.38  | 1.85  |
| 197 | 0.54 | 7.94 | 0.26  | 4.88  | 0.93  | -9.66 | 28.25  | -0.89 |
| 198 | 0.48 | 7.63 | -0.16 | 6.08  | 1.97  | -9.63 | 26.78  | -1.78 |
| 199 | 0.50 | 7.86 | -1.26 | 5.43  | 1.97  | -9.74 | 29.21  | -1.97 |

| k   | a48k   | a49k | a50k | a51k   | a52k  | a53k  | a54k  | a55k  |
|-----|--------|------|------|--------|-------|-------|-------|-------|
| 160 | -15.09 | 1.43 | 8.38 | 2.09   | 67.10 | 50.59 | -4.05 | -2.61 |
| 161 | -15.99 | 1.74 | 8.50 | -0.09  | 74.46 | 49.89 | -3.80 | -2.33 |
| 162 | -15.23 | 1.34 | 8.48 | -1.80  | 66.68 | 50.94 | -3.90 | -2.70 |
| 163 | -14.71 | 1.41 | 6.35 | -3.76  | 65.56 | 52.11 | -3.89 | -3.17 |
| 164 | -15.64 | 1.35 | 8.94 | -11.87 | 69.07 | 50.02 | -3.94 | -2.36 |
| 165 | -15.55 | 1.51 | 9.37 | 4.37   | 68.67 | 53.08 | -3.92 | -2.80 |
| 166 | -15.40 | 1.37 | 8.38 | 2.83   | 69.57 | 52.07 | -3.76 | -1.98 |
| 167 | -15.25 | 1.65 | 7.20 | -1.24  | 69.80 | 49.58 | -4.03 | -3.15 |
| 168 | -15.21 | 1.29 | 7.64 | -5.76  | 65.67 | 51.80 | -3.99 | -4.23 |
| 169 | -15.65 | 1.44 | 7.37 | 0.61   | 67.24 | 51.04 | -3.94 | -3.61 |
| 170 | -15.53 | 1.54 | 7.44 | -3.07  | 72.23 | 52.70 | -4.07 | -2.45 |
| 171 | -14.94 | 1.54 | 9.01 | -0.48  | 66.04 | 51.64 | -4.11 | -2.67 |
| 172 | -15.34 | 1.52 | 8.43 | 2.06   | 71.91 | 50.70 | -4.02 | -2.48 |

|     |        |      |       |       |        |       |       |       |
|-----|--------|------|-------|-------|--------|-------|-------|-------|
| 173 | -14.78 | 1.57 | 7.45  | -2.85 | 62.87  | 50.65 | -3.87 | -1.81 |
| 174 | -15.51 | 1.51 | 9.59  | 0.14  | 65.71  | 49.51 | -4.02 | -1.82 |
| 175 | -15.48 | 1.26 | 6.97  | 7.15  | 69.96  | 53.52 | -3.95 | -3.26 |
| 176 | -12.35 | 1.49 | 9.88  | -9.25 | 57.89  | 46.03 | -3.59 | -1.89 |
| 177 | -12.31 | 1.61 | 9.42  | -3.73 | 67.80  | 45.21 | -3.64 | -2.45 |
| 178 | -12.14 | 1.49 | 10.23 | -0.63 | 58.49  | 45.33 | -3.73 | -2.72 |
| 179 | -12.13 | 1.61 | 10.38 | 0.24  | 55.05  | 46.72 | -3.63 | -2.00 |
| 180 | -11.85 | 1.51 | 9.89  | -9.35 | 58.31  | 44.55 | -3.63 | -3.61 |
| 181 | -11.39 | 1.68 | 10.30 | -3.55 | 55.82  | 44.55 | -3.69 | -0.74 |
| 182 | -11.71 | 1.58 | 9.81  | -3.65 | 60.66  | 45.97 | -3.65 | -2.52 |
| 183 | -11.84 | 1.62 | 7.87  | -3.91 | 62.48  | 45.42 | -3.52 | -2.08 |
| 184 | -12.30 | 1.57 | 9.70  | 8.09  | 55.78  | 46.36 | -3.52 | -2.01 |
| 185 | -11.78 | 1.63 | 8.33  | -0.93 | 55.84  | 45.39 | -3.58 | -2.29 |
| 186 | -11.46 | 1.71 | 9.94  | -4.37 | 60.19  | 44.13 | -3.66 | -3.12 |
| 187 | -12.17 | 1.61 | 8.29  | -5.53 | 58.65  | 44.83 | -3.78 | -2.80 |
| 188 | -11.84 | 1.68 | 7.48  | -2.86 | 65.02  | 45.54 | -3.61 | -2.11 |
| 189 | -11.77 | 1.64 | 9.14  | 2.22  | 55.82  | 44.24 | -3.61 | -2.37 |
| 190 | -12.38 | 1.71 | 7.76  | -4.37 | 63.61  | 45.04 | -3.67 | -2.30 |
| 191 | -11.86 | 1.57 | 8.30  | 0.81  | 62.45  | 45.01 | -3.51 | -2.05 |
| 192 | -14.53 | 1.99 | 13.18 | 1.33  | 106.76 | 38.07 | -5.34 | -5.15 |
| 193 | -13.82 | 2.08 | 13.41 | 1.91  | 104.39 | 37.64 | -5.16 | -4.44 |
| 194 | -14.37 | 2.15 | 13.46 | -0.20 | 110.70 | 38.01 | -5.34 | -5.04 |
| 195 | -13.91 | 2.12 | 13.46 | 0.42  | 102.96 | 38.66 | -5.33 | -4.93 |
| 196 | -14.24 | 2.07 | 14.12 | 2.08  | 97.99  | 38.20 | -5.46 | -5.31 |
| 197 | -14.21 | 2.13 | 13.13 | -1.64 | 100.89 | 37.80 | -5.44 | -4.59 |
| 198 | -14.39 | 2.09 | 13.96 | -2.45 | 103.11 | 38.34 | -5.33 | -5.41 |
| 199 | -13.74 | 2.08 | 13.50 | -2.52 | 105.29 | 38.24 | -5.41 | -4.78 |

| k   | a56k  | a57k  | a58k  | a59k | a60k | a61k  | a62k   | a63k   |
|-----|-------|-------|-------|------|------|-------|--------|--------|
| 160 | 4.80  | -3.26 | -0.21 | 3.57 | 2.42 | -0.68 | -12.67 | -24.28 |
| 161 | -3.84 | -4.72 | -0.18 | 2.94 | 2.29 | -0.37 | 1.69   | 35.79  |
| 162 | 2.38  | -4.79 | -0.18 | 3.21 | 2.09 | -0.33 | 4.88   | -13.73 |
| 163 | -0.25 | -2.02 | -0.17 | 2.96 | 2.59 | -0.43 | 27.32  | 49.32  |
| 164 | 3.73  | -5.32 | -0.15 | 3.77 | 2.77 | -0.50 | -9.12  | 57.33  |
| 165 | 0.28  | -2.82 | -0.17 | 3.31 | 2.63 | -0.50 | 22.97  | 31.74  |
| 166 | -4.59 | -3.20 | -0.18 | 3.25 | 3.45 | -0.72 | -15.61 | 33.03  |
| 167 | -1.01 | -4.60 | -0.17 | 4.16 | 2.50 | -0.33 | -27.43 | -11.64 |
| 168 | -4.66 | -1.98 | -0.19 | 4.12 | 2.54 | -0.22 | -1.92  | -10.90 |
| 169 | 3.96  | -7.01 | -0.20 | 2.86 | 2.56 | -0.62 | 17.28  | -12.73 |
| 170 | 2.95  | -4.19 | -0.21 | 3.43 | 2.63 | -0.19 | -14.50 | -1.16  |
| 171 | -7.27 | -3.21 | -0.20 | 3.49 | 2.62 | -0.49 | 1.07   | -12.08 |
| 172 | 7.69  | -6.61 | -0.16 | 2.74 | 2.46 | -0.33 | 12.29  | -21.91 |
| 173 | 0.38  | -3.77 | -0.20 | 3.10 | 2.67 | -0.35 | -8.43  | 25.41  |
| 174 | 1.04  | -2.66 | -0.18 | 3.00 | 2.98 | -0.48 | -6.63  | -62.55 |
| 175 | 1.60  | -1.83 | -0.18 | 3.19 | 2.77 | -0.62 | -10.72 | -93.17 |
| 176 | -0.52 | -7.31 | -0.07 | 2.13 | 2.59 | -0.40 | -6.00  | 14.91  |
| 177 | -0.77 | -6.06 | -0.12 | 1.95 | 3.14 | -0.16 | -9.91  | -11.31 |
| 178 | -1.38 | -6.04 | -0.11 | 1.95 | 2.90 | -0.10 | -14.84 | -5.31  |
| 179 | 3.62  | -5.16 | -0.12 | 2.88 | 2.42 | -0.29 | -2.65  | 24.60  |
| 180 | -0.82 | -5.22 | -0.08 | 2.50 | 2.59 | -0.01 | -9.33  | 68.11  |
| 181 | 1.07  | -4.69 | -0.09 | 1.89 | 2.66 | -0.23 | 5.36   | 72.62  |
| 182 | 1.54  | -7.98 | -0.10 | 2.35 | 2.91 | -0.19 | -19.95 | -12.65 |

|     |       |        |       |       |       |       |        |        |
|-----|-------|--------|-------|-------|-------|-------|--------|--------|
| 183 | 0.38  | -6.18  | -0.10 | 2.44  | 2.23  | -0.51 | 30.27  | 7.42   |
| 184 | -1.62 | -7.10  | -0.09 | 1.94  | 2.99  | -0.47 | -3.11  | -34.92 |
| 185 | 2.42  | -5.25  | -0.08 | 2.10  | 2.45  | -0.43 | 3.54   | 3.01   |
| 186 | -5.80 | -6.76  | -0.10 | 1.44  | 2.25  | -0.49 | -21.05 | 17.21  |
| 187 | 7.44  | -5.48  | -0.12 | 2.46  | 2.44  | -0.27 | 23.77  | 30.52  |
| 188 | -5.44 | -6.90  | -0.11 | 2.41  | 2.25  | -0.51 | 10.61  | -0.47  |
| 189 | 0.88  | -6.94  | -0.09 | 2.24  | 2.12  | -0.02 | 20.18  | -19.77 |
| 190 | 8.28  | -6.66  | -0.10 | 2.54  | 2.34  | -0.33 | 21.36  | 57.54  |
| 191 | -3.19 | -5.28  | -0.09 | 2.05  | 2.63  | -0.25 | 7.47   | 31.17  |
| 192 | 5.63  | -31.18 | 0.32  | -1.71 | 12.41 | 0.20  | -0.23  | 7.04   |
| 193 | 2.70  | -30.14 | 0.28  | -2.06 | 12.10 | 0.16  | -29.14 | 20.81  |
| 194 | -1.45 | -29.85 | 0.28  | -1.81 | 12.41 | 0.08  | 17.53  | 16.41  |
| 195 | -2.76 | -29.39 | 0.31  | -1.97 | 12.28 | 0.17  | -0.96  | 15.02  |
| 196 | -3.77 | -29.74 | 0.29  | -2.19 | 12.65 | 0.09  | -9.79  | 31.38  |
| 197 | -3.05 | -29.86 | 0.30  | -1.86 | 12.48 | 0.13  | -18.07 | 9.58   |
| 198 | -2.17 | -30.42 | 0.30  | -1.83 | 12.21 | 0.18  | -5.19  | 14.38  |
| 199 | -2.93 | -30.44 | 0.29  | -1.97 | 12.62 | 0.04  | 16.43  | -0.26  |

| k   | a0k     | a1k  | a2k  | a3k    | a4k   | a5k   | a6k   | a7k   |
|-----|---------|------|------|--------|-------|-------|-------|-------|
| 200 | -84.84  | 1.13 | 5.88 | -40.53 | -7.81 | 13.76 | 5.16  | -2.61 |
| 201 | -86.46  | 1.17 | 5.92 | -41.73 | -7.49 | 14.37 | 4.96  | -2.65 |
| 202 | -85.65  | 0.92 | 5.61 | -41.84 | -7.73 | 14.39 | 4.88  | -3.04 |
| 203 | -83.58  | 0.92 | 5.57 | -41.07 | -7.79 | 14.00 | 5.78  | -2.87 |
| 204 | -84.03  | 0.90 | 5.59 | -40.04 | -8.07 | 15.05 | 5.36  | -2.80 |
| 205 | -88.01  | 1.00 | 5.82 | -40.27 | -7.37 | 14.65 | 5.64  | -2.95 |
| 206 | -91.14  | 1.09 | 5.67 | -38.94 | -7.49 | 13.90 | 5.00  | -3.05 |
| 207 | -87.84  | 1.13 | 5.79 | -39.92 | -7.62 | 14.06 | 4.61  | -3.12 |
| 208 | -139.32 | 2.88 | 8.38 | -29.54 | -7.87 | 11.87 | 0.06  | 1.37  |
| 209 | -136.35 | 2.65 | 8.20 | -29.63 | -7.90 | 11.88 | 0.14  | 1.88  |
| 210 | -136.39 | 2.88 | 8.19 | -29.62 | -7.91 | 11.54 | 0.09  | 1.74  |
| 211 | -140.86 | 2.90 | 8.40 | -28.91 | -8.02 | 11.83 | 1.40  | 1.81  |
| 212 | -141.28 | 2.64 | 8.02 | -30.90 | -8.11 | 11.87 | 0.91  | 1.38  |
| 213 | -147.44 | 3.03 | 8.55 | -29.96 | -8.18 | 12.28 | 0.27  | 0.89  |
| 214 | -137.06 | 2.83 | 8.21 | -28.00 | -7.95 | 11.69 | -0.13 | 1.43  |
| 215 | -136.69 | 2.81 | 8.31 | -31.18 | -8.32 | 12.24 | -0.51 | 1.76  |
| 216 | -144.43 | 3.02 | 8.72 | -30.56 | -7.94 | 11.59 | 0.02  | 1.81  |
| 217 | -137.63 | 2.89 | 8.13 | -30.66 | -7.76 | 11.71 | 0.40  | 1.39  |
| 218 | -132.91 | 2.90 | 8.74 | -30.41 | -8.04 | 11.61 | 0.24  | 2.17  |
| 219 | -146.30 | 2.70 | 8.31 | -32.57 | -8.00 | 12.00 | 0.33  | 1.31  |
| 220 | -141.57 | 2.69 | 8.20 | -30.05 | -8.02 | 11.79 | 0.65  | 1.73  |
| 221 | -143.00 | 2.79 | 8.25 | -29.94 | -7.79 | 11.74 | -0.08 | 1.70  |
| 222 | -138.77 | 2.83 | 8.47 | -28.20 | -7.64 | 11.82 | 0.14  | 1.48  |
| 223 | -140.99 | 2.63 | 8.27 | -30.91 | -7.93 | 11.74 | 0.18  | 1.55  |
| 224 | -54.29  | 0.60 | 4.08 | -17.30 | -4.02 | 6.24  | 2.53  | -0.47 |
| 225 | -49.96  | 0.52 | 4.11 | -17.63 | -3.57 | 6.25  | 2.57  | -0.65 |
| 226 | -53.86  | 0.63 | 4.07 | -16.88 | -3.88 | 6.15  | 2.39  | -0.17 |
| 227 | -53.38  | 0.55 | 4.12 | -16.78 | -3.97 | 6.27  | 2.64  | -0.44 |
| 228 | -52.04  | 0.58 | 4.25 | -17.54 | -3.97 | 6.26  | 2.50  | -0.44 |
| 229 | -49.03  | 0.59 | 4.08 | -16.65 | -3.89 | 6.26  | 2.41  | -0.68 |
| 230 | -51.57  | 0.47 | 3.92 | -17.80 | -4.05 | 6.13  | 2.83  | -0.45 |
| 231 | -51.91  | 0.73 | 4.07 | -16.88 | -3.72 | 6.31  | 2.68  | -0.76 |
| 232 | -53.06  | 0.54 | 4.08 | -17.42 | -3.88 | 6.13  | 2.80  | -0.66 |



|     |        |      |      |        |       |      |      |       |
|-----|--------|------|------|--------|-------|------|------|-------|
| 233 | -55.52 | 0.57 | 4.00 | -17.43 | -3.79 | 6.01 | 2.67 | -0.74 |
| 234 | -52.68 | 0.62 | 3.76 | -16.64 | -3.85 | 6.32 | 2.48 | -0.51 |
| 235 | -57.68 | 0.48 | 4.06 | -16.80 | -3.86 | 5.94 | 2.56 | -0.56 |
| 236 | -53.28 | 0.60 | 4.17 | -17.07 | -3.56 | 6.08 | 2.93 | -0.65 |
| 237 | -50.64 | 0.50 | 4.15 | -16.55 | -3.98 | 5.93 | 2.86 | -0.63 |
| 238 | -54.90 | 0.57 | 3.85 | -17.16 | -3.98 | 6.12 | 2.89 | -0.43 |
| 239 | -54.82 | 0.69 | 3.85 | -16.27 | -3.91 | 6.14 | 2.75 | -0.44 |

| k   | a8k    | a9k   | a10k  | a11k  | a12k  | a13k   | a14k | a15k   |
|-----|--------|-------|-------|-------|-------|--------|------|--------|
| 200 | 66.06  | 57.96 | -0.06 | 20.15 | 25.87 | -9.12  | 2.49 | -40.56 |
| 201 | 68.69  | 59.17 | -0.05 | 18.98 | 26.20 | -9.42  | 2.57 | -38.06 |
| 202 | 65.93  | 58.25 | -0.06 | 17.31 | 26.06 | -9.06  | 2.45 | -37.70 |
| 203 | 66.02  | 57.57 | -0.06 | 17.54 | 24.57 | -9.56  | 2.48 | -37.48 |
| 204 | 62.62  | 60.56 | -0.05 | 18.59 | 26.34 | -9.47  | 2.50 | -37.12 |
| 205 | 69.42  | 59.25 | -0.07 | 18.44 | 25.76 | -9.14  | 2.51 | -38.09 |
| 206 | 69.43  | 58.91 | -0.08 | 17.90 | 25.60 | -9.19  | 2.56 | -38.37 |
| 207 | 66.76  | 59.12 | -0.06 | 17.58 | 23.78 | -9.08  | 2.57 | -36.53 |
| 208 | 105.25 | 45.16 | 0.14  | 30.65 | 17.19 | -11.84 | 1.41 | -56.05 |
| 209 | 100.97 | 49.11 | 0.17  | 33.24 | 19.20 | -11.96 | 1.36 | -54.82 |
| 210 | 110.23 | 47.17 | 0.14  | 32.19 | 18.96 | -11.95 | 1.33 | -57.81 |
| 211 | 99.46  | 47.41 | 0.17  | 34.11 | 19.23 | -11.51 | 1.40 | -58.91 |
| 212 | 100.89 | 47.64 | 0.20  | 32.97 | 18.91 | -11.64 | 1.29 | -55.22 |
| 213 | 105.89 | 47.15 | 0.17  | 34.74 | 19.38 | -11.71 | 1.35 | -54.68 |
| 214 | 100.66 | 48.39 | 0.12  | 33.05 | 18.56 | -11.70 | 1.47 | -55.39 |
| 215 | 105.40 | 48.05 | 0.14  | 35.47 | 19.87 | -11.73 | 1.47 | -56.98 |
| 216 | 107.76 | 46.39 | 0.15  | 32.78 | 18.55 | -11.55 | 1.40 | -56.60 |
| 217 | 102.67 | 46.58 | 0.16  | 34.43 | 19.08 | -11.72 | 1.35 | -56.71 |
| 218 | 106.31 | 45.50 | 0.13  | 35.30 | 18.10 | -11.87 | 1.42 | -56.77 |
| 219 | 103.64 | 46.85 | 0.14  | 34.19 | 18.24 | -11.49 | 1.25 | -56.12 |
| 220 | 105.64 | 46.90 | 0.17  | 34.57 | 17.92 | -11.69 | 1.39 | -57.90 |
| 221 | 104.76 | 45.29 | 0.14  | 35.97 | 21.36 | -11.91 | 1.36 | -56.05 |
| 222 | 103.79 | 46.27 | 0.14  | 33.43 | 19.41 | -11.53 | 1.41 | -56.73 |
| 223 | 101.88 | 45.23 | 0.17  | 34.42 | 19.64 | -12.02 | 1.45 | -53.76 |
| 224 | 44.53  | 25.82 | -0.06 | 11.68 | 10.35 | -5.06  | 0.95 | -25.85 |
| 225 | 45.43  | 25.95 | -0.04 | 11.59 | 10.33 | -4.95  | 1.01 | -27.25 |
| 226 | 42.76  | 25.29 | -0.04 | 11.55 | 10.68 | -4.99  | 1.00 | -26.64 |
| 227 | 44.83  | 25.97 | -0.05 | 12.62 | 10.83 | -5.07  | 0.98 | -26.88 |
| 228 | 43.61  | 25.36 | -0.03 | 12.76 | 10.30 | -4.92  | 1.10 | -27.08 |
| 229 | 46.60  | 25.62 | -0.05 | 11.49 | 10.21 | -5.09  | 1.00 | -26.17 |
| 230 | 43.98  | 25.50 | -0.06 | 11.29 | 10.13 | -5.14  | 0.85 | -26.86 |
| 231 | 43.86  | 25.36 | -0.04 | 11.30 | 10.41 | -4.94  | 1.01 | -25.41 |
| 232 | 46.44  | 25.68 | -0.05 | 10.66 | 10.53 | -5.00  | 0.95 | -27.32 |
| 233 | 47.20  | 25.26 | -0.02 | 10.75 | 11.18 | -5.01  | 0.97 | -26.35 |
| 234 | 45.19  | 26.43 | -0.02 | 11.44 | 10.41 | -5.05  | 0.94 | -25.67 |
| 235 | 46.18  | 25.03 | -0.03 | 11.83 | 10.34 | -4.91  | 1.00 | -26.54 |
| 236 | 45.28  | 25.60 | -0.04 | 11.79 | 10.74 | -5.00  | 0.99 | -27.22 |
| 237 | 45.71  | 26.50 | -0.06 | 11.87 | 10.86 | -5.00  | 0.99 | -28.24 |
| 238 | 44.83  | 25.30 | -0.04 | 11.05 | 10.70 | -5.05  | 0.95 | -26.50 |
| 239 | 45.99  | 25.08 | -0.06 | 11.38 | 10.68 | -4.93  | 0.88 | -26.89 |

| k   | a16k  | a17k  | a18k   | a19k   | a20k  | a21k  | a22k  | a23k  |
|-----|-------|-------|--------|--------|-------|-------|-------|-------|
| 200 | -0.14 | -3.90 | -16.42 | 18.94  | -3.52 | -2.90 | 2.97  | -1.12 |
| 201 | -0.01 | -3.17 | -16.60 | 20.24  | -3.57 | -2.91 | 4.30  | -1.14 |
| 202 | 0.14  | -3.90 | -16.67 | 6.48   | -3.59 | -2.84 | -0.10 | -1.11 |
| 203 | -0.24 | -3.89 | -16.69 | 4.00   | -3.46 | -3.00 | -2.50 | -1.12 |
| 204 | -0.10 | -3.70 | -16.54 | 0.10   | -3.60 | -2.94 | -1.46 | -1.13 |
| 205 | -0.10 | -3.49 | -16.60 | 1.65   | -3.53 | -2.91 | 1.37  | -1.12 |
| 206 | 0.11  | -3.34 | -16.64 | 5.16   | -3.55 | -2.70 | -1.01 | -1.15 |
| 207 | 0.02  | -4.27 | -16.57 | -0.97  | -3.50 | -2.94 | 9.24  | -1.12 |
| 208 | 0.27  | -7.69 | -16.18 | 9.78   | -3.44 | -1.72 | 2.71  | -1.02 |
| 209 | -0.04 | -7.39 | -15.94 | -9.32  | -3.51 | -1.63 | 1.56  | -1.04 |
| 210 | -0.07 | -7.15 | -16.15 | 14.88  | -3.47 | -1.77 | -0.45 | -1.06 |
| 211 | 0.08  | -7.65 | -16.49 | 5.04   | -3.52 | -1.64 | -0.63 | -1.05 |
| 212 | -0.03 | -7.56 | -16.13 | -45.52 | -3.43 | -1.85 | -3.97 | -1.04 |
| 213 | 0.07  | -6.72 | -16.27 | -34.23 | -3.53 | -1.73 | -1.11 | -1.03 |
| 214 | 0.17  | -7.68 | -16.15 | 2.33   | -3.41 | -1.95 | 2.65  | -1.04 |
| 215 | 0.23  | -7.81 | -16.12 | 8.35   | -3.41 | -1.63 | -1.63 | -1.05 |
| 216 | -0.06 | -7.14 | -16.07 | -6.69  | -3.47 | -1.52 | 1.36  | -1.04 |
| 217 | -0.18 | -7.44 | -16.05 | -10.69 | -3.52 | -1.66 | 4.16  | -1.04 |
| 218 | -0.16 | -7.64 | -16.02 | 4.38   | -3.55 | -1.58 | -0.43 | -1.03 |
| 219 | -0.09 | -7.83 | -16.27 | 51.83  | -3.50 | -1.72 | 2.93  | -1.05 |
| 220 | -0.08 | -7.89 | -16.36 | -26.86 | -3.50 | -1.95 | 1.99  | -1.04 |
| 221 | -0.24 | -7.62 | -16.20 | -13.42 | -3.47 | -1.61 | -0.73 | -1.07 |
| 222 | 0.22  | -7.37 | -16.13 | -6.98  | -3.48 | -1.67 | 5.77  | -1.05 |
| 223 | -0.05 | -8.48 | -16.17 | 7.62   | -3.47 | -1.88 | 1.22  | -1.02 |
| 224 | 0.08  | -2.72 | -7.93  | -0.84  | -2.21 | -1.51 | 1.68  | -0.53 |
| 225 | 0.06  | -2.53 | -7.92  | 0.54   | -2.21 | -1.54 | 1.26  | -0.55 |
| 226 | -0.03 | -2.53 | -8.02  | 3.16   | -2.19 | -1.67 | 1.70  | -0.53 |
| 227 | -0.03 | -2.64 | -7.78  | 3.03   | -2.22 | -1.61 | -2.87 | -0.53 |
| 228 | -0.07 | -2.56 | -7.82  | -6.44  | -2.22 | -1.52 | -0.46 | -0.53 |
| 229 | 0.06  | -2.67 | -7.95  | 7.79   | -2.17 | -1.65 | -0.30 | -0.54 |
| 230 | 0.03  | -2.67 | -7.73  | -7.24  | -2.21 | -1.63 | -0.68 | -0.53 |
| 231 | 0.08  | -2.32 | -7.83  | -8.27  | -2.18 | -1.51 | 0.09  | -0.52 |
| 232 | -0.01 | -2.60 | -7.85  | -3.83  | -2.19 | -1.56 | -0.25 | -0.54 |
| 233 | 0.14  | -2.88 | -7.84  | 2.24   | -2.19 | -1.51 | 0.55  | -0.54 |
| 234 | -0.05 | -2.42 | -7.90  | -4.40  | -2.19 | -1.60 | 0.13  | -0.54 |
| 235 | -0.02 | -2.55 | -7.79  | -11.70 | -2.22 | -1.55 | -0.60 | -0.53 |
| 236 | 0.19  | -2.48 | -7.84  | 12.76  | -2.20 | -1.54 | 3.63  | -0.53 |
| 237 | -0.02 | -2.83 | -7.95  | 2.25   | -2.23 | -1.51 | -0.31 | -0.53 |
| 238 | -0.12 | -2.61 | -7.96  | 13.30  | -2.23 | -1.60 | -0.31 | -0.51 |
| 239 | 0.10  | -2.52 | -7.69  | 18.92  | -2.22 | -1.58 | 0.44  | -0.54 |

| k   | a24k  | a25k  | a26k  | a27k  | a28k   | a29k | a30k  | a31k   |
|-----|-------|-------|-------|-------|--------|------|-------|--------|
| 200 | 3.99  | -5.95 | 11.60 | -5.28 | -9.88  | 1.58 | 8.77  | -45.86 |
| 201 | 0.55  | -5.96 | 10.24 | -5.13 | -9.75  | 1.70 | 8.73  | -46.07 |
| 202 | -1.77 | -6.09 | 11.31 | -4.91 | -9.69  | 1.63 | 8.59  | -45.79 |
| 203 | 1.56  | -5.98 | 10.38 | -5.02 | -10.51 | 1.79 | 8.69  | -45.10 |
| 204 | 1.56  | -6.08 | 10.34 | -5.40 | -10.12 | 1.67 | 8.71  | -45.35 |
| 205 | 2.13  | -6.31 | 10.69 | -5.05 | -10.29 | 1.70 | 8.93  | -45.65 |
| 206 | 2.79  | -6.28 | 10.42 | -5.44 | -10.01 | 1.75 | 8.49  | -46.18 |
| 207 | -0.59 | -6.19 | 11.15 | -5.23 | -10.73 | 1.85 | 8.51  | -46.18 |
| 208 | -0.86 | -7.96 | -3.95 | -3.60 | -9.52  | 0.68 | 11.12 | -49.79 |

|     |       |       |       |       |       |      |       |        |
|-----|-------|-------|-------|-------|-------|------|-------|--------|
| 209 | 0.81  | -8.49 | -5.59 | -3.29 | -8.62 | 0.65 | 11.07 | -50.16 |
| 210 | -1.82 | -8.57 | -4.86 | -3.74 | -8.55 | 0.75 | 11.01 | -50.88 |
| 211 | -0.18 | -8.60 | -5.31 | -3.43 | -8.14 | 0.70 | 11.15 | -50.58 |
| 212 | 2.61  | -8.70 | -6.29 | -3.63 | -8.86 | 0.61 | 10.85 | -50.54 |
| 213 | -3.08 | -8.69 | -3.95 | -3.63 | -9.03 | 0.68 | 11.10 | -50.64 |
| 214 | -1.21 | -8.67 | -3.89 | -3.44 | -8.30 | 0.57 | 10.92 | -50.48 |
| 215 | -0.63 | -8.21 | -5.17 | -3.56 | -8.01 | 0.58 | 10.75 | -51.30 |
| 216 | -2.33 | -8.45 | -6.57 | -3.49 | -8.58 | 0.62 | 10.57 | -49.36 |
| 217 | -1.75 | -8.63 | -4.28 | -3.72 | -9.07 | 0.52 | 11.17 | -51.38 |
| 218 | -1.16 | -8.33 | -3.96 | -3.48 | -8.78 | 0.65 | 11.05 | -52.15 |
| 219 | -1.30 | -8.25 | -5.48 | -3.26 | -8.99 | 0.67 | 10.97 | -50.97 |
| 220 | -2.13 | -8.82 | -5.83 | -3.54 | -8.76 | 0.61 | 10.94 | -50.33 |
| 221 | 0.26  | -8.28 | -5.24 | -4.00 | -8.47 | 0.67 | 10.96 | -50.23 |
| 222 | 1.12  | -8.32 | -6.96 | -3.18 | -9.03 | 0.48 | 11.17 | -52.19 |
| 223 | 2.79  | -8.82 | -5.29 | -3.80 | -8.83 | 0.67 | 10.90 | -50.60 |
| 224 | 0.34  | -4.36 | 2.59  | -2.85 | -4.74 | 0.74 | 5.72  | -22.17 |
| 225 | -0.31 | -4.18 | 2.99  | -3.02 | -4.17 | 0.66 | 5.46  | -22.03 |
| 226 | -0.12 | -4.41 | 2.85  | -2.84 | -4.73 | 0.61 | 5.85  | -21.93 |
| 227 | -0.33 | -4.36 | 3.11  | -2.91 | -4.70 | 0.71 | 5.44  | -21.77 |
| 228 | 0.44  | -4.44 | 3.35  | -2.94 | -4.84 | 0.57 | 5.52  | -22.25 |
| 229 | 0.33  | -4.39 | 3.32  | -2.77 | -4.87 | 0.60 | 5.59  | -22.24 |
| 230 | -0.06 | -4.13 | 2.39  | -2.85 | -4.78 | 0.58 | 5.64  | -21.99 |
| 231 | 0.24  | -4.30 | 4.62  | -2.89 | -4.63 | 0.62 | 5.66  | -22.01 |
| 232 | 0.96  | -4.38 | 4.07  | -3.03 | -4.86 | 0.60 | 5.48  | -22.08 |
| 233 | -0.65 | -4.41 | 3.56  | -2.99 | -4.53 | 0.68 | 5.63  | -21.70 |
| 234 | 0.36  | -4.36 | 3.33  | -2.77 | -4.45 | 0.63 | 5.65  | -21.98 |
| 235 | -0.71 | -4.32 | 3.30  | -2.85 | -4.91 | 0.59 | 5.67  | -22.12 |
| 236 | -0.29 | -4.26 | 3.75  | -2.72 | -4.75 | 0.66 | 5.69  | -22.65 |
| 237 | 0.22  | -4.19 | 2.49  | -2.91 | -4.69 | 0.71 | 5.68  | -22.50 |
| 238 | -1.44 | -4.36 | 3.56  | -2.73 | -5.08 | 0.70 | 5.66  | -22.30 |
| 239 | -0.57 | -4.42 | 2.22  | -2.74 | -4.73 | 0.58 | 5.58  | -22.22 |

| k   | a32k  | a33k | a34k | a35k  | a36k  | a37k   | a38k  | a39k  |
|-----|-------|------|------|-------|-------|--------|-------|-------|
| 200 | 14.32 | 3.22 | 0.58 | -6.43 | -0.50 | -53.19 | -0.36 | -0.36 |
| 201 | 14.99 | 3.34 | 0.54 | -6.74 | -0.49 | -54.35 | -0.39 | -0.36 |
| 202 | 14.82 | 3.28 | 0.68 | -7.03 | -0.52 | -54.80 | -0.41 | -0.36 |
| 203 | 14.71 | 3.23 | 0.63 | -6.98 | -0.46 | -54.42 | -0.37 | -0.36 |
| 204 | 14.96 | 3.30 | 0.60 | -7.13 | -0.47 | -55.09 | -0.35 | -0.36 |
| 205 | 14.23 | 3.28 | 0.66 | -6.96 | -0.49 | -55.95 | -0.36 | -0.36 |
| 206 | 15.18 | 3.28 | 0.69 | -7.34 | -0.50 | -55.62 | -0.37 | -0.35 |
| 207 | 14.69 | 3.23 | 0.61 | -6.88 | -0.52 | -53.88 | -0.36 | -0.35 |
| 208 | 9.47  | 3.48 | 1.57 | -4.65 | -0.96 | -45.31 | -1.00 | -0.74 |
| 209 | 9.10  | 3.48 | 1.40 | -3.70 | -0.97 | -43.24 | -1.04 | -0.68 |
| 210 | 9.37  | 3.51 | 1.54 | -3.31 | -0.90 | -42.54 | -0.99 | -0.72 |
| 211 | 9.83  | 3.51 | 1.58 | -3.74 | -0.92 | -42.03 | -0.94 | -0.72 |
| 212 | 9.76  | 3.52 | 1.42 | -3.73 | -0.93 | -43.64 | -0.96 | -0.71 |
| 213 | 9.51  | 3.50 | 1.54 | -4.16 | -0.90 | -44.19 | -1.03 | -0.72 |
| 214 | 9.23  | 3.38 | 1.59 | -3.62 | -0.89 | -43.99 | -1.02 | -0.71 |
| 215 | 9.61  | 3.50 | 1.54 | -3.80 | -0.93 | -44.32 | -1.01 | -0.74 |
| 216 | 8.98  | 3.46 | 1.50 | -3.79 | -0.98 | -44.82 | -0.98 | -0.72 |
| 217 | 9.21  | 3.52 | 1.53 | -3.92 | -0.92 | -44.28 | -0.94 | -0.73 |
| 218 | 9.94  | 3.55 | 1.52 | -4.54 | -0.91 | -41.51 | -1.02 | -0.72 |

|     |      |      |      |       |       |        |       |       |
|-----|------|------|------|-------|-------|--------|-------|-------|
| 219 | 9.63 | 3.53 | 1.40 | -3.79 | -0.93 | -43.59 | -0.99 | -0.73 |
| 220 | 9.23 | 3.46 | 1.49 | -3.60 | -0.89 | -45.16 | -0.99 | -0.73 |
| 221 | 8.94 | 3.48 | 1.46 | -3.51 | -0.91 | -44.60 | -0.97 | -0.72 |
| 222 | 9.57 | 3.47 | 1.57 | -3.09 | -0.87 | -43.51 | -1.02 | -0.73 |
| 223 | 9.36 | 3.49 | 1.43 | -3.84 | -0.88 | -43.84 | -0.98 | -0.72 |
| 224 | 6.36 | 1.72 | 0.50 | -2.51 | -0.38 | -24.29 | -0.56 | -0.24 |
| 225 | 6.20 | 1.80 | 0.47 | -1.97 | -0.35 | -25.26 | -0.57 | -0.23 |
| 226 | 6.18 | 1.80 | 0.46 | -2.30 | -0.37 | -24.28 | -0.51 | -0.24 |
| 227 | 6.37 | 1.78 | 0.42 | -2.20 | -0.34 | -23.88 | -0.53 | -0.23 |
| 228 | 6.45 | 1.78 | 0.51 | -1.83 | -0.37 | -24.04 | -0.54 | -0.23 |
| 229 | 7.00 | 1.71 | 0.46 | -1.80 | -0.35 | -24.35 | -0.53 | -0.23 |
| 230 | 6.17 | 1.71 | 0.53 | -2.11 | -0.33 | -25.11 | -0.55 | -0.21 |
| 231 | 6.09 | 1.76 | 0.48 | -2.36 | -0.36 | -23.07 | -0.53 | -0.23 |
| 232 | 5.82 | 1.77 | 0.43 | -2.38 | -0.37 | -24.22 | -0.56 | -0.22 |
| 233 | 5.85 | 1.74 | 0.46 | -2.27 | -0.32 | -24.46 | -0.55 | -0.25 |
| 234 | 6.48 | 1.79 | 0.52 | -2.15 | -0.37 | -23.83 | -0.56 | -0.25 |
| 235 | 6.59 | 1.79 | 0.45 | -2.00 | -0.36 | -24.26 | -0.56 | -0.23 |
| 236 | 6.53 | 1.80 | 0.54 | -2.13 | -0.35 | -24.30 | -0.53 | -0.25 |
| 237 | 6.32 | 1.77 | 0.46 | -2.42 | -0.32 | -24.11 | -0.54 | -0.23 |
| 238 | 6.42 | 1.76 | 0.48 | -2.25 | -0.34 | -24.32 | -0.51 | -0.22 |
| 239 | 6.80 | 1.81 | 0.45 | -2.12 | -0.34 | -25.08 | -0.53 | -0.24 |

| k   | a40k | a41k | a42k  | a43k | a44k  | a45k   | a46k  | a47k  |
|-----|------|------|-------|------|-------|--------|-------|-------|
| 200 | 0.46 | 7.85 | -0.13 | 6.00 | -0.66 | -9.63  | 27.73 | -0.71 |
| 201 | 0.49 | 7.88 | 1.56  | 5.01 | 5.66  | -9.86  | 37.21 | -0.38 |
| 202 | 0.50 | 7.82 | -0.21 | 5.62 | 3.39  | -9.78  | 36.01 | -3.02 |
| 203 | 0.47 | 7.87 | 1.17  | 5.57 | 6.66  | -9.76  | 28.22 | -1.35 |
| 204 | 0.47 | 7.73 | -1.49 | 4.73 | -0.17 | -9.85  | 34.81 | -0.69 |
| 205 | 0.51 | 7.78 | 0.66  | 4.81 | 0.90  | -9.79  | 27.69 | -1.68 |
| 206 | 0.48 | 7.68 | 0.46  | 5.55 | 0.83  | -9.68  | 33.16 | 0.42  |
| 207 | 0.49 | 7.68 | 1.72  | 5.43 | -3.05 | -9.64  | 30.82 | 2.49  |
| 208 | 0.15 | 6.36 | -2.14 | 8.22 | -2.43 | -9.98  | 99.11 | -0.05 |
| 209 | 0.19 | 6.61 | 0.32  | 8.42 | 3.47  | -9.67  | 83.51 | -0.45 |
| 210 | 0.14 | 6.61 | -1.32 | 7.86 | 6.02  | -10.02 | 86.68 | 1.50  |
| 211 | 0.16 | 6.66 | 2.90  | 7.31 | 0.41  | -9.81  | 87.51 | 1.78  |
| 212 | 0.22 | 6.31 | -1.30 | 7.70 | -0.04 | -9.87  | 87.42 | -0.67 |
| 213 | 0.15 | 6.49 | -1.79 | 7.85 | 5.42  | -9.54  | 85.96 | -0.59 |
| 214 | 0.22 | 6.46 | 0.30  | 8.60 | 2.92  | -9.67  | 90.41 | 0.31  |
| 215 | 0.20 | 6.38 | -0.86 | 8.71 | 4.58  | -9.76  | 83.46 | -0.82 |
| 216 | 0.19 | 6.15 | 0.25  | 7.76 | -2.22 | -9.68  | 91.93 | -0.68 |
| 217 | 0.20 | 6.57 | -1.20 | 8.05 | -1.02 | -9.91  | 86.11 | 1.29  |
| 218 | 0.17 | 6.27 | -1.14 | 7.68 | 2.32  | -10.04 | 91.11 | -1.23 |
| 219 | 0.15 | 6.39 | -0.80 | 7.44 | 5.12  | -9.89  | 88.36 | 0.45  |
| 220 | 0.15 | 6.49 | -3.00 | 8.07 | 4.47  | -9.78  | 95.29 | -0.28 |
| 221 | 0.19 | 6.40 | 0.75  | 7.85 | 6.16  | -9.63  | 90.57 | -0.67 |
| 222 | 0.18 | 6.41 | 1.30  | 8.33 | 0.11  | -9.71  | 81.50 | 3.06  |
| 223 | 0.22 | 6.24 | -1.01 | 8.04 | 5.86  | -9.95  | 86.36 | -0.02 |
| 224 | 0.27 | 4.44 | -0.04 | 3.08 | 2.50  | -6.38  | 20.25 | -1.03 |
| 225 | 0.24 | 4.52 | -0.34 | 3.01 | -1.29 | -6.25  | 16.20 | -0.21 |
| 226 | 0.25 | 4.55 | -0.39 | 3.51 | -0.60 | -6.37  | 18.03 | -1.33 |
| 227 | 0.26 | 4.52 | 0.03  | 3.32 | -1.50 | -6.28  | 24.04 | -0.27 |
| 228 | 0.26 | 4.70 | -0.29 | 3.18 | -0.44 | -6.27  | 17.92 | -1.90 |

|     |      |      |       |      |       |       |       |       |
|-----|------|------|-------|------|-------|-------|-------|-------|
| 229 | 0.25 | 4.57 | 0.92  | 3.34 | 1.93  | -6.40 | 22.77 | -0.80 |
| 230 | 0.28 | 4.73 | -0.66 | 2.98 | 4.13  | -6.28 | 15.11 | -0.47 |
| 231 | 0.24 | 4.58 | 0.43  | 3.54 | -0.58 | -6.26 | 25.20 | 0.44  |
| 232 | 0.27 | 4.64 | 0.84  | 3.11 | 0.14  | -6.38 | 22.13 | -0.19 |
| 233 | 0.25 | 4.51 | 0.22  | 3.69 | -2.72 | -6.35 | 21.97 | 0.80  |
| 234 | 0.27 | 4.51 | -0.11 | 3.55 | 1.86  | -6.42 | 25.28 | -0.06 |
| 235 | 0.26 | 4.67 | 1.15  | 3.35 | 0.05  | -6.42 | 23.04 | -1.52 |
| 236 | 0.27 | 4.61 | -0.75 | 3.40 | -1.85 | -6.07 | 16.17 | 1.21  |
| 237 | 0.26 | 4.65 | -2.07 | 3.25 | 1.95  | -6.27 | 18.98 | -0.86 |
| 238 | 0.22 | 4.73 | -0.73 | 3.27 | -0.72 | -6.28 | 20.99 | 0.14  |
| 239 | 0.24 | 4.67 | 1.08  | 3.59 | -0.16 | -6.19 | 18.79 | -0.29 |

| k   | a48k   | a49k | a50k  | a51k  | a52k   | a53k  | a54k  | a55k  |
|-----|--------|------|-------|-------|--------|-------|-------|-------|
| 200 | -14.58 | 2.15 | 13.52 | 1.39  | 98.94  | 38.13 | -5.24 | -4.83 |
| 201 | -14.85 | 2.12 | 13.25 | -1.46 | 105.91 | 38.10 | -5.40 | -4.96 |
| 202 | -14.77 | 2.04 | 13.42 | -0.83 | 108.07 | 37.94 | -5.43 | -4.34 |
| 203 | -14.29 | 2.04 | 13.65 | 1.22  | 104.62 | 38.41 | -5.42 | -4.76 |
| 204 | -14.52 | 2.02 | 13.38 | -2.48 | 103.69 | 38.64 | -5.38 | -5.42 |
| 205 | -13.83 | 2.00 | 13.13 | -1.90 | 100.76 | 38.11 | -5.22 | -5.16 |
| 206 | -14.72 | 2.09 | 13.54 | 1.63  | 99.44  | 36.76 | -5.42 | -5.11 |
| 207 | -14.24 | 2.13 | 13.38 | -2.68 | 106.04 | 38.62 | -5.44 | -5.01 |
| 208 | -18.04 | 2.04 | 17.33 | -4.13 | 80.49  | 45.11 | -4.33 | -0.42 |
| 209 | -18.35 | 1.98 | 18.32 | 0.33  | 75.08  | 44.55 | -4.26 | -0.29 |
| 210 | -18.34 | 1.91 | 17.45 | -5.83 | 78.06  | 45.51 | -4.19 | 0.29  |
| 211 | -17.86 | 2.06 | 18.77 | -1.57 | 76.22  | 44.80 | -4.37 | -0.99 |
| 212 | -18.15 | 2.07 | 17.42 | 3.49  | 68.79  | 44.77 | -4.29 | -0.51 |
| 213 | -17.90 | 1.96 | 18.27 | 2.48  | 78.99  | 45.19 | -4.26 | -1.08 |
| 214 | -18.00 | 2.17 | 16.74 | 1.35  | 81.10  | 45.50 | -4.36 | 0.34  |
| 215 | -18.07 | 2.03 | 18.09 | -0.23 | 80.85  | 45.97 | -4.42 | -0.63 |
| 216 | -17.82 | 2.01 | 17.98 | -1.08 | 76.05  | 44.86 | -4.26 | 0.54  |
| 217 | -18.26 | 1.97 | 17.85 | 2.39  | 76.52  | 45.77 | -4.35 | -0.24 |
| 218 | -18.22 | 1.93 | 17.46 | -2.42 | 78.00  | 44.64 | -4.34 | -0.75 |
| 219 | -18.12 | 2.02 | 17.20 | -2.55 | 79.55  | 45.31 | -4.45 | -0.57 |
| 220 | -18.32 | 2.05 | 18.01 | -0.15 | 73.51  | 44.83 | -4.23 | -1.39 |
| 221 | -18.65 | 1.85 | 17.38 | 1.55  | 75.38  | 45.01 | -4.24 | 0.04  |
| 222 | -18.24 | 2.01 | 17.33 | 3.21  | 79.54  | 44.83 | -4.33 | -0.20 |
| 223 | -17.86 | 2.09 | 17.70 | -2.31 | 78.34  | 45.15 | -4.28 | 0.51  |
| 224 | -9.08  | 1.04 | 9.54  | -0.38 | 45.23  | 20.55 | -2.26 | -2.70 |
| 225 | -8.99  | 1.11 | 9.50  | -2.62 | 47.46  | 20.68 | -2.31 | -2.31 |
| 226 | -8.93  | 1.09 | 9.73  | -1.20 | 48.83  | 20.40 | -2.28 | -2.68 |
| 227 | -8.58  | 1.14 | 9.31  | -0.97 | 45.76  | 20.14 | -2.25 | -2.35 |
| 228 | -8.73  | 1.14 | 9.72  | 1.59  | 49.19  | 20.34 | -2.30 | -2.72 |
| 229 | -9.14  | 1.21 | 9.50  | 0.67  | 47.88  | 20.41 | -2.25 | -2.50 |
| 230 | -8.91  | 1.09 | 9.32  | 0.39  | 49.39  | 20.28 | -2.25 | -2.31 |
| 231 | -9.06  | 1.07 | 9.52  | 1.36  | 48.38  | 20.82 | -2.31 | -2.84 |
| 232 | -8.97  | 1.08 | 9.76  | 0.66  | 47.42  | 20.49 | -2.30 | -2.82 |
| 233 | -8.95  | 1.16 | 9.71  | -0.24 | 48.65  | 20.33 | -2.32 | -2.82 |
| 234 | -8.79  | 1.05 | 9.83  | 0.41  | 50.36  | 20.50 | -2.29 | -2.90 |
| 235 | -8.84  | 1.09 | 9.42  | 0.66  | 45.98  | 20.80 | -2.30 | -2.95 |
| 236 | -9.18  | 1.08 | 9.80  | -2.83 | 45.60  | 19.75 | -2.31 | -2.90 |
| 237 | -9.04  | 1.19 | 9.56  | 1.14  | 49.97  | 20.36 | -2.42 | -2.38 |
| 238 | -9.17  | 1.08 | 9.84  | 2.18  | 49.18  | 19.80 | -2.27 | -2.61 |

239 | -8.69 | 1.11 | 9.76 | 2.64 | 47.25 | 20.56 | -2.31 | -2.38

| k   | a56k  | a57k   | a58k | a59k  | a60k  | a61k  | a62k   | a63k   |
|-----|-------|--------|------|-------|-------|-------|--------|--------|
| 200 | -1.36 | -29.64 | 0.29 | -1.93 | 12.05 | 0.24  | 6.43   | 22.33  |
| 201 | 3.10  | -29.13 | 0.30 | -1.91 | 12.36 | 0.07  | -7.30  | -0.91  |
| 202 | -5.52 | -29.28 | 0.29 | -2.07 | 12.24 | 0.09  | -9.28  | 8.15   |
| 203 | -4.47 | -29.11 | 0.30 | -1.96 | 12.32 | 0.14  | -6.20  | 8.77   |
| 204 | -2.98 | -29.56 | 0.32 | -1.95 | 12.15 | 0.15  | 7.99   | 31.91  |
| 205 | -1.35 | -29.64 | 0.30 | -1.68 | 12.64 | 0.24  | -18.47 | -13.38 |
| 206 | 1.26  | -30.90 | 0.30 | -1.91 | 12.30 | 0.25  | 6.12   | 4.06   |
| 207 | 2.52  | -29.07 | 0.29 | -1.96 | 12.47 | 0.22  | 3.27   | -13.81 |
| 208 | -0.08 | -22.82 | 0.23 | -3.30 | 8.90  | -1.48 | 9.45   | -22.79 |
| 209 | -0.46 | -23.73 | 0.20 | -3.31 | 9.41  | -1.13 | 3.74   | -10.82 |
| 210 | 3.20  | -22.42 | 0.22 | -3.21 | 9.23  | -1.46 | -3.21  | 9.83   |
| 211 | -1.51 | -22.56 | 0.20 | -3.15 | 8.81  | -1.43 | -20.84 | -1.75  |
| 212 | 3.07  | -23.26 | 0.20 | -3.21 | 9.20  | -1.39 | 0.80   | -30.30 |
| 213 | -2.33 | -23.08 | 0.23 | -2.69 | 8.98  | -1.30 | -19.53 | 8.55   |
| 214 | -6.54 | -23.51 | 0.22 | -3.41 | 8.76  | -1.14 | 27.04  | 6.01   |
| 215 | -3.79 | -21.32 | 0.21 | -3.00 | 8.93  | -1.33 | -17.66 | -2.79  |
| 216 | -0.24 | -23.24 | 0.22 | -3.03 | 9.09  | -1.24 | 6.07   | 24.80  |
| 217 | 3.44  | -24.06 | 0.20 | -3.67 | 8.63  | -1.29 | -8.15  | 19.18  |
| 218 | -1.92 | -22.05 | 0.20 | -3.29 | 9.00  | -1.18 | 6.54   | 10.41  |
| 219 | 0.30  | -22.68 | 0.21 | -3.36 | 8.67  | -1.31 | -10.55 | 1.54   |
| 220 | -2.80 | -23.71 | 0.20 | -3.40 | 8.54  | -1.20 | 16.87  | -20.57 |
| 221 | -4.02 | -23.59 | 0.23 | -3.23 | 8.89  | -1.33 | -29.66 | -8.53  |
| 222 | 3.71  | -22.32 | 0.22 | -3.65 | 9.42  | -1.27 | -7.45  | 20.01  |
| 223 | 4.01  | -22.74 | 0.20 | -3.23 | 9.05  | -1.11 | -0.92  | 8.73   |
| 224 | -0.07 | -13.13 | 0.10 | -1.31 | 4.94  | 0.10  | -2.81  | 9.80   |
| 225 | 3.00  | -13.34 | 0.13 | -1.17 | 5.09  | 0.13  | 9.49   | 1.20   |
| 226 | 1.83  | -13.14 | 0.12 | -1.12 | 4.86  | 0.09  | -6.43  | 7.09   |
| 227 | 2.44  | -12.89 | 0.11 | -1.27 | 4.92  | 0.14  | 11.11  | 3.43   |
| 228 | -1.82 | -12.71 | 0.11 | -1.30 | 5.04  | 0.23  | 5.82   | 2.18   |
| 229 | 2.60  | -13.68 | 0.12 | -1.18 | 4.89  | 0.13  | -13.54 | -2.96  |
| 230 | -0.58 | -12.59 | 0.13 | -1.32 | 5.16  | 0.14  | 5.68   | 2.50   |
| 231 | -1.66 | -13.08 | 0.12 | -1.37 | 5.20  | 0.05  | 0.12   | 4.41   |
| 232 | 4.51  | -13.15 | 0.12 | -1.50 | 4.62  | 0.10  | -3.69  | -7.06  |
| 233 | 1.30  | -12.99 | 0.10 | -1.29 | 5.15  | 0.15  | -8.75  | -0.68  |
| 234 | 3.05  | -13.52 | 0.12 | -1.26 | 5.30  | 0.06  | -5.26  | -12.29 |
| 235 | 1.63  | -13.65 | 0.11 | -1.20 | 5.12  | 0.00  | -9.46  | 4.53   |
| 236 | 4.38  | -13.38 | 0.12 | -1.23 | 5.11  | 0.11  | -0.87  | -0.28  |
| 237 | -2.09 | -14.06 | 0.12 | -1.23 | 4.84  | 0.01  | -7.24  | -4.51  |
| 238 | 2.60  | -13.88 | 0.11 | -1.47 | 4.69  | 0.15  | -10.41 | -5.46  |
| 239 | 1.90  | -12.57 | 0.11 | -1.22 | 5.22  | 0.13  | 12.39  | 8.11   |

For all Gaussian functions  $g_{lk}$  for all images  $f_k$ , the exponential factor  $\alpha = 1/\epsilon$  is given by  
5  $\epsilon = 2.0615528128088303$ .

Subsequently, for each image  $f_k$ , the feature value  $\varphi_k$  corresponding to the filter  $\Phi$  may be calculated,  $\Phi(f_k(x, y), z_k) = \varphi_k$ , wherein  $z_k$  is a distance value corresponding to the

image  $f_k$  from the predefined data set. This yields a dataset with corresponding generated feature values  $\varphi_k$ . The hypothesis testing may use a Null-hypothesis that the filter does not describe the material classifier. The Null-Hypothesis may be given by  $H_0: \mu_1 = \mu_2 = \dots = \mu_J$ , wherein  $\mu_m$  is the expectation value of each material-group corresponding to the feature values  $\varphi_k$ . The hypothesis testing may use as alternative hypothesis that the filter does describe the material classifier. The alternative hypothesis may be given by  $H_1: \exists m, m': \mu_m \neq \mu_{m'}$ . The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each material, i.e. in total  $J$  mean values,  $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$ , for  $m \in [0, 1, \dots, J-1]$ , wherein  $N_m$  gives the number of feature values for material for each of the  $J$  materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all feature values  $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$ . The hypothesis testing may comprise determining a Mean Sum Squares within:  $mssw = \left( \sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J)$ . The hypothesis testing may comprise determining a Mean Sum of Squares between,  $mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1)$ . The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left( \frac{d_1}{2}, \frac{d_2}{2} \right)$ , where  $d_1 = N - J, d_2 = J - 1$ ,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein,  $I$  is the regularized incomplete Beta-Function,  $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$ , with the Euler Beta-Function  $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$  and  $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$  being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if  $p \leq 0.075$ , preferably  $p \leq 0.05$ , more preferably  $p \leq 0.025$ , and most preferably  $p \leq 0.01$ . For example, the image filter may pass the hypothesis testing if the p-value is smaller than  $\alpha = 0.05$ . In this case the Null-hypothesis  $H_0$  can be rejected and the alternative hypothesis  $H_1$  can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

Figure 2 shows experimental results of using the spot shape filter to calculate the material feature for different material groups. Specifically, the material feature  $\phi_{2m}$  is shown for different materials, namely from left to right for a wooden board denoted board, a fabric, a dark skin, a pale skin, and a highly translucent skin denoted trans, and

illustrates the separation of skin materials (right) from other materials (left). The spot shape filter passes the hypothesis testing. The p-value computed from the F-test is 0.0076. Thus, the Null-hypothesis  $H_0$  can be rejected with respect to a significance level of 0.01. Further, the alternative hypothesis  $H_1$  can be accepted with respect to a

5 significance level of 0.01. Thus, the image filter is distinguishes at least two material classifiers. In addition, Figure 2 shows the mean value of the feature per material,  $\bar{\varphi}_m$ , as the line within the respective box. Vertical bars mark the Q1-1.5 interquartile range (IQR) and  $Q3+1.5$  IQR, wherein  $IQR=Q3 - Q1$ , with Q1 and Q3 being the first and third quartiles. The box marks the Q1 and Q3 quartiles. Outliers are plotted as points. Figure 2

10 shows, that this material feature separates all skin samples from the non-skin samples. In other words, the expectation values of the Skin Features for skin and non-skin materials differ.

Figure 1 moreover shows, in a highly schematic illustration, an exemplary embodiment of

15 the detector 110, wherein the detector 110 specifically may be embodied as a camera 136 and/or may be part of a camera 136. The camera 136 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 138, which, besides the at least one

20 detector 110, comprises one or more beacon devices 140, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 142, which comprises the at least one detector system 138 and, further, an entertainment device 144, which comprises the human-machine interface

25 142. The figure further shows an embodiment of a tracking system 146 for tracking a position of the object 112, which comprises the detector system 138. The components of the devices and systems shall be explained in further detail below.

Fig. 1 further shows an exemplary embodiment of a scanning system 148 for scanning a

30 scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 148 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well as, optionally, at least one further illumination source, not depicted here. The illumination source 124, generally, is configured for emitting at

35 least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 140 and/or on a surface of the object 112. The scanning system 148 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at

40 least one dot and the scanning system 148, specifically the detector 110, by using the at least one detector 110.



The detector 110, besides the optical sensors 120, comprises the at least one evaluation device 132, having e.g. at least one image analysis device 150 and/or the at least one position evaluation device 152, as symbolically depicted in Fig. 1. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 154 and/or by one or more interfaces, as symbolically depicted in Fig. 1. Further, the one or more connectors 154 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 154, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 156 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 158, the position of which may be manipulated by a user 160. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 160 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 156. Further, the detector 110 may comprise the at least one transfer device 129 such as one or more optical systems, preferably comprising one or more lenses. An opening 162 inside the housing 156, which, preferably, is located concentrically with regard to the optical axis 128 of the detector 110, preferably defines a direction of view 164 of the detector 110. A coordinate system 166 may be defined, in which a direction parallel or anti-parallel to the optical axis 128 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 128 may be defined as transversal directions. In the coordinate system 166, symbolically depicted in Fig. 1, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 142, in order to provide at least one item of information to a machine 168. In the embodiments schematically depicted in Fig. 1, the machine 168 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 168, such as into the computer.

As outlined above, Fig. 1 also depicts an example of a tracking system 146, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 146 comprises the detector 110 and at least one track controller 170.

5 The track controller 170 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 170 may be an independent device and/or may be fully or partially integrated into the machine 168, specifically the computer, as indicated in Fig. 1 and/or into the evaluation device 132.

10 Similarly, as outlined above, the human-machine interface 142 may form part of an entertainment device 144. The machine 168, specifically the computer, may also form part of the entertainment device 144. Thus, by means of the user 160 functioning as the object 112 and/or by means of the user 160 handling a control device functioning as the object 112, the user 160 may input at least one item of information, such as at least one  
15 control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Fig. 1 also depicts an example of an inertial measurement unit 172 for use in an electronic device. The inertial measurement unit 172 is adapted to receive data  
20 determined by the detector 110. The inertial measurement unit 172 further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an  
25 accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

30 The inertial measurement unit 172 may comprise the detector 110 and/or may be connected to the detector 110, via at least one data connection. The evaluation device 132 and/or at least one processing device of the inertial measurement unit 172 may be configured for determining at least one combined distance information, in particular  
35 using at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further sensor of the inertial measurement unit 172.

40

## List of reference numbers

|     |                            |
|-----|----------------------------|
| 110 | detector                   |
| 112 | object                     |
| 116 | sensor element             |
| 118 | matrix                     |
| 120 | optical sensor             |
| 122 | light-sensitive area       |
| 124 | illumination source        |
| 126 | illumination light beam    |
| 128 | optical axis               |
| 129 | transfer device            |
| 130 | reflection light beam      |
| 132 | evaluation device          |
| 134 | divider                    |
| 136 | camera                     |
| 138 | detector system            |
| 140 | beacon device              |
| 142 | human-machine-interface    |
| 144 | entertainment device       |
| 146 | tracking system            |
| 148 | scanning system            |
| 150 | image analysis device      |
| 152 | position evaluation device |
| 154 | connector                  |
| 156 | housing                    |
| 158 | control device             |
| 160 | user                       |
| 162 | opening                    |
| 164 | direction of view          |
| 166 | coordinate system          |
| 168 | machine                    |
| 170 | track controller           |
| 172 | inertial measurement unit  |

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- US 2016/0206216 A1  
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- 5 “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979  
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- 10 WO 2018/091649 A1 and WO 2018/091638 A2  
chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997  
R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349
- 15 WO 2012/110924 A1  
DE 198 46 619 A1  
CN 108 363 482 A  
US 2018/033146 A1

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Friedrich SCHICK : Art Unit: TBD  
Serial No.: TBD :  
Filed: 2024-06-14 : Examiner: TBD  
Confirmation No.: TBD :  
For: DETECTOR FOR IDENTIFYING AT :  
LEAST ONE MATERIAL PROPERTY :

June 14, 2024

**CERTIFICATION STATEMENT**

In accordance with 37 C.F.R. 1.97 and 1.98 and MPEP 609, and in compliance with the duty of disclosure set forth in 37 C.F.R. 1.56, Applicants submit the attached PTO/SB/08A for consideration by the Patent and Trademark Office in the above-entitled application and to be made of record therein. Copies of the references that have not been provided in accordance with 37 C.F.R. § 1.98(d) because they are of record in a prior application (Serial Number 18/425,144) relied upon for an earlier filing date under 35 U.S.C. § 120.

Respectfully submitted,

/Christopher M. Goff/  
Christopher M. Goff, Reg. No. 41,785  
Armstrong Teasdale LLP  
7700 Forsyth Blvd., Suite 1800  
St. Louis, MO 63105

CMG/EMF/ZSW

Via EFS

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION  
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

|                       |                                                         |                                               |  |
|-----------------------|---------------------------------------------------------|-----------------------------------------------|--|
| First Named Inventor: | Friedrich SCHICK                                        | Nonprovisional Application Number (if known): |  |
| Title of Invention:   | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                                               |  |

**APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.**

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:
  - I. ☒ **Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)**
    - i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via EFS-Web.  
---OR---  
(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.
    - ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.
  - II. ☐ **Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)**
    - i. A request for continued examination has been filed with, or prior to, this form.
    - ii. If the application is a utility application, this certification and request is being filed via EFS-Web.
    - iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
    - iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
    - v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

|                    |                       |                                  |            |
|--------------------|-----------------------|----------------------------------|------------|
| Signature          | /Christopher M. Goff/ | Date                             | 2024-06-14 |
| Name (Print/Typed) | Christopher M. Goff   | Practitioner Registration Number | 41,785     |

**Note:** This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.\*

☒ \*Total of 1 forms are submitted.

## Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

UNITED STATES  
PATENT AND TRADEMARK OFFICEP.O. Box 1450  
Alexandria, VA 22313 - 1450  
www.uspto.gov

## ELECTRONIC PAYMENT RECEIPT

APPLICATION #  
18/743,352RECEIPT DATE / TIME  
06/14/2024 11:48:31 AM Z ETATTORNEY DOCKET #  
27843-2603 182037US03

## Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

## Application Information

APPLICATION TYPE    Utility - Nonprovisional Application  
under 35 USC 111(a)

PATENT #    -

CONFIRMATION #    2279

FILED BY    Barbara McGregor

PATENT CENTER #    65969003

AUTHORIZED BY    Christopher Goff

CUSTOMER #    76656

FILING DATE    -

CORRESPONDENCE  
ADDRESS    -FIRST NAMED  
INVENTOR    Friedrich SCHICK

## Payment Information

PAYMENT METHOD  
DA / 012384PAYMENT TRANSACTION ID  
E20246DC16269165PAYMENT AUTHORIZED BY  
Barbara McGregor

| FEE CODE | DESCRIPTION                                                                                     | ITEM PRICE(\$) | QUANTITY | ITEM TOTAL(\$) |
|----------|-------------------------------------------------------------------------------------------------|----------------|----------|----------------|
| 1817     | REQUEST FOR PRIORITIZED EXAMINATION                                                             | 4200.00        | 1        | 4200.00        |
| 1111     | UTILITY PATENT APPL. SEARCH FEE                                                                 | 700.00         | 1        | 700.00         |
| 1011     | BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T)) | 320.00         | 1        | 320.00         |
| 1311     | PATENT APPL. EXAMINATION FEE                                                                    | 800.00         | 1        | 800.00         |
| 1081     | UTILITY APPL. SIZE FEE > THAN 100 PAGES (FEE X PAGES >= 101 IN 50                               | 420.00         | 1        | 420.00         |



## PAGE INCREMENTS)

|                          |                            |        |    |                   |
|--------------------------|----------------------------|--------|----|-------------------|
| 1202                     | EACH CLAIM IN EXCESS OF 20 | 100.00 | 10 | 1000.00           |
| <b>TOTAL<br/>AMOUNT:</b> |                            |        |    | <b>\$7,440.00</b> |

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

#### **New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

#### **National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

#### **New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                         |                        |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                         | Application Number     |                       |
| Title of Invention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |
| <p>The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.</p> <p>This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.</p> |                                                         |                        |                       |

**Secrecy Order 37 CFR 5.2:**

|                          |                                                                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.) |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Inventor Information:**

|                                                                                                                                                                                 |                   |                                          |                    |               |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------|--------------------|---------------|------------------------|
| <b>Inventor 1</b>                                                                                                                                                               |                   |                                          |                    |               | <a href="#">Remove</a> |
| <b>Legal Name</b>                                                                                                                                                               |                   |                                          |                    |               |                        |
| <b>Prefix</b>                                                                                                                                                                   | <b>Given Name</b> | <b>Middle Name</b>                       | <b>Family Name</b> | <b>Suffix</b> |                        |
|                                                                                                                                                                                 | Friedrich         |                                          | SCHICK             |               |                        |
| <b>Residence Information (Select One)</b> <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |                   |                                          |                    |               |                        |
| <b>City</b>                                                                                                                                                                     | Ludwigshafen      | <b>Country of Residence <sup>i</sup></b> | DE                 |               |                        |
| <b>Mailing Address of Inventor:</b>                                                                                                                                             |                   |                                          |                    |               |                        |
| <b>Address 1</b>                                                                                                                                                                |                   | Industriestrasse 35                      |                    |               |                        |
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| <b>City</b>                                                                                                                                                                     | Ludwigshafen      | <b>State/Province</b>                    |                    |               |                        |
| <b>Postal Code</b>                                                                                                                                                              | 67063             | <b>Country <sup>i</sup></b>              | DE                 |               |                        |
| <b>Inventor 2</b>                                                                                                                                                               |                   |                                          |                    |               | <a href="#">Remove</a> |
| <b>Legal Name</b>                                                                                                                                                               |                   |                                          |                    |               |                        |
| <b>Prefix</b>                                                                                                                                                                   | <b>Given Name</b> | <b>Middle Name</b>                       | <b>Family Name</b> | <b>Suffix</b> |                        |
|                                                                                                                                                                                 | Peter             |                                          | SCHILLEN           |               |                        |
| <b>Residence Information (Select One)</b> <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |                   |                                          |                    |               |                        |
| <b>City</b>                                                                                                                                                                     | Ludwigshafen      | <b>Country of Residence <sup>i</sup></b> | DE                 |               |                        |
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| <b>Address 1</b>                                                                                                                                                                |                   | Industriestrasse 35                      |                    |               |                        |
| <b>Address 2</b>                                                                                                                                                                |                   |                                          |                    |               |                        |
| <b>City</b>                                                                                                                                                                     | Ludwigshafen      | <b>State/Province</b>                    |                    |               |                        |
| <b>Postal Code</b>                                                                                                                                                              | 67063             | <b>Country <sup>i</sup></b>              | DE                 |               |                        |
| <b>Inventor 3</b>                                                                                                                                                               |                   |                                          |                    |               | <a href="#">Remove</a> |
| <b>Legal Name</b>                                                                                                                                                               |                   |                                          |                    |               |                        |

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

| Prefix                                                                                                                                                                          | Given Name | Middle Name | Family Name | Suffix |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|--------|
|                                                                                                                                                                                 | Patrick    |             | SCHINDLER   |        |
| <b>Residence Information (Select One)</b> <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |            |             |             |        |

|      |              |                                   |    |
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**Mailing Address of Inventor:**

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| Inventor 4  | <a href="#">Remove</a> |                      |    |
| Legal Name  |                        |                      |    |

| Prefix                                                                                                                                                                          | Given Name | Middle Name | Family Name | Suffix |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|--------|
|                                                                                                                                                                                 | Andre      |             | SCHMIDT     |        |
| <b>Residence Information (Select One)</b> <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |            |             |             |        |

|      |              |                                   |    |
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**Mailing Address of Inventor:**

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| Address 1   | Industriestrasse 35    |                      |    |
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| Inventor 5  | <a href="#">Remove</a> |                      |    |
| Legal Name  |                        |                      |    |

| Prefix                                                                                                                                                                          | Given Name | Middle Name | Family Name | Suffix |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|--------|
|                                                                                                                                                                                 | Michael    |             | EBERSPACH   |        |
| <b>Residence Information (Select One)</b> <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |            |             |             |        |

|      |              |                                   |    |
|------|--------------|-----------------------------------|----|
| City | Ludwigshafen | Country of Residence <sup>i</sup> | DE |
|------|--------------|-----------------------------------|----|

**Mailing Address of Inventor:**

|             |                     |                      |    |
|-------------|---------------------|----------------------|----|
| Address 1   | Industriestrasse 35 |                      |    |
| Address 2   |                     |                      |    |
| City        | Ludwigshafen        | State/Province       |    |
| Postal Code | 67063               | Country <sup>i</sup> | DE |

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

|                                                                                                                                                                          |              |                                   |             |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------|-------------|--------|
| <b>Inventor 6</b> <span style="float: right;">Remove</span>                                                                                                              |              |                                   |             |        |
| Legal Name                                                                                                                                                               |              |                                   |             |        |
| Prefix                                                                                                                                                                   | Given Name   | Middle Name                       | Family Name | Suffix |
|                                                                                                                                                                          | Christian    |                                   | LENNARTZ    |        |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |              |                                   |             |        |
| City                                                                                                                                                                     | Ludwigshafen | Country of Residence <sup>i</sup> | DE          |        |

|                                     |              |                      |    |  |
|-------------------------------------|--------------|----------------------|----|--|
| <b>Mailing Address of Inventor:</b> |              |                      |    |  |
| Address 1                           |              | Industriestrasse 35  |    |  |
| Address 2                           |              |                      |    |  |
| City                                | Ludwigshafen | State/Province       |    |  |
| Postal Code                         | 67063        | Country <sup>i</sup> | DE |  |

|                                                                                                                                                                          |            |                                   |             |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------|-------------|--------|
| <b>Inventor 7</b> <span style="float: right;">Remove</span>                                                                                                              |            |                                   |             |        |
| Legal Name                                                                                                                                                               |            |                                   |             |        |
| Prefix                                                                                                                                                                   | Given Name | Middle Name                       | Family Name | Suffix |
|                                                                                                                                                                          | Robert     |                                   | SEND        |        |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |            |                                   |             |        |
| City                                                                                                                                                                     | Karlsruhe  | Country of Residence <sup>i</sup> | DE          |        |

|                                     |           |                      |    |  |
|-------------------------------------|-----------|----------------------|----|--|
| <b>Mailing Address of Inventor:</b> |           |                      |    |  |
| Address 1                           |           | Luisenstr. 25        |    |  |
| Address 2                           |           |                      |    |  |
| City                                | Karlsruhe | State/Province       |    |  |
| Postal Code                         | 76137     | Country <sup>i</sup> | DE |  |

|                                                                                                                                                                          |            |                                   |             |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------|-------------|--------|
| <b>Inventor 8</b> <span style="float: right;">Remove</span>                                                                                                              |            |                                   |             |        |
| Legal Name                                                                                                                                                               |            |                                   |             |        |
| Prefix                                                                                                                                                                   | Given Name | Middle Name                       | Family Name | Suffix |
|                                                                                                                                                                          | Lars       |                                   | DIESELBERG  |        |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |            |                                   |             |        |
| City                                                                                                                                                                     | Karlsruhe  | Country of Residence <sup>i</sup> | DE          |        |

|                                     |  |               |  |  |
|-------------------------------------|--|---------------|--|--|
| <b>Mailing Address of Inventor:</b> |  |               |  |  |
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| Address 2                           |  |               |  |  |

|                                           |  |                                                         |  |                       |  |
|-------------------------------------------|--|---------------------------------------------------------|--|-----------------------|--|
| <b>Application Data Sheet 37 CFR 1.76</b> |  | Attorney Docket Number                                  |  | 27843-2603 182037US03 |  |
|                                           |  | Application Number                                      |  |                       |  |
| Title of Invention                        |  | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |  |                       |  |

|                                                                                                                                                                          |              |                                   |             |                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------|-------------|-----------------------------------------------------------------------|
| City                                                                                                                                                                     | Karlsruhe    | State/Province                    |             |                                                                       |
| Postal Code                                                                                                                                                              | 76185        | Country <sup>i</sup>              | DE          |                                                                       |
| Inventor 9                                                                                                                                                               |              |                                   |             | <a href="#" style="border: 1px solid black; padding: 2px;">Remove</a> |
| Legal Name                                                                                                                                                               |              |                                   |             |                                                                       |
| Prefix                                                                                                                                                                   | Given Name   | Middle Name                       | Family Name | Suffix                                                                |
|                                                                                                                                                                          | Heiko        |                                   | HENGEN      |                                                                       |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |              |                                   |             |                                                                       |
| City                                                                                                                                                                     | Steinweiler  | Country of Residence <sup>i</sup> | DE          |                                                                       |
|                                                                                                                                                                          |              |                                   |             |                                                                       |
| <b>Mailing Address of Inventor:</b>                                                                                                                                      |              |                                   |             |                                                                       |
| Address 1                                                                                                                                                                |              | Siedlerweg 8                      |             |                                                                       |
| Address 2                                                                                                                                                                |              |                                   |             |                                                                       |
| City                                                                                                                                                                     | Steinweiler  | State/Province                    |             |                                                                       |
| Postal Code                                                                                                                                                              | 76872        | Country <sup>i</sup>              | DE          |                                                                       |
| Inventor 10                                                                                                                                                              |              |                                   |             | <a href="#" style="border: 1px solid black; padding: 2px;">Remove</a> |
| Legal Name                                                                                                                                                               |              |                                   |             |                                                                       |
| Prefix                                                                                                                                                                   | Given Name   | Middle Name                       | Family Name | Suffix                                                                |
|                                                                                                                                                                          | Ingmar       |                                   | BRUDER      |                                                                       |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |              |                                   |             |                                                                       |
| City                                                                                                                                                                     | Ludwigshafen | Country of Residence <sup>i</sup> | DE          |                                                                       |
|                                                                                                                                                                          |              |                                   |             |                                                                       |
| <b>Mailing Address of Inventor:</b>                                                                                                                                      |              |                                   |             |                                                                       |
| Address 1                                                                                                                                                                |              | Industriestrasse 35               |             |                                                                       |
| Address 2                                                                                                                                                                |              |                                   |             |                                                                       |
| City                                                                                                                                                                     | Ludwigshafen | State/Province                    |             |                                                                       |
| Postal Code                                                                                                                                                              | 67063        | Country <sup>i</sup>              | DE          |                                                                       |
| Inventor 11                                                                                                                                                              |              |                                   |             | <a href="#" style="border: 1px solid black; padding: 2px;">Remove</a> |
| Legal Name                                                                                                                                                               |              |                                   |             |                                                                       |
| Prefix                                                                                                                                                                   | Given Name   | Middle Name                       | Family Name | Suffix                                                                |
|                                                                                                                                                                          | Jakob        |                                   | UNGER       |                                                                       |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |              |                                   |             |                                                                       |
| City                                                                                                                                                                     | Ludwigshafen | Country of Residence <sup>i</sup> | DE          |                                                                       |
|                                                                                                                                                                          |              |                                   |             |                                                                       |
| <b>Mailing Address of Inventor:</b>                                                                                                                                      |              |                                   |             |                                                                       |

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

|                                                                                                                                                                          |                     |                                   |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------|---------------------------------------|
| Address 1                                                                                                                                                                | Industriestrasse 35 |                                   |                                       |
| Address 2                                                                                                                                                                |                     |                                   |                                       |
| City                                                                                                                                                                     | Ludwigshafen        | State/Province                    |                                       |
| Postal Code                                                                                                                                                              | 67063               | Country <sup>i</sup>              | DE                                    |
| Inventor 12                                                                                                                                                              |                     |                                   | <input type="button" value="Remove"/> |
| Legal Name                                                                                                                                                               |                     |                                   |                                       |
| Prefix                                                                                                                                                                   | Given Name          | Middle Name                       | Family Name                           |
|                                                                                                                                                                          | Christian           |                                   | BONSIGNORE                            |
| Residence Information (Select One) <input type="radio"/> US Residency <input checked="" type="radio"/> Non US Residency <input type="radio"/> Active US Military Service |                     |                                   |                                       |
| City                                                                                                                                                                     | Ludwigshafen        | Country of Residence <sup>i</sup> | DE                                    |

|                                                                                                                                                                                |                     |                      |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|----|
| <b>Mailing Address of Inventor:</b>                                                                                                                                            |                     |                      |    |
| Address 1                                                                                                                                                                      | Industriestrasse 35 |                      |    |
| Address 2                                                                                                                                                                      |                     |                      |    |
| City                                                                                                                                                                           | Ludwigshafen        | State/Province       |    |
| Postal Code                                                                                                                                                                    | 67063               | Country <sup>i</sup> | DE |
| All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the <b>Add</b> button. <input type="button" value="Add"/> |                     |                      |    |

## Correspondence Information:

|                                                                                                                                       |                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Enter either Customer Number or complete the Correspondence Information section below.<br>For further information see 37 CFR 1.33(a). |                                                                                                          |
| <input type="checkbox"/> An Address is being provided for the correspondence Information of this application.                         |                                                                                                          |
| Customer Number                                                                                                                       | 76656                                                                                                    |
| Email Address                                                                                                                         | uspatents@atllp.com <input type="button" value="Add Email"/> <input type="button" value="Remove Email"/> |

## Application Information:

|                                         |                                                         |                                           |                          |
|-----------------------------------------|---------------------------------------------------------|-------------------------------------------|--------------------------|
| Title of the Invention                  | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                                           |                          |
| Attorney Docket Number                  | 27843-2603 182037US03                                   | Small Entity Status Claimed               | <input type="checkbox"/> |
| Application Type                        | Nonprovisional                                          |                                           |                          |
| Subject Matter                          | Utility                                                 |                                           |                          |
| Total Number of Drawing Sheets (if any) | 2                                                       | Suggested Figure for Publication (if any) | 1                        |

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

**Filing By Reference:**

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

|                                                        |                          |                                            |
|--------------------------------------------------------|--------------------------|--------------------------------------------|
| Application number of the previously filed application | Filing date (YYYY-MM-DD) | Intellectual Property Authority or Country |
|                                                        |                          |                                            |

**Publication Information:**

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

**Representative Information:**

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

|                    |                                                  |                                              |                                                         |
|--------------------|--------------------------------------------------|----------------------------------------------|---------------------------------------------------------|
| Please Select One: | <input checked="" type="radio"/> Customer Number | <input type="radio"/> US Patent Practitioner | <input type="radio"/> Limited Recognition (37 CFR 11.9) |
| Customer Number    | 76656                                            |                                              |                                                         |

**Domestic Benefit/National Stage Information:**

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

|                          |                 |                          |                                    |
|--------------------------|-----------------|--------------------------|------------------------------------|
| Prior Application Status | Pending         | <a href="#">Remove</a>   |                                    |
| Application Number       | Continuity Type | Prior Application Number | Filing or 371(c) Date (YYYY-MM-DD) |
|                          | Continuation of | 18425144                 | 2024-01-29                         |

|                                                                                                                                           |                        |                                                         |                                    |                        |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------|------------------------------------|------------------------|-------------------------|
| <b>Application Data Sheet 37 CFR 1.76</b>                                                                                                 |                        | Attorney Docket Number                                  |                                    | 27843-2603 182037US03  |                         |
|                                                                                                                                           |                        | Application Number                                      |                                    |                        |                         |
| Title of Invention                                                                                                                        |                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                                    |                        |                         |
| Prior Application Status                                                                                                                  |                        | Patented                                                |                                    | <a href="#">Remove</a> |                         |
| Application Number                                                                                                                        | Continuity Type        | Prior Application Number                                | Filing Date (YYYY-MM-DD)           | Patent Number          | Issue Date (YYYY-MM-DD) |
| 18425144                                                                                                                                  | Continuation of        | 17439492                                                | 2021-09-15                         | 11947013               | 2024-04-02              |
| Prior Application Status                                                                                                                  |                        | Pending                                                 |                                    | <a href="#">Remove</a> |                         |
| Application Number                                                                                                                        | Continuity Type        | Prior Application Number                                | Filing or 371(c) Date (YYYY-MM-DD) |                        |                         |
| 17439492                                                                                                                                  | a 371 of international | PCT/EP20/56759                                          | 2020-03-13                         |                        |                         |
| Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the <b>Add</b> button. <a href="#">Add</a> |                        |                                                         |                                    |                        |                         |

## Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)<sup>i</sup> the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

|                                                                                                                            |  |                      |                          |                                          |
|----------------------------------------------------------------------------------------------------------------------------|--|----------------------|--------------------------|------------------------------------------|
| Application Number                                                                                                         |  | Country <sup>i</sup> | Filing Date (YYYY-MM-DD) | Access Code <sup>i</sup> (if applicable) |
| 19163250.4                                                                                                                 |  | EP                   | 2019-03-15               |                                          |
| Additional Foreign Priority Data may be generated within this form by selecting the <b>Add</b> button. <a href="#">Add</a> |  |                      |                          |                                          |

## Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.



|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

## Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

**NOTE:** This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

### 1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

**A. Priority Document Exchange (PDX)** - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

**B. Search Results from U.S. Application to EPO** - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

### 2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

**NOTE:** Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

## Applicant Information:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                |                                                                        |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------|------------------------|
| Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                |                                                                        |                        |
| <b>Applicant 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                |                                                                        | <a href="#">Remove</a> |
| <p>If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.</p> |                                                                |                                                                        |                        |
| <a href="#">Clear</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                |                                                                        |                        |
| <input checked="" type="radio"/> Assignee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <input type="radio"/> Legal Representative under 35 U.S.C. 117 | <input type="radio"/> Joint Inventor                                   |                        |
| <input type="radio"/> Person to whom the inventor is obligated to assign.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                | <input type="radio"/> Person who shows sufficient proprietary interest |                        |
| If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                |                                                                        |                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                |                                                                        |                        |
| Name of the Deceased or Legally Incapacitated Inventor: <input type="text"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                        |                        |
| If the Applicant is an Organization check here. <input checked="" type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                |                                                                        |                        |
| Organization Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | trinamiX GmbH                                                  |                                                                        |                        |
| <b>Mailing Address Information For Applicant:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                |                                                                        |                        |
| Address 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Industriestrasse 35                                            |                                                                        |                        |
| Address 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                |                                                                        |                        |
| City                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Ludwigshafen am Rhein                                          | State/Province                                                         |                        |
| Country                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | DE                                                             | Postal Code                                                            | 67063                  |
| Phone Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                | Fax Number                                                             |                        |
| Email Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                |                                                                        |                        |
| Additional Applicant Data may be generated within this form by selecting the Add button. <a href="#">Add</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                        |                        |

## Assignee Information including Non-Applicant Assignee Information:

|                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

**Assignee 1**

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

[Remove](#)

If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

|        |            |             |             |        |
|--------|------------|-------------|-------------|--------|
| Prefix | Given Name | Middle Name | Family Name | Suffix |
|        |            |             |             |        |

**Mailing Address Information For Assignee including Non-Applicant Assignee:**

|                      |  |  |                |  |
|----------------------|--|--|----------------|--|
| Address 1            |  |  |                |  |
| Address 2            |  |  |                |  |
| City                 |  |  | State/Province |  |
| Country <sup>i</sup> |  |  | Postal Code    |  |
| Phone Number         |  |  | Fax Number     |  |
| Email Address        |  |  |                |  |

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

[Add](#)**Signature:**[Remove](#)

**NOTE:** This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

|            |                       |           |      |                     |            |
|------------|-----------------------|-----------|------|---------------------|------------|
| Signature  | /Christopher M. Goff/ |           |      | Date (YYYY-MM-DD)   | 2024-06-14 |
| First Name | Christopher M.        | Last Name | Goff | Registration Number | 41785      |

Additional Signature may be generated within this form by selecting the Add button.

[Add](#)

|                                           |                                                         |                        |                       |
|-------------------------------------------|---------------------------------------------------------|------------------------|-----------------------|
| <b>Application Data Sheet 37 CFR 1.76</b> |                                                         | Attorney Docket Number | 27843-2603 182037US03 |
|                                           |                                                         | Application Number     |                       |
| Title of Invention                        | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                        |                       |

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

# Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

## TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA/82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| Application Number     |                                                         |
| Filing Date            | 2024-06-14                                              |
| First Named Inventor   | Friedrich SCHICK                                        |
| Title                  | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |
| Art Unit               |                                                         |
| Examiner Name          |                                                         |
| Attorney Docket Number | 27843-2603 182037US03                                   |

### SIGNATURE of Applicant or Patent Practitioner

|                                                    |                       |                     |            |
|----------------------------------------------------|-----------------------|---------------------|------------|
| Signature                                          | /Christopher M. Goff/ | Date (Optional)     | 2024-06-14 |
| Name                                               | Christopher M. Goff   | Registration Number | 41785      |
| Title (if Applicant is a juristic entity)          | Attorney              |                     |            |
| Applicant Name (if Applicant is a juristic entity) |                       |                     |            |

**NOTE:** This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



\*Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

*If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.*

Doc Code: PA..

Document Description: Power of Attorney

PTO/AIA/82B (07-13)

Approved for use through 03/31/2021. OMB 0651-0035

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## POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

76656

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number:

OR



Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

trinamiX GmbH



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

### SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

2020/06/09

Name

Dr. Hans Graf

Title

Patent Attorney

**NOTE:** Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.



Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

|                                                                                                     |                        |                       |            |
|-----------------------------------------------------------------------------------------------------|------------------------|-----------------------|------------|
| <b>INFORMATION DISCLOSURE<br/>STATEMENT BY APPLICANT</b><br>( Not for submission under 37 CFR 1.99) | Application Number     |                       |            |
|                                                                                                     | Filing Date            |                       | 2024-06-14 |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK      |            |
|                                                                                                     | Art Unit               |                       |            |
|                                                                                                     | Examiner Name          |                       |            |
|                                                                                                     | Attorney Docket Number | 27843-2603 182037US03 |            |

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|                                                                                                            | 1       | 20160155006        | A1                     | 2016-06-02       | MAKKAPATI et al.                                |                                                                        |        |
|                                                                                                            | 2       | 20160206216        | A1                     | 2016-07-21       | KIRENKO                                         |                                                                        |        |
|                                                                                                            | 3       | 20170363465        | A1                     | 2017-12-21       | SEND et al.                                     |                                                                        |        |
|                                                                                                            | 4       | 20180033146        | A1                     | 2018-02-01       | BLEYER et al.                                   |                                                                        |        |
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| <b>INFORMATION DISCLOSURE<br/>STATEMENT BY APPLICANT</b><br>( Not for submission under 37 CFR 1.99) | Application Number     |                  |                       |
|                                                                                                     | Filing Date            |                  | 2024-06-14            |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK |                       |
|                                                                                                     | Art Unit               |                  |                       |
|                                                                                                     | Examiner Name          |                  |                       |
|                                                                                                     | Attorney Docket Number |                  | 27843-2603 182037US03 |

| Examiner Initial* | Cite No | Foreign Document Number <sup>3</sup> | Country Code <sup>2</sup> | Kind Code <sup>4</sup> | Publication Date | Name of Patentee or Applicant of cited Document | Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear | T <sup>5</sup>           |
|-------------------|---------|--------------------------------------|---------------------------|------------------------|------------------|-------------------------------------------------|--------------------------------------------------------------------------|--------------------------|
|                   | 1       | 108363482                            | CN                        | A                      | 2018-08-03       | JIANGSU SIDIANLING ROBOT CO LTD.                |                                                                          | <input type="checkbox"/> |
|                   | 2       | 19846619                             | DE                        | A1                     | 2000-04-27       | BYK-GARDNER GMBH                                |                                                                          | <input type="checkbox"/> |
|                   | 3       | 2012110924                           | WO                        | A1                     | 2012-08-23       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 4       | 2014097181                           | WO                        | A1                     | 2014-06-26       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 5       | 2014198629                           | WO                        | A1                     | 2014-12-18       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 6       | 2018091638                           | WO                        | A1                     | 2018-05-24       | TRINAMIX GMBH                                   |                                                                          | <input type="checkbox"/> |
|                   | 7       | 2018091640                           | WO                        | A2                     | 2018-08-24       | TRINAMIX GMBH                                   |                                                                          | <input type="checkbox"/> |
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| <b>INFORMATION DISCLOSURE<br/>STATEMENT BY APPLICANT</b><br>( Not for submission under 37 CFR 1.99) | Application Number     |                       |            |
|                                                                                                     | Filing Date            |                       | 2024-06-14 |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK      |            |
|                                                                                                     | Art Unit               |                       |            |
|                                                                                                     | Examiner Name          |                       |            |
|                                                                                                     | Attorney Docket Number | 27843-2603 182037US03 |            |

| Examiner Initials* | Cite No | Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published. | T <sup>5</sup>           |
|--------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|                    | 1       | EICHLER et al., "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 171 to 266, Springer Verlag, ISBN 0939-0979. Cited in specification, no English translation available                     | <input type="checkbox"/> |
|                    | 2       | International Preliminary Report on Patentability for corresponding PCT/EP2020/056759 mailed 30 Sept. 2021, 9 Pages                                                                                                                                             | <input type="checkbox"/> |
|                    | 3       | International Search Report and Written Opinion for corresponding PCT/EP2020/056759 mailed May 29, 2021, 11 Pages                                                                                                                                               | <input type="checkbox"/> |
|                    | 4       | JIANG et al., "Dreidimensionales Computersehen" Springer, Berlin Heidelberg, 1997, Chapter 2. 32 pages. Cited in specification, no English translation available                                                                                                | <input type="checkbox"/> |
|                    | 5       | R.A. STREET: Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349                                                                                                                                                    | <input type="checkbox"/> |

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#### EXAMINER SIGNATURE

|                    |  |                 |  |
|--------------------|--|-----------------|--|
| Examiner Signature |  | Date Considered |  |
|--------------------|--|-----------------|--|

\*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

<sup>1</sup> See Kind Codes of USPTO Patent Documents at [www.USPTO.GOV](http://www.USPTO.GOV) or MPEP 901.04. <sup>2</sup> Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). <sup>3</sup> For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. <sup>4</sup> Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. <sup>5</sup> Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE  
STATEMENT BY APPLICANT**  
( Not for submission under 37 CFR 1.99)

|                        |                       |
|------------------------|-----------------------|
| Application Number     |                       |
| Filing Date            | 2024-06-14            |
| First Named Inventor   | Friedrich SCHICK      |
| Art Unit               |                       |
| Examiner Name          |                       |
| Attorney Docket Number | 27843-2603 182037US03 |

**CERTIFICATION STATEMENT**

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

**OR**

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

**SIGNATURE**

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

|            |                       |                     |            |
|------------|-----------------------|---------------------|------------|
| Signature  | /Christopher M. Goff/ | Date (YYYY-MM-DD)   | 2024-06-14 |
| Name/Print | Christopher M. Goff   | Registration Number | 41785      |

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

## Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

**Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)**  
*The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.*

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))  
 for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

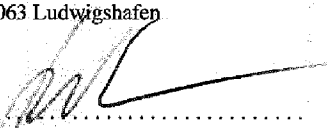
This declaration is directed to this international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. PCT/..... (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

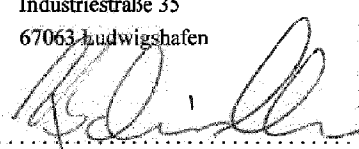
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

**Name:** SCHICK, Friedrich  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
 67063 Ludwigshafen

**Inventor's Signature:**   
 (The signature must be that of the inventor, not that of the agent)

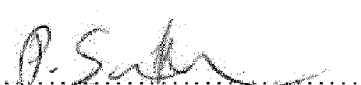
**Date:** 09.05.19

**Name:** SCHILLEN, Peter  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
 67063 Ludwigshafen

**Inventor's Signature:**   
 (The signature must be that of the inventor, not that of the agent)

**Date:** 09.04.19

**Name:** SCHINDLER, Patrick  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
 67063 Ludwigshafen

**Inventor's Signature:**   
 (The signature must be that of the inventor, not that of the agent)

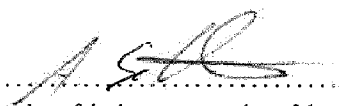
**Date:** 09.04.19

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

**Continuation of Box No. VIII (i) to (v) DECLARATION**

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

**Name:** SCHMIDT, Andre  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

**Inventor's Signature:**   
(The signature must be that of the inventor, not that of the agent)

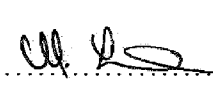
**Date:** 11.04.2019

**Name:** EBERSPACH, Michael  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

**Inventor's Signature:**   
(The signature must be that of the inventor, not that of the agent)

**Date:** 11.04.2019

**Name:** LENNARTZ, Christian  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

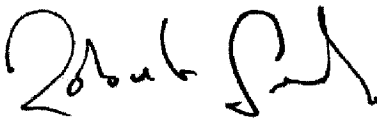
**Inventor's Signature:**   
(The signature must be that of the inventor, not that of the agent)

**Date:** 11.4.2019

**Continuation of Box No. VIII (i) to (v) DECLARATION**

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

**Name:** SEND, Robert  
**Residence:** Germany  
**Mailing Address:** Luisenstr. 25

Inventor's Signature:   
(The signature must be that of the inventor, not that of the agent)

Date: ... 2019/09/11 .....

**Name:** DIESELBERG, Lars  
**Residence:** Germany  
**Mailing Address:** Geibelstr. 22  
76185 Karlsruhe

Inventor's Signature: .....  
(The signature must be that of the inventor, not that of the agent)

Date: .....

**Name:** HENGEN, Heiko  
**Residence:** Germany  
**Mailing Address:** Siedlerweg 8  
76872 Steinweiler

Inventor's Signature: .....  
(The signature must be that of the inventor, not that of the agent)

Date: .....

**Continuation of Box No. VIII (i) to (v) DECLARATION**

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

**Name:** SEND, Robert  
**Residence:** Germany  
**Mailing Address:** Luisenstr. 25  
 76137 Karlsruhe

**Inventor's Signature:** .....  
 (The signature must be that of the inventor, not that of the agent)

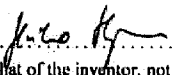
**Date:** .....

**Name:** DIESSELBERG, Lars  
**Residence:** Germany  
**Mailing Address:** Geibelstr. 22  
 76185 Karlsruhe

**Inventor's Signature:**  .....  
 (The signature must be that of the inventor, not that of the agent)

**Date:** 2019-09-05

**Name:** HENGEN, Heiko  
**Residence:** Germany  
**Mailing Address:** SIEDLERWEG 8  
 76872 STEINWEILER

**Inventor's Signature:**  .....  
 (The signature must be that of the inventor, not that of the agent)

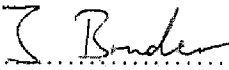
**Date:** 2019-09-05



**Continuation of Box No. VIII (i) to (v) DECLARATION**

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: BRUDER, Ingmar  
Residence: Germany  
Mailing Address: Industriestraße 35  
67063 Ludwigshafen

Inventor's Signature:  .....

Date: 12.09.2019 .....

(The signature must be that of the inventor, not that of the agent)

Name: UNGER, Jakob  
Residence: Germany  
Mailing Address: Industriestraße 35  
67063 Ludwigshafen

Inventor's Signature: .....

Date: .....

(The signature must be that of the inventor, not that of the agent)

Name: BONSIGNORE, Christian  
Residence: Germany  
Mailing Address: Industriestraße 35  
67063 Ludwigshafen

Inventor's Signature: .....

Date: .....

(The signature must be that of the inventor, not that of the agent)

**Continuation of Box No. VIII (i) to (v) DECLARATION**

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

**Name:** BRUDER, Ingmar  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

**Inventor's Signature:** .....  
(The signature must be that of the inventor, not that of the agent)

**Date:** .....

**Name:** UNGER, Jakob  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

**Inventor's Signature:** .....  
(The signature must be that of the inventor, not that of the agent)

**Date:** 13.8.19

**Name:** BONSIGNORE, Christian  
**Residence:** Germany  
**Mailing Address:** Industriestraße 35  
67063 Ludwigshafen

**Inventor's Signature:** .....  
(The signature must be that of the inventor, not that of the agent)

**Date:** 14.08.19

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FIG. 1

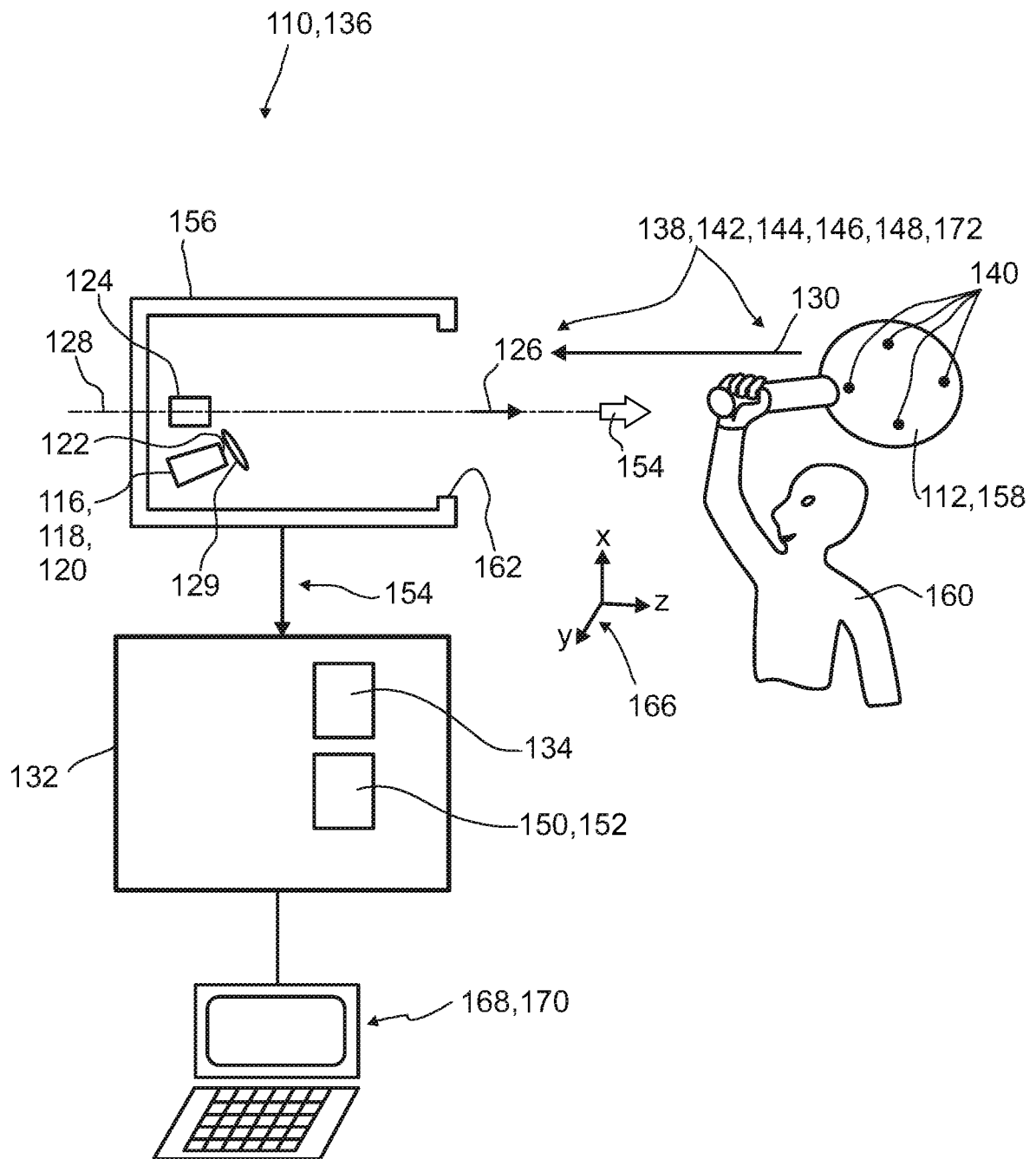


FIG.2

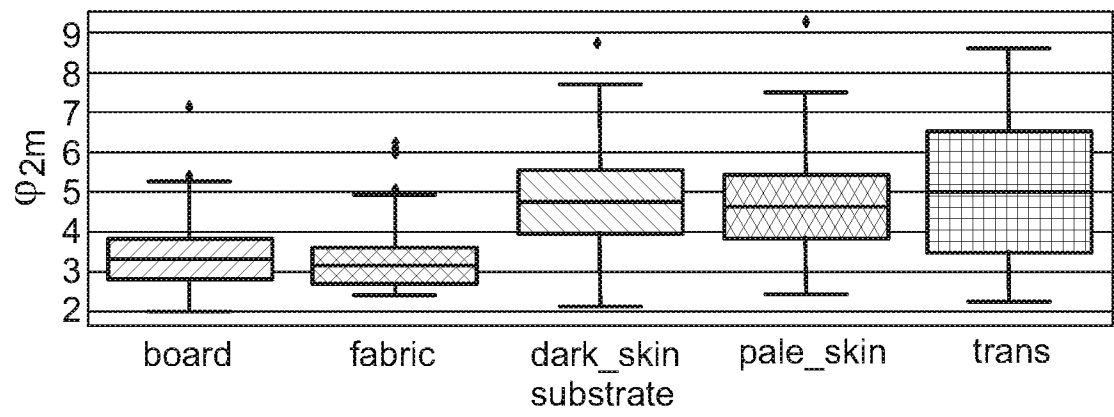
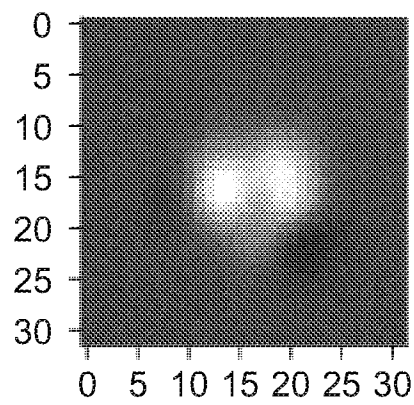


FIG.3





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United States Patent and Trademark Office  
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www.uspto.gov

| APPLICATION<br>NUMBER | FILING of<br>371(c) DATE | GRP ART<br>UNIT | FIL FEE REC'D | ATTY. DOCKET NO       | TOT CLAIMS | IND CLAIMS |
|-----------------------|--------------------------|-----------------|---------------|-----------------------|------------|------------|
| 18/743,352            | 06/14/2024               |                 | 3240          | 27843-2603 182037US03 | 30         | 1          |

CONFIRMATION NO. 2279

76656  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

## FILING RECEIPT



0000000075296240

Date Mailed: 07/16/2024

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

**Please verify the accuracy of the data presented on this receipt.** If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

### Inventor(s)

Friedrich SCHICK, Ludwigshafen, GERMANY;  
Peter SCHILLEN, Ludwigshafen, GERMANY;  
Patrick SCHINDLER, Ludwigshafen, GERMANY;  
Andre SCHMIDT, Ludwigshafen, GERMANY;  
Michael EBERSPACH, Ludwigshafen, GERMANY;  
Christian LENNARTZ, Ludwigshafen, GERMANY;  
Robert SEND, Karlsruhe, GERMANY;  
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Heiko HENGEL, Steinweiler, GERMANY;  
Ingmar BRUDER, Ludwigshafen, GERMANY;  
Jakob UNGER, Ludwigshafen, GERMANY;  
Christian BONSIGNORE, Ludwigshafen, GERMANY;

### Applicant(s)

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

**Power of Attorney:** The patent practitioners associated with Customer Number 76656

### Domestic Priority data as claimed by applicant

This application is a CON of 18/425,144 01/29/2024  
which is a CON of 17/439,492 09/15/2021 PAT 11,947,013  
which is a 371 of PCT/EP20/56759 03/13/2020

Foreign Applications (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.)

EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019 No Access Code Provided

**Permission to Access Application via Priority Document Exchange:** Yes

**Permission to Access Search Results:** Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

**If Required, Foreign Filing License Granted:** 07/15/2024

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/743,352**

**Projected Publication Date:** 10/24/2024

**Non-Publication Request:** No

**Early Publication Request:** No

**Title**

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

**Preliminary Class**

**Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications:** No

## **PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES**

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign

patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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### **Title 37, Code of Federal Regulations, 5.11 & 5.15**

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# PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number  
18/743,352

## APPLICATION AS FILED - PART I

| (Column 1)                                                                |                                                                                                                                                                                                                               | (Column 2)   | SMALL ENTITY |         | OR | OTHER THAN SMALL ENTITY |         |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|---------|----|-------------------------|---------|
| FOR                                                                       | NUMBER FILED                                                                                                                                                                                                                  | NUMBER EXTRA | RATE(\$)     | FEE(\$) |    | RATE(\$)                | FEE(\$) |
| BASIC FEE<br>(37 CFR 1.16(a), (b), or (c))                                | N/A                                                                                                                                                                                                                           | N/A          | N/A          |         |    | N/A                     | 320     |
| SEARCH FEE<br>(37 CFR 1.16(k), (l), or (m))                               | N/A                                                                                                                                                                                                                           | N/A          | N/A          |         |    | N/A                     | 700     |
| EXAMINATION FEE<br>(37 CFR 1.16(o), (p), or (q))                          | N/A                                                                                                                                                                                                                           | N/A          | N/A          |         |    | N/A                     | 800     |
| TOTAL CLAIMS<br>(37 CFR 1.16(i))                                          | 30 minus 20 =                                                                                                                                                                                                                 | * 10         |              |         | OR | x 100 =                 | 1000    |
| INDEPENDENT CLAIMS<br>(37 CFR 1.16(h))                                    | 1 minus 3 =                                                                                                                                                                                                                   | *            |              |         |    | x =                     | 0       |
| APPLICATION SIZE FEE<br>(37 CFR 1.16(s))                                  | If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s). |              |              |         |    |                         | 420     |
| MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))                         |                                                                                                                                                                                                                               |              |              |         |    |                         | 0       |
| * If the difference in column 1 is less than zero, enter "0" in column 2. |                                                                                                                                                                                                                               |              | TOTAL        |         |    | TOTAL                   | 3240    |

## APPLICATION AS AMENDED - PART II

|             |                                                                 | (Column 1)                       | (Column 2) | (Column 3)                         | SMALL ENTITY       |          | OR                 | OTHER THAN SMALL ENTITY |          |                    |
|-------------|-----------------------------------------------------------------|----------------------------------|------------|------------------------------------|--------------------|----------|--------------------|-------------------------|----------|--------------------|
| AMENDMENT A |                                                                 | CLAIMS REMAINING AFTER AMENDMENT |            | HIGHEST NUMBER PREVIOUSLY PAID FOR | PRESENT EXTRA      | RATE(\$) | ADDITIONAL FEE(\$) |                         | RATE(\$) | ADDITIONAL FEE(\$) |
|             | Total<br>(37 CFR 1.16(i))                                       | *                                | Minus      | **                                 | =                  | x        | =                  | OR                      | x        | =                  |
|             | Independent<br>(37 CFR 1.16(h))                                 | *                                | Minus      | ***                                | =                  | x        | =                  | OR                      | x        | =                  |
|             | Application Size Fee (37 CFR 1.16(s))                           |                                  |            |                                    |                    |          |                    | OR                      |          |                    |
|             | FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                  |            |                                    |                    |          |                    | OR                      |          |                    |
|             |                                                                 |                                  |            |                                    | TOTAL<br>ADD'L FEE |          | OR                 | TOTAL<br>ADD'L FEE      |          |                    |
| AMENDMENT B |                                                                 | CLAIMS REMAINING AFTER AMENDMENT |            | HIGHEST NUMBER PREVIOUSLY PAID FOR | PRESENT EXTRA      | RATE(\$) | ADDITIONAL FEE(\$) |                         | RATE(\$) | ADDITIONAL FEE(\$) |
|             | Total<br>(37 CFR 1.16(i))                                       | *                                | Minus      | **                                 | =                  | x        | =                  | OR                      | x        | =                  |
|             | Independent<br>(37 CFR 1.16(h))                                 | *                                | Minus      | ***                                | =                  | x        | =                  | OR                      | x        | =                  |
|             | Application Size Fee (37 CFR 1.16(s))                           |                                  |            |                                    |                    |          |                    | OR                      |          |                    |
|             | FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                  |            |                                    |                    |          |                    | OR                      |          |                    |
|             |                                                                 |                                  |            |                                    | TOTAL<br>ADD'L FEE |          | OR                 | TOTAL<br>ADD'L FEE      |          |                    |

\* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

\*\* If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

\*\*\* If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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A handwritten signature in black ink, reading "Kathi Vidal".

Kathi Vidal

Under Secretary of Commerce for Intellectual Property and  
Director of the United States Patent and Trademark Office

A handwritten signature in black ink, reading "Derrick Brent".

Derrick Brent

Deputy Under Secretary of Commerce for Intellectual Property and  
Deputy Director of the United States Patent and Trademark Office



# UNITED STATES PATENT AND TRADEMARK OFFICE

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| APPLICATION NO.          | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|--------------------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352               | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |
| 76656                    | 7590        | 08/07/2024           | EXAMINER              |                  |
| Patent Docket Department |             |                      | CENTRAL, DOCKET       |                  |
| Armstrong Teasdale LLP   |             |                      | ART UNIT              |                  |
| 7700 Forsyth Boulevard   |             |                      | PAPER NUMBER          |                  |
| Suite 1800               |             |                      | OPAP                  |                  |
| St. Louis, MO 63105      |             |                      | NOTIFICATION DATE     |                  |
|                          |             |                      | DELIVERY MODE         |                  |
|                          |             |                      | 08/07/2024            |                  |
|                          |             |                      | ELECTRONIC            |                  |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

USpatents@atllp.com

|                                                                                     |                                         |                                      |                                 |
|-------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------|---------------------------------|
| <b><i>Decision Dismissing Request for<br/>Prioritized Examination (Track I)</i></b> | <b>Application No.</b><br>18/743,352    | <b>Applicant(s)</b><br>SCHICK et al. |                                 |
|                                                                                     | <b>Examiner</b><br>FIKIRTE A<br>GEREMEW | <b>Art Unit</b><br>OMBL              | <b>AIA (FITF) Status</b><br>Yes |

1. THE REQUEST FILED 14 June 2024 IS **DISMISSED** BECAUSE:

- A. ☐ The application is not a utility application under 35 U.S.C. 111(a) filed by Patent Electronic System or a plant application under 35 U.S.C. 111(a):
- ☐ The application is a utility application that was not filed by Patent Electronic System.
  - ☐ The application is neither a utility application nor a plant application, but rather is a \_\_\_\_\_, which is excluded from the Track I program.
- B. ☐ The request was not filed with the application or on the same date of the application.
- C. ☒ One or more of the following fees were not filed with the application:
- ☐ Basic filing fee, as set forth in 37 CFR 1.16(a), or for a plant application, 37 CFR 1.16(c).
  - ☐ Search fee, as set forth in 37 CFR 1.16(k), or for a plant application, 37 CFR 1.16(m).
  - ☐ Examination fee, as set forth in 37 CFR 1.16(o), or for a plant application, 37 CFR 1.16(q).
  - ☐ Publication fee, as set forth in 37 CFR 1.18(d).
  - ☒ Track I processing fee, as set forth in 37 CFR 1.17(i).
  - ☐ Track I prioritized examination fee, as set forth in 37 CFR 1.17(c).
  - ☐ Application size fee, due because the specification and drawings exceed 100 sheets of paper, as set forth in 37 CFR 1.16(s).
  - ☐ Excess independent claim fee, due because the number of independent claims exceeds three, as set forth in 37 CFR 1.16(h).
  - ☐ Excess claim fee, due because the number of claims exceeds twenty, as set forth in 37 CFR 1.16(i).
- D. ☐ The executed inventor's oath or declaration, or an application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) was not filed with the application or on the same date the application was filed.
- E. ☐ The application contains or has been amended to contain:
- ☐ More than four independent claims.
  - ☐ More than thirty total claims.
  - ☐ One or more multiple dependent claims.
- F. ☐ The Track I program has exceeded its limit of 15,000 requests for the current fiscal year.
- G. ☐ Other: \_\_\_\_\_

2. CONCLUSION

- ☐ **ALL of the above defect(s) consist only of one or more of items E (i-iii). Applicant is hereby accorded a non-extendable ONE MONTH time period from the mailing date of this Decision to (1) correct the subject defect(s) and (2) file a petition under 37 CFR 1.181 requesting reconsideration of prioritized examination. If no such petition is timely filed, the application will not undergo prioritized examination.**

- OR -

- ☒ **The application will not undergo prioritized examination, because the request was not accompanied by all of the required items.**

Telephone inquiries with regard to this decision should be directed to FIKIRTE A GEREMEW at (703) 756-1930. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.

/FIKIRTE A GEREMEW/  
PROGRAM SUPPORT SPECIALIST, OMBL



IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

|                 |                                                         |                   |                |
|-----------------|---------------------------------------------------------|-------------------|----------------|
| First Inventor: | Friedrich Schick                                        | Art Unit:         | OPAP           |
| Serial No.:     | 18/743,352                                              | Confirmation No.: | 2279           |
| Date Filed:     | June 14, 2024                                           | Examiner:         | DOCKET CENTRAL |
| Title:          | Detector for identifying at least one material property |                   |                |

August 8, 2024

**PETITION UNDER 37 C.F.R. 1.181 FOR RECONSIDERATION OF DENIED TRACK I REQUEST**

Mail Stop Petitions  
Commissioner for Patents  
P.O. Box 1450  
Alexandria, Virginia 22313-1450

Dear Sir:

Applicant hereby petitions pursuant to 37 C.F.R. § 1.181 for reconsideration of denied Track I request. A dismissal of Track I request was received on August 7, 2024. It indicates that the Track I request filed June 14, 2024, was dismissed for failure to pay the Track I processing fee as set forth in 37 CFR 1.17(i). The omission of this processing fee was inadvertent and unintentional.

Accordingly, Applicant submits herewith the Track I processing fee and requests reconsideration of denied Track I request.

The Commissioner is hereby authorized to charge the fee set forth in 37 C.F.R. 1.17(i), and to charge any under payment or credit any over payment of fees to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP.

Respectfully Submitted,

/Christopher M. Goff/  
Christopher M. Goff  
Reg. No. 41,785  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard, Suite 1800  
St. Louis, Missouri 63105  
(314) 621-5070



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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

|               |                             |                       |
|---------------|-----------------------------|-----------------------|
| APPLICATION # | RECEIPT DATE / TIME         | ATTORNEY DOCKET #     |
| 18/743,352    | 08/08/2024 02:32:55 PM Z ET | 27843-2603 182037US03 |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                        |                                                          |                      |                  |
|------------------------|----------------------------------------------------------|----------------------|------------------|
| APPLICATION TYPE       | Utility - Nonprovisional Application under 35 USC 111(a) | PATENT #             | -                |
| CONFIRMATION #         | 2279                                                     | FILED BY             | Barbara McGregor |
| PATENT CENTER #        | 66709612                                                 | FILING DATE          | 06/14/2024       |
| CUSTOMER #             | 76656                                                    | FIRST NAMED INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE ADDRESS | -                                                        | AUTHORIZED BY        | Christopher Goff |

### Documents

**TOTAL DOCUMENTS: 1**

| DOCUMENT                                                                                            | PAGES | DESCRIPTION                                    | SIZE (KB) |
|-----------------------------------------------------------------------------------------------------|-------|------------------------------------------------|-----------|
| 2024-08-08 Petition Under 37 CFR 1.181 for Reconsideration of Denied Track 1 Request 27843-2603.pdf | 2     | Petition for review by the Office of Petitions | 98 KB     |

### Digest

| DOCUMENT                                                                                            | MESSAGE DIGEST(SHA-512)                                                                                                                  |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-08-08 Petition Under 37 CFR 1.181 for Reconsideration of Denied Track 1 Request 27843-2603.pdf | 45B61DA7627EFF0F7313D0F4BCD6009DC00DB6858736ED6F<br>C2FD8DE7EEB4A832871601CD344140D00260F6CC35E888675<br>CA46AAD65ACAC0684BA90F0D6C52A68 |

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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## ELECTRONIC PAYMENT RECEIPT

|                   |                                    |                              |
|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>08/08/2024 02:32:55 PM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Barbara McGregor |
| PATENT CENTER #           | 66709612                                                    | AUTHORIZED BY           | Christopher Goff |
| CUSTOMER #                | 76656                                                       | FILING DATE             | 06/14/2024       |
| CORRESPONDENCE<br>ADDRESS | -                                                           | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |

### Payment Information

|                |                        |                       |
|----------------|------------------------|-----------------------|
| PAYMENT METHOD | PAYMENT TRANSACTION ID | PAYMENT AUTHORIZED BY |
| DA / 012384    | E202488E34007003       | Barbara McGregor      |

| FEE CODE                 | DESCRIPTION                                           | ITEM PRICE(\$) | QUANTITY | ITEM TOTAL(\$)  |
|--------------------------|-------------------------------------------------------|----------------|----------|-----------------|
| 1830                     | PROCESSING FEE, EXCEPT IN<br>PROVISIONAL APPLICATIONS | 140.00         | 1        | 140.00          |
| <b>TOTAL<br/>AMOUNT:</b> |                                                       |                |          | <b>\$140.00</b> |

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#### New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

#### National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage

submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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| APPLICATION NO.          | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|--------------------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352               | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |
| 76656                    | 7590        | 08/21/2024           | EXAMINER              |                  |
| Patent Docket Department |             |                      | CENTRAL, DOCKET       |                  |
| Armstrong Teasdale LLP   |             |                      | ART UNIT              |                  |
| 7700 Forsyth Boulevard   |             |                      | PAPER NUMBER          |                  |
| Suite 1800               |             |                      | OPAP                  |                  |
| St. Louis, MO 63105      |             |                      | NOTIFICATION DATE     |                  |
|                          |             |                      | DELIVERY MODE         |                  |
|                          |             |                      | 08/21/2024            |                  |
|                          |             |                      | ELECTRONIC            |                  |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

USpatents@atllp.com



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Commissioner for Patents  
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Doc Code: TRACK1.GRANT

**Decision Granting Request for  
Prioritized Examination  
(Track I or After RCE)**

Application No.:18/743,352

The petition, filed August 8, 2024, under 37 CFR 1.181 is GRANTED. In view of the specific circumstances surrounding this application and, in particular, the filing of the Track 1 Request requirements including irregularities with Patent Center involving payment of the processing fee, the Office, in this specific instance, hereby waives, *sua sponte*, the processing fee date requirement of the Prioritized Examination, Track 1, program to the extent necessary to render the processing fee as timely paid and sufficient to fulfill such requirement. **Petitioner's attention is drawn to the first sentence of Item 1 of the signed Certification and Request for Prioritized Examination, filed June 14, 2024, which specifically identifies the processing fee.** Petitioner should note that because of the specific circumstances of this application, and in this specific instance, the processing fee was accepted as timely, **HOWEVER, IN ANY FUTURE REQUESTS FOR PRIORITIZED EXAMINATION, TRACK 1, ALL OF THE REQUIREMENTS, INCLUDING PAYMENT OF THE PROCESSING FEE, AS REQUIRED BY 37 CFR 1.102(e), MUST BE TIMELY SATISFIED FOR AN APPLICATION TO BE ACCEPTED INTO THE PRIORITIZED EXAMINATION, TRACK 1, PROGRAM.**

1. THE REQUEST FILED June 14, 2024 IS **GRANTED**.

The above-identified application has met the requirements for prioritized examination

- A. ☒ for an original nonprovisional application (Track I).  
B. ☐ for an application undergoing continued examination (RCE).

2. **The above-identified application will undergo prioritized examination.** The application will be accorded special status throughout its entire course of prosecution until one of the following occurs:

- A. filing a **petition for extension of time** to extend the time period for filing a reply;  
B. filing an **amendment to amend the application to contain more than four independent claims, more than thirty total claims**, or a multiple dependent claim;  
C. filing a **request for continued examination**;  
D. filing a notice of appeal;  
E. filing a request for suspension of action;  
F. mailing of a notice of allowance;  
G. mailing of a final Office action;  
H. completion of examination as defined in 37 CFR 41.102; or  
I. abandonment of the application.

Telephone inquiries with regard to this decision should be directed to Vanitha Elgart at 571-272-7395.

/VANITHA M ELGART/  
Petitions Examiner, OPET

|                                   |                                       |  |                                                             |             |
|-----------------------------------|---------------------------------------|--|-------------------------------------------------------------|-------------|
| <i>Notice of References Cited</i> | Application/Control No.<br>18/743,352 |  | Applicant(s)/Patent Under<br>Reexamination<br>SCHICK et al. |             |
|                                   | Examiner<br>FRANTZ B JEAN             |  | Art Unit<br>2454                                            | Page 1 of 1 |

U.S. PATENT DOCUMENTS

| * |   | Document Number<br>Country Code-Number-Kind Code | Date<br>YYYY-MM-DD | Name            | CPC Classification | US Classification |
|---|---|--------------------------------------------------|--------------------|-----------------|--------------------|-------------------|
| * | A | US-20180033146-A1                                | 2018-02-01         | Bleyer; Michael | G01B11/2513        | 1/1               |
| * | B | US-20180276843-A1                                | 2018-09-27         | SEND; Robert    | G02B3/14           | 1/1               |
| * | C | US-20170363465-A1                                | 2017-12-21         | SEND; Robert    | G01J1/08           | 1/1               |
|   | D |                                                  |                    |                 |                    |                   |
|   | E |                                                  |                    |                 |                    |                   |
|   | F |                                                  |                    |                 |                    |                   |
|   | G |                                                  |                    |                 |                    |                   |
|   | H |                                                  |                    |                 |                    |                   |
|   | I |                                                  |                    |                 |                    |                   |
|   | J |                                                  |                    |                 |                    |                   |
|   | K |                                                  |                    |                 |                    |                   |
|   | L |                                                  |                    |                 |                    |                   |
|   | M |                                                  |                    |                 |                    |                   |

FOREIGN PATENT DOCUMENTS

| * |   | Document Number<br>Country Code-Number-Kind Code | Date<br>YYYY-MM-DD | Country | Name       | CPC Classification |
|---|---|--------------------------------------------------|--------------------|---------|------------|--------------------|
|   | N | DE-19846619-A1                                   | 2000-04-27         | DE      | SCHWARTZ P | G01B11/303         |
|   | O |                                                  |                    |         |            |                    |
|   | P |                                                  |                    |         |            |                    |
|   | Q |                                                  |                    |         |            |                    |
|   | R |                                                  |                    |         |            |                    |
|   | S |                                                  |                    |         |            |                    |
|   | T |                                                  |                    |         |            |                    |

NON-PATENT DOCUMENTS

| * |   | Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages) |
|---|---|-------------------------------------------------------------------------------------------|
|   | U |                                                                                           |
|   | V |                                                                                           |
|   | W |                                                                                           |
|   | X |                                                                                           |

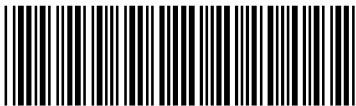
\*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)  
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.



|                                                                                                                    |                                              |                                                                 |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Index of Claims</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                    | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

|   |                 |   |                   |   |                     |   |                 |
|---|-----------------|---|-------------------|---|---------------------|---|-----------------|
| ✓ | <b>Rejected</b> | - | <b>Cancelled</b>  | N | <b>Non-Elected</b>  | A | <b>Appeal</b>   |
| = | <b>Allowed</b>  | ÷ | <b>Restricted</b> | I | <b>Interference</b> | O | <b>Objected</b> |

| CLAIMS                                                                                                                                                                            |          |            |  |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|--|--|--|--|--|--|--|--|
| <input type="checkbox"/> Claims renumbered in the same order as presented by applicant <input type="checkbox"/> CPA <input type="checkbox"/> T.D. <input type="checkbox"/> R.1.47 |          |            |  |  |  |  |  |  |  |  |
| CLAIM                                                                                                                                                                             |          | DATE       |  |  |  |  |  |  |  |  |
| Final                                                                                                                                                                             | Original | 09/20/2024 |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 1        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 2        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 3        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 4        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 5        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 6        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 7        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 8        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 9        | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 10       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 11       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 12       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 13       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 14       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 15       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 16       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 17       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 18       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 19       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 20       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 21       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 22       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 23       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 24       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 25       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 26       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 27       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 28       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 29       | ✓          |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 30       | ✓          |  |  |  |  |  |  |  |  |

|                                                                                                                 |                                              |                                                                 |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Search Notes</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                 | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

| CPC - Searched*                                                                                                                                  |            |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Symbol                                                                                                                                           | Date       | Examiner |
| G01S17/46 G06T7/521 G06T7/73 G06V10/60 G06V10/145<br>G06V10/764 H04N23/74 H04N23/56 G01B11/22 G01C21/16<br>G01S17/66 G06T5/002 G06T5/20 G06T7/20 | 09/20/2024 | FJ       |

| CPC Combination Sets - Searched* |      |          |
|----------------------------------|------|----------|
| Symbol                           | Date | Examiner |
|                                  |      |          |

| US Classification - Searched* |          |            |          |
|-------------------------------|----------|------------|----------|
| Class                         | Subclass | Date       | Examiner |
| 348                           | 135      | 09/20/2024 | FJ       |

\* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

| Search Notes                                         |            |          |
|------------------------------------------------------|------------|----------|
| Search Notes                                         | Date       | Examiner |
| Derwent, EPO, JPO, IBMTDBD, OCR, PGPUB, Google, East | 09/19/2024 | FJ       |

| Interference Search |                       |      |          |
|---------------------|-----------------------|------|----------|
| US Class/CPC Symbol | US Subclass/CPC Group | Date | Examiner |
|                     |                       |      |          |

|  |  |
|--|--|
|  |  |
|--|--|



a detector for identifying at least one material pr X



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Tools

"consisting" (and any subsequent words) was ignored because we limit queries to 32 words.



Google Patents

<https://patents.google.com/patent/>

## Detector for identifying at least one material property

Described herein is a detector for identifying at least one material property  $m$ . The detector includes at least one sensor element including a matrix of ...



Justia

<https://patents.justia.com/patent/>

## Enhanced material detection by stereo beam profile analysis

The invention relates to a detector for determining at least one material property, a method for determining a material property, a mobile device and various ...



Google Patents

<https://patents.google.com/patent/>

## Detector for optically detecting at least one object

As explained above, the detector can furthermore comprise at least one illumination source for generating electromagnetic rays. The illumination source can ...



PatentGuru

<https://www.patentguru.com/assignee/trinamix-gmbh/>

## Trinamix gmbhPatents

Abstract: Described herein is a detector for identifying at least one material property  $m$ . The detector includes at least one sensor element ...



trea.com

<https://trea.com/information/patentapplication/>

## OPTICAL SKIN DETECTION FOR FACE UNLOCK | TREA

Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a ...



googleapis.com

<https://patentimages.storage.googleapis.com/.../PDF/>

## Untitled - Googleapis.com

... object recognition comprising - at least one illumination source (114) configured for projecting at least one illumination pattern comprising a plurality of ...



Justia

<https://patents.justia.com/inventor/robert-schindler/>

## Robert Schindler Inventions, Patents and Patent Applications

Abstract: Described herein is a detector for identifying at least one material property  $m$ . The detector includes at least one sensor element including a ...

Missing: group | Show results with: group



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<https://patentimages.storage.googleapis.com/.../PDF/>

## Untitled - Googleapis.com

The projector may comprise at least one light source, also denoted illumination device or illumination source, configured for generating at least one light beam ...



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## ASML Netherlands B.V.

A lithographic apparatus includes an illumination system, a projection system, and a detection system. The illumination system illuminates a pattern of a ...



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## SPARSE UNDER DISPLAY LIDAR | TREAS

A series of illumination patterns A1 through Am may be used to determine the pattern of emitted light for a first repetition through an mth repetition ...

1 2 3 4 5 6 7 8 9 10      Next

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|                                                                                                     |                        |                       |            |
|-----------------------------------------------------------------------------------------------------|------------------------|-----------------------|------------|
| <b>INFORMATION DISCLOSURE<br/>STATEMENT BY APPLICANT</b><br>( Not for submission under 37 CFR 1.99) | Application Number     |                       | 18743352   |
|                                                                                                     | Filing Date            |                       | 2024-06-14 |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK      |            |
|                                                                                                     | Art Unit               | 2454                  |            |
|                                                                                                     | Examiner Name          | Frantz B Jean         |            |
|                                                                                                     | Attorney Docket Number | 27843-2603 182037US03 |            |

| U.S.PATENTS                                                                                                |         |                    |                        |                  |                                                 |                                                                        | Remove |
|------------------------------------------------------------------------------------------------------------|---------|--------------------|------------------------|------------------|-------------------------------------------------|------------------------------------------------------------------------|--------|
| Examiner Initial*                                                                                          | Cite No | Patent Number      | Kind Code <sup>1</sup> | Issue Date       | Name of Patentee or Applicant of cited Document | Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear |        |
|                                                                                                            | 1       |                    |                        |                  |                                                 |                                                                        |        |
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| Examiner Initial*                                                                                          | Cite No | Publication Number | Kind Code <sup>1</sup> | Publication Date | Name of Patentee or Applicant of cited Document | Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear |        |
|                                                                                                            | 1       | 20160155006        | A1                     | 2016-06-02       | MAKKAPATI et al.                                |                                                                        |        |
|                                                                                                            | 2       | 20160206216        | A1                     | 2016-07-21       | KIRENKO                                         |                                                                        |        |
|                                                                                                            | 3       | 20170363465        | A1                     | 2017-12-21       | SEND et al.                                     |                                                                        |        |
|                                                                                                            | 4       | 20180033146        | A1                     | 2018-02-01       | BLEYER et al.                                   |                                                                        |        |
|                                                                                                            | 5       | 20180276843        | A1                     | 2018-09-27       | SEND et al.                                     |                                                                        |        |
| If you wish to add additional U.S. Published Application citation information please click the Add button. |         |                    |                        |                  |                                                 |                                                                        | Add    |
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|                                                                                                     |                        |                  |                       |  |
|-----------------------------------------------------------------------------------------------------|------------------------|------------------|-----------------------|--|
| <b>INFORMATION DISCLOSURE<br/>STATEMENT BY APPLICANT</b><br>( Not for submission under 37 CFR 1.99) | Application Number     |                  | 18743352              |  |
|                                                                                                     | Filing Date            |                  | 2024-06-14            |  |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK |                       |  |
|                                                                                                     | Art Unit               |                  | 2454                  |  |
|                                                                                                     | Examiner Name          | Frantz B Jean    |                       |  |
|                                                                                                     | Attorney Docket Number |                  | 27843-2603 182037US03 |  |

| Examiner Initial* | Cite No | Foreign Document Number <sup>3</sup> | Country Code <sup>2</sup> | Kind Code <sup>4</sup> | Publication Date | Name of Patentee or Applicant of cited Document | Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear | T <sup>5</sup>           |
|-------------------|---------|--------------------------------------|---------------------------|------------------------|------------------|-------------------------------------------------|--------------------------------------------------------------------------|--------------------------|
|                   | 1       | 108363482                            | CN                        | A                      | 2018-08-03       | JIANGSU SIDIANLING ROBOT CO LTD.                |                                                                          | <input type="checkbox"/> |
|                   | 2       | 19846619                             | DE                        | A1                     | 2000-04-27       | BYK-GARDNER GMBH                                |                                                                          | <input type="checkbox"/> |
|                   | 3       | 2012110924                           | WO                        | A1                     | 2012-08-23       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 4       | 2014097181                           | WO                        | A1                     | 2014-06-26       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 5       | 2014198629                           | WO                        | A1                     | 2014-12-18       | BASF SE                                         |                                                                          | <input type="checkbox"/> |
|                   | 6       | 2018091638                           | WO                        | A1                     | 2018-05-24       | TRINAMIX GMBH                                   |                                                                          | <input type="checkbox"/> |
|                   | 7       | 2018091640                           | WO                        | A2                     | 2018-08-24       | TRINAMIX GMBH                                   |                                                                          | <input type="checkbox"/> |
|                   | 8       | 2018091649                           | WO                        | A1                     | 2018-05-24       | TRINAMIX GMBH                                   |                                                                          | <input type="checkbox"/> |
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|                                                                                                     | Filing Date            |                       | 2024-06-14 |
|                                                                                                     | First Named Inventor   | Friedrich SCHICK      |            |
|                                                                                                     | Art Unit               | 2454                  |            |
|                                                                                                     | Examiner Name          | Frantz B Jean         |            |
|                                                                                                     | Attorney Docket Number | 27843-2603 182037US03 |            |

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|--------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|                    | 1       | EICHLER et al., "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 171 to 266, Springer Verlag, ISBN 0939-0979. Cited in specification, no English translation available                     | <input type="checkbox"/> |
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|                                                                                                     | Examiner Name          | Frantz B Jean         |            |
|                                                                                                     | Attorney Docket Number | 27843-2603 182037US03 |            |

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

### SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

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| Signature  | /Christopher M. Goff/ | Date (YYYY-MM-DD)   | 2024-06-14 |
| Name/Print | Christopher M. Goff   | Registration Number | 41785      |

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⑫

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⑩

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⑤①

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**Die folgenden Angaben sind den vom Anmelder eingereichten Unterlagen entnommen**

⑤④

Vorrichtung und Verfahren zur Bestimmung der Qualität strukturierter Oberflächen

⑤⑦

Die Erfindung betrifft eine Vorrichtung zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen, die eine erste optische Einrichtung aufweist, welche Licht unter einem vorbestimmten Winkel auf die Meßfläche ausstrahlt, und eine zweite optische Einrichtung beinhaltet, welche mit wenigstens einem Fotosensor das von der Meßfläche reflektierte Licht aufnimmt. Die optischen Einrichtungen sind derart beschaffen, daß das reflektierte Licht durch die Struktur der Meßfläche beeinflußt wird, und in einer Auswerteeinrichtung, die zur Steuerung des Meßablaufs vorgesehen ist und die eine Prozessoreinrichtung und eine Speichereinrichtung aufweist, wird das reflektierte Licht ausgewertet und daraus wenigstens eine Strukturkennzahl abgeleitet, welche wenigstens eine strukturbedingte Eigenschaft der Oberfläche charakterisiert.

**DE 198 46 619 A 1**

## Beschreibung

Die vorliegende Erfindung betrifft eine Vorrichtung und ein Verfahren zur Bestimmung der Qualität strukturierter Oberflächen.

Unter der Qualität einer Oberfläche bzw. einer strukturierten Oberfläche sollen hier die physikalischen Eigenschaften einer Oberfläche verstanden werden, die das Aussehen einer Oberfläche für den menschlichen Betrachter bestimmen.

Insbesondere Eigenschaften wie Struktur, Farbe, Helligkeit der Farbe, Glanz, Abbildungsschärfe (engl. DOI), Glanzschleier, Oberflächentexturen und Oberflächenwelligkeiten (engl. orange peel) etc. kennzeichnen strukturierte Oberflächen.

Die Beschaffenheit der sichtbaren Oberflächen ist eine wesentliche Eigenschaft von Gegenständen des täglichen Lebens, wie beispielsweise Einrichtungsgegenständen und Gebrauchsgegenständen, wie Autos und dergleichen.

Im Bereich der Kraftfahrzeugtechnik wird viel Wert auf den visuellen Eindruck der sichtbaren Oberflächen gelegt. Im folgenden werden die technischen Probleme, die bei der Gestaltung von Oberflächen im Kraftfahrzeugbereich entstehen, näher erläutert, ohne jedoch die vorliegende Erfindung in ihrer Anwendung in irgendeiner Weise einzuschränken.

Kraftfahrzeugkarosserien werden üblicherweise mit einer Hochglanz- oder Metalllackierung versehen, deren Reflexionsvermögen oder Glanzkennwert den entsprechenden Werten von anderen Flächen, wie z. B. Möbeln, weit überlegen ist. Der hohe Glanz der verwendeten Lacke und die relativ großen, ebenen Flächen erfordern eine außerordentlich sorgfältige Vorbereitung der zu lackierenden Flächen und eine besonders sorgfältige Auftragung des Lacks.

Im Stand der Technik sind verschiedene Verfahren, Vorrichtungen und Apparate bekannt, wie z. B. in der DE 44 34 203 A1 beschrieben, mit denen die visuellen Eigenschaften und speziell das Reflexionsverhalten von Oberflächen bestimmt werden können.

Weiterhin werden in der Automobil-Industrie im Innenraum von Kraftfahrzeugen eine Vielzahl von strukturierten Oberflächen verarbeitet, die aus natürlichen Stoffen, wie z. B. Holz oder Kunststoff bestehen.

Der visuelle Eindruck strukturierter Oberflächen hängt wesentlich von der Ausgeprägtheit der Struktur der Oberfläche ab, wird jedoch auch von den anderen obengenannten optischen Kenngrößen bestimmt.

Die Automobil-Hersteller beschäftigen heute eine Vielzahl von Prüfern und verwenden eine Vielzahl von aufwendigen, großbauenden Meßapparaturen, die die Qualität der Flächen visuell oder automatisch prüfen, um Qualitätsmängel von Oberflächen schon bei der Produktion zu erkennen. Diese Methode bringt jedoch eine ganze Reihe von Nachteilen mit sich.

Die Arbeit der visuellen Prüfer ist sehr anstrengend und erfordert Werkräume, deren Lichtverhältnisse immer exakt definiert sind. Dennoch wurden große Unterschiede bei der Bewertung von gleichen Oberflächen durch verschiedene Prüfer festgestellt, da zum einen der jeweilige physiologische Eindruck von Prüfer zu Prüfer unterschiedlich ist, und da zum anderen das Sehvermögen des individuellen Prüfers auch von dessen jeweiliger physischer Verfassung abhängt.

Es bereitet somit große Schwierigkeiten, eine untere Qualitätsgrenze zu definieren, deren Unterschreiten nicht mehr hingenommen werden kann.

Weiterhin ist es auch für erfahrene Prüfer schwierig festzustellen, welche Ursachen ein ungenügender visueller Eindruck hat. Besonders bei strukturierten Oberflächen wird der visuelle Eindruck durch eine Kombination unterschiedlicher

Parameter bestimmt, so daß es schwierig ist, aufgrund der gemachten Beobachtungen und Messungen z. B. Steuerungswerte einer automatischen Lackiereinrichtung zur Verbesserung der Qualität zu verändern.

Strukturierte Kunststoffoberflächen werden häufig durch ein Prägeverfahren hergestellt, bei dem die verwendeten Formen bzw. Prägerollen etc. einem Verschleiß unterliegen. Mit zunehmender Anzahl geprägter Oberflächen wird die Struktur der Prägeform und somit auch die Struktur der Kunststoffoberfläche immer geringer. Durch einen regelmäßigen Austausch der Prägewerkzeuge kann die Produktion hinsichtlich Qualität und Quantität optimiert werden; allerdings ist die Lebensdauer der verwendeten Werkzeuge bei Schwankungen in der Materialzusammensetzung oder Schwankungen der Umgebungsbedingungen unterschiedlich, so daß die Werkzeuge im allgemeinen schon ausgetauscht werden, obwohl ihre Qualität zur Herstellung fertiger strukturierter Oberflächen noch ausreicht.

Werden zur Prüfung der Qualität strukturierter Oberflächen automatisch arbeitende Geräte verwendet, um z. B. den Glanz oder die Abbildungsschärfe der Oberfläche zu bestimmen, so können quantitative Zahlenwerte erhalten werden. Ein entscheidender Nachteil ist jedoch bei den visuellen und auch bei den automatischen Prüfungen, daß zwar Werte für die untersuchten, optischen Eigenschaften bestimmt werden, der Benutzer jedoch keinerlei Angaben über die Ursache hat.

Ein weiterer Nachteil herkömmlicher Meßapparaturen ist, daß sie einen erheblichen Raumbedarf aufweisen, und somit nicht vom Benutzer mitgenommen werden können.

Es ist deshalb die Aufgabe der vorliegenden Erfindung, ein Verfahren und eine Vorrichtung der eingangs genannten Art zur Verfügung zu stellen, so daß damit eine reproduzierbare Bewertung der Qualität von Oberflächen und insbesondere strukturierter Oberflächen erfolgen kann und daß derartige physikalische Größen der zu messenden Oberfläche bestimmt werden, daß eine charakteristische strukturbedingte Eigenschaft dieser Oberfläche bestimmt wird.

Ein weiterer Aspekt der Aufgabe ist es, eine Vorrichtung zu schaffen, die kleinbauend und leicht gestaltet und derart beschaffen ist, daß sie von einer Bedienungsperson leicht mitgenommen und ohne weitere Hilfsmittel zur Bewertung einer Oberfläche herangezogen werden kann.

Ein weiterer Aspekt der Aufgabe ist es, eine Vorrichtung zum Messen visueller Eigenschaften zu schaffen, bei welcher trotz des kompakten Aufbaus gemäß dem vorherigen Aspekt der Aufgabe der Erfindung die Möglichkeiten zur Messung gegenüber den im Stand der Technik bekannten Vorrichtungen erheblich erweitert ist.

Ein weiterer Aspekt der Aufgabe ist es, ein Verfahren anzugeben, welches eine vorteilhafte Erfassung einer charakteristischen strukturbedingten Eigenschaft von Oberflächen ermöglicht.

Diese Aufgabe wird erfindungsgemäß durch eine Vorrichtung gelöst, wie sie in Anspruch 1 definiert ist. Das erfindungsgemäße Verfahren ist Gegenstand des Anspruchs 45.

Zu bevorzugende Weiterbildungen der Erfindung sind Gegenstand der Unteransprüche.

Bei der erfindungsgemäßen Vorrichtung wird mit einer ersten optischen Einrichtung, die eine oder mehrere Lichtquellen aufweist, Licht unter einem vorbestimmten Winkel auf eine Meßfläche gerichtet, die Teil der zu messenden Oberfläche ist.

Mit einer unter einem zweiten, vorbestimmten Winkel zu dieser Meßfläche ausgerichteten, zweiten optischen Einrichtung, die einen oder mehrere lichtempfindliche Fotosensoren aufweist, wird das von der Meßfläche reflektierte Licht aufgenommen.

Dieser Fotosensor bzw. diese Fotosensoren, der bzw. die mindestens eine lichtempfindliche Fläche aufweisen, geben ein elektrisches Meßsignal aus, welches für das reflektierte Licht charakteristisch ist, wobei die erste und die zweite optische Einrichtung so beschaffen sind, daß das reflektierte Licht durch die Struktur dieser Meßfläche beeinflusst wird.

In der erfindungsgemäßen Vorrichtung ist eine Auswerteinrichtung angeordnet, die zur Steuerung des Meßablaufs vorgesehen ist und die wenigstens eine Prozessoreinrichtung und wenigstens eine Speichereinrichtung aufweist und die das reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche wenigstens eine der strukturbedingten Eigenschaften dieser Oberfläche charakterisiert.

Eine Ausgabereinrichtung, die in der erfindungsgemäßen Vorrichtung angeordnet ist, dient zur Ausgabe der Meßwerte.

Die Auswerteinrichtung wertet über ein in der Speichereinrichtung gespeichertes Programm die Meßsignale aus und speichert in einer bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung die Meßsignale und/oder die Meßwerte in dieser Speichereinrichtung.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind in der zweiten optischen Einrichtung eine Vielzahl von Fotosensoren angeordnet, welche in Reihen und/oder Spalten angeordnet sein können.

Im Unterschied zu konventionellen Vorrichtungen können bei der erfindungsgemäßen Vorrichtung die Fotosensoren auf einem gemeinsamen Substrat angeordnet werden. Die einzelnen Fotosensoren können auch durch Verschaltung lichtempfindlicher Flächen auf einem gemeinsamen Substrat gebildet werden, wobei die Fläche jedes lichtempfindlichen Sensors so gewählt oder wählbar ist, daß sie in der Meßvorrichtung einem vorgegebenen Winkelbereich des Reflexionswinkels entspricht.

Die erfindungsgemäße Vorrichtung hat erhebliche Vorteile:

Eine wesentliche Voraussetzung zur Steuerung des Fertigungsablaufs bei der Herstellung von strukturierten und nichtstrukturierten Oberflächen ist die Bestimmung der optischen Qualität oder des visuellen Eindrucks der Oberfläche. Durch die Analyse des an der Oberfläche reflektierten Lichts wird eine Strukturkennzahl abgeleitet, die die Qualität der Oberfläche beschreibt. Eine Kennzahl für die Struktur strukturierter Oberflächen wird von herkömmlichen visuellen oder automatischen Meßmethoden nicht bestimmt.

Dies ist besonders vorteilhaft, da das Profil der Struktur einer strukturierten Oberfläche wesentlich zum visuellen Gesamteindruck dieser Oberfläche beiträgt.

Bei der Verwendung von Präge-, Guß- oder anderen Verfahren unterliegen die verwendeten Formen einem Verschleiß, der u. a. von der Anzahl hergestellter, strukturierter Oberflächen abhängt. Neben der Anzahl hergestellter Bauteile mit strukturierten Oberflächen spielen aber auch Umweltbedingungen, wie Luftfeuchtigkeit, Material und Lufttemperatur eine entscheidende Rolle für die Qualität der herzustellenden Oberfläche.

Bei herkömmlichen Herstellungsverfahren müssen die verwendeten Formen nach einer bestimmten Anzahl geprägter oder gegossener Bauteile oder nach einer bestimmten Zeitdauer ausgetauscht werden, um eine gleichbleibend hohe Qualität der hergestellten Bauteile zu sichern.

Dieses Verfahren führt zu einem unnötig hohen Aufwand, da oftmals auch Werkzeuge mit noch ausreichender Qualität ausgewechselt werden, während es andererseits auch vorkommt, daß durch Umwelteinflüsse oder Materialinhomogenitäten bzw. -unterschiede die Werkzeuge vor dem vorgesehenen Wechselzeitpunkt verschlissen sind.

Eine Möglichkeit zur Vermeidung eines zu großen Ausschusses ist die regelmäßige Kontrolle. Dazu muß der Produktionsablauf zur Überprüfung der Werkzeuge angehalten werden, so daß sich kostspielige Stillstandszeiten ergeben.

Mit der erfindungsgemäßen Vorrichtung ist es möglich, einfach, schnell und zuverlässig während der laufenden Produktion die Qualität strukturierter Oberflächen zu bestimmen, so daß ein unzulässiger Verschleiß der Werkzeuge frühzeitig festgestellt werden kann und unnötige Stillstandszeiten der Produktionsanlage vermieden werden.

Ein weiterer Vorteil der Verwendung der erfindungsgemäßen Vorrichtung ist, daß unnötig häufige Wechsel der verwendeten Werkzeuge, Formen und Prägerollen und dergl. unterbleiben können, was zu geringeren Produktionskosten führt.

Ein weiterer wesentlicher Vorteil ist, daß die erfindungsgemäße Vorrichtung kleinbauend ist und so vom Benutzer jederzeit mitgenommen werden kann.

Deshalb eignet sich die erfindungsgemäße Vorrichtung auch für die Endkontrolle an schlecht zugänglichen Orten oder in Innenräumen von z. B. Kraftfahrzeugen, wo eine Vielzahl strukturierter Oberflächen eingesetzt wird.

Ein weiterer Vorteil einer solchen kleinbauenden Vorrichtung ist, daß auch an konkav oder konvex gekrümmten Oberflächen die Qualität einfach und zuverlässig bestimmt werden kann.

Vorteilhaft ist auch, daß durch die erfindungsgemäße Vorrichtung unterschiedlichste Stellen einer zu untersuchenden Oberfläche schnell und einfach gemessen werden können und so die Qualitätsverteilung über einer Oberfläche oder auch die Reproduzierbarkeit getestet werden kann.

In einer bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist wenigstens ein erster Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts ein Lichtmuster auf. Die erste optische Einrichtung kann eine Mustereinrichtung enthalten. Das von dieser wenigstens einen Lichtquelle dieser ersten optischen Einrichtung ausgestrahlte Licht wird bei dieser Weiterbildung der erfindungsgemäßen Vorrichtung wenigstens teilweise auf diese Mustereinrichtung gestrahlt. Das durch diese Mustereinrichtung transmittierte Licht wird entsprechend dem Muster der Mustereinrichtung in der Amplitude und/oder Phase beeinflusst.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist das von dieser Mustereinrichtung ausgestrahlte Lichtmuster wenigstens eine Hell-/Dunkel-Kante, oder eine Vielzahl von Hell-/Dunkel-Kanten auf, wobei in einer weiteren Weiterbildung der erfindungsgemäßen Vorrichtung wenigstens ein Teil dieser Vielzahl von Hell-/Dunkel-Kanten wenigstens abschnittsweise parallel zueinander verläuft.

Das Muster dieser Mustereinrichtung kann symmetrisch zur optischen Achse angeordnet sein. In dem Fall, daß das ausgestrahlte Lichtmuster eine Vielzahl von Hell-/Dunkel-Kanten aufweist, können die einzelnen Hell-/Dunkel-Kanten einen geraden, kreis- oder sinusförmigen Verlauf aufweisen.

In einer bevorzugten Weiterbildung ist wenigstens ein Teil dieser Vielzahl von Hell-/Dunkel-Kanten gitter-, kreuzgitter- oder kreisförmig angeordnet.

Die Projektion eines Lichtmusters auf die zu untersuchende Oberfläche ist besonders vorteilhaft, da die Auswertung des von der zu untersuchenden Oberfläche bzw. der Meßfläche reflektierten Lichtmusters Rückschlüsse auf die Qualität und die Struktur der zu untersuchenden Meßfläche erlaubt.

Es ist möglich, daß das Lichtmuster keine scharfen Übergänge von hell nach dunkel aufweist, sondern die Intensität

des projizierten Lichts sinus- oder sägezahnförmig beeinflußt.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist ein erster Teil eines von dieser ersten optischen Einrichtung ausgestrahlten Lichts dieses Lichtmuster auf, während ein zweiter Teil ohne Lichtmuster auf die Meßfläche gerichtet ist.

Es ist aber auch möglich, daß bei der erfindungsgemäßen Vorrichtung die Mustereinrichtung zuschaltbar ist, so daß ein abwechselndes Messen mit Lichtmuster und ohne Lichtmuster möglich ist.

Dazu ist es möglich, daß diese Mustereinrichtung z. B. eine LCD-Einrichtung aufweist, bei welcher einzelne Bereiche oder einzelne Pixel je nach Bedarf zu unterschiedlichen Lichtmustern verschaltet werden können, so daß das Lichtmuster schnell und einfach an die gegebenen Erfordernisse angepaßt bzw. abgeschaltet werden kann.

Durch Auswertung des Signals dieses wenigstens einen Fotosensors, welcher wenigstens einen Teil des reflektierten Lichtmusters aufnimmt, kann eine Strukturkennzahl von dieser Auswerteeinrichtung bestimmt werden, welche wenigstens für eine strukturbedingte Eigenschaft dieser Oberfläche charakteristisch ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung bestimmt diese Auswerteeinrichtung wenigstens für einen Teil dieser Vielzahl von Fotosensoren eine Steigung des elektrischen und/oder digitalen Meßsignals, indem eine Differenz des elektrischen und/oder digitalen Meßsignals des Fotosensors mit diesem elektrischen und/oder digitalen Meßsignal des nächsten Fotosensors in dieser Reihe und/oder Spalte gebildet wird, oder indem Differenzen des elektrischen und/oder digitalen Meßsignals dieses Fotosensors mit diesem elektrischen und/oder digitalen Meßsignal wenigstens eines Teils der direkt oder diagonal benachbarten Fotosensoren gebildet werden.

Dies ist besonders vorteilhaft, da die Steigungen der Meßsignale und insbesondere die Steigungen der Meßsignale bei Bestrahlung mit einem Lichtmuster dazu geeignet sind, eine Strukturkennzahl abzuleiten, welche für wenigstens eine strukturbedingte Eigenschaft der zu untersuchenden Meßfläche charakteristisch ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung bestimmt diese Auswerteeinrichtung wenigstens einen Durchschnitt wenigstens eines Teils dieser Steigungen der Meßsignale dieser Fotosensoren und bestimmt aus diesem wenigstens einen Durchschnitt diese charakteristische Strukturkennzahl für diese wenigstens eine strukturbedingte Eigenschaft dieser Oberfläche.

Aus diesen Steigungen kann aber auch ein Histogramm bestimmt werden, in welchem für vorbestimmte und/oder wählbare Bereiche die Steigung über der Anzahl dieser Steigungen in den entsprechenden Intervallen bzw. Bereichen aufgetragen wird. Mit der Bildung des Schwerpunkts und/oder des Mittelwerts der Steigungen kann eine charakteristische Strukturkennzahl in dieser Auswerteeinrichtung abgeleitet werden, welche die strukturbedingten Eigenschaften dieser Oberfläche charakterisiert.

Diese bevorzugte Weiterbildung der erfindungsgemäßen Vorrichtung ist besonders vorteilhaft, da durch die Bestimmung wenigstens eines Teils der Steigungen und durch die Bildung der Durchschnittssteigung schnell und zuverlässig eine charakteristische Strukturkennzahl für diese Meßfläche bestimmt werden kann.

Abgerundete Oberflächenstrukturen, wie sie z. B. bei Verwendung von abgenutzten oder verschlissenen Werkzeugen bei der Produktion strukturierter Oberflächen entstehen, reflektieren auf diese Oberfläche projiziertes Licht diffuser als Oberflächen mit schärfer ausgeprägtem Profil.

Durch die Verwendung einer solchen Weiterbildung der erfindungsgemäßen Vorrichtung ist es möglich, die strukturbedingten Eigenschaften zu bestimmen.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung werden als optische Kenngrößen neben dieser charakteristischen Strukturkennzahl noch wenigstens eine, zwei, drei oder mehr charakteristische, optische Kenngrößen dieser Meßfläche bestimmt.

In bevorzugten Weiterbildungen der Erfindung wird der Glanz (engl. gloss) oder der Schleierglanz (engl. haze) oder die Abbildungsschärfe (engl. distinction of image DOI) oder ein repräsentatives Maß für die typische Wellenlänge und deren Amplitude (engl. orange peel) der Topologie der Schichtdicke dieser Meßfläche bestimmt.

Die Bestimmung des orange peel erfolgt in einem vorbestimmten Wellenlängenintervall; die Auswertung zur Bestimmung des orange peel kann aber auch in zwei oder mehr Wellenlängenbereichen erfolgen. Weiterhin ist es möglich, die Farbe der Meßfläche oder die Helligkeit der Farbe zu bestimmen.

In einer weiteren bevorzugten Weiterbildung dieser Erfindung werden neben der Strukturkennzahl noch weitere der obengenannten charakteristischen optischen Kenngrößen bestimmt.

Eine solche Weiterbildung der erfindungsgemäßen Vorrichtung ist besonders vorteilhaft, da mit dieser kleinbauenden Vorrichtung u. U. alle interessierenden, visuellen und physikalischen Eigenschaften dieser Meßfläche bestimmt werden können.

Bei einer vorteilhaften Weiterbildung aller vorstehend beschriebenen Ausführungsbeispiele ist eine dritte optische Einrichtung angeordnet, welche wenigstens eine Lichtquelle aufweist und Licht mit einer vorgegebenen, spektralen Charakteristik ausstrahlt, welches in einem vorbestimmten Winkel auf diese Meßfläche gerichtet ist.

In den bisher genannten bevorzugten Weiterbildungen der erfindungsgemäßen Vorrichtung kann der vorbestimmte Winkel, in welchem das von der ersten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist und/oder der vorbestimmte Winkel, unter welchem die zweite optische Einrichtung das von dieser Meßfläche reflektierte Licht aufnimmt, und/oder der vorbestimmte Winkel, in welchem das von der dritten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, frei eingestellt werden und umfaßt insbesondere die Winkel 5°, 10°, 15°, 20°, 30°, 45°, 60°, 75°, 80° und 85° zwischen der Normalen der Meßfläche und dem entsprechenden Winkel. Es sei jedoch darauf hingewiesen, daß auch andere als die hier genannten Winkel möglich sind.

In einer weiteren bevorzugten Weiterbildung der Erfindung ist dieses von dieser dritten optischen Einrichtung ausgestrahlte Licht in einem derartigen Winkel auf die Oberfläche gerichtet, daß das unmittelbar von der Meßfläche gemäß der Fresnel'schen Reflexion gerichtet reflektierte Licht gegenüber dieser Meßfläche einen anderen Winkel aufweist als der Winkel zwischen dieser Meßfläche und dem von dieser Meßfläche gerichtet reflektierten Licht, welches von dieser ersten optischen Einrichtung ausgestrahlt wird, so daß die zweite optische Einrichtung bzw. die Fotosensoren im wesentlichen das von der dritten optischen Einrichtung ausgestrahlte Licht und diffus von dieser Meßfläche reflektierte Licht aufnimmt.

Diese wenigstens eine Lichtquelle dieser dritten optischen Einrichtung kann eine, zwei und besonders bevorzugt drei lichtausstrahlende Elemente aufweisen, deren Licht sich in der emittierten spektralen Charakteristik derart unterscheidet, daß die emittierten Wellenlängenbereiche unterschiedlich sind.

Die emittierten Wellenlängenbereiche dieser lichtausstrahlenden Elemente können sich im sichtbaren Bereich des Lichtspektrums zumindest teilweise überlappen, und in einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind die emittierten, spektralen Charakteristika dieser lichtausstrahlenden Elemente voneinander linear unabhängig.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind wenigstens zwei, vorzugsweise drei oder mehr Fotosensoren derart gestaltet, daß sie sich in ihrer spektralen Charakteristik unterscheiden, so daß als optische Kenngröße dieser Meßfläche die Farbe des reflektierten Lichts erfaßbar ist.

In einer anderen, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist wenigstens ein Fotosensor wenigstens zwei, vorzugsweise drei oder mehr fotosensitive Elemente auf, deren elektrische Ausgangssignale einzeln erfaßbar sind und die sich in ihrer spektralen Charakteristik unterscheiden, so daß die Farbe des von dieser Meßfläche reflektierten Lichts erfaßbar ist.

Eine Weiterbildung der erfindungsgemäßen Vorrichtung, die es ermöglicht, die Farbe des von dieser Meßfläche reflektierten Lichts zu erfassen, ist besonders vorteilhaft, da die Farbe und die Helligkeit der Farbe wesentliche Merkmale von Oberflächen und auch von strukturierten Oberflächen sind.

In weiteren bevorzugten Weiterbildungen der erfindungsgemäßen Vorrichtung sind in dieser ersten und/oder dieser dritten optischen Einrichtung eine oder mehrere Leuchtdioden oder Laserlichtquellen angeordnet.

Wenigstens ein Teil des von dieser ersten optischen Einrichtung und/oder dieser dritten optischen Einrichtung ausgestrahlten Lichts ist im wesentlichen parallel, divergent oder konvergent.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung strahlt diese erste und/oder dritte optische Einrichtung wenigstens einen Lichtfleck mit vorbestimmtem Durchmesser senkrecht zur Ausbreitungsrichtung des ausgestrahlten Lichts oder wenigstens einen Lichtstreifen mit vorbestimmter Länge und Breite senkrecht zur Ausbreitungsrichtung des ausgestrahlten Lichts aus.

Dieses von der ersten optischen Einrichtung projizierte Lichtmuster kann auch aus verschiedenen Lichtflecken oder Lichtstreifen bestehen.

Es sei darauf hingewiesen, daß diese wenigstens eine optische Kenngröße bzw. diese Strukturkennzahl auch aus den elektrischen und/oder digitalen Meßsignalen dieser Fotosensoren bestimmt werden kann, welche diesen von dieser Meßfläche reflektierten zweiten Teil ohne Lichtmuster dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts aufnehmen.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung ist in möglichst unmittelbarer Nähe wenigstens einer Lichtquelle dieser ersten optischen Einrichtung und/oder dieser wenigstens einen Lichtquelle bzw. dieses wenigstens einen lichtausstrahlenden Elementes dieser dritten optischen Einrichtung und/oder dieses wenigstens einen Fotosensors bzw. dieses wenigstens einen fotosensitiven Elementes wenigstens eine Temperaturmeßeinrichtung angeordnet, welcher zur Bestimmung der charakteristischen Temperatur der jeweiligen Lichtquelle, lichtausstrahlenden Elemente, Fotosensoren oder lichtsensitiven Elementen vorgesehen ist, damit eine temperaturkorrigierte Bestimmung dieses wenigstens einen Strukturkennwerts und/oder dieser wenigstens einen optischen Kenngröße erfolgen kann.

Die Bestimmung der charakteristischen Temperaturen der Lichtquellen und Fotosensoren ist besonders vorteilhaft, da

diese elektrischen Einrichtungen temperaturabhängig arbeiten und durch eine temperaturkorrigierte Bestimmung der Kenngrößen und Kennwerte die Reproduzierbarkeit von Messungen erhöht wird; denn mit zunehmender Meßdauer oder bei wechselndem Meßapparat kann sich die Temperatur und deren Verteilung innerhalb der erfindungsgemäßen Vorrichtung ändern.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird wenigstens ein Teil des Verlaufs des auf dieser Vielzahl von Fotosensoren abgebildeten Lichtmusters bestimmt. Durch eine Abweichung des gemessenen Verlaufs vom idealen Verlauf kann ein charakteristischer Profilhöhenkennwert dieser Meßfläche bestimmt werden.

In einer besonders bevorzugten Weiterbildung wird der Verlauf wenigstens einer Hell-/Dunkel-Kante abschnittsweise auf dieser Vielzahl von Fotosensoren bzw. auf dem CCD-Chip bestimmt.

Wird eine helle oder dunkle Linie auf eine zu untersuchende Oberfläche projiziert, welche ein Rechteckprofil mit einer Profilhöhe  $h$  aufweist, so findet eine räumliche Trennung des Reflexes dieser Linie auf dem CCD-Chip der zweiten optischen Einrichtung statt.

Die von den unteren Kanten des Rechteckprofils reflektierten Teile der dunklen oder hellen Linie werden räumlich parallel versetzt zu denen reflektiert, die an den oberen Kanten des Rechteckprofils reflektiert werden.

Das reflektierte Abbild einer idealen, rechteckförmig strukturierten Oberfläche sind zwei gestrichelte helle bzw. dunkle Linien auf dem CCD-Chip.

Mit dem bekannten Einstrahl- und Reflexionswinkel und dem Abstand dieser zwei gestrichelten Linien auf dem CCD-Chip kann die Höhe des Profils an jedem Ort der Meßfläche bestimmt werden.

Bei der Vermessung von nicht rechteckförmigen Oberflächenprofilen kann analog mit den entsprechenden geometrischen Beziehungen ein Profilhöhenkennwert dieser Meßfläche bestimmt werden.

Die Bestimmung eines Profilhöhenkennwertes ist besonders vorteilhaft, da beim Verschleifen der Herstellungswerkzeuge auch die Profilhöhe der hergestellten Bauteile abnehmen kann.

In einer weiteren bevorzugten Weiterbildung der Erfindung wird diese Strukturkennzahl mittels eines Triangulationsverfahrens bestimmt, wobei diese Bestimmung dieser Strukturkennzahl ausschließlich oder auch zusätzlich mittels dieses Triangulationsverfahrens erfolgen kann. Es sei darauf hingewiesen, daß eines der gängigen Triangulationsverfahren, wie sie im Stand der Technik bekannt geworden sind, eingesetzt werden kann.

Beispielhaft sei auf die aus den amerikanischen Patenten US 5,546,189 und US 5,744,793 dargestellten Triangulationsverfahren hingewiesen, die hiermit Teil der Offenbarung der vorliegenden Erfindung werden.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird im Gegensatz zu dem in dem US-Patent 5,546,189 hergestellten Verfahren kein bzw. nicht ausschließlich ein Grauwert bestimmt, welcher repräsentativ für die Höhe der Meßfläche am Meßort ist, sondern es wird die Oberflächenhöhe am Meßort in größtmöglicher Präzision bestimmt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung ist eine zusätzliche optische Einrichtung mit wenigstens einem Fotosensor vorgesehen, welche unter einem vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist. In einer besonders bevorzugten Weiterbildung ist diese zusätzliche optische Einrichtung in der Ebene angeordnet, die durch den Meßpunkt, die zweite optische Einrichtung und die dritte optische Einrichtung aufgespannt wird, und beson-

ders bevorzugt ist, daß der Winkel, unter dem diese zusätzliche optische Einrichtung das reflektierte Licht aufnimmt, gleich dem Winkel ist, unter dem die zweite optische Einrichtung das reflektierte Licht aufnimmt, so daß die zweite und die zusätzliche optische Einrichtung symmetrisch zu dem von der ersten optischen Einrichtung oder symmetrisch zu dem von der dritten optischen Einrichtung ausgestrahlten Licht angeordnet sind.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird die Intensität und der Ort des Auftreffpunktes der reflektierten Intensität auf dieser zweiten optischen Einrichtung und dieser zusätzlichen optischen Einrichtung bestimmt, und besonders bevorzugt wird von dieser Auswerteeinheit ein charakteristischer Ortskennwert für den Ort der Messung und die gemessene Intensität in dieser Speichereinrichtung gespeichert, und diese Auswerteeinheit bestimmt einen charakteristischen Abstandskennwert für den Abstand zum Meßpunkt auf der zu untersuchenden Oberfläche und speichert diesen in dieser Speichereinrichtung.

In einer weiteren bevorzugten Weiterbildung der Erfindung weist die erste optische Einrichtung eine Scanneinrichtung auf, welche die zu untersuchende Oberfläche derart abtastet, daß das von dieser ersten optischen Einrichtung ausgestrahlte Licht zeitlich nacheinander die zu untersuchende Oberfläche abtastet.

Besonders bevorzugt bestimmt diese Auswerteeinrichtung mit diesen Abstandskennwerten und diesen Ortskennwerten wenigstens eine dieser wenigstens einen Strukturkennzahl.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird wenigstens ein quantitativer Steilheitskennwert für die Steilheit der Struktur der zu untersuchenden Oberfläche bestimmt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung bestimmt diese Auswerteeinrichtung wenigstens einen quantitativen Schärfenkennwert für die Schärfe der Kanten der zu untersuchenden Oberfläche.

Diese Weiterbildungen sind besonders vorteilhaft, da mit der Abnutzung der Werkzeuge die Struktur der Oberfläche der herzustellenden Bauteile abnimmt und insbesondere durch Abnutzung und Verschleiß die Steilheit der Struktur abnimmt und insbesondere auch die Schärfe der Kanten abnimmt, da mit zunehmendem Verschleiß das Profil des Werkzeugs abrundet.

In einer weiteren, bevorzugten Weiterbildung wird aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der genannten charakteristischen optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) eine Bewertungsstrukturkennzahl abgeleitet, welche wenigstens für einen Teil der visuellen Eigenschaften dieser Oberfläche charakteristisch ist.

Bei der Bewertung von Oberflächen und insbesondere von strukturierten Oberflächen hängt der visuelle Eindruck nicht nur von der Ausprägung der Struktur, sondern auch von den anderen genannten charakteristischen, optischen Größen ab. Beispielsweise sind bei besonders glatten und hochreflektierenden Flächen die Anforderungen an den Herstellungsprozeß besonders hoch, da auf solchen Oberflächen das menschliche Auge auch kleinste Störungen, Unebenheiten und Veränderungen wahrnimmt. Bei Kraftfahrzeugen mit sogenannter "Metallic"-Lackierung ist es nahezu unmöglich, auf einer großen Fläche einen Lackschaden derart nachzubessern, daß dieser Schaden für den Betrachter unsichtbar bleibt.

Im Gegensatz dazu sind die Anforderungen bei matten und schwach reflektierenden Flächen an die Beschichtungsqualität des Lacks und der darunter liegenden Oberfläche erheblich geringer.

Während Oberflächenrauigkeit und orange peel schon im geringen Maß bei hochglänzenden Flächen das physiologische Empfinden eines Betrachters stören, wirken sich weitaus größere Rauigkeiten und Oberflächenwelligkeiten bei schwach reflektierenden und matten Flächen auf den visuellen Eindruck eines Betrachters nur wenig oder gar nicht aus. Bei strukturierten Oberflächen ist der visuelle Eindruck durch die Struktur bedingt weniger abhängig von der Oberflächenrauigkeit, dem Glanz oder den anderen optischen Kenngrößen.

Gemäß dieser bevorzugten Weiterbildung der Erfindung wird deshalb wenigstens eine Bewertungsstrukturkennzahl und wenigstens eine der genannten charakteristischen Größen gebildet, welche ein Maß für die Qualität der Meßfläche darstellt.

In einer bevorzugten Weiterbildung der Erfindung wird aus diesem Kennwert für den Glanz der Meßfläche und dieser Strukturkennzahl eine Bewertungsstrukturkennzahl derart gebildet, daß für Flächen mit geringem Glanz auch bei einer nicht idealen Strukturkennzahl eine ausreichende bzw. befriedigende Bewertungsstrukturkennzahl gebildet wird.

In einer weiteren, bevorzugten Weiterbildung ist vorgesehen, daß diese Bewertungsstrukturkennzahl derart gebildet wird, daß z. B. ein höherer Glanz eine schlechtere Strukturkennzahl ausgleicht, so daß diese Bewertungsstrukturkennzahl ein qualitativ ausreichendes Ergebnis erzielt.

In einer weiteren, bevorzugten Weiterbildung ist vorgesehen, daß für wenigstens zwei oder alle der genannten optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) eine Bewertungsstrukturkennzahl gebildet wird oder daß aus zwei oder mehr der genannten charakteristischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) und dieser Strukturkennzahl eine Bewertungsstrukturkennzahl abgeleitet wird, welche für den visuellen Eindruck dieser Meßfläche charakteristisch ist.

Die Bildung wenigstens einer Bewertungsstrukturkennzahl aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der charakteristischen optischen Größen, wie Glanz etc., ist besonders vorteilhaft, da eine besonders einfache und automatische bzw. automatisierte Bestimmung der Qualität der zu untersuchenden Oberfläche möglich ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung ist wenigstens ein unterer bzw. oberer Grenzwert für diese wenigstens eine Strukturkennzahl und/oder diese wenigstens eine charakteristische optische Kenngröße und/oder diese wenigstens eine Bewertungsstrukturkennzahl festlegbar, bei dessen Unter- bzw. Überschreitung eine Nachricht auf diese Ausgabereinrichtung oder ein Alarmsignal von akustischer oder optischer Art ausgegeben wird.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird dieser wenigstens ein Grenzwert durch Messung wenigstens einer Referenzoberfläche automatisch bestimmt. Es ist z. B. möglich, daß eine, zwei oder mehr bewertete Referenzoberflächen unterschiedlicher Qualität gemessen werden, wobei vorteilhafterweise wenigstens eine dieser Referenzoberflächen eine hohe Qualität und wenigstens eine weitere eine unzureichende Qualität aufweist.

Durch eine derartige teilautomatisierte Kalibrierung dieser Vorrichtung können schnell und zuverlässig verschiedene Meßpunkte auf einer Oberfläche oder verschiedenen zu untersuchenden Oberflächen durchgeführt werden.

In einer weiteren, bevorzugten Weiterbildung ist ein Ein- bzw. Ausschalter angeordnet.

Des weiteren kann an dieser erfindungsgemäßen Vorrichtung eine Eingabevorrichtung vorgesehen sein, wobei diese u. a. dafür vorgesehen sein kann, daß der Benutzer ein Typkennzeichen für den Typ der zu messenden Oberfläche ein-

geben kann, wobei nachfolgende Messungen mit Bezug auf dieses Typkennzeichen von dieser Auswerteeinrichtung ausgewertet und in dieser Speichereinrichtung abgelegt werden.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird der Benutzer beim Einschalten durch ein akustisches oder optisches Signal dazu aufgefordert werden, in dieser Eingabeeinrichtung ein Typkennzeichen für den zu messenden Oberflächentyp einzugeben, wobei in einer besonders bevorzugten Weiterbildung der Benutzer jederzeit durch Eingabe eines anderen Typkennzeichens den Oberflächentyp ändern kann, auf den sich die nachfolgenden Messungen beziehen.

Die Eingabe bzw. die Wahl eines Typkennzeichens vor der Messung eines bestimmten Oberflächentyps ist besonders vorteilhaft, da bei der Bewertung verschiedener Oberflächentypen die einzelnen Kennwerte eine unterschiedliche Bedeutung und bei Bildung einer Bewertungsstrukturkennzahl eine unterschiedliche Gewichtung haben.

In einer weiteren, bevorzugten Weiterbildung der Erfindung kann der Benutzer bei jeder Messung durch Eingabe der Benutzer-Beurteilungszahl in dieser Eingabeeinrichtung diese Meßfläche gemäß seinem physiologischen Eindruck bewerten. Diese Benutzer-Beurteilungszahl wird besonders bevorzugt zusammen mit dieser wenigstens einen charakteristischen, optischen Kenngröße und dieser wenigstens einen Strukturkennzahl und dieser wenigstens einen Bewertungsstrukturkennzahl in dieser Speichereinrichtung dauerhaft abgelegt, wobei der Wertebereich dieser Benutzer-Beurteilungszahl wenigstens zwei unterschiedliche Qualitätswerte, die "gut" und "schlecht" entsprechen, umfaßt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung werden wenigstens ein Teil dieser in dieser Speichereinrichtung abgelegten Benutzer-Beurteilungszahlen mit zugehörigen Kenngrößen und Kennwerte und entsprechenden Messungen zu automatischen Bestimmungen wenigstens eines dieses wenigstens eines Grenzwerts für dieses Typkennzeichen der zu untersuchenden Oberfläche verwendet.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung berücksichtigt diese Auswerteeinrichtung im wesentlichen alle in dieser Speichereinrichtung für dieses Typkennzeichen der zu messenden Oberfläche abgelegten Benutzer-Beurteilungszahlen und die entsprechenden Kenngrößen und Kennwerte und Strukturkennwerte zur Bestimmung dieses wenigstens einen Grenzwerts und zur Bestimmung der Gewichtungsfaktoren, welche zur Bestimmung der Bewertungsstrukturkennzahl verwendet werden, so daß diese Gewichtungsfaktoren und dieser wenigstens eine Grenzwert mit zunehmender Anzahl von Messungen derart angepaßt werden, daß diese Bewertungsstrukturkennzahl dem visuellen Eindruck des Benutzers angepaßt wird.

Diese bevorzugte Weiterbildung der Erfindung ist besonders vorteilhaft, da diese Vorrichtung bei der Bestimmung dieser Kenngrößen und Kennwerte und dieser Bewertungsstrukturkennzahl lernfähig ist und auf der Basis eines Expertensystems die Zuverlässigkeit bei der Bestimmung dieser wenigstens einen Bewertungsstrukturkennzahl mit der Anzahl der Messungen zunimmt.

Bei dieser bevorzugten Weiterbildung der Erfindung ist es weiterhin vorgesehen, daß bei der Bestimmung dieser Grenzwerte und Gewichtungsfaktoren nicht alle gemessenen und gespeicherten Werte für dieses Typkennzeichen verwendet werden, da sich der physiologische Zustand des Benutzers auf die Bewertung dieser Meßfläche auswirkt, und da durch irrtümliche Eingabe falscher Benutzer-Beurteilungszahlen ungünstige Grenzwerte und Gewich-

tungsfaktoren bestimmt werden können.

Möglich ist es z. B., daß bei einer Vielzahl von vorliegenden Meßergebnissen eine bestimmte Prozentzahl von z. B. 5% oder 10% bei der Bildung dieser Gewichtungsfaktoren und dieser Grenzwerte unberücksichtigt bleiben, wobei die Auswahl der unberücksichtigt bleibenden Werte nach bekannten, statistischen Verfahren erfolgt.

Diese Weiterbildung der Erfindung ist besonders vorteilhaft, da auch bei irrtümlicher Eingabe falscher Benutzer-Beurteilungszahlen diese im wesentlichen unberücksichtigt bleiben, so daß durch die Lernfähigkeit dieser Auswerteeinrichtung bedingt die Zuverlässigkeit der automatisch bestimmten Bewertungsstrukturkennzahlen mit zunehmender Anzahl von Messungen steigt.

Bei einer vorteilhaften Weiterbildung aller vorstehend beschriebenen Ausführungsbeispiele ist diese Vorrichtung relativ zur Meßfläche in einer im wesentlichen abstandsgleichen Richtung verschiebbar, und es ist eine Wegstrecken-Meßeinrichtung vorgesehen, welche diese relative Verschiebung quantitativ erfaßt. In dieser Speichereinrichtung werden die entlang vorgegebenen Meßschwellen auf dieser Oberfläche gemessenen Strukturkennwerte und/oder charakteristischen optischen Kenngrößen und/oder Bewertungsstrukturkennzahlen gespeichert.

In einer weiteren, bevorzugten Weiterbildung ist wenigstens ein Meßrad vorgesehen, welches während der Messung auf der zu messenden Oberfläche aufgesetzt ist und sich während der Relativbewegung zwischen der Vorrichtung und der zu messenden Oberfläche dreht, wobei dieses wenigstens eine Meßrad mit einem Drehwinkelgeber verbunden sein kann, welcher ein elektrisches Drehwinkelsignal ausgibt, welches für den vom Meßrad zurückgelegten Drehwinkel repräsentativ ist.

Diese bevorzugte Weiterbildung der Erfindung ist besonders vorteilhaft, da zu untersuchende Oberflächen z. B. systematisch über die gesamte Oberfläche gemessen werden können.

In der bevorzugten Weiterbildung der Erfindung, in welcher eine Vielzahl von Fotosensoren angeordnet sind, wird besonders bevorzugterweise ein CCD-Chip eingesetzt, welcher in einer weiteren, bevorzugten Weiterbildung der Erfindung als Farb-CCD-Chip ausgeführt sein kann.

Bei Einsatz eines zweidimensionalen CCD-Chips wird ein zweidimensionales Abbild dieser Meßfläche in der zweiten optischen Einrichtung auf den CCD-Chip abgebildet.

In einer bevorzugten Weiterbildung der Erfindung wird vor Beginn der Messungen oder vor Beginn jeder einzelnen Messung ein Dunkelbild aufgenommen, und die Werte werden gemittelt. Bei der Messung wird das Bild mit Beleuchtung aufgenommen, ebenfalls gemittelt und davon der Mittelwert des Dunkelbildes abgezogen. Es wird ein Wert für den Kontrast dieses Bildes bestimmt. Mit den Mittelwerten des Dunkelbildes, des beleuchteten Bildes und dem Wert für den Kontrast wird die Helligkeit des gemessenen Bildes normiert.

Die weitere Auswertung erfolgt mit den normierten Werten bzw. mit dem normierten Bild.

In einer weiteren, bevorzugten Weiterbildung der Erfindung erfolgt die Auswertung über eine oder mehrere Fouriertransformationen. Besonders bevorzugt werden für jede Zeile die Ortsfrequenzkoeffizienten aus einer Fouriertransformation dieser Zeile bestimmt und die Ortsfrequenzkoeffizienten für das gesamte Bild werden aus dem Mittelwert der Frequenzen aller Zeilen bestimmt.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird dieser Strukturkennwert über ein übliches Verfahren zur Rauheitsmessung von Oberflächen bestimmt, wie z. B. in DIN 4768 (Ausgabe 1990-



05), ISO 13565-1 (Ausgabe 1996-12) oder ISO 13565-2 (Ausgabe 1996-12) beschrieben, die somit Teil der Offenbarung werden.

Es sei darauf hingewiesen, daß das Zusammenwirken der einzelnen erfindungsgemäßen Merkmale in jeder beliebigen Kombination bevorzugt ist. Insbesondere sind auch die durch die unabhängigen und abhängigen Ansprüche offenbarten Merkmalskombinationen unter Weglassung eines oder mehrerer Merkmale bevorzugt. Für den Fachmann ist ersichtlich, daß über die hier dargestellten Ausführungsbeispiele hinaus eine Vielzahl weiterer Modifikationen und Ausführungen denkbar sind, die von der Erfindung erfaßt sind. Die Erfindung beschränkt sich insbesondere nicht nur auf die hier dargestellten Ausführungsformen.

Weitere Vorteile, Merkmale und Anwendungsmöglichkeiten der vorliegenden Erfindung ergeben sich aus der nachfolgenden Beschreibung von Ausführungsbeispielen im Zusammenhang mit den Zeichnungen.

Darin zeigen:

**Fig. 1** einen Schnitt eines ersten Ausführungsbeispiels gemäß der vorliegenden Erfindung;

**Fig. 2** ein weiteres Ausführungsbeispiel eines Sensorelements, bei welchem die zur Lichtmessung herangezogenen Flächen variiert werden können;

**Fig. 3** ein weiteres Ausführungsbeispiel eines Sensorelements, bei welchem sieben Fotosensoren mit den lichtempfindlichen Flächen gebildet werden;

**Fig. 4** ein Diagramm, welches das Reflexionsverhalten bei Beleuchtung ohne Lichtmuster zeigt, wobei auf der Ordinate die gemessene Lichtintensität und auf der Abszisse die Winkelabweichung bezüglich des idealen Reflexionswinkels aufgetragen ist;

**Fig. 5** ein Abbild des auf die Meßfläche projizierten Lichtmusters,

**Fig. 6** den prinzipiellen, schaltungstechnischen Aufbau einer Meßeinrichtung, wie er bei den Ausführungsbeispielen gemäß **Fig. 1–5** zu verwenden ist; und

**Fig. 7** ein weiteres Ausführungsbeispiel der Erfindung, welches auch zu weitergehenden Messungen geeignet ist.

Ein erstes Ausführungsbeispiel der Erfindung wird nun in bezug auf die **Fig. 1** beschrieben.

Die in der **Fig. 1** schematisch dargestellte Meßeinrichtung zur Bestimmung der Qualität strukturierter Oberflächen weist ein Gehäuse auf, in welchem ein erster Belichtungstubus 2 angeordnet ist.

In diesem Belichtungstubus ist, schematisch angedeutet, eine Lichtquelle 3 angeordnet, eine Blende 4, eine Lichtmustereinrichtung 6 und eine Linseneinrichtung 5.

Das von der punktförmigen Lichtquelle 3 ausgestrahlte Licht wird in seiner Apertur durch die Blende 4 begrenzt und trifft auf die Lichtmustereinrichtung 6. Je nach Ausführungsform kann in der Lichtmustereinrichtung 6 eine teilweise transparente Lichtmusterplatte 9 angeordnet sein, bei welcher auf einem Teil der ausgeleuchteten Fläche die Phase und/oder die Amplitude des auftreffenden Lichts derart beeinflusst wird, daß das transmittierte Licht ein charakteristisches Lichtmuster aufweist. In diesem Ausführungsbeispiel wird die Amplitude des von der Lichtquelle 3 ausgestrahlten Lichts durch Hell-/Dunkel-Kanten auf der Lichtmusterplatte 9 beeinflusst.

Weiterhin wird das ausgestrahlte Licht von der Linse 5 gebündelt und trifft durch die Öffnung 7 auf die Meßfläche 8 auf. Von der Meßfläche 8 wird das Licht reflektiert und tritt in den Meßtubus 10 ein, welcher ebenfalls eine Linse 11, eine Blende 12 und den eigentlichen Sensor 13 aufweist.

Die Reflexionsmeßvorrichtung weist weiterhin eine (nicht dargestellte) Steuereinrichtung auf, durch welche der Betrieb der Vorrichtung gesteuert wird und eine ebenfalls

nicht dargestellte Anzeigeeinrichtung, durch welche die gemessenen Strukturkennwerte und charakteristischen optischen Kenngrößen angezeigt werden.

In **Fig. 2** ist der Sensor eines weiteren Ausführungsbeispiels dargestellt. Dieses Ausführungsbeispiel ist mit Ausnahme des Sensors genauso aufgebaut wie in **Fig. 1**. **Fig. 2** zeigt die Sensoreinrichtung 20, bei welcher eine Vielzahl von fotosensitiven Elementen 21 in Reihen und Spalten angeordnet ist.

Bei diesem Beispiel wird ein CCD-Chip eingesetzt. Auf der Fläche des CCD-Chips werden einzelne Flächenelemente zu unterschiedlichen Fotosensoren 22, 23, 24 verschaltet, so daß bei der Messung nur die Signale dieser Fotosensoren 22, 23, 24 vorliegen.

Es sei darauf hingewiesen, daß sich durch die Verschaltung einzelner Flächenelemente zu Fotosensoren die unterschiedlichsten Meßgeometrien realisieren lassen. Weiterhin müssen zur Messung nicht alle Flächenelemente herangezogen werden.

In **Fig. 3** ist ein weiteres Ausführungsbeispiel der Sensoreinrichtung dargestellt. Der weitere Aufbau und der Betrieb dieser Vorrichtung erfolgt in gleicher Weise wie bei **Fig. 1**.

In **Fig. 3** ist die Sensoreinrichtung 30 dargestellt, welche wie bei dem Ausführungsbeispiel gemäß **Fig. 2** als CCD-Chip ausgeführt ist.

Auf der Fläche des CCD-Chips werden durch Verschaltung einzelner Flächenelemente unterschiedliche Fotosensoren 31, 32, 33 und 34 gebildet. Durch die Verschaltung einzelner Flächenelemente zu Fotosensoren kann schnell und einfach nach unterschiedlichen Meßnormen, wie z. B. der amerikanischen ASTM E 430 gemessen werden.

In einem weiteren Ausführungsbeispiel ist der prinzipielle Aufbau gleich dem des Ausführungsbeispiels nach **Fig. 1**, aber es wird kein Lichtmuster auf die Meßfläche projiziert. Als Sensoreinrichtung wird ein CCD-Chip eingesetzt, mit dem alle Flächenelemente einzeln gemessen werden.

Im Diagramm gemäß **Fig. 4** ist auf der Ordinate die gemessene Lichtintensität aufgetragen und auf der Abszisse die Winkelabweichung bezüglich des idealen Reflexionswinkels. Die gemessene Intensität ist im Bereich des idealen Reflexionswinkels am höchsten und nimmt dann mit zunehmenden Winkelabstand ab. Aus der so erfaßten Kurve kann auf einfache Weise das Reflexionsverhalten an der Oberfläche beurteilt werden.

Außerdem ist es möglich, aus der Kurve die Kennwerte abzuleiten, welche gemäß den verschiedenen Normen das Reflexionsverhalten von Oberflächen kennzeichnen.

In **Fig. 5** ist das von einer Meßfläche auf die Sensoreinrichtung reflektierte Bild dargestellt. Das Lichtmuster 50 weist dunkle Kanten 51 und helle Linien 52 auf. Ein zweiter Teil des Reflexes 53 weist kein Lichtmuster auf.

Mit zunehmender Rauigkeit nimmt der Kontrast im Lichtmuster ab, während ein schlechterer Glanz sich auf die Intensität auswirkt. Orange peel führt zu einer Verzerrung der einzelnen dunklen und hellen Linien, und eine strukturierte Oberfläche führt im Fall eines Rechteckprofils zum Versatz der einzelnen Linien im Bereich der Vertiefungen. Bei Sägezahn- oder Dreiecksprofilen können an den entsprechenden Abschnitten korrelierende Neigungen der Linien festgestellt werden.

Bei dem aufgenommenen Bild wird die Intensität und der Kontrast ausgewertet, und es werden die Gradienten im Bereich des Lichtmusters bestimmt. Durch Mittelwertbildung der Gradienten wird ein charakteristisches Maß für die Struktur der Meßfläche bestimmt.

In **Fig. 7** ist der prinzipielle, spaltungstechnische Aufbau einer Meßeinrichtung, wie sie bei den Ausführungsbeispielen

len gemäß **Fig. 1-6** zu verwenden ist, dargestellt.

Der prinzipielle Meßaufbau ist für alle gezeigten Ausführungsbeispiele derselbe; lediglich die Programmierung weicht je nach verwendeter Sensorart ab.

Die Meßeinrichtung weist insgesamt eine Steuereinrichtung **60** auf, welche einen handelsüblichen Mikroprozessor enthält, der durch ein Programm gesteuert wird, welches in einem Speicher **61** abgelegt ist. Der Kommunikation der Steuereinrichtung **60** mit dem Benutzer dient eine Eingabe-einrichtung **62**, die eine Anzahl von Schaltern enthält, um den Betrieb der Steuereinrichtung zu Starten und um (bei den entsprechenden Ausführungsbeispielen) zwischen einzelnen Betriebsarten umschalten zu können. Weiterhin kann der Benutzer ein Typkennzeichen für die zu messende, strukturierte Oberfläche in die Eingabeeinrichtung **61** eingeben, auf das sich nachfolgende Messungen beziehen.

Neben dem Mikroprozessor weist die Steuereinrichtung Ein-/Ausgabeeinrichtungen auf, welche der Verbindung der Steuereinrichtung mit den einzelnen Bauelementen der Vorrichtung dienen.

Die Steuereinrichtung ist mit der Lichtquelle **3** und dem Sensor **14** verbunden. Die Ergebnisse der Messung werden in einem Display **65** angezeigt, welcher vorzugsweise als LCD-Display ausgebildet ist. Für die weitere Auswertung der Messung ist eine Verbindung zu einem externen Computer **76** vorgesehen; die Meßergebnisse werden vorzugsweise auch in der Speichereinrichtung **61** abgelegt.

Die Meßvorrichtung wird durch eine (nicht dargestellte) Batterie mit Strom versorgt.

Die Meßvorrichtung wird insgesamt vorzugsweise in einem Gehäuse **1** angeordnet, welches in etwa die Abmessung eines Taschenbuchs aufweist.

Ein weiteres Ausführungsbeispiel der vorliegenden Erfindung wird nun in Bezug auf **Fig. 7** beschrieben.

Bei diesem Ausführungsbeispiel ist die Vorrichtung insgesamt in einem Gehäuse **100** angeordnet, welches eine Öffnung **101** aufweist, mit der das Gerät auf die zu messende Oberfläche aufgesetzt wird.

Im Unterschied zu den vorangegangenen Ausführungsbeispielen wird das Gerät jedoch nicht direkt auf die Oberfläche aufgesetzt, sondern mittels (schematisch angedeuteter) mindestens zwei Gummiwalzen **103**, **104** oder mindestens vier Gummirädern **103**, **104**, welche drehbeweglich (nicht gezeigt) im Gehäuse **100** gelagert sind.

Wenigstens eines der Gummiräder oder Walzen ist mit einer (nicht dargestellten) Wegstreckenmeßeinrichtung versehen, welche die Winkelbewegungen der Gummiräder **103**, **104** erfaßt und ein dafür repräsentatives elektrisches Signal ausgibt.

Die Vorrichtung weist weiterhin eine erste optische Einrichtung **110** auf, in welcher eine punktförmige Lichtquelle **111** und eine Linse **112** angeordnet sind. Diese erste optische Einrichtung **110** ist derart ausgerichtet, daß die optische Achse in einem vorbestimmten Winkel (im gezeigten Beispiel  $45^\circ$ ) zu der zu messenden Oberfläche **115** ausgerichtet ist.

Unter einem zweiten vorbestimmten Winkel (hier ebenfalls  $45^\circ$ ) ist eine zweite optische Einrichtung angeordnet, die eine Linseneinrichtung **121**, eine Blendeneinrichtung **122** und einen senkrecht zur optischen Achse ausgerichteten Meßsensor **123** aufweist, welcher in diesem Ausführungsbeispiel als Farb-CCD-Chip ausgeführt ist.

An einer dritten optischen Einrichtung **130** sind drei lichtausstrahlende Elemente **132**, **133** und **134** angeordnet, die hier als LEDs ausgeführt sind, welche jeweils eine unterschiedliche spektrale Charakteristik aufweisen, d. h., welche Licht mit unterschiedlichen Farbcharakteristiken ausstrahlen.

In dem hier dargestellten Ausführungsbeispiel fällt das von den LEDs ausgestrahlte Licht im wesentlichen senkrecht auf die zu untersuchende Oberfläche.

Das von der ersten und der dritten optischen Einrichtung ausgestrahlte Licht wird an der zu untersuchenden Oberfläche **115** reflektiert und fällt zum Teil auf den FOTOSensor bzw. Farb-CCD-Chip **125**.

Da die lichtausstrahlenden Elemente **132-134** Licht mit unterschiedlichen Farbcharakteristiken im sichtbaren Bereich des Spektrums ausstrahlen, ist es möglich, einen Farbkennwert für die zu untersuchende Oberfläche zu bestimmen.

#### Patentansprüche

1. Vorrichtung zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen mit:  
einer ersten optischen Einrichtung mit wenigstens einer Lichtquelle, deren Licht in einem vorbestimmten Winkel auf eine Meßfläche, die Teil der zu messenden Oberfläche ist, gerichtet ist,  
einer zweiten optischen Einrichtung, welche in einem vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist, und welche das von der Meßfläche reflektierte Licht aufnimmt, wobei diese zweite optische Einrichtung wenigstens einen FOTOSensor mit wenigstens einer lichtempfindlichen Fläche aufweist, wobei dieser wenigstens eine FOTOSensor ein elektrisches Meßsignal ausgibt, welches für das reflektierte Licht charakteristisch ist;  
einer Auswerteeinrichtung, die zur Steuerung des Meßablaufs vorgesehen ist und die wenigstens eine Prozessoreinrichtung und wenigstens eine Speichereinrichtung aufweist;  
einer Ausgabe-Einrichtung;  
wobei diese erste optische Einrichtung und diese zweite optische Einrichtung so beschaffen sind, daß dieses reflektierte Licht durch die Struktur dieser Meßfläche beeinflußt wird, und  
wobei diese Auswerteeinrichtung dieses reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche wenigstens eine strukturebedingte Eigenschaft dieser Oberfläche charakterisiert.
2. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung über ein in dieser Speichereinrichtung gespeichertes Programm diese Meßsignale auswertet und/oder in dieser Speichereinrichtung speichert.
3. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese zweite optische Einrichtung eine Vielzahl von FOTOSensoren beinhaltet, welche in Reihen und/oder Spalten angeordnet sind.
4. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein erster Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts ein Lichtmuster aufweist.
5. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieses von dieser ersten optischen Einrichtung ausgestrahlte Lichtmuster wenigstens eine Hell-/Dunkelkante oder eine Vielzahl von Hell-/Dunkelkanten aufweist.
6. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil dieser Vielzahl von Hell-/Dunkelkan-

ten wenigstens abschnittsweise parallel zueinander verläuft.

7. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil dieser Vielzahl von Hell-/Dunkelkanten gitter- oder kreuzgitter- oder kreisförmig angeordnet sind.

8. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieser erste Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts dieses Lichtmuster aufweist und daß ein zweiter Teil ohne Lichtmuster auf diese Meßfläche gerichtet ist.

9. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung für wenigstens einen Teil dieser Vielzahl von Fotosensoren eine Steigung des elektrischen und/oder digitalen Meßsignals bestimmt, indem eine Differenz des elektrischen und/oder digitalen Meßsignals dieses Fotosensors mit diesem elektrischen und/oder digitalen Meßsignals des nächsten Fotosensors in dieser Reihe und/oder Spalte gebildet wird.

10. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung für wenigstens einen Teil dieser Vielzahl von Fotosensoren die Steigungen des elektrischen und/oder digitalen Meßsignals bestimmt, indem die Differenzen des elektrischen und/oder digitalen Meßsignals dieser Fotosensoren mit den elektrischen und/oder digitalen Meßsignalen aller benachbarten Fotosensoren gebildet werden.

11. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen Durchschnitt wenigstens eines Teils der Steigungen bestimmt und daraus diese charakteristische Strukturkennzahl für diese wenigstens eine strukturbedingte Eigenschaft dieser Oberfläche bestimmt.

12. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß eine, zwei, drei oder mehr charakteristische optische Kenngrößen dieser Meßfläche bestimmt werden.

13. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese charakteristische optische Kenngröße der Glanz oder der Glanzschleier oder die Abbildungsschärfe (DOT) der Meßfläche ist.

14. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese charakteristische optische Kenngröße ein repräsentatives Maß für die typische Wellenlänge und deren Amplitude (Orange peel) der Topologie der Schichtdicke dieser Meßfläche in einem vorbestimmten Wellenlängenintervall ist, wobei diese Auswertung auch in zwei oder mehr Wellenlängenbereichen erfolgen kann.

15. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Vorrichtung eine dritte optische Einrichtung mit wenigstens einer Lichtquelle aufweist und Licht mit einer vorgegebenen spektralen Charakteristik ausstrahlt, welches in einem vorbestimmten Winkel auf diese Meßfläche gerichtet ist.

16. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieser vorbestimmte Winkel, in welchem das von der ersten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, und/oder daß dieser vorbestimmte Winkel, in welchem die zweite optische

Einrichtung das von dieser Meßfläche reflektierte Licht aufnimmt, und/oder daß dieser vorbestimmte Winkel, in welchem das von der dritten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, einstellbar ist/sind und insbesondere die Winkel 5°, 10°, 15°, 20°, 30°, 45°, 60°, 75°, 80° und 85° umfaßt.

17. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieses von dieser dritten optischen Einrichtung ausgestrahlte Licht in einem derartigen Winkel auf die Oberfläche gerichtet ist, daß das unmittelbar von der Meßfläche gemäß der Fresnel'schen Reflexion gerichtet reflektierte Licht gegenüber dieser Meßfläche einen anderen Winkel aufweist, als der Winkel zwischen dieser Meßfläche und dem von dieser Meßfläche gerichtet reflektierten Licht, welches von dieser ersten optischen Einrichtung ausgestrahlt wird.

18. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese wenigstens eine Lichtquelle dieser dritten optischen Einrichtung wenigstens zwei, vorzugsweise drei lichtausstrahlende Elemente aufweist, deren Licht sich in der emittierten spektralen Charakteristik derart unterscheidet, daß die emittierten Wellenlängenbereiche unterschiedlich sind.

19. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß sich diese emittierten Wellenlängenbereiche dieser lichtausstrahlenden Elemente im sichtbaren Bereich des Lichtspektrums zumindest teilweise überlappen, und daß diese emittierten spektralen Charakteristiken dieser lichtausstrahlenden Elemente voneinander linear unabhängig sind.

20. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Fotosensor wenigstens zwei, vorzugsweise drei oder mehr fotosensitive Elemente aufweist, deren elektrische Ausgangssignale einzeln erfaßbar sind und die sich in ihrer spektralen Charakteristik unterscheiden, so daß als optische Kenngröße dieser Meßfläche die Farbe des reflektierten Lichtes erfaßbar ist.

21. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese wenigstens eine Lichtquelle dieser ersten optischen und/oder dieser dritten optischen Einrichtung eine oder mehr Leuchtdioden und/oder Laserlichtquellen aufweist.

22. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder diese dritte optische Einrichtung im wesentlichen paralleles Licht ausstrahlt.

23. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder dritte optische Einrichtung im wesentlichen divergentes oder konvergentes Licht ausstrahlt.

24. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder dritte optische Einrichtung wenigstens einen Lichtfleck mit vorbestimmten Durchmesser senkrecht zur Ausbreitungsrichtung oder wenigstens einen Lichtstreifen mit vorbestimmter Länge und Breite senkrecht zur Ausbreitungsrichtung ausstrahlt.

25. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eine dieser wenigstens einen optischen Kenngrößen aus den elektrischen und/oder digitalen Meßsignalen dieser Fotosensoren bestimmt wird, welche diesen von der Meßfläche reflektierten zweiten Teil dieses

von dieser ersten optischen Einrichtung ausgestrahlten Lichts aufnehmen.

26. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß in möglichst unmittelbarer Nähe dieser Lichtquelle(n) dieser ersten optischen Einrichtung und/oder dieser Lichtquelle(n) dieser dritten optischen Einrichtung und/oder dieses wenigstens einen Fotosensors wenigstens eine Temperaturmeß-Einrichtung angeordnet ist, welche zur Bestimmung der charakteristischen Temperatur der jeweiligen Lichtquelle(n) und/oder der jeweiligen Fotosensoren vorgesehen ist, damit eine temperaturkorrigierte Bestimmung dieses wenigstens einen Strukturkennwerts und/oder dieser wenigstens einen optischen Kenngröße erfolgen kann.

27. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil des Verlaufs des Abbilds dieser wenigstens einen Hell-/Dunkelkante auf dieser Vielzahl von Fotosensoren bestimmt wird und durch eine Abweichung vom idealen Verlauf von diesem gemessenen Verlauf ein charakteristischer Profilhöhen-Kennwert dieser Meßfläche bestimmt wird.

28. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Strukturkennzahl mittels eines Triangulationsverfahrens bestimmt wird.

29. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens einer dieser wenigstens einen Sensors die Intensität des von dieser ersten optischen Einrichtung ausgestrahlten und von der zu untersuchenden Oberfläche reflektierten Lichts bestimmt, und daß diese Auswerteeinheit einen charakteristischen Orts-Kennwert für den Ort der Messung in dieser Speichereinrichtung speichert und daß diese Auswerteeinheit aus dieser gemessenen Intensität einen charakteristischen Abstands-Kennwert für den Abstand der zu untersuchenden Oberfläche von dieser Vorrichtung bestimmt und in dieser Speichereinrichtung speichert.

30. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste optische Einrichtung eine Scan-Einrichtung aufweist, welche diese zu untersuchende Oberfläche abtastet.

31. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerte-Einrichtung mit diesen Abstands-Kennwerten und diesen Orts-Kennwerten wenigstens eine dieser wenigstens einen Strukturkennzahl bestimmt.

32. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen quantitativen Steilheitskennwert für die Steilheit der Struktur der zu untersuchenden Oberfläche bestimmt.

33. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen quantitativen Schärfenkennwert für die Schärfe der Kanten der Struktur der zu untersuchenden Oberfläche bestimmt.

34. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eine Bewertungs-Strukturkennzahl aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der genannten charakteristischen optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, Orange Peel) gebildet wird, oder daß für jede der bestimmten charakteristischen optischen Kenngrößen (Glanz,

Schleierglanz, Abbildungsschärfe, Orange Peel) wenigstens eine Bewertungs-Strukturkennzahl bestimmt wird.

35. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß jeweils wenigstens ein unterer und/oder wenigstens ein oberer Grenzwert für diese wenigstens eine Strukturkennzahl und/oder diese wenigstens eine charakteristische optische Kenngröße und /oder diese wenigstens eine Bewertungs-Strukturkennzahl festlegbar ist, bei dessen Unter- bzw. Überschreitung eine Nachricht auf dieser Ausgabe-Einrichtung oder ein Alarmsignal ausgegeben wird.

36. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens einer dieser wenigstens einen Grenzwerts durch Messung wenigstens einer Referenzoberfläche automatisch bestimmt werden.

37. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß eine Eingabe-Einrichtung angeordnet ist, in welche der Benutzer ein Typkennzeichen für den zu messenden Oberflächentyp eingeben kann, und daß die nachfolgenden Messungen mit Bezug auf dieses Typkennzeichen in dieser Speichereinrichtung dauerhaft abgelegt werden.

38. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß zusätzlich ein Ein-/Ausschalter angeordnet ist, und daß der Benutzer beim Einschalten durch ein akustisches und/ oder optisches Signal aufgefordert wird, in diese Eingabe-Einrichtung ein Typkennzeichen für den zu messenden Oberflächentyp einzugeben.

39. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß der Benutzer durch Eingabe eines Typänderungszeichens oder durch Betätigen eines Typänderungsschalters diese Auswerte-Einrichtung in einen Typänderungsmodus umschaltet, und daß nach Eingabe eines weiteren Typkennzeichens die Ergebnisse nachfolgender Messungen mit Bezug auf dieses weitere Typkennzeichen in dieser Speichereinrichtung dauerhaft abgelegt werden.

40. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß der Benutzer nach jeder Messung durch Eingabe einer Benutzer-Beurteilungszahl in diese Eingabe-Einrichtung diese Meßfläche bewerten kann, und daß diese Benutzer-Beurteilungszahl zusammen mit wenigstens einer dieser Kenngrößen und -werte und dieser wenigstens einen Bewertungs-Strukturkennzahl in dieser Speichereinrichtung dauerhaft abgelegt wird, wobei diese Benutzer-Beurteilungszahl ein subjektives Maß des Benutzers für die Qualität dieser Meßfläche ist, und wobei der Wertebereich dieser Benutzer-Beurteilungszahl wenigstens zwei unterschiedliche Werte, die "gut" und "schlecht" entsprechen, umfaßt.

41. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerte-Einrichtung wenigstens ein Teil dieser in dieser Speichereinrichtung abgelegten Benutzer-Beurteilungszahlen und zugehörigen Kenngrößen und -werten zur automatischen Bestimmung wenigstens eines dieses wenigstens einen Grenzwerts für dieses Typkennzeichen verwendet wird.

42. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Vorrichtung relativ zur Meßfläche in einer im wesentli-

chen abstandsgleichen Richtung verschiebbar ist, und daß eine Wegstreckenmeßeinrichtung vorgesehen ist, welche diese relative Verschiebung quantitativ erfaßt, und daß weiterhin eine Speichereinrichtung vorgesehen ist, in welcher die entlang vorgegebener Meßstellen auf der Oberfläche gemessenen Strukturkennwerte und/oder optischen Kenngrößen abgespeichert werden.

43. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Meßrad vorgesehen ist, welches während der Messung auf der zu messenden Oberfläche aufgesetzt ist, und sich während der Relativbewegung zwischen der Vorrichtung und der zu messenden Oberfläche dreht.

44. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eines dieses wenigstens einen Meßrades mit einem Drehwinkel-Geber verbunden ist, der ein elektrisches Drehwinkelsignal ausgibt, welches für den vom Meßrad zurückgelegten Drehwinkel repräsentativ ist.

45. Verfahren zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen insbesondere unter Verwendung einer Vorrichtung gemäß mindestens einem vorhergehenden Ansprüche, bei welchem eine erste optische Einrichtung mit einer ersten Lichtquelle vorgesehen ist, um Licht unter einem vorbestimmten Winkel auf eine zu messende Oberfläche zu richten, und

eine zweite optische Einrichtung vorgesehen ist, welche wenigstens einen Fotosensor aufweist, welche in einem zweiten vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist, um das von der Meßfläche reflektierte Licht aufzunehmen, wobei dieser wenigstens eine Fotosensor ein elektrisches Meßsignal ausgibt, welches für das aufgenommene Licht charakteristisch ist, und

eine Auswerteeinrichtung angeordnet ist, die den Meßablauf steuert und die wenigstens eine Prozessoreinrichtung beinhaltet und welche diese Meßsignale in einer Speichereinrichtung speichert, und eine Ausgabe-Einrichtung die Meßergebnisse ausgibt, und diese Auswerteeinrichtung dieses reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche strukturbedingte Eigenschaften der Oberfläche charakterisiert.

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Hierzu 4 Seite(n) Zeichnungen

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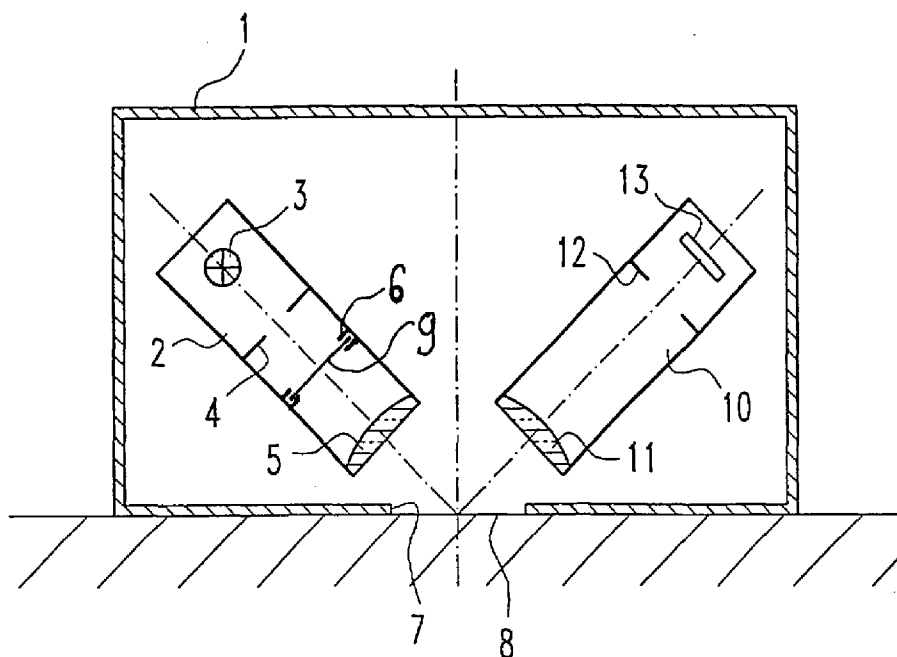


Fig.1

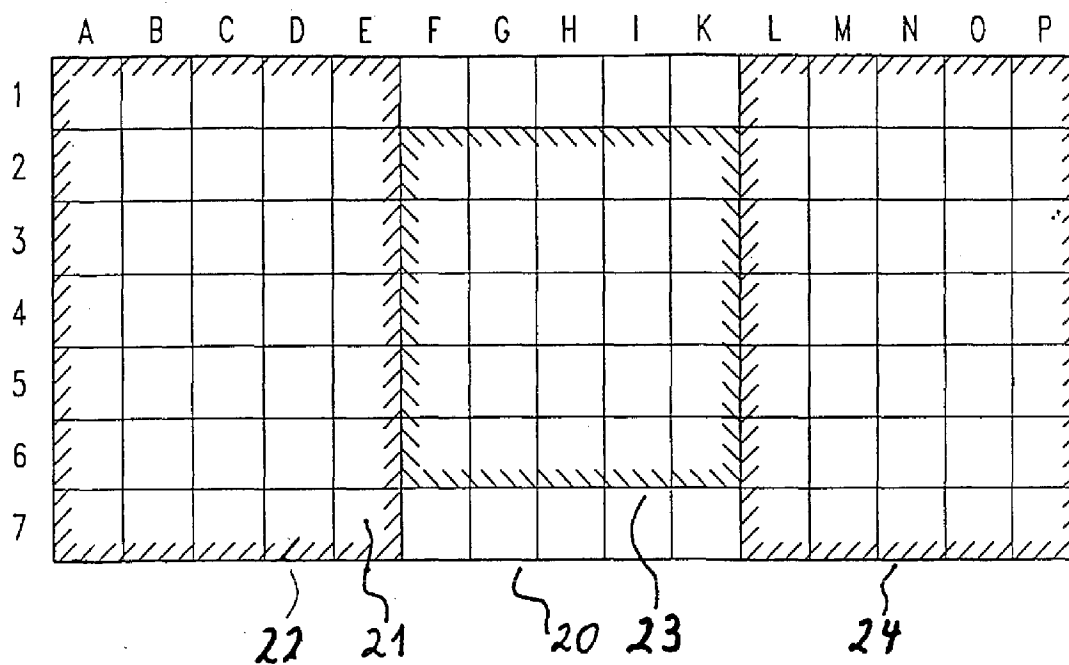


Fig. 2

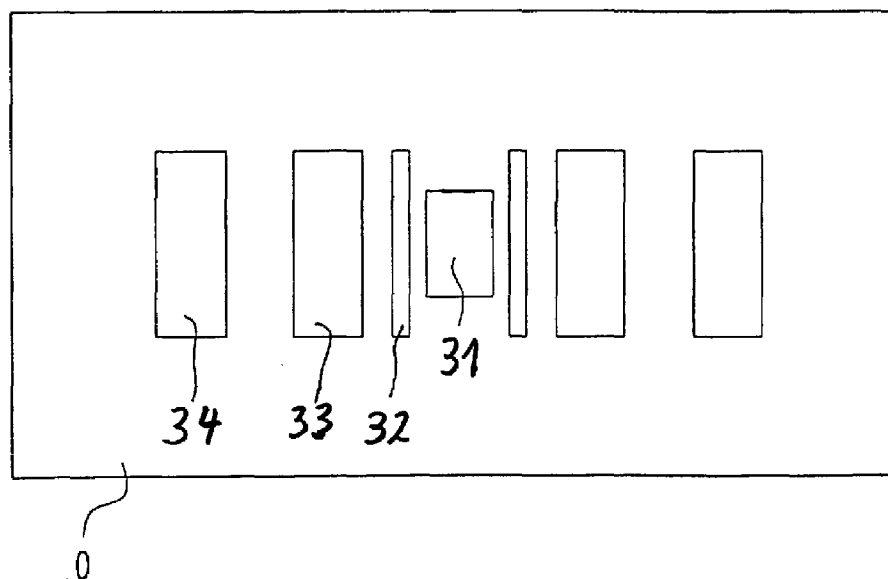


Fig. 3

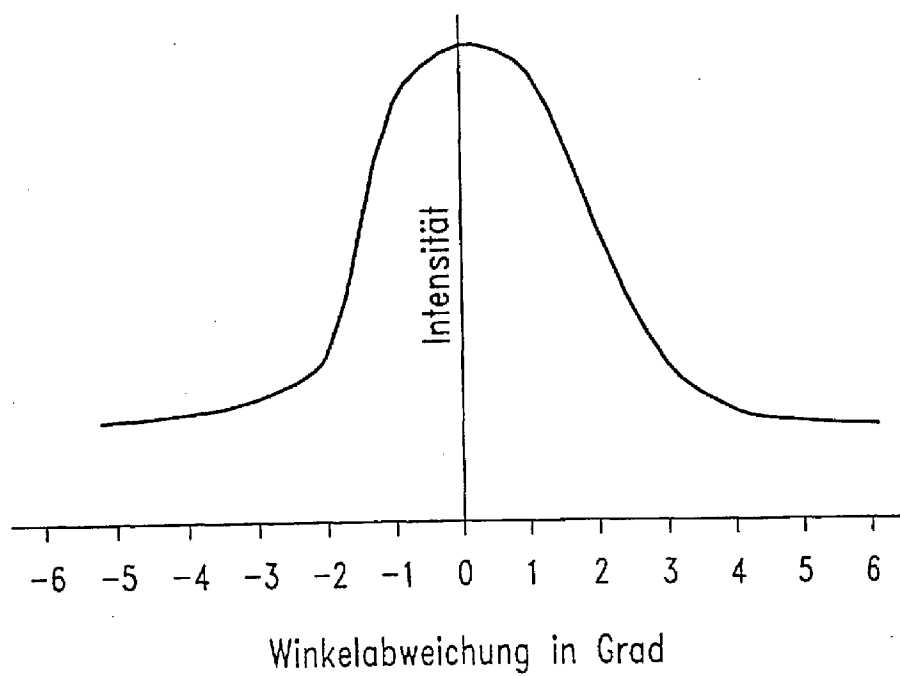


Fig. 4

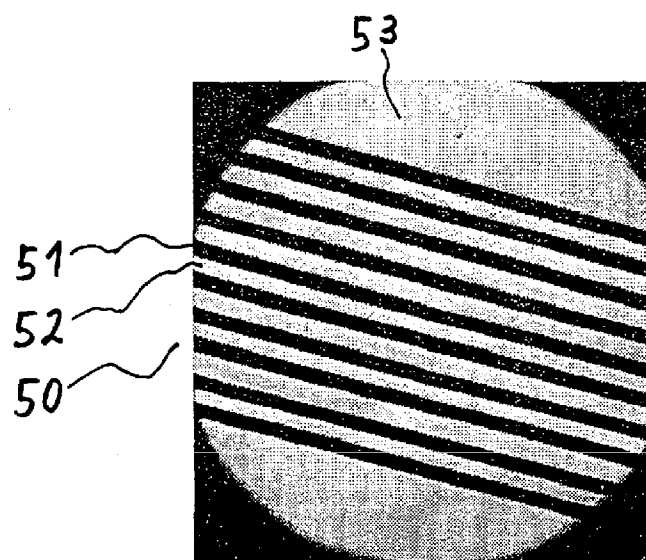


Fig. 5



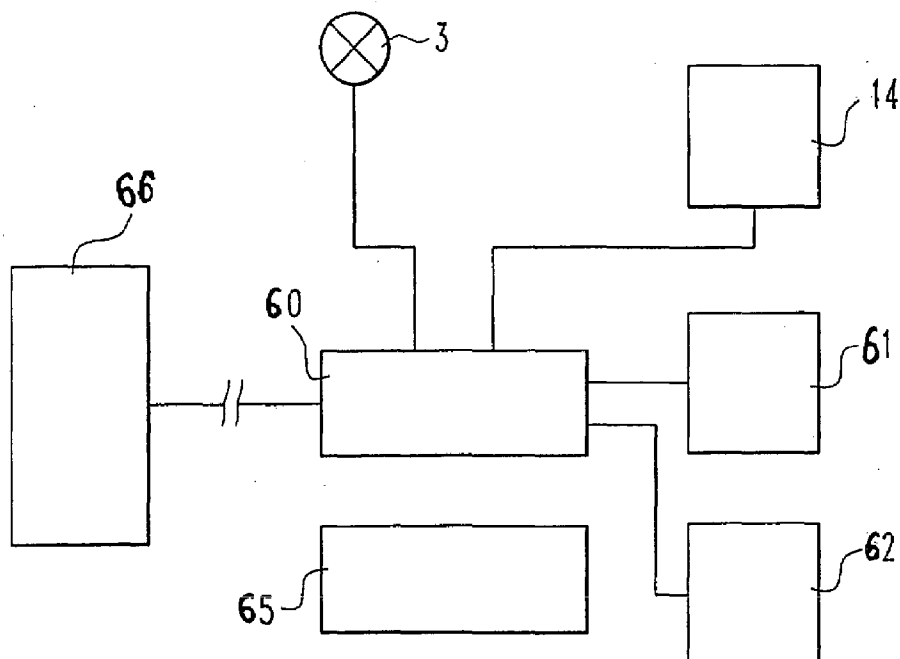


Fig. 6

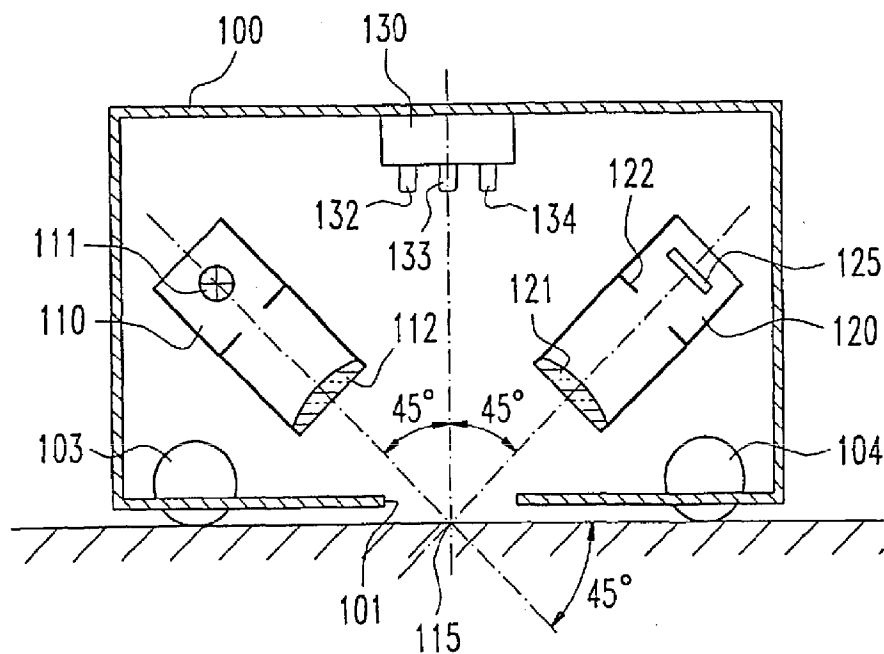
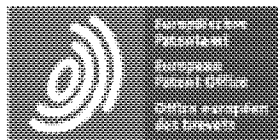


Fig. 7



## Notice

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## DESCRIPTION DE19846619A1

[0001]

<sup>13</sup> The present invention relates to a device and a method for determining the quality of structured surfaces.

[0002]

<sup>18</sup> The quality of a surface or a structured surface should be understood here to mean the physical properties of a surface that determine the appearance of a surface for the human observer.

[0003]

<sup>23</sup> In particular, properties such as structure, color, brightness of the color, gloss, sharpness of image

<sup>25</sup> DOI), glossy veils, surface textures and surface waviness (orange peel) etc. characterize structured surfaces.

[0004]

<sup>30</sup> The nature of the visible surfaces is an essential property of objects of everyday life, such as furnishings and everyday objects such as cars and the like.

[0005]

<sup>35</sup> In the field of automotive engineering, great importance is attached to the visual impression of the visible surfaces.

<sup>37</sup> The technical problems that arise in the design of surfaces in the motor vehicle sector are

explained in more detail below, without, however, restricting the application of the present invention in any way.

[0006]

<sup>43</sup> Motor vehicle bodies are usually provided with a high-gloss or metallic finish, the reflectivity or gloss characteristic of which the corresponding values of other surfaces, such as.

<sup>45</sup> B. furniture, is far superior.

<sup>46</sup> The high gloss of the paints used and the relatively large, flat surfaces require extremely careful preparation of the surfaces to be painted and particularly careful application of the paint.

[0007]

<sup>51</sup> In the prior art, various methods, devices and apparatus are known, such as.

<sup>52</sup> B. in DE 44 34 203 A1, with which the visual properties and especially the reflection behavior of surfaces can be determined.

[0008]

<sup>57</sup> Furthermore, a large number of structured surfaces are processed in the automotive industry in the interior of motor vehicles, which are made of natural materials, such as.

<sup>59</sup> B. consist of wood or plastic.

[0009]

<sup>63</sup> The visual impression of structured surfaces depends essentially on the distinctiveness of the structure of the surface, but is also determined by the other optical parameters mentioned above.

[0010]

<sup>69</sup> The automobile manufacturers now employ a large number of testers and use a large number of complex, large-scale measuring devices that check the quality of the surfaces visually or automatically in order to identify quality defects in surfaces during production.

<sup>72</sup> However, this method has a number of disadvantages.

[0011]

<sup>76</sup> The work of the visual inspectors is very strenuous and requires work rooms whose lighting conditions are always precisely defined.

<sup>78</sup> Nevertheless, large differences were found in the evaluation of the same surfaces by different testers, because on the one hand the respective physiological impression differs from testers to

testers, and on the other hand the eyesight of the individual testers also depends on their respective physical condition.

#### [0012]

<sup>85</sup> It is therefore very difficult to define a lower quality limit, below which it can no longer be accepted.

#### [0013]

<sup>90</sup> Furthermore, it is difficult even for experienced testers to determine what causes an inadequate visual impression.

<sup>92</sup> In the case of structured surfaces in particular, the visual impression is determined by a combination of different parameters, so that it is difficult to determine e.g.

<sup>94</sup> B. to change control values of an automatic painting device to improve the quality.

#### [0014]

<sup>98</sup> Structured plastic surfaces are often produced using an embossing process in which the shapes or shapes used are

<sup>100</sup> Embossing rollers etc. are subject to wear.

<sup>101</sup> With an increasing number of embossed surfaces, the structure of the embossing mold and thus also the structure of the plastic surface becomes smaller and smaller.

<sup>103</sup> By regularly exchanging the embossing tools, production can be optimized in terms of quality and quantity; However, the service life of the tools used differs in the event of fluctuations in the material composition or fluctuations in the ambient conditions, so that the tools are generally already exchanged, although their quality is still sufficient for the production of finished structured surfaces.

#### [0015]

<sup>111</sup> Are automatically working devices used to check the quality of structured surfaces, e.g.

<sup>112</sup> B. to determine the gloss or the sharpness of the image, quantitative numerical values can be obtained.

<sup>114</sup> However, a decisive disadvantage in the visual and also in the automatic tests is that although values are determined for the optical properties examined, the user has no information whatsoever about the cause.

#### [0016]

<sup>120</sup> Another disadvantage of conventional measuring apparatus is that they require a considerable amount of space and therefore cannot be taken along by the user.

[0017]

<sup>125</sup> It is therefore the object of the present invention to provide a method and a device of the type mentioned above, so that a reproducible assessment of the quality of surfaces and in particular structured surfaces can be carried out and that such physical quantities of the surface to be measured can be determined that a characteristic structure-related property of this surface is determined.

[0018]

<sup>133</sup> Another aspect of the object is to create a device which is compact and lightweight and is designed in such a way that it can be easily taken along by an operator and used to evaluate a surface without further aids.

[0019]

<sup>139</sup> Another aspect of the object is to create a device for measuring visual properties in which, despite the compact design according to the previous aspect of the object of the invention, the possibilities for measurement are considerably expanded compared to the devices known in the prior art.

[0020]

<sup>146</sup> Another aspect of the task is to specify a method which enables an advantageous detection of a characteristic structure-related property of surfaces.

[0021]

<sup>151</sup> According to the invention, this object is achieved by a device as defined in claim 1.

<sup>152</sup> The method according to the invention is the subject of claim 45.

[0022]

<sup>156</sup> Preferred developments of the invention are the subject matter of the subclaims.

[0023]

<sup>160</sup> In the device according to the invention, a first optical device, which has one or more light sources, directs light at a predetermined angle onto a measuring surface which is part of the surface to be measured.

[0024]

<sup>166</sup> The light reflected from the measuring surface is recorded with a second optical device which is oriented at a second, predetermined angle to this measuring surface and has one or more light-sensitive photosensors.

[0025]

<sup>172</sup> This photosensor or photosensors, which have at least one light-sensitive surface, emit an electrical measurement signal which is characteristic of the reflected light, the first and second optical devices being designed so that the reflected light passes through the structure this measuring surface is influenced.

[0026]

<sup>179</sup> An evaluation device is arranged in the device according to the invention, which is provided for controlling the measurement sequence and which has at least one processor device and at least one memory device and which evaluates the reflected light and derives at least one structural code from it, which characterizes at least one of the structural properties of this surface.

[0027]

<sup>187</sup> An output device which is arranged in the device according to the invention is used to output the measured values.

[0028]

<sup>192</sup> The evaluation device evaluates the measurement signals via a program stored in the memory device and, in a preferred development of the device according to the invention, stores the measurement signals and / or the measured values in this memory device.

[0029]

<sup>198</sup> In a further preferred development of the device according to the invention, a multiplicity of photo sensors are arranged in the second optical device, which photo sensors can be arranged in rows and / or columns.

[0030]

<sup>204</sup> In contrast to conventional devices, in the device according to the invention the photosensors can be arranged on a common substrate.

206 The individual photosensors can also be formed by interconnecting photosensitive surfaces on a common substrate, the surface of each photosensitive sensor being selected or selectable such that it corresponds to a predetermined angular range of the angle of reflection in the measuring device.

#### [0031]

213 The device according to the invention has considerable advantages: An essential prerequisite for controlling the production sequence in the production of structured and non-structured surfaces is the determination of the optical quality or the visual impression of the surface.

216 By analyzing the light reflected on the surface, a structural index is derived that describes the quality of the surface.

218 A key figure for the structure of structured surfaces is not determined by conventional visual or automatic measuring methods.

#### [0032]

223 This is particularly advantageous since the profile of the structure of a structured surface makes a significant contribution to the overall visual impression of this surface.

#### [0033]

228 When embossing, casting or other processes are used, the molds used are subject to wear, including depends on the number of structured surfaces produced.

230 In addition to the number of manufactured components with structured surfaces, environmental conditions such as air humidity, material and air temperature also play a decisive role in the quality of the surface to be manufactured.

#### [0034]

236 In conventional manufacturing processes, the molds used must be replaced after a certain number of embossed or cast components or after a certain period of time in order to ensure a consistently high quality of the components produced.

#### [0035]

242 This method leads to an unnecessarily high effort, since tools of sufficient quality are often exchanged, while on the other hand it also happens that the tools are worn out before the intended change time due to environmental influences or material inhomogeneities or differences.

[0036]

<sup>249</sup> One way of avoiding excessive rejects is to check them regularly.

<sup>250</sup> For this purpose, the production process must be stopped to check the tools, so that costly downtimes result.

[0037]

<sup>255</sup> With the device according to the invention it is possible to determine the quality of structured surfaces easily, quickly and reliably during ongoing production, so that impermissible wear of the tools can be detected early and unnecessary downtimes of the production plant can be avoided.

[0038]

<sup>262</sup> Another advantage of using the device according to the invention is that unnecessarily frequent changes of the tools, molds and stamping rollers and the like used can be omitted, which leads to lower production costs.

[0039]

<sup>268</sup> Another important advantage is that the device according to the invention is compact and can therefore be taken along by the user at any time.

[0040]

<sup>273</sup> Therefore, the device according to the invention is also suitable for the final inspection in poorly accessible places or in interiors of z.

<sup>275</sup> B. motor vehicles, where a variety of structured surfaces is used.

[0041]

<sup>279</sup> A further advantage of such a small device is that the quality can be determined simply and reliably even on concavely or convexly curved surfaces.

[0042]

<sup>284</sup> It is also advantageous that the device according to the invention can be used to measure a wide variety of points on a surface to be examined quickly and easily and thus the quality distribution over a surface or also the reproducibility can be tested.

[0043]



290 In a preferred development of the device according to the invention, at least a first part of this light emitted by this first optical device has a light pattern.

292 The first optical device may include a pattern device.

293 The light emitted by this at least one light source of this first optical device is at least partially radiated onto this pattern device in this development of the device according to the invention.

295 The light transmitted through this pattern device is influenced in amplitude and / or phase in accordance with the pattern of the pattern device.

#### [0044]

300 In a further preferred development of the device according to the invention, the light pattern emitted by this pattern device has at least one light / dark edge, or a multiplicity of light / dark edges, in a further development of the device according to the invention at least part of this multiplicity of Light / dark edges run parallel to one another at least in sections.

#### [0045]

307 The pattern of this pattern device can be arranged symmetrically to the optical axis.

308 In the event that the emitted light pattern has a plurality of light / dark edges, the individual light / dark edges can have a straight, circular or sinusoidal course.

#### [0046]

313 In a preferred development, at least a part of this plurality of light / dark edges is arranged in a grid, cross-grid or circular shape.

#### [0047]

318 The projection of a light pattern onto the surface to be examined is particularly advantageous since the evaluation of the light pattern reflected from the surface to be examined or the measuring surface allows conclusions to be drawn about the quality and the structure of the measuring surface to be examined.

#### [0048]

325 It is possible that the light pattern does not have any sharp transitions from light to dark, but rather influences the intensity of the projected light in a sinusoidal or sawtooth shape.

#### [0049]

330 In a further, preferred development of the device according to the invention, a first part of a light emitted by this first optical device has this light pattern, while a second part without a light

pattern is directed onto the measuring surface.

[0050]

<sup>336</sup> However, it is also possible that the pattern device can be switched on in the device according to the invention, so that an alternating measurement with a light pattern and without a light pattern is possible.

[0051]

<sup>342</sup> For this purpose, it is possible that this pattern device z.

<sup>343</sup> B. has an LCD device in which individual areas or individual pixels can be connected to different light patterns as required, so that the light pattern can be quickly and easily adapted to the given requirements or switched off.

[0052]

<sup>349</sup> By evaluating the signal from this at least one photosensor, which records at least part of the reflected light pattern, this evaluation device can determine a structural code number which is characteristic of at least one structure-related property of this surface.

[0053]

<sup>355</sup> In a further, preferred development of the device according to the invention, this evaluation device determines a slope of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by adding a difference between the electrical and / or digital measurement signal of the photosensor and this electrical and / or digital measurement signal of the next photosensor in this row and / or column is formed, or by forming differences in the electrical and / or digital measurement signal of this photosensor with this electrical and / or digital measurement signal of at least a part of the directly or diagonally adjacent photosensors.

[0054]

<sup>366</sup> This is particularly advantageous since the gradients of the measurement signals and in particular the gradients of the measurement signals when irradiated with a light pattern are suitable for deriving a structural code which is characteristic of at least one structure-related property of the measurement surface to be examined.

[0055]

<sup>373</sup> In a further preferred development of the device according to the invention, this evaluation

device determines at least an average of at least part of these slopes of the measurement signals of these photosensors and from this at least one average determines this characteristic structural number for this at least one structure-related property of this surface.

#### [0056]

380 From these slopes, however, a histogram can also be determined, in which the slope over the number of these slopes in the corresponding intervals or ranges for predetermined and / or selectable areas.

383 Areas is applied.

384 With the formation of the center of gravity and / or the mean value of the gradients, a characteristic structural number can be derived in this evaluation device, which characterizes the structure-related properties of this surface.

#### [0057]

390 This preferred development of the device according to the invention is particularly advantageous because a characteristic structural number for this measuring surface can be determined quickly and reliably by determining at least some of the slopes and by forming the average slope.

#### [0058]

397 Rounded surface structures, as they are e.g.

398 B. when using worn or worn tools in the production of structured surfaces, reflect light projected onto this surface more diffusely than surfaces with a sharper profile.

#### [0059]

403 By using such a development of the device according to the invention, it is possible to determine the structure-related properties.

#### [0060]

408 In a further preferred development of the device according to the invention, in addition to this characteristic structural number, at least one, two, three or more characteristic, optical parameters of this measuring surface are determined as optical parameters.

#### [0061]

414 In preferred developments of the invention, the gloss or haze or the distinction of image DOI or a representative measure of the typical wavelength and its amplitude (orange peel) is used

The topology of the layer thickness of this measuring surface is determined.

[0062]

<sup>420</sup> The orange peel is determined in a predetermined wavelength interval; however, the evaluation to determine the orange peel can also be carried out in two or more wavelength ranges.

<sup>422</sup> It is also possible to determine the color of the measuring surface or the brightness of the color.

[0063]

<sup>426</sup> In a further preferred development of this invention, in addition to the structural code, further of the above-mentioned characteristic optical parameters are determined.

[0064]

<sup>431</sup> Such a development of the device according to the invention is particularly advantageous because this small device u.

<sup>433</sup> U. all interesting, visual and physical properties of this measuring surface can be determined.

[0065]

<sup>437</sup> In an advantageous further development of all the exemplary embodiments described above, a third optical device is arranged which has at least one light source and emits light with a predetermined spectral characteristic which is directed at this measuring surface at a predetermined angle.

[0066]

<sup>444</sup> In the previously mentioned preferred developments of the device according to the invention, the predetermined angle at which the light emitted by the first optical device is aligned with this measuring surface and / or the predetermined angle at which the second optical device receives the light reflected from this measuring surface, and / or the predetermined angle at which the light emitted by the third optical device is aligned with this measuring surface can be freely set and includes in particular the angles 5 °, 10 °, 15 °, 20 °, 30 °, 45 °, 60 °, 75 °, 80 ° and 85 ° between the normal of the measuring surface and the corresponding angle.

<sup>451</sup> It should be noted, however, that angles other than those mentioned here are also possible.

[0067]

<sup>455</sup> In a further preferred development of the invention, this light emitted by this third optical device is directed onto the surface at an angle such that the light reflected directly from the measuring surface according to the Fresnel reflection has a different angle to this measuring surface than

the angle between this measuring surface and the light reflected by this measuring surface, which is emitted by this first optical device, so that the second optical device or the photosensors essentially receive the light emitted by the third optical device and light reflected diffusely from this measuring surface.

[0068]

<sup>465</sup> This at least one light source of these third optical devices can have one, two and particularly preferably three light-emitting elements, the light of which differs in the emitted spectral characteristic in such a way that the emitted wavelength ranges are different.

[0069]

<sup>471</sup> The emitted wavelength ranges of these light-emitting elements can at least partially overlap in the visible range of the light spectrum, and in a further preferred development of the device according to the invention, the emitted, spectral characteristics of these light-emitting elements are linearly independent of one another.

[0070]

<sup>478</sup> In a further preferred development of the device according to the invention, at least two, preferably three or more photosensors are designed in such a way that they differ in their spectral characteristics, so that the color of the reflected light can be detected as the optical parameter of this measuring surface.

[0071]

<sup>485</sup> In another preferred development of the device according to the invention, at least one photosensor has at least two, preferably three or more photosensitive elements, the electrical output signals of which can be detected individually and which differ in their spectral characteristics, so that the color of the light reflected from this measuring surface can be detected is.

[0072]

<sup>493</sup> A further development of the device according to the invention, which makes it possible to detect the color of the light reflected from this measuring surface, is particularly advantageous since the color and the brightness of the color are essential features of surfaces and also of structured surfaces.

[0073]

500 In further preferred developments of the device according to the invention, one or more light-emitting diodes or laser light sources are arranged in this first and / or this third optical device.

[0074]

505 At least part of the light emitted by this first optical device and / or this third optical device is essentially parallel, divergent or convergent.

[0075]

510 In a further, preferred development of the device according to the invention, this first and / or third optical device emits at least one light spot with a predetermined diameter perpendicular to the direction of propagation of the emitted light or at least one light strip with a predetermined length and width perpendicular to the direction of propagation of the emitted light.

[0076]

517 This light pattern projected by the first optical device can also consist of different light spots or light strips.

[0077]

522 It should be noted that this at least one optical parameter or this structural code can also be determined from the electrical and / or digital measurement signals of these photosensors, which record this second part reflected by this measurement surface without a light pattern of this light emitted by this first optical device.

[0078]

529 In a further preferred development of the device according to the invention, at least one light source of this first optical device and / or this at least one light source or this at least one light-emitting element of this third optical device and / or this at least one photo sensor or this at least one is in the immediate vicinity Photosensitive element arranged at least one temperature measuring device, which is provided for determining the characteristic temperature of the respective light source, light emitting elements, photosensors or light-sensitive elements so that a temperature-corrected determination of this at least one structural parameter and / or this at least one optical parameter can take place.

[0079]

540 The determination of the characteristic temperatures of the light sources and photosensors is particularly advantageous because these electrical devices work as a function of temperature

and the reproducibility of measurements is increased by a temperature-corrected determination of the parameters and parameters; because with increasing measurement duration or with changing measuring apparatus, the temperature and its distribution within the device according to the invention can change.

[0080]

549 In a further, preferred development of the invention, at least part of the course of the light pattern imaged on this plurality of photo sensors is determined.

551 A characteristic profile height parameter of this measuring surface can be determined by a deviation of the measured profile from the ideal profile.

[0081]

556 In a particularly preferred development, the course of at least one light / dark edge is determined in sections on this plurality of photo sensors or on the CCD chip.

[0082]

561 If a light or dark line is projected onto a surface to be examined, which has a rectangular profile with a profile height  $h$ , the reflection of this line is spatially separated on the CCD chip of the second optical device.

[0083]

567 The parts of the dark or light line that are reflected by the lower edges of the rectangular profile are reflected spatially offset parallel to those that are reflected at the upper edges of the rectangular profile.

[0084]

573 The reflected image of an ideal, rectangular structured surface are two dashed light or dark lines on the CCD chip.

[0085]

578 With the known angle of incidence and reflection and the distance between these two dashed lines on the CCD chip, the height of the profile can be determined at any location on the measuring surface.

[0086]

584 When measuring non-rectangular surface profiles, a profile height characteristic value of this measuring surface can be determined analogously with the corresponding geometric relationships.

#### [0087]

590 The determination of a profile height parameter is particularly advantageous because when the manufacturing tools wear out, the profile height of the manufactured components can also decrease.

#### [0088]

596 In a further preferred development of the invention, this structural code number is determined by means of a triangulation method, it being possible for this structural code number to be determined exclusively or additionally by means of this triangulation method.

599 It should be pointed out that one of the common triangulation methods, as they have become known in the prior art, can be used.

#### [0089]

604 As an example, reference should be made to the triangulation methods shown in the American patents US Pat. No. 5,546,189 and US Pat. No. 5,744,793, which hereby become part of the disclosure of the present invention.

#### [0090]

610 In a further preferred development of the device according to the invention, in contrast to the method produced in US Pat. No. 5,546,189, not only a gray value is determined which is representative of the height of the measuring surface at the measuring location, but the surface height at the measuring location is as large as possible Precision determined.

#### [0091]

617 In a further, preferred development of the invention, an additional optical device with at least one photo sensor is provided, which is oriented at a predetermined angle to this measuring surface.

620 In a particularly preferred development, this additional optical device is arranged in the plane spanned by the measuring point, the second optical device and the third optical device, and it is particularly preferred that the angle at which this additional optical device absorbs the reflected light is equal to the angle at which the second optical device receives the reflected light, so that the second and the additional optical device are arranged symmetrically to the light emitted by the first optical device or symmetrically to the light emitted by the third optical



device.

[0092]

<sup>630</sup> In a further, preferred development of the invention, the intensity and the location of the point of impact of the reflected intensity on this second optical device and this additional optical device is determined, and a characteristic location value for the location of the measurement and the measured intensity is particularly preferably provided by this evaluation unit stored in this memory device, and this evaluation unit determines a characteristic distance value for the distance to the measuring point on the surface to be examined and stores this in this memory device.

[0093]

<sup>640</sup> In a further preferred development of the invention, the first optical device has a scanning device which scans the surface to be examined in such a way that the light emitted by this first optical device scans the surface to be examined one after the other.

[0094]

<sup>646</sup> Particularly preferably, this evaluation device uses these distance parameters and these location parameters to determine at least one of these at least one structure code.

[0095]

<sup>651</sup> In a further, preferred development of the device according to the invention, at least one quantitative steepness parameter is determined for the steepness of the structure of the surface to be examined.

[0096]

<sup>657</sup> In a further, preferred development of the invention, this evaluation device determines at least one quantitative sharpness parameter for the sharpness of the edges of the surface to be examined.

[0097]

<sup>663</sup> These developments are particularly advantageous because with the wear of the tools, the structure of the surface of the components to be manufactured decreases and, in particular, the steepness of the structure decreases due to wear and tear and, in particular, the sharpness of the edges decreases, since the profile of the tool rounds off with increasing wear.

#### [0098]

<sup>670</sup> In a further, preferred development, an evaluation structure index is derived from this at least one structural index and at least one of the aforementioned characteristic optical quantities (gloss, haze, image sharpness, orange peel), which is characteristic of at least some of the visual properties of this surface.

#### [0099]

<sup>677</sup> When evaluating surfaces and, in particular, structured surfaces, the visual impression depends not only on the development of the structure, but also on the other characteristic, optical parameters mentioned.

<sup>680</sup> For example, in the case of particularly smooth and highly reflective surfaces, the demands on the manufacturing process are particularly high, since the human eye perceives even the smallest disturbances, unevenness and changes on such surfaces.

<sup>683</sup> In motor vehicles with so-called "metallic" paintwork, it is almost impossible to touch up paint damage over a large area in such a way that this damage remains invisible to the viewer.

#### [0100]

<sup>688</sup> In contrast to this, the requirements for the coating quality of the paint and the underlying surface are considerably lower for matt and weakly reflective surfaces.

#### [0101]

<sup>693</sup> While surface roughness and orange peel disturb the physiological perception of a viewer even to a small extent on high-gloss surfaces, far greater roughness and surface waviness on weakly reflective and matt surfaces have little or no effect on the viewer's visual impression.

<sup>696</sup> In the case of structured surfaces, the visual impression due to the structure is less dependent on the surface roughness, the gloss or the other optical parameters.

#### [0102]

<sup>701</sup> According to this preferred development of the invention, therefore, at least one evaluation structure index and at least one of the named characteristic variables are formed, which represents a measure of the quality of the measuring surface.

#### [0103]

<sup>707</sup> In a preferred development of the invention, this characteristic value for the gloss of the measuring surface and this structural characteristic number is used to form an evaluation structure characteristic number in such a way that a sufficient or satisfactory evaluation

structure characteristic number is formed for areas with low gloss even if the structure characteristic number is not ideal.

#### [0104]

715 In a further, preferred development it is provided that this evaluation structure code is formed such that, for.

717 B. a higher gloss compensates for a poorer structural index, so that this evaluation structural index achieves a qualitatively sufficient result.

#### [0105]

722 In a further, preferred development it is provided that an evaluation structure index is formed for at least two or all of the mentioned optical variables gloss, haze, image sharpness, orange peel or that from two or more of the mentioned characteristic variables (gloss, haze, image sharpness, orange peel ) and from this structural index an evaluation structure index is derived, which is characteristic of the visual impression of this measuring surface.

#### [0106]

730 The formation of at least one evaluation structure index from this at least one structural index and at least one of the characteristic optical variables, such as gloss, etc., is particularly advantageous since a particularly simple and automatic or automated determination of the quality of the surface to be examined is possible.

#### [0107]

737 In a further, preferred development of the device according to the invention, at least one lower or upper limit value can be determined for this at least one structural code number and / or this at least one characteristic optical parameter and / or this at least one evaluation structure code number.

741 If a message is exceeded on this output device or an alarm signal of an acoustic or optical type is output.

#### [0108]

746 In a further, preferred development of the invention, this at least one limit value is automatically determined by measuring at least one reference surface.

748 It is Z.

749 B. possible that one, two or more evaluated reference surfaces of different quality are measured, wherein advantageously at least one of these reference surfaces has a high quality and at least one other has an inadequate quality.

#### [0109]

<sup>755</sup> Such a partially automated calibration of this device enables different measuring points to be carried out quickly and reliably on a surface or on different surfaces to be examined.

#### [0110]

<sup>760</sup> In a further, preferred development, an entry or

<sup>761</sup> Arranged on-off switch.

#### [0111]

<sup>765</sup> Furthermore, an input device can be provided on this device according to the invention, this being inter alia. it can be provided that the user can enter a type identifier for the type of surface to be measured, with subsequent measurements being evaluated with reference to this type identifier by this evaluation device and stored in this memory device.

#### [0112]

<sup>772</sup> In a further, preferred development of the invention, the user will be prompted by an acoustic or optical signal when switching on to enter a type identifier for the surface type to be measured in this input device, whereby in a particularly preferred development the user can enter another type identifier at any time Can change the surface type to which the following measurements refer.

#### [0113]

<sup>780</sup> Entering or selecting a type identifier before measuring a specific surface type is particularly advantageous because the individual characteristic values have a different meaning when evaluating different surface types and a different weighting when forming an evaluation structure characteristic number.

#### [0114]

<sup>787</sup> In a further, preferred development of the invention, the user can evaluate this measurement area according to his physiological impression during each measurement by entering the user assessment number in this input device.

<sup>790</sup> This user assessment number is particularly preferably stored permanently in this memory device together with this at least one characteristic, optical parameter and this at least one structure code and this at least one assessment structure code, the range of values of this user assessment number at least two different quality values, the "good" and " bad "includes.

[0115]

<sup>797</sup> In a further, preferred development of the invention, at least some of these user assessment numbers stored in this memory device with associated parameters and parameters of the corresponding measurements are used to automatically determine at least one of this at least one limit value for this type code of the surface to be examined.

[0116]

<sup>804</sup> In a further, preferred development of the device according to the invention, this evaluation device takes into account essentially all of the user assessment numbers stored in this memory device for this type code of the surface to be measured and the corresponding parameters and parameters and structural parameters for determining this at least one limit value and for determining the weighting factors which are used to determine the evaluation structure index, so that these weighting factors and this at least one limit value are adapted with an increasing number of measurements in such a way that this evaluation structure index is adapted to the visual impression of the user.

[0117]

<sup>815</sup> This preferred development of the invention is particularly advantageous since this device can learn when determining these parameters and parameters and this evaluation structure code and, on the basis of an expert system, the reliability in determining this at least one evaluation structure code increases with the number of measurements.

[0118]

<sup>822</sup> In this preferred development of the invention, it is also provided that when determining these limit values and weighting factors, not all measured and stored values are used for this type identifier, since the physiological condition of the user affects the evaluation of this measurement area, and because of erroneous input incorrect user rating numbers, undesired limit values and weighting factors can be determined.

[0119]

<sup>830</sup> It is possible e.g.

<sup>831</sup> B. that a certain percentage of z.

<sup>832</sup> B. 5% or 10% remain unconsidered in the formation of these weighting factors and these limit values, the selection of the unconsidered values is made according to known statistical methods.

#### [0120]

<sup>838</sup> This development of the invention is particularly advantageous because even if incorrect user assessment numbers are entered by mistake, they are essentially ignored, so that the learning capability of this evaluation device increases the reliability of the automatically determined evaluation structure parameters with an increasing number of measurements.

#### [0121]

<sup>845</sup> In an advantageous further development of all of the exemplary embodiments described above, this device can be displaced relative to the measuring surface in a direction essentially equidistant from one another, and a distance measuring device is provided which quantitatively detects this relative displacement.

<sup>849</sup> The structural parameters and / or characteristic optical parameters and / or evaluation structure parameters measured along predetermined measurement thresholds on this surface are stored in this memory device.

#### [0122]

<sup>855</sup> In a further, preferred development, at least one measuring wheel is provided which is placed on the surface to be measured during the measurement and rotates during the relative movement between the device and the surface to be measured, whereby this at least one measuring wheel can be connected to a rotary encoder, which outputs an electrical angle of rotation signal which is representative of the angle of rotation covered by the measuring wheel.

#### [0123]

<sup>863</sup> This preferred development of the invention is particularly advantageous because surfaces to be examined z.

<sup>865</sup> B. can be measured systematically over the entire surface.

#### [0124]

<sup>869</sup> In the preferred development of the invention, in which a plurality of photo sensors are arranged, a CCD chip is particularly preferably used, which in a further preferred development of the invention can be designed as a color CCD chip.

#### [0125]

<sup>875</sup> When using a two-dimensional CCD chip, a two-dimensional image of this measuring surface is imaged on the CCD chip in the second optical device.

#### [0126]

880 In a preferred development of the invention, a dark image is recorded before the start of the measurements or before the start of each individual measurement, and the values are averaged.

883 During the measurement, the image is recorded with illumination, also averaged and the mean value of the dark image subtracted from this.

885 A value is determined for the contrast of this image.

886 The brightness of the measured image is normalized with the mean values of the dark image, the illuminated image and the value for the contrast.

#### [0127]

891 The further evaluation takes place with the normalized values or with the normalized image.

#### [0128]

895 In a further, preferred development of the invention, the evaluation takes place via one or more Fourier transformations.

897 The spatial frequency coefficients are particularly preferably determined for each line from a Fourier transformation of this line and the spatial frequency coefficients for the entire image are determined from the mean value of the frequencies of all lines.

#### [0129]

903 In a further, preferred development of the device according to the invention, this structural parameter is determined using a conventional method for measuring the roughness of surfaces, such as, for.

906 B. in DIN 4768 (edition 1990-05), ISO 13565-1 (edition 1996-12) or ISO 13565-2 (edition 1996-12), which thus become part of the disclosure.

#### [0130]

911 It should be pointed out that the interaction of the individual features according to the invention in any combination is preferred.

913 In particular, the combinations of features disclosed by the independent and dependent claims are also preferred, with one or more features being omitted.

915 It is evident to the person skilled in the art that, in addition to the exemplary embodiments shown here, a large number of further modifications and designs are conceivable which are covered by the invention.

918 In particular, the invention is not limited to the embodiments shown here.

[0131]

<sup>922</sup> Further advantages, features and possible applications of the present invention emerge from the following description of exemplary embodiments in conjunction with the drawings.

[0132]

<sup>927</sup> Show in it:

[0133]

<sup>931</sup> 1 shows a section of a first embodiment according to the present invention;

[0134]

<sup>935</sup> 2 shows a further exemplary embodiment of a sensor element in which the areas used for light measurement can be varied;

[0135]

<sup>940</sup> 3 shows a further exemplary embodiment of a sensor element in which seven photosensors are formed with the light-sensitive surfaces;

[0136]

<sup>945</sup> 4 shows a diagram which shows the reflection behavior when illuminated without a light pattern, the measured light intensity being plotted on the ordinate and the angular deviation with respect to the ideal reflection angle being plotted on the abscissa;

[0137]

<sup>951</sup> 5 shows an image of the light pattern projected onto the measuring surface,

[0138]

<sup>955</sup> 6 shows the basic circuit design of a measuring device as is to be used in the exemplary embodiments according to FIGS. 1-5; and

[0139]

<sup>960</sup> 7 shows a further exemplary embodiment of the invention, which is also suitable for further measurements.



#### [0140]

*965* A first embodiment of the invention will now be described with reference to FIG.

#### [0141]

*969* The measuring device shown schematically in the figure for determining the quality of structured surfaces has a housing in which a first exposure tube 2 is arranged.

#### [0142]

*974* In this exposure tube, indicated schematically, a light source 3, a diaphragm 4, a light pattern device 6 and a lens device 5 are arranged.

#### [0143]

*979* The light emitted by the point-shaped light source 3 is limited in its aperture by the diaphragm 4 and strikes the light pattern device 6.

*981* Depending on the embodiment, a partially transparent light pattern plate 9 can be arranged in the light pattern device 6, in which the phase and / or the amplitude of the incident light is influenced on part of the illuminated surface in such a way that the transmitted light has a characteristic light pattern.

*985* In this exemplary embodiment, the amplitude of the light emitted by the light source 3 is influenced by light / dark edges on the light pattern plate 9.

#### [0144]

*990* Furthermore, the emitted light is bundled by the lens 5 and impinges on the measuring surface 8 through the opening 7.

*992* The light is reflected from the measuring surface 8 and enters the measuring tube 10, which also has a lens 11, a diaphragm 12 and the actual sensor 13.

#### [0145]

*997* The reflection measuring device also has a control device (not shown) by which the operation of the device is controlled and a display device, also not shown, by which the measured structural parameters and characteristic optical parameters are displayed.

#### [0146]

*1003* In Fig. 2, the sensor of a further embodiment is shown.

1004 With the exception of the sensor, this exemplary embodiment is constructed in the same way as in FIG. 1.

1006 FIG. 2 shows the sensor device 20 in which a multiplicity of photosensitive elements 21 are arranged in rows and columns.

#### [0147]

1011 A CCD chip is used in this example.

1012 On the surface of the CCD chip, individual surface elements are interconnected to form different photosensors 22, 23, 24, so that only the signals from these photosensors 22, 23, 24 are present during the measurement.

#### [0148]

1018 It should be pointed out that the most varied of measurement geometries can be implemented by interconnecting individual surface elements to form photosensors.

1020 Furthermore, not all surface elements have to be used for the measurement.

#### [0149]

1024 In Fig. 3, a further embodiment of the sensor device is shown.

1025 The further construction and operation of this device takes place in the same way as in FIG. 1.

#### [0150]

1029 In FIG. 3, the sensor device 30 is shown, which, as in the exemplary embodiment according to FIG. 2, is designed as a CCD chip.

#### [0151]

1034 Different photo sensors 31, 32, 33 and 34 are formed on the surface of the CCD chip by interconnecting individual surface elements.

1036 By interconnecting individual surface elements to form photosensors, different measuring standards, such as.

1038 B. the American ASTM E 430 can be measured.

#### [0152]

1042 In a further exemplary embodiment, the basic structure is the same as that of the exemplary embodiment according to FIG. 1, but no light pattern is projected onto the measuring surface.

1044 A CCD chip is used as the sensor device, with which all surface elements are measured individually.

### [0153]

- 1049 In the diagram according to FIG. 4, the measured light intensity is plotted on the ordinate and the angular deviation with respect to the ideal reflection angle is plotted on the abscissa.
- 1051 The measured intensity is highest in the area of the ideal reflection angle and then decreases with increasing angular distance.
- 1053 From the curve recorded in this way, the reflection behavior on the surface can be assessed in a simple manner.

### [0154]

- 1058 It is also possible to derive the characteristic values from the curve, which characterize the reflection behavior of surfaces according to the various standards.

### [0155]

- 1063 In FIG. 5, the image reflected from a measuring surface onto the sensor device is shown.
- 1064 The light pattern 50 has dark edges 51 and light lines 52.
- 1065 A second part of the reflex 53 has no light pattern.

### [0156]

- 1069 With increasing roughness, the contrast in the light pattern decreases, while a poorer gloss affects the intensity.
- 1071 Orange peel leads to a distortion of the individual dark and light lines, and in the case of a rectangular profile, a structured surface leads to the displacement of the individual lines in the area of the depressions.
- 1074 In the case of sawtooth or triangular profiles, correlating inclinations of the lines can be determined at the corresponding sections.

### [0157]

- 1079 The intensity and contrast of the recorded image are evaluated and the gradients in the area of the light pattern are determined.
- 1081 A characteristic measure for the structure of the measuring surface is determined by averaging the gradients.

### [0158]

- 1086 In Fig. 7, the basic, cleavage-related structure of a measuring device, as is to be used in the embodiments according to FIGS. 1-6, is shown.

#### [0159]

*1091* The basic measurement setup is the same for all shown exemplary embodiments; only the programming differs depending on the type of sensor used.

#### [0160]

*1096* The measuring device overall has a control device 60 which contains a commercially available microprocessor which is controlled by a program which is stored in a memory 61.

*1098* The communication between the control device 60 and the user is provided by an input device 62 which contains a number of switches in order to start the operation of the control device and to be able to switch between individual operating modes (in the corresponding exemplary embodiments).

*1102* Furthermore, the user can enter a type identifier for the structured surface to be measured into the input device 61, to which subsequent measurements refer.

#### [0161]

*1107* In addition to the microprocessor, the control device has input / output devices which serve to connect the control device to the individual components of the device.

#### [0162]

*1112* The control device is connected to the light source 3 and the sensor 14.

*1113* The results of the measurement are displayed in a display 65, which is preferably designed as an LCD display.

*1115* A connection to an external computer 76 is provided for further evaluation of the measurement; the measurement results are preferably also stored in the memory device 61.

#### [0163]

*1120* The measuring device is powered by a battery (not shown).

#### [0164]

*1124* The measuring device as a whole is preferably arranged in a housing 1 which has approximately the dimensions of a pocket book.

#### [0165]

*1129* Another embodiment of the present invention will now be described with reference to FIG.

[0166]

*1133* In this exemplary embodiment, the device is arranged as a whole in a housing 100 which has an opening 101 with which the device is placed on the surface to be measured.

[0167]

*1138* In contrast to the previous exemplary embodiments, however, the device is not placed directly on the surface, but by means of (schematically indicated) at least two rubber rollers 103, 104 or at least four rubber wheels 103, 104, which are rotatably mounted (not shown) in the housing 100.

[0168]

*1145* At least one of the rubber wheels or rollers is provided with a distance measuring device (not shown) which detects the angular movements of the rubber wheels 103, 104 and outputs a representative electrical signal.

[0169]

*1151* The device also has a first optical device 110 in which a point light source 111 and a lens 112 are arranged.

*1153* This first optical device 110 is aligned such that the optical axis is aligned at a predetermined angle (45 ° in the example shown) to the surface 115 to be measured.

[0170]

*1158* A second optical device is arranged at a second predetermined angle (here also 45 °), which has a lens device 121, a diaphragm device 122 and a measuring sensor 123 oriented perpendicular to the optical axis, which in this embodiment is designed as a color CCD chip.

[0171]

*1164* Arranged on a third optical device 130 are three light-emitting elements 132, 133 and 134, which are embodied here as LEDs, each of which has a different spectral characteristic, i. that is, which light emit with different color characteristics.

[0172]

*1170* In the exemplary embodiment shown here, the light emitted by the LEDs falls essentially perpendicularly onto the surface to be examined.

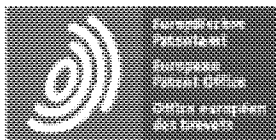
[0173]

<sup>1175</sup> The light emitted by the first and the third optical device is reflected on the surface 115 to be examined and is partly incident on the photosensor or the photo sensor.

<sup>1177</sup> Color CCD chip 125.

[0174]

<sup>1181</sup> Since the light-emitting elements 132-134 emit light with different color characteristics in the visible region of the spectrum, it is possible to determine a color characteristic for the surface to be examined.



## Notice

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## CLAIMS DE19846619A1

1.

<sup>13</sup> Apparatus for the quantified determination of the quality of structured surfaces with: a first optical device with at least one light source, the light of which is directed at a predetermined angle onto a measuring surface which is part of the surface to be measured, a second optical device which is directed at a predetermined angle is aligned with this measuring surface and which receives the light reflected from the measuring surface, this second optical device having at least one photosensor with at least one light-sensitive surface, this at least one photosensor outputting an electrical measurement signal which is characteristic of the reflected light; an evaluation device which is provided for controlling the measurement sequence and which has at least one processor device and at least one memory device; an output device; This first optical device and this second optical device are designed so that this reflected light is influenced by the structure of this measuring surface, and this evaluation device evaluates this reflected light and derives at least one structural code from it, which characterizes at least one structure-related property of this surface.

2.

<sup>29</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device evaluates these measurement signals via a program stored in this memory device and / or stores them in this memory device.

3.

<sup>35</sup> Device according to at least one of the preceding claims, characterized in that this second optical device contains a plurality of photo sensors which are arranged in rows and / or columns.

4.

<sup>41</sup> Device according to at least one of the preceding claims, characterized in that at least a first part of this light emitted by this first optical device has a light pattern.

5.

<sup>46</sup> Device according to at least one of the preceding claims, characterized in that this light pattern emitted by this first optical device has at least one light / dark edge or a plurality of light / dark edges.

6.

<sup>52</sup> Device according to at least one of the preceding claims, characterized in that at least some of this plurality of light / dark edges run parallel to one another at least in sections.

7.

<sup>57</sup> Device according to at least one of the preceding claims, characterized in that at least some of this multiplicity of light / dark edges are arranged in a grid or cross grid or circular manner.

8.

<sup>62</sup> Device according to at least one of the preceding claims, characterized in that this first part of this light emitted by this first optical device has this light pattern and that a second part without a light pattern is directed onto this measuring surface.

9.

<sup>68</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device determines a slope of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by comparing a difference between the electrical and / or digital measurement signal of this photosensor with this electrical and / or or digital measurement signal of the next photo sensor in this row and / or column is formed.

10.

<sup>76</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device determines the gradients of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by comparing the differences in the electrical and / or digital measurement signals of these photosensors with the electrical and / or or digital



measurement signals from all neighboring photo sensors are formed.

11.

<sup>84</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least an average of at least part of the slopes and from this determines this characteristic structural number for this at least one structure-related property of this surface.

12.

<sup>91</sup> Device according to at least one of the preceding claims, characterized in that one, two, three or more characteristic optical parameters of this measuring surface are determined.

13.

<sup>96</sup> Device according to at least one of the preceding claims, characterized in that this characteristic optical parameter is the gloss or the gloss haze or the image sharpness (DOI) of the measuring surface.

14.

<sup>102</sup> Device according to at least one of the preceding claims, characterized in that this characteristic optical parameter is a representative measure for the typical wavelength and its amplitude (orange peel) of the topology of the layer thickness of this measuring surface in a predetermined wavelength interval, this evaluation also in two or more Wavelength ranges can be done.

15.

<sup>110</sup> Device according to at least one of the preceding claims, characterized in that this device has a third optical device with at least one light source and emits light with a predetermined spectral characteristic which is directed at this measuring surface at a predetermined angle.

16.

<sup>116</sup> Device according to at least one of the preceding claims, characterized in that this predetermined angle at which the light emitted by the first optical device is aligned with this measuring surface, and / or that this predetermined angle at which the second optical device is directed from this measuring surface absorbs reflected light, and / or that this predetermined angle at which the light emitted by the third optical device is aligned with this measuring surface is / are adjustable and in particular the angles 5 °, 10 °, 15 °, 20 °, 30 °, 45 °, 60 °, 75

°, 80 ° and 85 °.

17.

<sup>126</sup> Device according to at least one of the preceding claims, characterized in that this light emitted by this third optical device is directed onto the surface at an angle such that the light reflected directly from the measuring surface according to the Fresnel reflection is different from this measuring surface Has angle, as the angle between this measuring surface and the directed reflected light from this measuring surface, which is emitted by this first optical device.

18.

<sup>134</sup> Device according to at least one of the preceding claims, characterized in that this at least one light source of this third optical device has at least two, preferably three light-emitting elements, the light of which differs in the emitted spectral characteristic in such a way that the emitted wavelength ranges are different.

19.

<sup>141</sup> Device according to at least one of the preceding claims, characterized in that these emitted wavelength ranges of these light-emitting elements at least partially overlap in the visible range of the light spectrum, and that these emitted spectral characteristics of these light-emitting elements are linearly independent of one another.

20.

<sup>148</sup> Device according to at least one of the preceding claims, characterized in that at least one photosensor has at least two, preferably three or more photosensitive elements, the electrical output signals of which can be recorded individually and which differ in their spectral characteristics, so that the optical characteristic of this measuring surface is the color of the reflected light can be detected.

21.

<sup>156</sup> Device according to at least one of the preceding claims, characterized in that this at least one light source of this first optical and / or this third optical device has one or more light-emitting diodes and / or laser light sources.

22.

<sup>162</sup> Device according to at least one of the preceding claims, characterized in that this first and / or this third optical device emits substantially parallel light.

23.

<sup>167</sup> Device according to at least one of the preceding claims, characterized in that this first and / or third optical device emits essentially divergent or convergent light.

24.

<sup>172</sup> Device according to at least one of the preceding claims, characterized in that this first and / or third optical device emits at least one light spot with a predetermined diameter perpendicular to the direction of propagation or at least one light strip with a predetermined length and width perpendicular to the direction of propagation.

25.

<sup>179</sup> Device according to at least one of the preceding claims, characterized in that at least one of these at least one optical parameters is determined from the electrical and / or digital measurement signals of these photosensors, which pick up this second part of this light emitted by this first optical device, reflected by the measurement surface.

26.

<sup>186</sup> Device according to at least one of the preceding claims, characterized in that at least one temperature measuring device is arranged in the immediate vicinity of this light source (s), this first optical device and / or this light source (s), this third optical device and / or this at least one photosensor which is provided for determining the characteristic temperature of the respective light source (s) and / or the respective photosensors so that a temperature-corrected determination of this at least one structural parameter and / or this at least one optical parameter can take place.

27.

<sup>196</sup> Device according to at least one of the preceding claims, characterized in that at least part of the course of the image of this at least one light / dark edge is determined on this plurality of photosensors and a characteristic profile height parameter is determined by a deviation from the ideal course of this measured course Measurement area is determined.

28.

<sup>203</sup> Device according to at least one of the preceding claims, characterized in that this structural code number is determined by means of a triangulation method.

29.

<sup>208</sup> Device according to at least one of the preceding claims, characterized in that at least one of said at least one sensor determines the intensity of the light emitted by this first optical device and reflected by the surface to be examined, and that this evaluation unit has a characteristic location value for the location of the The measurement is stored in this memory device and that this evaluation unit determines a characteristic distance value for the distance of the surface to be examined from this device from this measured intensity and stores it in this memory device.

30.

<sup>218</sup> Device according to at least one of the preceding claims, characterized in that this first optical device has a scanning device which scans this surface to be examined.

31.

<sup>223</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device uses these distance parameters and these location parameters to determine at least one of these at least one structure code.

32.

<sup>229</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least one quantitative steepness characteristic value for the steepness of the structure of the surface to be examined.

33.

<sup>235</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least one quantitative sharpness parameter for the sharpness of the edges of the structure of the surface to be examined.

34.

<sup>241</sup> Device according to at least one of the preceding claims, characterized in that at least one evaluation structural code is formed from this at least one structural code and at least one of the said characteristic optical variables (gloss, haze, image sharpness, orange peel), or that for each of the determined characteristic optical parameters (gloss, haze, image sharpness, orange peel) at least one evaluation structure index is determined.

35.

<sup>249</sup> Device according to at least one of the preceding claims, characterized in that in each case at least one lower and / or at least one upper limit value can be determined for this at least one structural code and / or this at least one characteristic optical parameter and / or this at least one evaluation structural code its lower resp.

<sup>253</sup> Exceeding a message on this output device or an alarm signal is issued.

36.

<sup>257</sup> Device according to at least one of the preceding claims, characterized in that at least one of these at least one limit value is automatically determined by measuring at least one reference surface.

37.

<sup>263</sup> Device according to at least one of the preceding claims, characterized in that an input device is arranged into which the user can enter a type identifier for the surface type to be measured, and that the subsequent measurements with reference to this type identifier are permanently stored in this memory device.

38.

<sup>270</sup> Device according to at least one of the preceding claims, characterized in that an on / off switch is additionally arranged, and that the user is prompted by an acoustic and / or optical signal when switching on to enter a type identifier for the type of surface to be measured in this input device to enter.

39.

<sup>277</sup> Device according to at least one of the preceding claims, characterized in that the user switches this evaluation device to a type change mode by entering a type change code or by actuating a type change switch, and that after entering a further type code the results of subsequent measurements with reference to this further type code in stored permanently in this storage device.

40.

<sup>285</sup> Device according to at least one of the preceding claims, characterized in that the user can evaluate this measuring surface after each measurement by entering a user assessment number into this input device, and that this user assessment number together with at least one of these parameters and values and this at least one structural evaluation number is permanently stored in this memory device, this user evaluation number being a subjective

measure of the user for the quality of this measurement area, and the range of values of this user evaluation number at least two different values, the "good" and "bad" correspond, includes.

41.

<sup>296</sup> Device according to at least one of the preceding claims, characterized in that this evaluation device uses at least part of these user assessment numbers and associated parameters and values stored in this memory device for the automatic determination of at least one of said at least one limit value for this type identifier.

42.

<sup>303</sup> Device according to at least one of the preceding claims, characterized in that this device is displaceable relative to the measuring surface in a substantially equidistant direction, and that a distance measuring device is provided which quantitatively detects this relative displacement, and that a storage device is also provided in which the structural parameters and / or optical parameters measured along predetermined measuring points on the surface are stored.

43.

<sup>312</sup> Device according to at least one of the preceding claims, characterized in that at least one measuring wheel is provided which is placed on the surface to be measured during the measurement and rotates during the relative movement between the device and the surface to be measured.

44.

<sup>319</sup> Device according to at least one of the preceding claims, characterized in that at least one of this at least one measuring wheel is connected to a rotation angle transmitter which outputs an electrical rotation angle signal which is representative of the rotation angle covered by the measuring wheel.

45.

<sup>326</sup> Method for the quantified determination of the quality of structured surfaces, in particular using a device according to at least one of the preceding claims, in which a first optical device with a first light source is provided to direct light at a predetermined angle onto a surface to be measured, and a second optical device Device is provided which has at least one photo sensor which is aligned at a second predetermined angle to this measuring surface in order to receive the light reflected from the measuring surface, this at least one photo sensor emitting

an electrical measurement signal which is characteristic of the recorded light, and an evaluation device is arranged which controls the measurement sequence and which contains at least one processor device and which stores these measurement signals in a memory device, and an output device outputs the measurement results, and this evaluation device The device evaluates this reflected light and derives at least one structural number from it, which characterizes the structure-related properties of the surface.

## PE2E SEARCH - Search History (Prior Art)

| Ref # | Hits   | Search Query                                                                                                                                                                                                                                                                                                                              | DBs                                                                                                                                                                                                                                                                                                                  | Default Operator | Plurals | British Equivalents | Time Stamp          |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|---------------------|---------------------|
| L1    | 2516   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:39 AM |
| L2    | 440018 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:49 AM |
| L3    | 757    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 02:01 AM |
| L4    | 183885 | (detect\$3 OR sen\$3)                                                                                                                                                                                                                                                                                                                     | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                    | OR               | ON      | ON                  | 2023/09/09          |



|    |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|    |        | near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                           | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 02:34 AM               |
| L5 | 249451 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L6 | 2430   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L7 | 11     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:38 AM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L8  | 1872340 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 09:57 AM |
| L9  | 1341757 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:00 AM |
| L10 | 113287  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:01 AM |
| L11 | 67      | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/09/09 10:02 AM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
|-----|----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L12 | 0  | L11 AND I1 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L13 | 35 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L14 | 11 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L15 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 06:13 PM |
| L16 | 2541   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L17 | 454819 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L18 | 791    | (G01S17/46 OR                                                                                                                                                                                                                                                                                                                             | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/11/18          |

|     |        |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:50 PM               |
| L19 | 188776 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L20 | 259070 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L21 | 2529   | L4 AND I5                                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L22 | 12      | L6 AND I3                                                                                                         | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L23 | 1891255 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L24 | 1363696 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L25 | 114665  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
|-----|----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L26 | 70 | L10 AND I6 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L27 | 27 | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L28 | 36 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L29 | 12     | L11 AND I3                                                                                                                                                                                                                                                                                                                                | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/18 08:50 PM |
| L30 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L31 | 2596   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19 08:07 PM |
| L32 | 468828 | (348/135).ccls. OR                                                                                                                                                                                                                                                                                                                        | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/09/19          |



|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73 OR<br>G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56 OR<br>G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                 | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:07 PM               |
| L33 | 903    | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L34 | 208447 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L35 | 296319 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L36 | 2914    | L4 AND I5                                                                                                  | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L37 | 18      | L6 AND I3                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L38 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L39 | 1447259 | filter near6 (image OR<br>median OR contrast OR                                                            | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
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| L40 | 119738 | energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L41 | 92     | L8 AND I9                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L42 | 39     | L10 AND I6                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
|     |        | L11 AND I1                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |      |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L43 | 48   | L11 AND I2                                               | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L44 | 18   | L11 AND I3                                               | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19 08:07 PM |
| L45 | 4    | "20180033146".pn.                                        | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19 08:07 PM |
| L46 | 2596 | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/09/19 08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
|     |        | ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           |    |    |    |                     |
| L47 | 468828 | (348/135).cls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                       | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L48 | 903    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L49 | 208447 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                              | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/09/19 08:07 PM |

|     |         |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L50 | 296319  | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L51 | 2914    | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L52 | 18      | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L53 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3                                               | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |         | OR distance) near60<br>beam near5 (light OR<br>profile)                                                           | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L54 | 1447259 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L55 | 119738  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L56 | 92      | L10 AND I6                                                                                                        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|----|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L57 | 39 | L11 AND I1        | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L58 | 48 | L11 AND I2        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L59 | 18 | L11 AND I3        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L60 | 4  | "20180033146".pn. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |



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|  |  |  | CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) |  |  |  |  |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|

**PE2E SEARCH - Search History (Interference)**

There are no Interference searches to show.



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## BIB DATA SHEET

CONFIRMATION NO. 2279

| SERIAL NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FILING or<br>371(c) DATE                                                     | CLASS                                  | GROUP ART UNIT                                                                   | ATTORNEY<br>DOCKET NO.        |                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|-------------------------------|------------------------------------|
| 18/743,352                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 06/14/2024                                                                   | 348                                    | 2454                                                                             | 27843-2603<br>182037US03      |                                    |
| <b>APPLICANTS</b><br>trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                              |                                        |                                                                                  |                               |                                    |
| <b>INVENTORS</b><br>Friedrich SCHICK, Ludwigshafen, GERMANY;<br>Peter SCHILLEN, Ludwigshafen, GERMANY;<br>Patrick SCHINDLER, Ludwigshafen, GERMANY;<br>Andre SCHMIDT, Ludwigshafen, GERMANY;<br>Michael EBERSPACH, Ludwigshafen, GERMANY;<br>Christian LENNARTZ, Ludwigshafen, GERMANY;<br>Robert SEND, Karlsruhe, GERMANY;<br>Lars DIESSELBERG, Karlsruhe, GERMANY;<br>Heiko HENGEN, Steinweiler, GERMANY;<br>Ingmar BRUDER, Ludwigshafen, GERMANY;<br>Jakob UNGER, Ludwigshafen, GERMANY;<br>Christian BONSIGNORE, Ludwigshafen, GERMANY; |                                                                              |                                        |                                                                                  |                               |                                    |
| <b>** CONTINUING DATA *****</b><br>This application is a CON of 18/425,144 01/29/2024<br>which is a CON of 17/439,492 09/15/2021 PAT 11947013<br>which is a 371 of PCT/EP20/56759 03/13/2020                                                                                                                                                                                                                                                                                                                                                |                                                                              |                                        |                                                                                  |                               |                                    |
| <b>** FOREIGN APPLICATIONS *****</b><br>EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                              |                                        |                                                                                  |                               |                                    |
| <b>** IF REQUIRED, FOREIGN FILING LICENSE GRANTED **</b><br>07/15/2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                              |                                        |                                                                                  |                               |                                    |
| Foreign Priority claimed<br><input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>35 USC 119(a-d) conditions met <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>Verified and<br>Acknowledged                                                                                                                                                                                                                                                                                                       | <input checked="" type="checkbox"/> Met after<br>Allowance<br>FJ<br>Initials | <b>STATE OR<br/>COUNTRY</b><br>GERMANY | <b>SHEETS<br/>DRAWINGS</b><br>2                                                  | <b>TOTAL<br/>CLAIMS</b><br>30 | <b>INDEPENDENT<br/>CLAIMS</b><br>1 |
| <b>ADDRESS</b><br>Patent Docket Department<br>Armstrong Teasdale LLP<br>7700 Forsyth Boulevard<br>Suite 1800<br>St. Louis, MO 63105<br>UNITED STATES                                                                                                                                                                                                                                                                                                                                                                                        |                                                                              |                                        |                                                                                  |                               |                                    |
| <b>TITLE</b><br>DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                              |                                        |                                                                                  |                               |                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                              |                                        | <input type="checkbox"/> All Fees<br><input type="checkbox"/> 1.16 Fees (Filing) |                               |                                    |

|                                                              |                                                                                                                   |                                                                                                                                                                                                                                                                             |                                                              |                                            |                                      |                                 |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------|--------------------------------------|---------------------------------|
| <b>FILING FEE<br/>RECEIVED</b><br>0.00                       | FEES: Authority has been given in Paper<br>No. _____ to charge/credit DEPOSIT ACCOUNT<br>No. _____ for following: | <table border="1"><tr><td><input type="checkbox"/> 1.17 Fees (Processing Ext. of time)</td></tr><tr><td><input type="checkbox"/> 1.18 Fees (Issue)</td></tr><tr><td><input type="checkbox"/> Other _____</td></tr><tr><td><input type="checkbox"/> Credit</td></tr></table> | <input type="checkbox"/> 1.17 Fees (Processing Ext. of time) | <input type="checkbox"/> 1.18 Fees (Issue) | <input type="checkbox"/> Other _____ | <input type="checkbox"/> Credit |
| <input type="checkbox"/> 1.17 Fees (Processing Ext. of time) |                                                                                                                   |                                                                                                                                                                                                                                                                             |                                                              |                                            |                                      |                                 |
| <input type="checkbox"/> 1.18 Fees (Issue)                   |                                                                                                                   |                                                                                                                                                                                                                                                                             |                                                              |                                            |                                      |                                 |
| <input type="checkbox"/> Other _____                         |                                                                                                                   |                                                                                                                                                                                                                                                                             |                                                              |                                            |                                      |                                 |
| <input type="checkbox"/> Credit                              |                                                                                                                   |                                                                                                                                                                                                                                                                             |                                                              |                                            |                                      |                                 |



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| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |

76656 7590 09/24/2024  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

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| EXAMINER |
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JEAN, FRANTZ B

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| ART UNIT | PAPER NUMBER |
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2454

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|-------------------|---------------|
| NOTIFICATION DATE | DELIVERY MODE |
|-------------------|---------------|

09/24/2024

ELECTRONIC

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

USpatents@atllp.com



## **DETAILED ACTION**

### ***Notice of Pre-AIA or AIA Status***

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This is a first office action in response to an application for letters patent filed on 14 June 2024. Claims 1-30 are presented for examination.

### ***Information Disclosure Statement***

The information disclosure statement (IDS) submitted on 06/14/2024 was filed before the mailing date of the first office action on the merit. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

### ***Specification***

The disclosure is objected to because of the following informalities: The cross-reference to related applications must be updated to reflect the current status of the related applications.

Appropriate correction is required.

### ***Double Patenting***

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious

over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit [www.uspto.gov/patent/patents-forms](http://www.uspto.gov/patent/patents-forms). The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission.

For more information about eTerminal Disclaimers, refer to [www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer](http://www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer).

Claims 1 and 7 of the instant application are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 14-15, and 24 of U.S. Patent No. 11947013. Although the claims at issue are not identical, they are not patentably distinct from each other because the claims are arguably broader than claims 1, 14-15, and 24 of patent Number "013" which encompass the same metes, bounds, and limitations. Therefore, it would be obvious to eliminate the limitations of the narrower claims, since it has been held omission of an element and its function and a combination where the remaining elements perform the same functions as before involves only routine skill in the art. See *in re Karlson*, 136 USPQ184.

Patent Number: 11947013

Exemplary claim 1

1. **A detector for identifying at least one material property  $m$  comprising at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object ; and at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature  $\Phi_{1z}$  by applying at least one distance dependent image filter  $D_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; and a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|p_{o_{tot}h_{er},o_z}| > 0.40$  with ... being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device at least one material feature ( $2_m$  by applying at least one material**



dependent image filter  $O_z$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: **a luminance filter; a spot shape filter;** a squared norm gradient; a standard deviation; a smoothness filter; a Gaussian filter or median filter; **a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof;** and a further material dependent image filter  $O_{z_{other}}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|p_{O_z, O_{z_{other}}}| \geq 0.40$  with ( $D_m$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, **and wherein the evaluation device is configured for determining** a longitudinal coordinate  $z$  and **the material property  $m$  by evaluating the distance feature ( $p_z$  and the material feature ( $2_m$ .**

2. (Currently Amended) The detector according claim 1, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance.

3. (Currently Amended) The detector according to claim 1, wherein  $p < 0.075$ .

4. (Currently Amended) The detector according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, and a speckle.

5. (Currently Amended) The detector according to claim 1, wherein the further distance dependent image filter  $M_{O_z}$  correlates to one or more of the distance dependent image filters  $C_z$  by  $|p_{O_z, C_z}| \geq 0.60$ .

6. (Currently Amended) The detector according to claim 1, wherein the further material dependent image filter  $O_{2,other}$  correlates to one or more of the material dependent image filters  $D_m$  by  $|\rho_{O_{2,other}, \Phi_m}| \geq 0.60$ .

7. (Currently Amended) The detector according to claim 1, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by using a predetermined relationship between  $(p_i, (p_{2m}$  and  $z, m$ .

8. (Currently Amended) The detector according to claim 1, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by a function  $z(p_i, (2_m)$  and/or  $m(p(2m)$ .  $\Phi_{1z}$ ,

9. (Currently Amended) The detector according to claim 1, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

10. (Currently Amended) The detector according to claim 1, wherein the evaluation device is configured for determining whether at least one of  $D_1$  or  $O_2$  is a function of the feature  $(z$  or  $p_{im}$   $2m$  of the other image filter or whether at least one of  $(D_1$  or  $O_2$  is a function of the at least one other image filter, wherein the evaluation device  $[(132)]$  is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

11. (Currently Amended) The detector  $[(110)]$  according to claim 10, wherein the evaluation device  $[(132)]$  is configured for determining at least one of  $z$  and/or  $m$  by applying at least one further filter depending on at least one of  $ep_{iz}$  and  $(_{2m}$  to the reflection image.

12. (Currently Amended) The detector  $[(110)]$  according to claim 1, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal  $Q$  from at least two sensor signals of the sensor element  $[(116)]$ , wherein the evaluation device  $[(132)]$  is configured for deriving the combined signal  $Q$  by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device  $[(132)]$  is configured for using at least one predetermined relationship between the combined signal  $Q$  and the distance feature  $(p_{iz}$  for determining the distance feature  $(iz$ .

13. (Currently Amended) The detector  $[[[110)]]$  according to claim 1, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature  $\Phi_{12}$  by optimizing at least one blurring function  $f_a$ , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

14. **The detector according to claim 1, wherein the sensor element comprises at least one CMOS sensor.**

15. The detector according to claim 1, **wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines.**

16. (Currently Amended) The detector (110) according to claim 15, wherein the illumination source  $[[[124)]]$  comprises at least one laser source and at least one diffractive optical element.

17. (Currently Amended) A detector system  $[[[138)]]$ , the detector system  $[[[138)]]$  comprising at least one detector  $[[[110)]]$  according to claim 1, the detector system  $[[[138)]]$  further comprising at least one beacon device  $[[[140)]]$  adapted to direct at least one light beam towards the detector  $[[[110)]]$ , wherein the beacon device  $[[[140)]]$  is at least one of attachable to the object  $[[[112)]]$ , holdable by the object  $[[[112)]]$  and integratable into the object  $[[[112)]]$ .

18. (Currently Amended) A human-machine interface  $[[[142)]]$  for exchanging at least one item of information between a user  $[[[160)]]$  and a machine  $[[[168)]]$ , wherein the human-machine interface  $[[[142)]]$  comprises at least one detector system  $[[[138)]]$  according to claim 17, wherein the at least one beacon device  $[[[140)]]$  is adapted to be at least one of directly or

indirectly attached to the user [(160)] and held by the user [(160)], wherein the human-machine interface [(142)] is designed to determine at least one position of the user [(160)] by means of the detector system [(138)], wherein the human-machine interface [(142)] is designed to assign to the position at least one item of information.

19. (Currently Amended) An entertainment device [(144)] for carrying out at least one entertainment function, wherein the entertainment device [(144)] comprises at least one human-machine interface [(142)] according to claim 18, wherein the entertainment device [(144)] is designed to enable at least one item of information to be input by a player by means of the human-machine interface [(142)], wherein the entertainment device [(144)] is designed to vary the entertainment function in accordance with the information.

20. (Currently Amended) A tracking system [(146)] for tracking a position of at least one movable object, the tracking system [(146)] comprising at least one detector system [(138)] according to claim 17 referring to a detector system, the tracking system [(146)] further comprising at least one track controller [(170)], wherein the track controller [(170)] is adapted to track a series of positions of the object [(112)] at specific points in time.

21. (Currently Amended) A scanning system [(148)] for determining a depth profile of a scenery, the scanning system [(148)] comprising at least one detector [(110)] according to claim 1, the scanning system [(148)] further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

22. (Currently Amended) A camera [(136)] for imaging at least one object [(112)], the camera [(136)] comprising at least one detector [(110)] according to claim 1.

23. (Currently Amended) An inertial measurement unit [(172)] for use in an electronic device, wherein the inertial measurement unit [(172)] is adapted to receive data determined by at least one detector [(110)] according to claim 1, wherein the inertial measurement unit [(172)] further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheelspeed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; and a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space,

relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, and speed.

24. (Currently Amended) A method for determining at least one material property of at least one object  $[(112)]$  by using at least one detector  $[(110)]$  according to claim 1, the method comprising the following steps: a) determining at least one reflection image of the object  $[(112)]$  by using at least one sensor element  $[(116)]$  having a matrix  $[(118)]$  of optical sensors  $[(120)]$ , the optical sensors  $[(120)]$  each having a light-sensitive area  $[(122)]$ ; and b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device  $[(132)]$ , wherein the evaluating comprises: b1) determining at least one distance feature  $p_{1z}$  by applying at least one distance dependent image filter  $(D_1)$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof and a further distance dependent image filter  $\Phi_{1other}$  which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho\Phi_{1other}, \Phi_z| \geq 0.40$  with  $C_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, b2) determining at least one material feature  $(p_{2m})$  by applying at least one material dependent image filter  $C_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; and a further material dependent image filter  $C_{2other}$  which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|\rho\Phi_{2other}, \Phi_m| \geq 0.40$  with  $\Phi_{mm}$  being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, and b3) determining a longitudinal coordinate  $z$  and the material property  $m$  by evaluating the distance feature  $eplz$  and the material feature  $(p_{2m})$ .

25. (Currently Amended) A method of using the detector [((110))] according to claim 1, the method comprising using the detector [((110))] for a purpose of use selected from the group consisting of: a position measurement in traffic technology; an entertainment application, a security application; a surveillance application, a safety application, a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; and a manufacturing application.

Claims 1, 7, 10, 15, 21, 23, and 25 of the instant application are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 11, 13, 15, 17-20 of copending Application No. 18/425,144 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other because the claims are arguably broader than claims 1, 14-15, and 24 of patent Number "013" which encompass the same metes, bounds, and limitations. Therefore, it would be obvious to eliminate the limitations of the narrower claims, since it has been held omission of an element and its function and a combination where the remaining elements perform the same functions as before involves only routine skill in the art. See *in re Karlson*, 136 USPQ184.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Application Number: 18/425,144

Exemplary claim 1

1. (Original) **A detector for identifying at least one material property m, comprising at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection**

**image of a light beam originating from at least one object;** and at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, **wherein the evaluation device is configured for determining at least one material feature  $2m$  by applying at least one material dependent image filter 02 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof;** and a further material dependent image filter 02other which correlates to one or more of the spot shape filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|pD2other, Dm| > 0.40$  with ( $Dm$  being one of the spot shape filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, and a linear combination thereof, and **wherein the evaluation device is configured for determining the material property  $m$  by evaluating the material feature  $\varphi 2m$ .**

2. (Original) The detector according claim 1, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value,  $p$ , is smaller or equal than a pre-defined level of significance.
3. (Original) The detector according to claim 2, wherein  $p < 0.075$ .

4. (Original) The detector according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, and a speckle.
5. (Original) The detector according to claim 1, wherein the further material dependent image filter  $\Phi_{2other}$  correlates to one or more of the material dependent image filters  $\Phi_m$  by  $|\rho\Phi_{2other}, \Phi_m| \geq 0.60$ .
6. (Original) The detector according to claim 1, wherein the evaluation device is configured for determining whether  $F_2$  is a function of the feature ( $P_{2m}$  of the other image filter or whether  $F_2$  is a function of the at least one other image filter.
7. (Original) The detector according to claim 6, wherein the evaluation device is configured for determining  $m$  by applying at least one further filter depending on at least one of  $\rho_{2m}$  to the reflection image.
8. (Original) The detector according to claim 1, wherein the evaluation device is configured for determining at least one distance feature  $\rho_{1z}$  by applying at least one distance dependent image filter  $C_1$  to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; and a further distance dependent image filter  $G_1$  other which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by  $|\rho\Phi_{1other}, \Phi_z| \geq 0.40$  with  $\Phi_z$  being one of the depth-from-photon-ratio filter or the depth-from-defocus filter and a linear combination thereof and wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  by evaluating the distance feature  $\rho_{1z}$ .
9. (Original) The detector according to claim 8, wherein the material property  $m$  and/or the longitudinal coordinate  $z$  are determined by a function  $z(\rho_{1z}, \Phi_{2m})$  and/or  $m(q_{1z}, \rho_{2m})$ .
10. (Original) The detector according to claim 8, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal  $Q$  from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal  $Q$  by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at



least one predetermined relationship between the combined signal Q and the distance feature cp1 z for determining the distance feature p1z.

11. (Original) The detector according to claim 1, wherein the sensor element comprises at least one CMOS sensor.

12. (Original) The detector according to claim 1, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines.

13. (Original) The detector according to claim 12, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

14. (Original) The detector according to claim 12, wherein the illumination source is configured for generating at least one regular point pattern for illumination of the object.

15. (Original) The detector according to claim 12, wherein the illumination source comprises at least one vertical cavity surface-emitting laser.

16. (Original) The detector according to claim 12, wherein the illumination source emits light having a wavelength of 780 nm to 3.0  $\mu$ m.

17. (Original) The detector according to claim 1, wherein the detector is configured for identification and/or classification of the material.

18. (Original) The detector according to claim 1, wherein the detector is configured for detection of human skin.

19. (Original) The detector according to claim 1, wherein the evaluation device is configured for performing at least one image analysis and/or image processing in order to identify a reflection feature in the reflection image.

20. (Original) A method for determining at least one material property of at least one object by using at least one detector, the method comprising the following steps: a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area; and b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises: b1) determining at least one material feature  $p_{2m}$  by applying at least one material dependent image filter  $F_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; and a further material dependent image filter  $F_{2other}$  which correlates to one or more of the spot shape filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by  $|p_{F_{2other}}, \Phi_m| \geq 0.40$  with  $G_m$  being one of the spot shape filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, and a linear combination thereof, and b3) determining the material property  $m$  by evaluating the material feature  $p_{2m}$ .

Claims 1-30 will be allowed upon the submission of a Terminal Disclaimer to obviate the double patenting rejection.

The prior art of record does not teach and suggest as a whole the combination of a detector for identifying at least one material property  $m$  comprising at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group

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consisting of: at least one point pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines; at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object; and at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one material feature  $cp_{2m}$  by applying at least one material dependent image filter  $CD_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; and a linear combination thereof; wherein the evaluation device is configured for determining the material property  $m$  by evaluating the material feature  $cp_{2m}$ .

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANTZ B JEAN whose telephone number is (571)272-3937. The examiner can normally be reached 8-5 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Glenton B Burgess can be reached on 5712723949. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/FRANTZ B JEAN/  
Primary Examiner, Art Unit 2454

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| PATENT APPLICATION FEE DETERMINATION RECORD                                                                                     |                                                                                          |                                                                                                                                                                                                                               |                            | Application or Docket Number<br>18/743,352 |               | Filing Date<br>06/14/2024 |  | <input type="checkbox"/> To be Mailed |  |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------|---------------|---------------------------|--|---------------------------------------|--|
| Substitute for Form PTO-875                                                                                                     |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| ENTITY: <input checked="" type="checkbox"/> LARGE <input type="checkbox"/> SMALL <input type="checkbox"/> MICRO                 |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| APPLICATION AS FILED - PART I                                                                                                   |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| FOR                                                                                                                             |                                                                                          | (Column 1)<br>NUMBER FILED                                                                                                                                                                                                    | (Column 2)<br>NUMBER EXTRA | RATE (\$)                                  |               | FEE (\$)                  |  |                                       |  |
| <input type="checkbox"/> BASIC FEE<br>(37 CFR 1.16(a), (b), or (c))                                                             |                                                                                          | N/A                                                                                                                                                                                                                           | N/A                        | N/A                                        |               |                           |  |                                       |  |
| <input type="checkbox"/> SEARCH FEE<br>(37 CFR 1.16(k), (i), or (m))                                                            |                                                                                          | N/A                                                                                                                                                                                                                           | N/A                        | N/A                                        |               |                           |  |                                       |  |
| <input type="checkbox"/> EXAMINATION FEE<br>(37 CFR 1.16(o), (p), or (q))                                                       |                                                                                          | N/A                                                                                                                                                                                                                           | N/A                        | N/A                                        |               |                           |  |                                       |  |
| TOTAL CLAIMS<br>(37 CFR 1.16(i))                                                                                                |                                                                                          | minus 20 =                                                                                                                                                                                                                    | *                          | x \$100 =                                  |               |                           |  |                                       |  |
| INDEPENDENT CLAIMS<br>(37 CFR 1.16(h))                                                                                          |                                                                                          | minus 3 =                                                                                                                                                                                                                     | *                          | x \$480 =                                  |               |                           |  |                                       |  |
| <input type="checkbox"/> APPLICATION SIZE FEE (37 CFR 1.16(s))                                                                  |                                                                                          | If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s). |                            |                                            |               |                           |  |                                       |  |
| <input type="checkbox"/> MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))                                                      |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| * If the difference in column 1 is less than zero, enter "0" in column 2.                                                       |                                                                                          |                                                                                                                                                                                                                               |                            | TOTAL                                      |               |                           |  |                                       |  |
| APPLICATION AS AMENDED - PART II                                                                                                |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| AMENDMENT                                                                                                                       | 10/16/2024                                                                               | CLAIMS REMAINING AFTER AMENDMENT                                                                                                                                                                                              |                            | HIGHEST NUMBER PREVIOUSLY PAID FOR         | PRESENT EXTRA | RATE (\$)                 |  | ADDITIONAL FEE (\$)                   |  |
|                                                                                                                                 | Total<br>(37 CFR 1.16(i))                                                                | * 30                                                                                                                                                                                                                          | Minus                      | ** 20                                      | = 10          | x \$100 =                 |  | 1000                                  |  |
|                                                                                                                                 | Independent<br>(37 CFR 1.16(h))                                                          | * 1                                                                                                                                                                                                                           | Minus                      | *** 3                                      | = 0           | x \$480 =                 |  | 0                                     |  |
|                                                                                                                                 | <input type="checkbox"/> Application Size Fee (37 CFR 1.16(s))                           |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
|                                                                                                                                 | <input type="checkbox"/> FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
|                                                                                                                                 |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               | TOTAL ADD'L FEE           |  | 1000                                  |  |
| AMENDMENT                                                                                                                       |                                                                                          | CLAIMS REMAINING AFTER AMENDMENT                                                                                                                                                                                              |                            | HIGHEST NUMBER PREVIOUSLY PAID FOR         | PRESENT EXTRA | RATE (\$)                 |  | ADDITIONAL FEE (\$)                   |  |
|                                                                                                                                 | Total<br>(37 CFR 1.16(i))                                                                | *                                                                                                                                                                                                                             | Minus                      | **                                         | =             | x \$0 =                   |  |                                       |  |
|                                                                                                                                 | Independent<br>(37 CFR 1.16(h))                                                          | *                                                                                                                                                                                                                             | Minus                      | ***                                        | =             | x \$0 =                   |  |                                       |  |
|                                                                                                                                 | <input type="checkbox"/> Application Size Fee (37 CFR 1.16(s))                           |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
|                                                                                                                                 | <input type="checkbox"/> FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
|                                                                                                                                 |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               | TOTAL ADD'L FEE           |  |                                       |  |
| * If the entry in column 1 is less than the entry in column 2, write "0" in column 3.                                           |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               | PSA                       |  |                                       |  |
| ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".                                       |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               | /TAMMY D MCBETH/          |  |                                       |  |
| *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".                                        |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |
| The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1. |                                                                                          |                                                                                                                                                                                                                               |                            |                                            |               |                           |  |                                       |  |

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

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|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>10/16/2024 12:34:59 PM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Zachary Wiersma  |
| PATENT CENTER #           | 67590787                                                    | FILING DATE             | 06/14/2024       |
| CUSTOMER #                | 76656                                                       | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE<br>ADDRESS | -                                                           | AUTHORIZED BY           | -                |

### Documents

**TOTAL DOCUMENTS: 2**

| DOCUMENT             | PAGES | DESCRIPTION                                 | SIZE<br>(KB) |
|----------------------|-------|---------------------------------------------|--------------|
| petition-request.pdf | 3     | Terminal Disclaimer-Filed<br>(Electronic)   | 44 KB        |
| grantLetter.pdf      | 1     | Terminal Disclaimer-Electronic-<br>Approved | 18 KB        |

### Digest

| DOCUMENT             | MESSAGE DIGEST(SHA-512)                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| petition-request.pdf | C268DC35D07BFC18B77C8CD489FB05B3EAFDE89663EE3E5<br>AF3C45D42A6EF87D2027F2C73EE30CEC6BEC88956377DC9C<br>8D9C6627013F1359CF5328AA8A7DB2639 |
| grantLetter.pdf      | EF1742FD00A39FD8123CCFCAADF31EE51F464DF53FBD9F31                                                                                         |

7199BA975D4D623E90426F1ACC9A2F1036B07F9741DDC3317  
BBAD8F01B26DBBD436C47BE920390EA

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**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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## TERMINAL DISCLAIMER TO OBVIATE A DOUBLE PATENTING REJECTION OVER A PRIOR PATENT

|               |             |                      |                       |
|---------------|-------------|----------------------|-----------------------|
| APPLICATION # | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET #     |
| 18743352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY



Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action



This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.

| Owner         | Percent interest |
|---------------|------------------|
| trinamiX GmbH | 100%             |
| <b>Total</b>  | <b>100%</b>      |

The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of any patent granted on pending reference Application Number(s)

| Application # | Filing Date |
|---------------|-------------|
| 18425144      | 01/29/2024  |



as the term of any patent granted on said reference application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending reference application. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and any patent granted on the reference application are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the owner does not disclaim the terminal part of any patent granted on the instant application that would extend to the expiration date of the full statutory term of any patent granted on said reference application, "as the term of any patent granted on said reference application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending reference application," in the event that any such patent granted on the pending reference application: expires for failure to pay a maintenance fee, is held unenforceable, is found invalid by a court of competent jurisdiction, is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321, has all claims canceled by a reexamination certificate, is reissued, or is in any manner terminated prior to the expiration of its full statutory term as shortened by any terminal disclaimer filed prior to its grant.

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The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of prior patent number(s)

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**Patent #**

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as the term of said prior patent is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the prior patent are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

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- expires for failure to pay a maintenance fee;
- is held unenforceable;
- is found invalid by a court of competent jurisdiction;
- is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321;
- has all claims canceled by a reexamination certificate;
- is reissued; or
- is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer.



Terminal disclaimer fee under 37 CFR 1.20(d) included with Electronic Terminal Disclaimer request.

**Applicant claims the following entity status:**

Regular Undiscounted

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I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

---

I certify, in accordance with 37 CFR 1.4(d)(4) that I am: An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application

**Signature**

**Name**

**Registration #**

/Zachary Wiersma/

Zachary Wiersma

77808

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\* Statement under 37 CFR 3.73(b) is required if terminal disclaimer is signed by the assignee (owner). Form PTO/SB/96 may be used for making this certification. See MPEP 324.



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## **APPROVAL LETTER**

**APPLICATION #  
18/743,352**

**FILING DATE  
06/14/2024**

**APPLICANT/PATENT UNDER REEXAMINATION  
Friedrich SCHICK**

### **Title of Invention**

**DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY**

**Electronic terminal disclaimer filed on 10/16/2024**



**Approved**

**This patent is subject to a Terminal Disclaimer**

**Approved / Disapproved by: Electronic Terminal Disclaimer automatically approved**

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## ELECTRONIC PAYMENT RECEIPT

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|---------------|-----------------------------|-----------------------|
| APPLICATION # | RECEIPT DATE / TIME         | ATTORNEY DOCKET #     |
| 18/743,352    | 10/16/2024 12:34:59 PM Z ET | 27843-2603 182037US03 |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                        |                                                          |                      |                  |
|------------------------|----------------------------------------------------------|----------------------|------------------|
| APPLICATION TYPE       | Utility - Nonprovisional Application under 35 USC 111(a) | PATENT #             | -                |
| CONFIRMATION #         | 2279                                                     | FILED BY             | Zachary Wiersma  |
| PATENT CENTER #        | 67590787                                                 | AUTHORIZED BY        | -                |
| CUSTOMER #             | 76656                                                    | FILING DATE          | 06/14/2024       |
| CORRESPONDENCE ADDRESS | -                                                        | FIRST NAMED INVENTOR | Friedrich SCHICK |

### Payment Information

|                        |                                                                                                                                                                                                                                                                 |                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| PAYMENT METHOD         | PAYMENT TRANSACTION ID                                                                                                                                                                                                                                          | PAYMENT AUTHORIZED BY |
| DA / 012384            | E20240FC35075010                                                                                                                                                                                                                                                | Zachary Wiersma       |
| PRE-AUTHORIZED ACCOUNT | PRE-AUTHORIZED CATEGORY                                                                                                                                                                                                                                         |                       |
| 012384                 | 37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges) |                       |

| FEE CODE      | DESCRIPTION                                         | ITEM PRICE(\$) | QUANTITY | ITEM TOTAL(\$) |
|---------------|-----------------------------------------------------|----------------|----------|----------------|
| 1814          | STATUTORY DISCLAIMER, INCLUDING TERMINAL DISCLAIMER | 170.00         | 1        | 170.00         |
| TOTAL AMOUNT: |                                                     |                |          | \$170.00       |

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**IN the United States Patent & Trademark Office**

|                 |                                                            |                   |               |
|-----------------|------------------------------------------------------------|-------------------|---------------|
| First Inventor: | Friedrich Schick                                           | Art Unit:         | 2454          |
| Serial No.:     | 18/743,352                                                 | Confirmation No.: | 2279          |
| Date Filed:     | June 14, 2024                                              | Examiner:         | Frantz B Jean |
| Title:          | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL<br>PROPERTY |                   |               |

October 16, 2024

**RESPONSE TO NON-FINAL OFFICE ACTION**

TO THE COMMISSIONER FOR PATENTS:

In response to the Non-Final Office Action dated September 24, 2024, please consider the following remarks.

**Amendments to the Specification** are on page 2.

**Amendments to the Claims** begin on page 3.

**Remarks** begin on page 7.

**Conclusion** is on page 8.

**AMENDMENTS TO THE SPECIFICATION**

Please amend Page 1, Lines 4-9 of the specification as follows:

This application is a continuation application of U.S. Patent Application No. 18/425,144, filed January 29, 2024, which is a continuation application of U.S. Patent Application No. 17/439,492, filed September 15, 2021, and which issued as U.S. Patent No. 11,947,013 on April 2, 2024, which is a U.S. National Phase Application of International Patent Application No. PCT/EP20/56759, filed March 13, 2020, which claims priority to European Patent Application No. 19163250.4, filed March 15, 2019, the entire contents of which are hereby incorporated by reference herein.

## **AMENDMENTS TO THE CLAIMS**

1. (Original) A detector for identifying at least one material property  $m$  comprising  
  
at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines;  
  
at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object; and  
  
at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,  
  
wherein the evaluation device is configured for determining at least one material feature  $\phi 2m$  by applying at least one material dependent image filter  $\Phi 2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; and a linear combination thereof;  
  
wherein the evaluation device is configured for determining the material property  $m$  by evaluating the material feature  $\phi 2m$ .



2. (Original) The detector according claim 1, wherein the material dependent image filter is a spot shape filter.
3. (Original) The detector according to claim 1, wherein the material dependent image filter is Law's energy filter.
4. (Original) The detector according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, and a deviation from a Lambertian surface reflection.
5. (Original) The detector according to claim 1, wherein the evaluation device is configured for determining at least one distance feature  $\phi 1z$  by applying at least one distance dependent image filter  $\Phi 1$  to the reflection image, and  
  
wherein the evaluation device is configured for determining a longitudinal coordinate  $z$  by evaluating the distance feature  $\phi 1z$ .
6. (Original) The detector according to claim 5, wherein the distance dependent image filter is a structured-light-filter.
7. (Original) The detector according to claim 1, wherein the sensor element comprises at least one CMOS sensor.
8. (Original) The detector according to claim 7, wherein the CMOS sensor comprises at least 100000 optical sensors.
9. (Original) The detector according to claim 8, wherein the CMOS sensor is sensitive in the range of 700 nm to 1000 nm.
10. (Original) The detector according to claim 1, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.
11. (Original) The detector according to claim 10, wherein the illumination source is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.

12. (Original) The detector according to claim 11, wherein the illumination source emits light having a wavelength of 700 nm to 1000 nm.
13. (Original) The detector according to claim 12, wherein the illumination source comprises at least one vertical cavity surface-emitting laser.
14. (Original) The detector according to claim 13, wherein the illumination source is integrated into a smartphone.
15. (Original) The detector according to claim 1, wherein the illumination source emits light having a wavelength of 700 nm to 1000 nm.
16. (Original) The detector according to claim 15, wherein the illumination source is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.
17. (Original) The detector according to claim 1, wherein the illumination source is integrated into a smartphone.
18. (Original) The detector according to claim 17, wherein the illumination source and the optical sensors is arranged in a common plane.
19. (Original) The detector according to claim 18, wherein the illumination source emits light having a wavelength of 700 nm to 1000 nm.
20. (Original) The detector according to claim 1, wherein the object is a human.
21. (Original) The detector according to claim 1, wherein the detector is configured for identification and/or classification of the material.
22. (Original) The detector according to claim 20, wherein the detector is configured for determining and/or validating whether a surface to be examined is or comprises human skin.
23. (Original) The detector according to claim 1, wherein the detector is configured for detection of human skin.

24. (Original) The detector according to claim 23, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, and a deviation from a Lambertian surface reflection.
25. (Original) The detector according to claim 1, wherein the evaluation device is configured for performing at least one image analysis and/or image processing in order to identify a reflection feature in the reflection image.
26. (Original) The detector according to claim 25, wherein the evaluation device is adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature.
27. (Original) The detector according to claim 25, wherein the evaluation device is adapted to determine at least one reference feature in at least one reference image corresponding to the at least one reflection feature.
28. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises selection of at least one region of interest.
29. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises a background correction.
30. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises applying a principle curvature-based region detector.

## **REMARKS**

### **I. Status of the Claims**

Claims 1-30 are pending, unamended, and under examination. No new matter is added herein.

### **II. Objection to the Specification**

The Office has based the objection on, “The cross-reference to related applications must be updated to reflect the current status of the related applications” (see Page 2 of the Office Action dated 09/24/24).

Without commenting on the appropriateness of the objection, the specification has been amended herein consistent with the comments of the Office. It is believed that these amendments overcome the objection.

### **III. Double Patenting Rejections**

*i. Claims 1 and 7 stand rejected on the ground of nonstatutory double patenting as allegedly being unpatentable over claims 1, 14-15, and 24 of U.S. Patent No. 11,947,013.*

Without commenting on the appropriateness of the rejections, submitted herewith is a terminal disclaimer over U.S. Patent No. 11,947,013. It is believed that this submission overcomes the rejections.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

*ii. Claims 1, 7, 10, 15, 21, 23, and 25 stand rejected on the ground of nonstatutory double patenting as allegedly being unpatentable over claims 1, 11, 13, 15, 17-20 of U.S. Patent Application No. 18/425,144.*

Without commenting on the appropriateness of the rejections, submitted herewith is a terminal disclaimer over U.S. Patent No. 11,947,013. It is believed that this submission overcomes the rejections.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

**CONCLUSION**

Applicant requests reconsideration and allowance of all pending claims. Applicant authorizes the Commissioner to charge any fees due in connection with this response to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP. Please reference docket number 27843-2603.

Respectfully submitted,

/Christopher M. Goff/

Christopher M. Goff  
USPTO Registration 41,785  
ARMSTRONG TEASDALE LLP  
7700 Forsyth Blvd., Suite 1800  
St. Louis, Missouri 63105  
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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

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|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>10/16/2024 12:37:27 PM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Zachary Wiersma  |
| PATENT CENTER #           | 67590955                                                    | FILING DATE             | 06/14/2024       |
| CUSTOMER #                | 76656                                                       | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE<br>ADDRESS | -                                                           | AUTHORIZED BY           | -                |

### Documents

**TOTAL DOCUMENTS: 4**

| DOCUMENT                                                                 |       | PAGES | DESCRIPTION                                                            | SIZE<br>(KB) |
|--------------------------------------------------------------------------|-------|-------|------------------------------------------------------------------------|--------------|
| 2024-10-16 Response to<br>Non-Final Office Action<br>27843-2603.pdf      |       | 8     | -                                                                      | 198 KB       |
| 2024-10-16 Response to<br>Non-Final Office Action<br>27843-2603-A....pdf | (1-1) | 1     | Amendment/Request for<br>Reconsideration-After Non-<br>Final Rejection | 103 KB       |
| 2024-10-16 Response to<br>Non-Final Office Action<br>27843-2603-SPEC.pdf | (2-2) | 1     | Specification                                                          | 102 KB       |
| 2024-10-16 Response to<br>Non-Final Office Action<br>27843-2603-CLM.pdf  | (3-6) | 4     | Claims                                                                 | 149 KB       |
| 2024-10-16 Response to<br>Non-Final Office Action                        | (7-8) | 2     | Applicant Arguments/Remarks<br>Made in an Amendment                    | 143 KB       |

27843-2603-REM.pdf

**Digest**

| <b>DOCUMENT</b>                                                    | <b>MESSAGE DIGEST(SHA-512)</b>                                                                                                    |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2024-10-16 Response to Non-Final Office Action 27843-2603.pdf      | F8C918E77DF3BA74CAE37160AF9669DE5F9BAEFA753139716B1F6904ACBAF30E64E0338028686CBB8BDE4D14C06C6B46328294E527932ED4B73A6B84026B5D20  |
| 2024-10-16 Response to Non-Final Office Action 27843-2603-A....pdf | 0D0CC616601D9C5E9A110AD96B876A901D7F6323E2CC61B06B6F4A1F09C4430DAE77FD5DC9C7C7A204A9263CE266318B3D2F55EA9B24F80E0A0BBBD1089964093 |
| 2024-10-16 Response to Non-Final Office Action 27843-2603-SPEC.pdf | 96C9B384456ABEC9DE42D31672310317DBA74965F717996797F0C41DF582D05D171008AC9A9EB3147211608868CA354E4A199D1570C45DA272EF5D6A267EEFB1  |
| 2024-10-16 Response to Non-Final Office Action 27843-2603-CLM.pdf  | 3B0061CF59DF821CF8C2679B6B87840173864004EC2846487E054F6E5D4164E28259C41D6A2BD907F3E0D7F6F9839779CEF91EFA4457453A19748BB07ACBA6BD  |
| 2024-10-16 Response to Non-Final Office Action 27843-2603-REM.pdf  | B019FF22978F0C88B471E26720415865C81FF0E707D6C4E528F737B61E858B58B886D61A958D3EACEEB0E1D4FF651B42744DB091E77D1D431289885B07C0C816  |

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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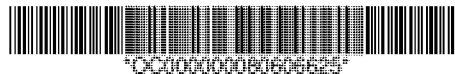
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| APPLICATION NUMBER | FILING OR 371(C) DATE | FIRST NAMED APPLICANT | ATTY. DOCKET NO./TITLE |
|--------------------|-----------------------|-----------------------|------------------------|
| 18/743,352         | 06/14/2024            | Friedrich SCHICK      | 27843-2603 182037US03  |

**CONFIRMATION NO. 2279**

76656  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

## PUBLICATION NOTICE



Date Mailed: 11/14/2024

**Title:**DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

**Publication No.**US-2024-0377533-A1

**Publication Date:**11/14/2024

## NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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|                                   |                                       |  |                                                             |             |
|-----------------------------------|---------------------------------------|--|-------------------------------------------------------------|-------------|
| <b>Notice of References Cited</b> | Application/Control No.<br>18/743,352 |  | Applicant(s)/Patent Under<br>Reexamination<br>SCHICK et al. |             |
|                                   | Examiner<br>FRANTZ B JEAN             |  | Art Unit<br>2454                                            | Page 1 of 1 |

U.S. PATENT DOCUMENTS

| * |   | Document Number<br>Country Code-Number-Kind Code | Date<br>YYYY-MM-DD | Name         | CPC Classification | US Classification |
|---|---|--------------------------------------------------|--------------------|--------------|--------------------|-------------------|
| * | A | US-20180276843-A1                                | 2018-09-27         | SEND; Robert | G02B3/14           | 1/1               |
|   | B |                                                  |                    |              |                    |                   |
|   | C |                                                  |                    |              |                    |                   |
|   | D |                                                  |                    |              |                    |                   |
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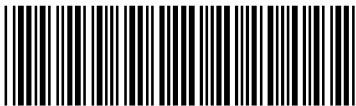
FOREIGN PATENT DOCUMENTS

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|   | N |                                                  |                    |         |      |                    |
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NON-PATENT DOCUMENTS

| * |   | Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages) |
|---|---|-------------------------------------------------------------------------------------------|
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\*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)  
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|                                                                                                                 |                                              |                                                                 |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Search Notes</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                 | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

| CPC - Searched*                                                                                                                                  |            |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Symbol                                                                                                                                           | Date       | Examiner |
| G01S17/46 G06T7/521 G06T7/73 G06V10/60 G06V10/145<br>G06V10/764 H04N23/74 H04N23/56 G01B11/22 G01C21/16<br>G01S17/66 G06T5/002 G06T5/20 G06T7/20 | 09/20/2024 | FJ       |

| CPC Combination Sets - Searched* |      |          |
|----------------------------------|------|----------|
| Symbol                           | Date | Examiner |
|                                  |      |          |

| US Classification - Searched* |          |            |          |
|-------------------------------|----------|------------|----------|
| Class                         | Subclass | Date       | Examiner |
| 348                           | 135      | 09/20/2024 | FJ       |

\* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

| Search Notes                                         |            |          |
|------------------------------------------------------|------------|----------|
| Search Notes                                         | Date       | Examiner |
| Derwent, EPO, JPO, IBMTDBD, OCR, PGPUB, Google, East | 09/19/2024 | FJ       |
| updated search                                       | 11/14/2024 | FJ       |

| Interference Search |                       |      |          |
|---------------------|-----------------------|------|----------|
| US Class/CPC Symbol | US Subclass/CPC Group | Date | Examiner |
|                     |                       |      |          |

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|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Index of Claims</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                    | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

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|---|-----------------|---|-------------------|---|---------------------|---|-----------------|
| ✓ | <b>Rejected</b> | - | <b>Cancelled</b>  | N | <b>Non-Elected</b>  | A | <b>Appeal</b>   |
| = | <b>Allowed</b>  | ÷ | <b>Restricted</b> | I | <b>Interference</b> | O | <b>Objected</b> |

| CLAIMS                                                                                                                                                                            |          |            |            |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|------------|--|--|--|--|--|--|--|
| <input type="checkbox"/> Claims renumbered in the same order as presented by applicant <input type="checkbox"/> CPA <input type="checkbox"/> T.D. <input type="checkbox"/> R.1.47 |          |            |            |  |  |  |  |  |  |  |
| CLAIM                                                                                                                                                                             |          | DATE       |            |  |  |  |  |  |  |  |
| Final                                                                                                                                                                             | Original | 09/20/2024 | 11/14/2024 |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 1        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 2        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 3        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 4        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 5        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 6        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 7        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 8        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 9        | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 10       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 11       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 12       | ✓          | ✓          |  |  |  |  |  |  |  |
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|                                                                                                                                                                                   | 14       | ✓          | ✓          |  |  |  |  |  |  |  |
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|                                                                                                                                                                                   | 18       | ✓          | ✓          |  |  |  |  |  |  |  |
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|                                                                                                                                                                                   | 22       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 23       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 24       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 25       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 26       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 27       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 28       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 29       | ✓          | ✓          |  |  |  |  |  |  |  |
|                                                                                                                                                                                   | 30       | ✓          | ✓          |  |  |  |  |  |  |  |

## PE2E SEARCH - Search History (Prior Art)

| Ref # | Hits   | Search Query                                                                                                                                                                                                                                                                                                                              | DBs                                                                                                                                                                                                                                                                                                                  | Default Operator | Plurals | British Equivalents | Time Stamp          |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|---------------------|---------------------|
| L1    | 2516   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:39 AM |
| L2    | 440018 | (348/135).cccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                              | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:49 AM |
| L3    | 757    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 02:01 AM |
| L4    | 183885 | (detect\$3 OR sen\$3)                                                                                                                                                                                                                                                                                                                     | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                    | OR               | ON      | ON                  | 2023/09/09          |

|    |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|    |        | near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                           | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 02:34 AM               |
| L5 | 249451 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L6 | 2430   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L7 | 11     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:38 AM |

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| L8  | 1872340 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 09:57 AM |
| L9  | 1341757 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:00 AM |
| L10 | 113287  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:01 AM |
| L11 | 67      | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/09/09 10:02 AM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
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| L12 | 0  | L11 AND I1 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L13 | 35 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L14 | 11 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L15 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 06:13 PM |
| L16 | 2541   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L17 | 454819 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L18 | 791    | (G01S17/46 OR                                                                                                                                                                                                                                                                                                                             | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/11/18          |



|     |        |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:50 PM               |
| L19 | 188776 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L20 | 259070 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L21 | 2529   | L4 AND I5                                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L22 | 12      | L6 AND I3                                                                                                         | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L23 | 1891255 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L24 | 1363696 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L25 | 114665  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
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| L26 | 70 | L10 AND I6 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L27 | 27 | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L28 | 36 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
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| L29 | 12     | L11 AND I3                                                                                                                                                                                                                                                                                                                                | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L30 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L31 | 2596   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L32 | 468828 | (348/135).ccls. OR                                                                                                                                                                                                                                                                                                                        | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/09/19             |

|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73 OR<br>G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56 OR<br>G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                 | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:07 PM               |
| L33 | 903    | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L34 | 208447 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L35 | 296319 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L36 | 2914    | L4 AND I5                                                                                                  | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L37 | 18      | L6 AND I3                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L38 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L39 | 1447259 | filter near6 (image OR<br>median OR contrast OR                                                            | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L40 | 119738 | energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L41 | 92     | L8 AND I9                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L42 | 39     | L10 AND I6                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
|     |        | L11 AND I1                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |      |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L43 | 48   | L11 AND I2                                                     | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L44 | 18   | L11 AND I3                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L45 | 4    | "20180033146".pn.                                              | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L46 | 2596 | ( Friedrichr near3<br>Schoick) OR (Peter<br>near3 Schillen) OR | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |



|     |        |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
|     |        | ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           |    |    |    |                     |
| L47 | 468828 | (348/135).cls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                       | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L48 | 903    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L49 | 208447 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                              | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/09/19 08:07 PM |

|     |         |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L50 | 296319  | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L51 | 2914    | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L52 | 18      | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L53 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3                                               | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |         | OR distance) near60<br>beam near5 (light OR<br>profile)                                                           | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L54 | 1447259 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L55 | 119738  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L56 | 92      | L10 AND I6                                                                                                        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |

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| L57 | 39 | L11 AND I1        | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L58 | 48 | L11 AND I2        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L59 | 18 | L11 AND I3        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L60 | 4  | "20180033146".pn. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L61 | 2      | "11947013".pn.                                                                                                                                                                                                                                                                                                                            | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/09/19 09:42 PM |
| L62 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L63 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR                                                                                                                                                                                                 | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L64 | 1005   | G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                                                                                                                                                                   | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L65 | 219410 | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L66 | 318145 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | L4 AND I5                                                                                                                                                                                                                                       | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L68 | 18      | L6 AND I3                                                                                          | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L69 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L70 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L71 | 122964 | L8 AND I9  | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L72 | 102    | L10 AND I6 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L73 | 41     | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L74 | 56     | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |



|     |      |                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L75 | 18   | L11 AND I3                                                                                                                                                                                                                    | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L76 | 4    | "20180033146".pn.                                                                                                                                                                                                             | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L77 | 2657 | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L78 | 496753 | (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv.<br>(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L79 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                   | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L80 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                                                                        | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L81 | 318145 | longitudinal WITH coordinate AND                                                                                                                                                                                                                                                                                           | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |         |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                     |
|-----|---------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L82 | 3134    | (material OR property OR evaluat\$3 OR distance)<br><br>L4 AND I5                              | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br><br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L83 | 18      | L6 AND I3                                                                                      | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L84 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE,                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/10/23 11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L85 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L86 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L87 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L88 | 41      | L11 AND I1                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |    |                   |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
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| L89 | 56 | L11 AND I2        | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L90 | 18 | L11 AND I3        | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L91 | 4  | "20180033146".pn. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK,                                                            | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L92 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L93 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L94 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L95 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | classify\$3) WITH<br>(material OR property)                                                   | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L96 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L97 | 3134   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L98 | 18     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|------|---------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L99  | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L100 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L101 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L102 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |



|      |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |                        |
|------|----|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L103 | 41 | L11 AND I1 | CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L104 | 56 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L105 | 18 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L106 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L107 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L108 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L109 | 1005   | (G01S17/46 OR G06T7/521 OR                                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|      |        |                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |    |    |                        |
|------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L110 | 219410 | G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.<br><br>(detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property) | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L111 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L112 | 3134   | L4 AND I5                                                                                                                                                                                                                                                                                                          | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                                                                                                                                                                                                                                                                                                                                                            | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|------|---------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L113 | 18      | L6 AND I3                                                                                                         | RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L114 | 2008062 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L115 | 1497001 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L116 | 122964  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |     |            |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|------|-----|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L117 | 102 | L10 AND I6 | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L118 | 41  | L11 AND I1 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L119 | 56  | L11 AND I2 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|      |    |                   |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|----|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L120 | 18 | L11 AND I3        | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L121 | 4  | "20180033146".pn. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L122 | 2  | "11947013".pn.    | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |

## PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

## PE2E SEARCH - Search History (Prior Art)

| Ref # | Hits   | Search Query                                                                                                                                                                                                                                                                                                                              | DBs                                                                                                                                                                                                                                                                                                                  | Default Operator | Plurals | British Equivalents | Time Stamp          |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|---------------------|---------------------|
| L1    | 2516   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:39 AM |
| L2    | 440018 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:49 AM |
| L3    | 757    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 02:01 AM |
| L4    | 183885 | (detect\$3 OR sen\$3)                                                                                                                                                                                                                                                                                                                     | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                    | OR               | ON      | ON                  | 2023/09/09          |

|    |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|    |        | near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                           | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 02:34 AM               |
| L5 | 249451 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L6 | 2430   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L7 | 11     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:38 AM |



|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L8  | 1872340 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 09:57 AM |
| L9  | 1341757 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:00 AM |
| L10 | 113287  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:01 AM |
| L11 | 67      | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/09/09 10:02 AM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
|-----|----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L12 | 0  | L11 AND I1 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L13 | 35 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L14 | 11 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
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| L15 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 06:13 PM |
| L16 | 2541   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L17 | 454819 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L18 | 791    | (G01S17/46 OR                                                                                                                                                                                                                                                                                                                             | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/11/18          |

|     |        |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:50 PM               |
| L19 | 188776 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L20 | 259070 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L21 | 2529   | L4 AND I5                                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L22 | 12      | L6 AND I3                                                                                                         | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L23 | 1891255 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L24 | 1363696 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L25 | 114665  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |    |    |                        |
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| L26 | 70 | L10 AND I6 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L27 | 27 | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L28 | 36 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L29 | 12     | L11 AND I3                                                                                                                                                                                                                                                                                                                                | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/18 08:50 PM |
| L30 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L31 | 2596   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19 08:07 PM |
| L32 | 468828 | (348/135).ccls. OR                                                                                                                                                                                                                                                                                                                        | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/09/19          |

|     |        |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
|     |        | (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                 | USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                   |    |    |    | 08:07 PM            |
| L33 | 903    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L34 | 208447 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                      | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L35 | 296319 | longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)                                                                                                                        | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/09/19 08:07 PM |



|     |         |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L36 | 2914    | L4 AND I5                                                                                                  | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L37 | 18      | L6 AND I3                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L38 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L39 | 1447259 | filter near6 (image OR<br>median OR contrast OR                                                            | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L40 | 119738 | energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L41 | 92     | L8 AND I9                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L42 | 39     | L10 AND I6                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
|     |        | L11 AND I1                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |      |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
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| L43 | 48   | L11 AND I2                                                     | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L44 | 18   | L11 AND I3                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L45 | 4    | "20180033146".pn.                                              | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L46 | 2596 | ( Friedrichr near3<br>Schoick) OR (Peter<br>near3 Schillen) OR | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
|     |        | ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           |    |    |    |                     |
| L47 | 468828 | (348/135).cls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                       | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L48 | 903    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L49 | 208447 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                              | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/09/19 08:07 PM |

|     |         |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L50 | 296319  | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L51 | 2914    | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L52 | 18      | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L53 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3                                               | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
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|     |         | OR distance) near60<br>beam near5 (light OR<br>profile)                                                           | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L54 | 1447259 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L55 | 119738  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L56 | 92      | L10 AND I6                                                                                                        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|----|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L57 | 39 | L11 AND I1        | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L58 | 48 | L11 AND I2        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L59 | 18 | L11 AND I3        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L60 | 4  | "20180033146".pn. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L61 | 2      | "11947013".pn.                                                                                                                                                                                                                                                                                                                            | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/09/19 09:42 PM |
| L62 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L63 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR                                                                                                                                                                                                 | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/10/23 11:54 PM |



|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L64 | 1005   | G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                                                                                                                                                                   | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L65 | 219410 | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L66 | 318145 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | L4 AND I5                                                                                                                                                                                                                                       | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L68 | 18      | L6 AND I3                                                                                          | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L69 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L70 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L71 | 122964 | L8 AND I9  | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L72 | 102    | L10 AND I6 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L73 | 41     | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L74 | 56     | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |      |                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L75 | 18   | L11 AND I3                                                                                                                                                                                                                    | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L76 | 4    | "20180033146".pn.                                                                                                                                                                                                             | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L77 | 2657 | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

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|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L78 | 496753 | (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv.<br>(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L79 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                   | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L80 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                                                                        | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L81 | 318145 | longitudinal WITH coordinate AND                                                                                                                                                                                                                                                                                           | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

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|-----|---------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L82 | 3134    | (material OR property OR evaluat\$3 OR distance)<br><br>L4 AND I5                              | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br><br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L83 | 18      | L6 AND I3                                                                                      | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L84 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE,                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/10/23 11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L85 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L86 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L87 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L88 | 41      | L11 AND I1                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |    |                   |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|----|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L89 | 56 | L11 AND I2        | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L90 | 18 | L11 AND I3        | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L91 | 4  | "20180033146".pn. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK,                                                            | OR | ON | ON | 2024/10/23 11:54 PM |



|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L92 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L93 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L94 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L95 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | classify\$3) WITH<br>(material OR property)                                                   | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L96 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L97 | 3134   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L98 | 18     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|------|---------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L99  | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L100 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L101 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L102 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|      |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |                        |
|------|----|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L103 | 41 | L11 AND I1 | CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L104 | 56 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L105 | 18 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L106 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L107 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L108 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L109 | 1005   | (G01S17/46 OR G06T7/521 OR                                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|      |        |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|      |        | G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L110 | 219410 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L111 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L112 | 3134   | L4 AND I5                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|------|---------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L113 | 18      | L6 AND I3                                                                                                         | RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L114 | 2008062 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L115 | 1497001 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L116 | 122964  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |     |            |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
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| L117 | 102 | L10 AND I6 | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L118 | 41  | L11 AND I1 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L119 | 56  | L11 AND I2 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |



|      |    |                   |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|----|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L120 | 18 | L11 AND I3        | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L121 | 4  | "20180033146".pn. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L122 | 2  | "11947013".pn.    | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |

## PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



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## BIB DATA SHEET

CONFIRMATION NO. 2279

| SERIAL NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FILING or<br>371(c) DATE                                          | CLASS                                  | GROUP ART UNIT                                                                   | ATTORNEY<br>DOCKET NO.        |                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|-------------------------------|------------------------------------|
| 18/743,352                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 06/14/2024                                                        | 348                                    | 2454                                                                             | 27843-2603<br>182037US03      |                                    |
| <b>APPLICANTS</b><br>trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                   |                                        |                                                                                  |                               |                                    |
| <b>INVENTORS</b><br>Friedrich SCHICK, Ludwigshafen, GERMANY;<br>Peter SCHILLEN, Ludwigshafen, GERMANY;<br>Patrick SCHINDLER, Ludwigshafen, GERMANY;<br>Andre SCHMIDT, Ludwigshafen, GERMANY;<br>Michael EBERSPACH, Ludwigshafen, GERMANY;<br>Christian LENNARTZ, Ludwigshafen, GERMANY;<br>Robert SEND, Karlsruhe, GERMANY;<br>Lars DIESSELBERG, Karlsruhe, GERMANY;<br>Heiko HENGEN, Steinweiler, GERMANY;<br>Ingmar BRUDER, Ludwigshafen, GERMANY;<br>Jakob UNGER, Ludwigshafen, GERMANY;<br>Christian BONSIGNORE, Ludwigshafen, GERMANY; |                                                                   |                                        |                                                                                  |                               |                                    |
| <b>** CONTINUING DATA *****</b><br>This application is a CON of 18/425,144 01/29/2024<br>which is a CON of 17/439,492 09/15/2021 PAT 11947013<br>which is a 371 of PCT/EP20/56759 03/13/2020                                                                                                                                                                                                                                                                                                                                                |                                                                   |                                        |                                                                                  |                               |                                    |
| <b>** FOREIGN APPLICATIONS *****</b><br>EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                   |                                        |                                                                                  |                               |                                    |
| <b>** IF REQUIRED, FOREIGN FILING LICENSE GRANTED **</b><br>07/15/2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                   |                                        |                                                                                  |                               |                                    |
| Foreign Priority claimed<br><input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>35 USC 119(a-d) conditions met <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>Verified and<br>Acknowledged                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/> Met after<br>Allowance<br>FJ<br>Initials | <b>STATE OR<br/>COUNTRY</b><br>GERMANY | <b>SHEETS<br/>DRAWINGS</b><br>2                                                  | <b>TOTAL<br/>CLAIMS</b><br>30 | <b>INDEPENDENT<br/>CLAIMS</b><br>1 |
| <b>ADDRESS</b><br>Patent Docket Department<br>Armstrong Teasdale LLP<br>7700 Forsyth Boulevard<br>Suite 1800<br>St. Louis, MO 63105<br>UNITED STATES                                                                                                                                                                                                                                                                                                                                                                                        |                                                                   |                                        |                                                                                  |                               |                                    |
| <b>TITLE</b><br>DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                   |                                        |                                                                                  |                               |                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                   |                                        | <input type="checkbox"/> All Fees<br><input type="checkbox"/> 1.16 Fees (Filing) |                               |                                    |

|                                        |                                                                                                                   |                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FILING FEE<br/>RECEIVED</b><br>0.00 | FEES: Authority has been given in Paper<br>No. _____ to charge/credit DEPOSIT ACCOUNT<br>No. _____ for following: | <input type="checkbox"/> 1.17 Fees (Processing Ext. of time)<br><input type="checkbox"/> 1.18 Fees (Issue)<br><input type="checkbox"/> Other _____<br><input type="checkbox"/> Credit _____ |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## AI Overview

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This describes a detector system designed to identify a material property "m" by analyzing the reflection pattern of a structured light beam illuminating the object, where the illumination pattern is carefully chosen from a variety of options like point patterns, line patterns, or even complex geometric patterns, and the reflected image is then processed using specific image filters to extract material-dependent features for identification. 

Key components

Detector for identifying at least one material property  
The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is r...

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Detector for optically detecting at least one object

This classification concerns a method for determining the shape of an object by illuminating it with a structured light beam and detecting the reflected light.

Show more 

"consisting" (and any subsequent words) was ignored because we limit queries to 32 words.

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<https://patents.google.com/patent/> 

### Detector for identifying at least one material property

Described herein is a detector for identifying at least one material property m. The detector includes at least one sensor element including a matrix of ...

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<https://patents.google.com/patent/> 

### Detector for optically detecting at least one object

As explained above, the optical sensor is designed to generate at least one sensor signal in a manner dependent on the illumination of the sensor region, ...

 [Nweon Patent](#)  
<https://patent.nweon.com/...> 

### Meta Patent | Determining and reconstructing a shape and a ...

8 days ago -- A system, comprising: a light source to transmit an original beam of light; a first grating to receive and diffract the original beam of light ...

 [trea.com](#)  
<https://trea.com/information/patentgrant/> 

### Enhanced material detection by stereo beam profile analysis | TREA

WO 2020/187719 describes a detector for identifying at least one material property m. The detector comprises at least one sensor element comprising a matrix of ...

 [googleapis.com](#)  
<https://patentimages.storage.googleapis.com/...> [PDF](#) 

### Untitled - Googleapis.com

The projector may comprise at least one light source, also denoted illumination device or illumination source, configured for generating at least one light beam ...

 [googleapis.com](#)  
<https://patentimages.storage.googleapis.com/...> [PDF](#) 

### Untitled - googleapis.com

... detector for identifying at least one material property m. The de- tector comprises at least one sensor element comprising a matrix of optical sensors. The ...

 **FreePatentsOnline**  
<https://www.freepatentsonline.com> > ... {

## MACHINE VISION SYSTEMS, ILLUMINATION SOURCES ...

Apr 30, 2020 --- A machine vision system, comprising: at least one patterned area light source oriented to emit light toward an object to be inspected; at least ...

 **Justia**  
<https://patents.justia.com> > patents-by-us-classification {

## Utilizing A Spectrophotometer (i.e., Plural Beam) Patents and ...

The detector can measure either incoherent inelastically scattered light (e.g., using a spectrometer) or coherent inelastically scattered light (e.g., using an ...

Missing: group | Show results with: group

 **Nature**  
<https://www.nature.com> > nature biotechnology > articles {

## Three-dimensional structured illumination microscopy with ...

by X Li · 2023 · Cited by 50 --- Here we present two distinct, complementary methods to improve axial resolution in 3D SIM with minimal or no modification to the optical system.

Missing: comprising comprises

 **Justia**  
<https://patents.justia.com> > patents-by-us-classification {

## Document Pattern Analysis Or Verification Patents and ...

Abstract: Device for detecting reactive luminescent particles embedded in a substrate or surface having an infrared or ultraviolet illuminator; ...

1 2 3 4 5 6 7 8 9 10      Next

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| APPLICATION NO.          | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|--------------------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352               | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |
| 76656                    | 7590        | 11/19/2024           | EXAMINER              |                  |
| Patent Docket Department |             |                      | JEAN, FRANTZ B        |                  |
| Armstrong Teasdale LLP   |             |                      | ART UNIT              |                  |
| 7700 Forsyth Boulevard   |             |                      | PAPER NUMBER          |                  |
| Suite 1800               |             |                      | 2454                  |                  |
| St. Louis, MO 63105      |             |                      | NOTIFICATION DATE     |                  |
|                          |             |                      | DELIVERY MODE         |                  |
|                          |             |                      | 11/19/2024            |                  |
|                          |             |                      | ELECTRONIC            |                  |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com



## DETAILED ACTION

### *Notice of Pre-AIA or AIA Status*

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This office action is in response to applicant's correspondence filed on 16 October 2024. Claims 1-30 are still pending in the application.

### *Terminal Disclaimer*

The Terminal Disclaimer filed on 16 October 2024 has been approved by the USPTO.

### *Claim Interpretation*

The following is a quotation of 35 U.S.C. 112(f):

(f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of AIA 35 U.S.C. 112, sixth paragraph:

An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

This application includes one or more claim limitations that do not use the word “means,” but are nonetheless being interpreted under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are: **“illumination source configured for, sensor**



**element configured for, evaluation device configured for, detector configured for, evaluation device adapted to” in claims 1, 5, 11, 21-23, and 25-27.**

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph.

Claim limitation “**illumination source configured for, sensor element configured for, evaluation device configured for, detector configured for, evaluation device adapted to” in claims 1, 5, 11, 21-23, and 25-27**” invokes 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph. However, the written description fails to disclose the corresponding structure, material, or acts for performing the entire claimed function and to clearly link the structure, material, or acts to the function. Therefore, the claim is indefinite and is rejected under 35 U.S.C. 112(b) or AIA 35 U.S.C. 112, second paragraph.

Applicant may:

- (a) Amend the claim so that the claim limitation will no longer be interpreted as a limitation under 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph;
- (b) Amend the written description of the specification such that it expressly recites what structure, material, or acts perform the entire claimed function, without introducing any new matter (35 U.S.C. 132(a)); or

- (c) Amend the written description of the specification such that it clearly links the structure, material, or acts disclosed therein to the function recited in the claim, without introducing any new matter (35 U.S.C. 132(a)).

If applicant is of the opinion that the written description of the specification already implicitly or inherently discloses the corresponding structure, material, or acts and clearly links them to the function so that one of ordinary skill in the art would recognize what structure, material, or acts perform the claimed function, applicant should clarify the record by either:

- (a) Amending the written description of the specification such that it expressly recites the corresponding structure, material, or acts for performing the claimed function and clearly links or associates the structure, material, or acts to the claimed function, without introducing any new matter (35 U.S.C. 132(a)); or
- (b) Stating on the record what the corresponding structure, material, or acts, which are implicitly or inherently set forth in the written description of the specification, perform the claimed function.

For more information, see 37 CFR 1.75(d) and MPEP §§ 608.01(o) and 2181.

Claims 1-30 will be allowed upon correcting the deficiencies stated above.

A reason for allowance is included in the previous non-Final Office action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANTZ B JEAN whose telephone number is (571)272-3937. The examiner can normally be reached 8-5 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Glenton B Burgess can be reached on 5712723949. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/FRANTZ B JEAN/  
Primary Examiner, Art Unit 2454

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|                                                                                                                                 |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------|-----|---------------------------|--|---------------------------------------|--|---------------------|--|
| <b>PATENT APPLICATION FEE DETERMINATION RECORD</b><br>Substitute for Form PTO-875                                               |                                                                                          |                                                                                                                                                                                                                               |       | Application or Docket Number<br>18/743,352 |     | Filing Date<br>06/14/2024 |  | <input type="checkbox"/> To be Mailed |  |                     |  |
| <b>ENTITY:</b> <input checked="" type="checkbox"/> LARGE <input type="checkbox"/> SMALL <input type="checkbox"/> MICRO          |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
| <b>APPLICATION AS FILED - PART I</b>                                                                                            |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 |                                                                                          | (Column 1)                                                                                                                                                                                                                    |       | (Column 2)                                 |     |                           |  |                                       |  |                     |  |
| FOR                                                                                                                             |                                                                                          | NUMBER FILED                                                                                                                                                                                                                  |       | NUMBER EXTRA                               |     | RATE (\$)                 |  | FEE (\$)                              |  |                     |  |
| <input type="checkbox"/> BASIC FEE<br>(37 CFR 1.16(a), (b), or (c))                                                             |                                                                                          | N/A                                                                                                                                                                                                                           |       | N/A                                        |     | N/A                       |  |                                       |  |                     |  |
| <input type="checkbox"/> SEARCH FEE<br>(37 CFR 1.16(k), (i), or (m))                                                            |                                                                                          | N/A                                                                                                                                                                                                                           |       | N/A                                        |     | N/A                       |  |                                       |  |                     |  |
| <input type="checkbox"/> EXAMINATION FEE<br>(37 CFR 1.16(o), (p), or (q))                                                       |                                                                                          | N/A                                                                                                                                                                                                                           |       | N/A                                        |     | N/A                       |  |                                       |  |                     |  |
| TOTAL CLAIMS<br>(37 CFR 1.16(i))                                                                                                |                                                                                          | minus 20 = *                                                                                                                                                                                                                  |       |                                            |     | x \$100 =                 |  |                                       |  |                     |  |
| INDEPENDENT CLAIMS<br>(37 CFR 1.16(h))                                                                                          |                                                                                          | minus 3 = *                                                                                                                                                                                                                   |       |                                            |     | x \$480 =                 |  |                                       |  |                     |  |
| <input type="checkbox"/> APPLICATION SIZE FEE (37 CFR 1.16(s))                                                                  |                                                                                          | If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s). |       |                                            |     |                           |  |                                       |  |                     |  |
| <input type="checkbox"/> MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))                                                      |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
| * If the difference in column 1 is less than zero, enter "0" in column 2.                                                       |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     | TOTAL                     |  |                                       |  |                     |  |
| <b>APPLICATION AS AMENDED - PART II</b>                                                                                         |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 |                                                                                          | (Column 1)                                                                                                                                                                                                                    |       | (Column 2)                                 |     | (Column 3)                |  |                                       |  |                     |  |
| <b>AMENDMENT</b>                                                                                                                | 02/05/2025                                                                               | CLAIMS REMAINING AFTER AMENDMENT                                                                                                                                                                                              |       | HIGHEST NUMBER PREVIOUSLY PAID FOR         |     | PRESENT EXTRA             |  | RATE (\$)                             |  | ADDITIONAL FEE (\$) |  |
|                                                                                                                                 | Total<br>(37 CFR 1.16(i))                                                                | * 30                                                                                                                                                                                                                          | Minus | ** 30                                      | = 0 |                           |  | x \$200 =                             |  | 0                   |  |
|                                                                                                                                 | Independent<br>(37 CFR 1.16(h))                                                          | * 1                                                                                                                                                                                                                           | Minus | *** 3                                      | = 0 |                           |  | x \$600 =                             |  | 0                   |  |
|                                                                                                                                 | <input type="checkbox"/> Application Size Fee (37 CFR 1.16(s))                           |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 | <input type="checkbox"/> FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  | TOTAL ADD'L FEE                       |  | 0                   |  |
|                                                                                                                                 |                                                                                          | (Column 1)                                                                                                                                                                                                                    |       | (Column 2)                                 |     | (Column 3)                |  |                                       |  |                     |  |
| <b>AMENDMENT</b>                                                                                                                |                                                                                          | CLAIMS REMAINING AFTER AMENDMENT                                                                                                                                                                                              |       | HIGHEST NUMBER PREVIOUSLY PAID FOR         |     | PRESENT EXTRA             |  | RATE (\$)                             |  | ADDITIONAL FEE (\$) |  |
|                                                                                                                                 | Total<br>(37 CFR 1.16(i))                                                                | *                                                                                                                                                                                                                             | Minus | **                                         | =   |                           |  | x \$0 =                               |  |                     |  |
|                                                                                                                                 | Independent<br>(37 CFR 1.16(h))                                                          | *                                                                                                                                                                                                                             | Minus | ***                                        | =   |                           |  | x \$0 =                               |  |                     |  |
|                                                                                                                                 | <input type="checkbox"/> Application Size Fee (37 CFR 1.16(s))                           |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 | <input type="checkbox"/> FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j)) |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
|                                                                                                                                 |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  | TOTAL ADD'L FEE                       |  |                     |  |
| * If the entry in column 1 is less than the entry in column 2, write "0" in column 3.                                           |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  | Program Support Specialist            |  |                     |  |
| ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".                                       |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  | /AMANDA N FORD/                       |  |                     |  |
| *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".                                        |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |
| The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1. |                                                                                          |                                                                                                                                                                                                                               |       |                                            |     |                           |  |                                       |  |                     |  |

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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

|                   |                                    |                              |
|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>02/05/2025 09:47:10 AM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Barbara McGregor |
| PATENT CENTER #           | 69054069                                                    | FILING DATE             | 06/14/2024       |
| CUSTOMER #                | 76656                                                       | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE<br>ADDRESS | -                                                           | AUTHORIZED BY           | Christopher Goff |

### Documents

**TOTAL DOCUMENTS: 3**

| DOCUMENT                                                                 |       | PAGES | DESCRIPTION                                                            | SIZE<br>(KB) |
|--------------------------------------------------------------------------|-------|-------|------------------------------------------------------------------------|--------------|
| 2025-02-05 Response to<br>Non-Final Office Action<br>27843-2603.pdf      |       | 9     | -                                                                      | 212 KB       |
| 2025-02-05 Response to<br>Non-Final Office Action<br>27843-2603-A....pdf | (1-1) | 1     | Amendment/Request for<br>Reconsideration-After Non-<br>Final Rejection | 108 KB       |
| 2025-02-05 Response to<br>Non-Final Office Action<br>27843-2603-CLM.pdf  | (2-5) | 4     | Claims                                                                 | 156 KB       |
| 2025-02-05 Response to<br>Non-Final Office Action<br>27843-2603-REM.pdf  | (6-9) | 4     | Applicant Arguments/Remarks<br>Made in an Amendment                    | 160 KB       |

## Digest

| DOCUMENT                                                           | MESSAGE DIGEST(SHA-512)                                                                                                           |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2025-02-05 Response to Non-Final Office Action 27843-2603.pdf      | CB1893C0BEBFCF52CF7FE217B510434FA856C2728B3A55E8C0DD6C428441D306EDC3C9FD9C572B8FB35011F8E0D119464074A7D19EBF49E65CF2A0BA674260E0C |
| 2025-02-05 Response to Non-Final Office Action 27843-2603-A....pdf | 95C8BFF10495FEAC6EA79ED5CBEE3442E9D6619478FEE6A5B0EB46EE37A76BEDB167D1179768A700AC021F7E3C643ACE24555A2AF6E9B085362E9050DA1AA549  |
| 2025-02-05 Response to Non-Final Office Action 27843-2603-CLM.pdf  | CA62E9FF11147E523625AF49FD895674EB602AD66D349D9A4F2FE07DCCB2FD7D5F8F803887D82CA4B66F26C602A6B41509410C4AF8C668CD58EF445C7F3B8436  |
| 2025-02-05 Response to Non-Final Office Action 27843-2603-REM.pdf  | B133C2AFF630C0BF9508836398D7271B04A835A223698B63403F534B4302F1F54F3180B80B11032928850B9257C90298771A67E07F1F94612189593B27BC62B   |

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### **New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

### **National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

### **New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**IN the United States Patent & Trademark Office**

|                 |                                                            |                   |               |
|-----------------|------------------------------------------------------------|-------------------|---------------|
| First Inventor: | Friedrich Schick                                           | Art Unit:         | 2454          |
| Serial No.:     | 18/743,352                                                 | Confirmation No.: | 2279          |
| Date Filed:     | June 14, 2024                                              | Examiner:         | Frantz B Jean |
| Title:          | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL<br>PROPERTY |                   |               |

February 5, 2025

**RESPONSE TO NON-FINAL OFFICE ACTION**

TO THE COMMISSIONER FOR PATENTS:

In response to the Non-Final Office Action dated November 19, 2024, please consider the following remarks.

**Amendments to the Claims** begin on page 2.

**Remarks** begin on page 6.

**Conclusion** is on page 9.

## AMENDMENTS TO THE CLAIMS

1. (Currently Amended) A detector for identifying at least one material property m comprising

at least one ~~illumination source~~ light projector, wherein the ~~illumination source~~ light projector is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines;

at least one ~~sensor element~~ camera comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the ~~sensor element~~ camera is configured for recording at least one reflection image of a light beam originating from at least one object; and

at least one ~~evaluation device~~ processor configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the ~~evaluation device~~ processor is configured for determining at least one material feature  $\phi_2m$  by applying at least one material dependent image filter  $\Phi_2$  to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; and a linear combination thereof;



wherein the ~~evaluation device~~ processor is configured for determining the material property  $m$  by evaluating the material feature  $\phi_{2m}$ .

2. (Original) The detector according claim 1, wherein the material dependent image filter is a spot shape filter.
3. (Original) The detector according to claim 1, wherein the material dependent image filter is Law's energy filter.
4. (Original) The detector according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, and a deviation from a Lambertian surface reflection.
5. (Currently Amended) The detector according to claim 1, wherein the ~~evaluation device~~ processor is configured for determining at least one distance feature  $\phi_{1z}$  by applying at least one distance dependent image filter  $\Phi_1$  to the reflection image, and  
  
wherein the ~~evaluation device~~ processor is configured for determining a longitudinal coordinate  $z$  by evaluating the distance feature  $\phi_{1z}$ .
6. (Original) The detector according to claim 5, wherein the distance dependent image filter is a structured-light-filter.
7. (Currently Amended) The detector according to claim 1, wherein the ~~sensor element~~ camera comprises at least one CMOS sensor.
8. (Original) The detector according to claim 7, wherein the CMOS sensor comprises at least 100000 optical sensors.
9. (Original) The detector according to claim 8, wherein the CMOS sensor is sensitive in the range of 700 nm to 1000 nm.
10. (Currently Amended) The detector according to claim 1, wherein the ~~illumination source~~ light projector comprises at least one laser source and at least one diffractive optical element.

11. (Currently Amended) The detector according to claim 10, wherein the **illumination source light projector** is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.
12. (Currently Amended) The detector according to claim 11, wherein the **illumination source light projector** emits light having a wavelength of 700 nm to 1000 nm.
13. (Currently Amended) The detector according to claim 12, wherein the **illumination source light projector** comprises at least one vertical cavity surface-emitting laser.
14. (Currently Amended) The detector according to claim 13, wherein the **illumination source light projector** is integrated into a smartphone.
15. (Currently Amended) The detector according to claim 1, wherein the **illumination source light projector** emits light having a wavelength of 700 nm to 1000 nm.
16. (Currently Amended) The detector according to claim 15, wherein the **illumination source light projector** is configured for generating at least one random point pattern or a quasi-random pattern for illumination of the object.
17. (Currently Amended) The detector according to claim 1, wherein the **illumination source light projector** is integrated into a smartphone.
18. (Currently Amended) The detector according to claim 17, wherein the **illumination source light projector** and the optical sensors is arranged in a common plane.
19. (Currently Amended) The detector according to claim 18, wherein the **illumination source light projector** emits light having a wavelength of 700 nm to 1000 nm.
20. (Original) The detector according to claim 1, wherein the object is a human.
21. (Original) The detector according to claim 1, wherein the detector is configured for identification and/or classification of the material.
22. (Original) The detector according to claim 20, wherein the detector is configured for determining and/or validating whether a surface to be examined is or comprises

human skin.

23. (Original) The detector according to claim 1, wherein the detector is configured for detection of human skin.
24. (Original) The detector according to claim 23, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, and a deviation from a Lambertian surface reflection.
25. (Currently Amended) The detector according to claim 1, wherein the **evaluation device processor** is configured for performing at least one image analysis and/or image processing in order to identify a reflection feature in the reflection image.
26. (Currently Amended) The detector according to claim 25, wherein the **evaluation device processor** is adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature.
27. (Currently Amended) The detector according to claim 25, wherein the **evaluation device processor** is adapted to determine at least one reference feature in at least one reference image corresponding to the at least one reflection feature.
28. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises selection of at least one region of interest.
29. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises a background correction.
30. (Original) The detector according to claim 25, wherein the image analysis and/or image processing comprises applying a principle curvature-based region detector.

## **REMARKS**

### **I. Status of the Claims**

Claims 1, 5, 7, 10-19, and 25-27 are amended for clarity. Support can be found at least in Paragraphs [0033], [0038], [0103], and [0124], of the published application (US 2024/0377533).

Claims 1-30 are pending and under examination. No new matter is added herein.

### **II. Rejections under 35 U.S.C. 112(b)**

*Claims 1, 5, 11, 21-23, and 25-27 stand rejected under 35 U.S.C. 112(b) as allegedly being indefinite.*

The Office based the rejections in part on, “Claim limitation ‘**illumination source configured for, sensor element configured for, evaluation device configured for, detector configured for, evaluation device adapted to**’ in claims 1, 5, 11, 21-23, and 25-27’ invokes 35 U.S.C. 112(f) or AIA 35 U.S.C. 112, sixth paragraph. However, the written description fails to disclose the corresponding structure, material, or acts for performing the entire claimed function and to clearly link the structure, material, or acts to the function. Therefore, the claim is indefinite and is rejected under 35 U.S.C. 112(b) or AIA 35 U.S.C. 112, second paragraph” (see Page 3 of the Office Action dated 11/19/24).

Applicant respectfully disagrees. Initially, Applicant respectfully notes that the rejections relate to terms that include the phrases “configured for” or “adapted to”. However, such phrases are not indefinite. As set forth in MPEP §2111.04:

“The determination of whether each of these clauses is a limitation in a claim depends on the specific facts of the case. See, e.g., *Griffin v. Bertina*, 285 F.3d 1029, 1034, 62 USPQ2d 1431 (Fed. Cir. 2002) (finding that a “wherein” clause limited a process claim where the clause gave “meaning and purpose to the manipulative steps”). In In re

Giannelli, 739 F.3d 1375, 1378, 109 USPQ2d 1333, 1336 (Fed. Cir. 2014), the court found that an **"adapted to"** clause limited a machine claim where "the written description makes clear that 'adapted to,' as used in the [patent] application, has a narrower meaning, viz., that the claimed machine is **designed or constructed to be used** as a rowing machine whereby a pulling force is exerted on the handles." In Hoffer v. Microsoft Corp., 405 F.3d 1326, 1329, 74 USPQ2d 1481, 1483 (Fed. Cir. 2005), the court held that when a "'whereby' clause states a condition that is **material to patentability**, it cannot be ignored in order to change the substance of the invention." MPEP §2111.04, emphasis added.

In other words, MPEP §2111.04 establishes that the phrase "adapted to" is material to patentability and sufficient to limit a claim. That is, further structure is **not required** for definiteness when the phrase "adapted to" or its analog "configured for" is used. Applicant respectfully submits that the present claims are definite at least for this reason.

In addition, Applicant respectfully notes that the present claims recite structural terms: "illumination source", "sensor element", "evaluation device", and "detector". That is, the present claims do not merely a "means" for achieving a goal, but rather, recite a particular type of means (i.e., structure). Thus, the present claims and the specification include relevant structures.

To the extent that the Office asserts that the claimed structures are not sufficiently specific, Applicant respectfully notes that the relevant standards of examination are set forth in MPEP 2181:

"If persons of ordinary skill in the art reading the specification understand the term to have a sufficiently definite meaning as the name for the structure that performs the function, even when the term covers a **broad class of structures or identifies the structures by their function** (e.g.,

‘filters,’ ‘brakes,’ ‘clamp,’ ‘screwdriver,’ and ‘locks’) 35 U.S.C. 112(f) will not apply. *Apex Inc. v. Raritan Computer, Inc.*, 325 F.3d 1364, 1372-73, 66 USPQ2d 1444, 1451-52 (Fed. Cir. 2003); *CCS Fitness*, 288 F.3d at 1369, 62 USPQ2d at 1664; *Watts v. XL Sys. Inc.*, 232 F.3d 877, 880-81, 56 USPQ2d 1836, 1839 (Fed. Cir. 2000); *Personalized Media*, 161 F.3d at 704, 48 USPQ2d at 1888; *Greenberg v. Ethicon Endo-Surgery, Inc.*, 91 F.3d 1580, 1583, 39 USPQ2d 1783, 1786 (Fed. Cir. 1996) (**‘Many devices take their names from the functions they perform.’**)... The following are examples of structural terms that have been found **not** to invoke 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, paragraph 6: ‘circuit,’ ‘detent mechanism,’ **‘digital detector,’** ‘reciprocating member,’ ‘connector assembly,’ ‘perforation,’ ‘sealingly connected joints,’ and ‘eyeglass hanger member.’ See *Mass. Inst. of Tech.*, 462 F.3d at 1355-1356, 80 USPQ2d at 1332 (the court found the recitation of ‘aesthetic correction circuitry’ sufficient to avoid pre-AIA 35 U.S.C. 112, paragraph 6, treatment because **the term circuit, combined with a description of the function of the circuit, connoted sufficient structure to one of ordinary skill in the art.**.)” MPEP §2181, emphasis added.

Here, the present claims recite terms that encompass a broad class of structures that are defined by their functionalities. These recitations are fully consistent with the definite terms defined in MPEP §2181. For example, the present claims recite a “detector” and MPEP §2181 states that the term “digital detector” is definite. The Office has not stated that the claim term “detector” is indefinite merely for not specifically reciting an analog detector or a digital detector (and such a statement would be improper). The same logic applies to the other claim terms. Applicant respectfully submits that the present claims are further definite at least for this reason.

Finally, solely for the sake of advancing prosecution, the present claims have been amended to more precisely recite the claimed structures. Applicant respectfully submits that the present claims are still further definite at least for this reason.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

**CONCLUSION**

Applicant requests reconsideration and allowance of all pending claims. Applicant authorizes the Commissioner to charge any fees due in connection with this response to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP. Please reference docket number 27843-2603.

Respectfully submitted,

/Christopher M. Goff/

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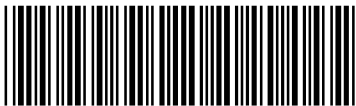
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|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Index of Claims</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                    | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

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|---|-----------------|---|-------------------|---|---------------------|---|-----------------|
| ✓ | <b>Rejected</b> | - | <b>Cancelled</b>  | N | <b>Non-Elected</b>  | A | <b>Appeal</b>   |
| = | <b>Allowed</b>  | ÷ | <b>Restricted</b> | I | <b>Interference</b> | O | <b>Objected</b> |

| CLAIMS                                                                                                                                                                                                  |          |            |            |            |  |  |  |  |  |  |
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| <input checked="" type="checkbox"/> Claims renumbered in the same order as presented by applicant <input type="checkbox"/> CPA <input checked="" type="checkbox"/> T.D. <input type="checkbox"/> R.1.47 |          |            |            |            |  |  |  |  |  |  |
| CLAIM                                                                                                                                                                                                   |          | DATE       |            |            |  |  |  |  |  |  |
| Final                                                                                                                                                                                                   | Original | 09/20/2024 | 11/14/2024 | 02/12/2025 |  |  |  |  |  |  |
|                                                                                                                                                                                                         | 1        | ✓          | ✓          | =          |  |  |  |  |  |  |
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|                                                                                                                                                                                                         | 5        | ✓          | ✓          | =          |  |  |  |  |  |  |
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|                                                                                                                                                                                                         | 24       | ✓          | ✓          | =          |  |  |  |  |  |  |
|                                                                                                                                                                                                         | 25       | ✓          | ✓          | =          |  |  |  |  |  |  |
|                                                                                                                                                                                                         | 26       | ✓          | ✓          | =          |  |  |  |  |  |  |
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|                                                                                                                                                                                                         | 28       | ✓          | ✓          | =          |  |  |  |  |  |  |
|                                                                                                                                                                                                         | 29       | ✓          | ✓          | =          |  |  |  |  |  |  |
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|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Search Notes</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                 | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

| CPC - Searched*                                                                                                                                  |            |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Symbol                                                                                                                                           | Date       | Examiner |
| G01S17/46 G06T7/521 G06T7/73 G06V10/60 G06V10/145<br>G06V10/764 H04N23/74 H04N23/56 G01B11/22 G01C21/16<br>G01S17/66 G06T5/002 G06T5/20 G06T7/20 | 09/20/2024 | FJ       |

| CPC Combination Sets - Searched* |      |          |
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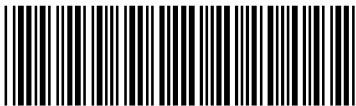
| US Classification - Searched* |          |            |          |
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| Class                         | Subclass | Date       | Examiner |
| 348                           | 135      | 09/20/2024 | FJ       |

\* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

| Search Notes                                         |            |          |
|------------------------------------------------------|------------|----------|
| Search Notes                                         | Date       | Examiner |
| Derwent, EPO, JPO, IBMTDBD, OCR, PGPUB, Google, East | 09/19/2024 | FJ       |
| updated search                                       | 11/14/2024 | FJ       |
| Updated Search                                       | 02/12/2025 | FJ       |

| Interference Search |                       |            |          |
|---------------------|-----------------------|------------|----------|
| US Class/CPC Symbol | US Subclass/CPC Group | Date       | Examiner |
| G01S                | 17/46                 | 02/12/2025 | FJ       |
| H04N                | 23/56                 | 02/12/2025 | FJ       |
| G06T                | 7/521                 | 02/12/2025 | FJ       |

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| <b><i>Search Notes</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                 | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

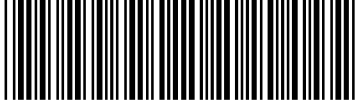
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|------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b>Issue Classification</b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                  | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

| CPC    |   |    |   |     |      |            |
|--------|---|----|---|-----|------|------------|
| Symbol |   |    |   |     | Type | Version    |
| G01S   | / | 17 | / | 46  | F    | 2013-01-01 |
| G06T   | / | 7  | / | 521 | I    | 2017-01-01 |
| G06T   | / | 7  | / | 73  | I    | 2017-01-01 |
| G06V   | / | 10 | / | 764 | I    | 2022-01-01 |
| G01B   | / | 11 | / | 22  | I    | 2013-01-01 |
| G01C   | / | 21 | / | 16  | I    | 2013-01-01 |
| G01S   | / | 17 | / | 66  | I    | 2013-01-01 |
| G06T   | / | 5  | / | 20  | I    | 2013-01-01 |
| G06T   | / | 7  | / | 20  | I    | 2013-01-01 |
| H04N   | / | 23 | / | 56  | I    | 2023-01-01 |
| H04N   | / | 23 | / | 74  | I    | 2023-01-01 |
| G06V   | / | 10 | / | 60  | I    | 2022-01-01 |
| G06V   | / | 10 | / | 145 | I    | 2022-01-01 |
| G06T   | / | 5  | / | 70  | I    | 2024-01-01 |

| CPC Combination Sets |   |  |   |      |     |         |         |
|----------------------|---|--|---|------|-----|---------|---------|
| Symbol               |   |  |   | Type | Set | Ranking | Version |
|                      | / |  | / |      |     |         |         |

|                                                    |                              |                     |                   |
|----------------------------------------------------|------------------------------|---------------------|-------------------|
| NONE                                               | <b>Total Claims Allowed:</b> |                     |                   |
| (Assistant Examiner)                               | (Date)                       | 30                  |                   |
| /FRANTZ B JEAN/<br>Primary Examiner, Art Unit 2454 | 12 February 2025             | O.G. Print Claim(s) | O.G. Print Figure |
| (Primary Examiner)                                 | (Date)                       | 1                   | 1                 |

|                                                                                                                         |                                              |                                                                 |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Issue Classification</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                         | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

|                                     |   |    |   |     |  |
|-------------------------------------|---|----|---|-----|--|
| <b>INTERNATIONAL CLASSIFICATION</b> |   |    |   |     |  |
| <b>CLAIMED</b>                      |   |    |   |     |  |
| G01S17/46                           | / | 17 | / | 46  |  |
| G06T7/521                           | / | 7  | / | 521 |  |
| H04N23/56                           | / | 23 | / | 56  |  |
| <b>NON-CLAIMED</b>                  |   |    |   |     |  |
|                                     | / |    | / |     |  |

|                                   |                 |
|-----------------------------------|-----------------|
| <b>US ORIGINAL CLASSIFICATION</b> |                 |
| <b>CLASS</b>                      | <b>SUBCLASS</b> |
|                                   |                 |

|                            |                                          |  |  |  |  |  |
|----------------------------|------------------------------------------|--|--|--|--|--|
| <b>CROSS REFERENCES(S)</b> |                                          |  |  |  |  |  |
| <b>CLASS</b>               | <b>SUBCLASS (ONE SUBCLASS PER BLOCK)</b> |  |  |  |  |  |
|                            |                                          |  |  |  |  |  |

|                                 |  |                              |                   |
|---------------------------------|--|------------------------------|-------------------|
| NONE                            |  | <b>Total Claims Allowed:</b> |                   |
| (Assistant Examiner)            |  | 30                           |                   |
| (Date)                          |  |                              |                   |
| /FRANTZ B JEAN/                 |  | 12 February 2025             |                   |
| Primary Examiner, Art Unit 2454 |  | O.G. Print Claim(s)          | O.G. Print Figure |
| (Primary Examiner)              |  | 1                            | 1                 |
| (Date)                          |  |                              |                   |

|                                                                                                                         |                                              |                                                                 |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|
| <b><i>Issue Classification</i></b><br> | <b>Application/Control No.</b><br>18/743,352 | <b>Applicant(s)/Patent Under Reexamination</b><br>SCHICK et al. |
|                                                                                                                         | <b>Examiner</b><br>FRANTZ B JEAN             | <b>Art Unit</b><br>2454                                         |

|                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> Claims renumbered in the same order as presented by applicant <input type="checkbox"/> CPA <input checked="" type="checkbox"/> T.D. <input type="checkbox"/> R.1.47 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| CLAIMS |          |       |          |       |          |       |          |       |          |       |          |       |          |       |          |
|--------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|
| Final  | Original | Final | Original | Final | Original | Final | Original | Final | Original | Final | Original | Final | Original | Final | Original |
|        | 1        |       | 10       |       | 19       |       | 28       |       |          |       |          |       |          |       |          |
|        | 2        |       | 11       |       | 20       |       | 29       |       |          |       |          |       |          |       |          |
|        | 3        |       | 12       |       | 21       |       | 30       |       |          |       |          |       |          |       |          |
|        | 4        |       | 13       |       | 22       |       |          |       |          |       |          |       |          |       |          |
|        | 5        |       | 14       |       | 23       |       |          |       |          |       |          |       |          |       |          |
|        | 6        |       | 15       |       | 24       |       |          |       |          |       |          |       |          |       |          |
|        | 7        |       | 16       |       | 25       |       |          |       |          |       |          |       |          |       |          |
|        | 8        |       | 17       |       | 26       |       |          |       |          |       |          |       |          |       |          |
|        | 9        |       | 18       |       | 27       |       |          |       |          |       |          |       |          |       |          |

|                                                                                                        |  |                                    |                        |
|--------------------------------------------------------------------------------------------------------|--|------------------------------------|------------------------|
| NONE<br><br>(Assistant Examiner) _____ (Date) _____                                                    |  | <b>Total Claims Allowed:</b><br>30 |                        |
| /FRANTZ B JEAN/<br>Primary Examiner, Art Unit 2454<br>(Primary Examiner) _____ (Date) 12 February 2025 |  | O.G. Print Claim(s)<br>1           | O.G. Print Figure<br>1 |

## AI Overview

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This describes a **material property detector** that uses a structured light approach to analyze the reflection patterns from an object when illuminated with specific light patterns, allowing for the identification of material properties by evaluating the characteristics of the reflected image using specialized image filters tailored to different materials.

### Key components:

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### Detector for identifying at least one material property

Described herein is a **detector** for **identifying at least one material property m**. The **detector includes at least one** sensor element including a matrix of ...

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<https://patents.google.com/patent>

### Detector with a projector for illuminating at least one object

Described herein is a **projector** for **illuminating at least one object** with at **least two illumination patterns**.

The **projector includes at least one** tunable ...

Missing: material m

Justia  
<https://patents.justia.com/patents-by-us-classification>

### Lamp Control Patents and Patent Applications (Class 353/85)

Abstract: Described herein is a method for calibrating a **detector** including at **least one** camera and at **least one projector**. The **projector is configured** for ...

Justia  
<https://patents.justia.com/patents-by-us-classification>

### Projected Image Combined With Real Object Patents and ...

**Projectors** provide full spectrum, amplitude modulated, and controllably divergent beams of **light** that can be steered with **one** or more degrees of freedom.

gassets.com  
<https://phwt2.gassets.com/uploads/2024/10>

### Artec Europe S.À.R.L v. Shenzhen Creativity 3D Tech. Co., No. 1 ...

by E Handy · 2024 — "It defines the scope of the **property** right being enforced, and is often the difference between infringement and non-infringement, or validity.

48 pages

Opto Engineering  
<https://www.opto-e.com/products/lighting>

### Machine Vision Lighting, LED illuminators, pattern projectors

We offer various **illumination** geometries including collimated telecentric illuminators, **pattern projectors**, backlights, ring **light** units, coaxial, dome, bar and ...

 Nweon Patent  
<https://patent.nweon.com> > ...

## Meta Patent | Addressable vertical cavity surface emitting laser ...

Jun 15, 2023 -- A method **comprising: selecting** a first dot **pattern, of a plurality of dot patterns**, based in part on a target density and a target distance, ...

 IPqwerly  
<https://www.ipqwerly.com> > [ipowner](#) > [owner](#) > 1136163...

## Rodenstock GmbH

A method for **determining** the sensitivity of at **least one eye of a** test subject based on at **least two** provided pairs of values for visual acuity and refraction, ...

 ResearchGate  
<https://www.researchgate.net> > [Phillips](#)

## Frederik DE BRUIJN | Senior Researcher | Master of Science

An apparatus for estimating a **property of a surface of an object comprises** a coherent **light source** (105) which generates at **least one light spot** on the surface.

 Nanoscience Instruments  
<https://www.nanoscience.com> > [Techniques](#)

## Transmission Electron Microscopy

Transmission electron microscopy (TEM) is **an** analytical technique used to visualize the smallest structures in matter.

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## BIB DATA SHEET

CONFIRMATION NO. 2279

| SERIAL NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FILING or<br>371(c) DATE | CLASS                                                          | GROUP ART UNIT                         | ATTORNEY<br>DOCKET NO.                                                           |                               |                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|-------------------------------|------------------------------------|
| 18/743,352                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 06/14/2024               | 348                                                            | 2454                                   | 27843-2603<br>182037US03                                                         |                               |                                    |
| <b>RULE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>APPLICANTS</b><br>trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>INVENTORS</b><br>Friedrich SCHICK, Ludwigshafen, GERMANY;<br>Peter SCHILLEN, Ludwigshafen, GERMANY;<br>Patrick SCHINDLER, Ludwigshafen, GERMANY;<br>Andre SCHMIDT, Ludwigshafen, GERMANY;<br>Michael EBERSPACH, Ludwigshafen, GERMANY;<br>Christian LENNARTZ, Ludwigshafen, GERMANY;<br>Robert SEND, Karlsruhe, GERMANY;<br>Lars DIESSELBERG, Karlsruhe, GERMANY;<br>Heiko HENGEN, Steinweiler, GERMANY;<br>Ingmar BRUDER, Ludwigshafen, GERMANY;<br>Jakob UNGER, Ludwigshafen, GERMANY;<br>Christian BONSIGNORE, Ludwigshafen, GERMANY; |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>** CONTINUING DATA *****</b><br>This application is a CON of 18/425,144 01/29/2024<br>which is a CON of 17/439,492 09/15/2021 PAT 11947013<br>which is a 371 of PCT/EP20/56759 03/13/2020                                                                                                                                                                                                                                                                                                                                                |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>** FOREIGN APPLICATIONS *****</b><br>EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>** IF REQUIRED, FOREIGN FILING LICENSE GRANTED **</b><br>07/15/2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |                                                                |                                        |                                                                                  |                               |                                    |
| Foreign Priority claimed <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>35 USC 119(a-d) conditions met <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No<br>Verified and /FRANTZ B JEAN/<br>Acknowledged Examiner's Signature                                                                                                                                                                                                                                                                     |                          | <input type="checkbox"/> Met after Allowance<br>FJ<br>Initials | <b>STATE OR<br/>COUNTRY</b><br>GERMANY | <b>SHEETS<br/>DRAWINGS</b><br>2                                                  | <b>TOTAL<br/>CLAIMS</b><br>30 | <b>INDEPENDENT<br/>CLAIMS</b><br>1 |
| <b>ADDRESS</b><br>Patent Docket Department<br>Armstrong Teasdale LLP<br>7700 Forsyth Boulevard<br>Suite 1800<br>St. Louis, MO 63105<br>UNITED STATES                                                                                                                                                                                                                                                                                                                                                                                        |                          |                                                                |                                        |                                                                                  |                               |                                    |
| <b>TITLE</b><br>DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                                                                |                                        |                                                                                  |                               |                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |                                                                |                                        | <input type="checkbox"/> All Fees<br><input type="checkbox"/> 1.16 Fees (Filing) |                               |                                    |



|                                        |                                                                                                                   |                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FILING FEE<br/>RECEIVED</b><br>0.00 | FEES: Authority has been given in Paper<br>No. _____ to charge/credit DEPOSIT ACCOUNT<br>No. _____ for following: | <input type="checkbox"/> 1.17 Fees (Processing Ext. of time)<br><input type="checkbox"/> 1.18 Fees (Issue)<br><input type="checkbox"/> Other _____<br><input type="checkbox"/> Credit _____ |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## PE2E SEARCH - Search History (Prior Art)

| Ref # | Hits   | Search Query                                                                                                                                                                                                                                                                                                                              | DBs                                                                                                                                                                                                                                                                                                                  | Default Operator | Plurals | British Equivalents | Time Stamp          |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|---------------------|---------------------|
| L1    | 2516   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:39 AM |
| L2    | 440018 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 01:49 AM |
| L3    | 757    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR               | ON      | ON                  | 2023/09/09 02:01 AM |
| L4    | 183885 | (detect\$3 OR sen\$3)                                                                                                                                                                                                                                                                                                                     | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                    | OR               | ON      | ON                  | 2023/09/09          |

|    |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|    |        | near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                           | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 02:34 AM               |
| L5 | 249451 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L6 | 2430   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:37 AM |
| L7 | 11     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>02:38 AM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L8  | 1872340 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 09:57 AM |
| L9  | 1341757 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:00 AM |
| L10 | 113287  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                       | OR | ON | ON | 2023/09/09 10:01 AM |
| L11 | 67      | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/09/09 10:02 AM |

|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
|-----|----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L12 | 0  | L11 AND I1 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L13 | 35 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |
| L14 | 11 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/09/09<br>10:03 AM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L15 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/09/09 06:13 PM |
| L16 | 2541   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L17 | 454819 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L18 | 791    | (G01S17/46 OR                                                                                                                                                                                                                                                                                                                             | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2023/11/18          |

|     |        |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:50 PM               |
| L19 | 188776 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L20 | 259070 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L21 | 2529   | L4 AND I5                                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L22 | 12      | L6 AND I3                                                                                                         | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L23 | 1891255 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile)        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L24 | 1363696 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L25 | 114665  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2023/11/18<br>08:50 PM |



|     |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                        |
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| L26 | 70 | L10 AND I6 | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L27 | 27 | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |
| L28 | 36 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2023/11/18<br>08:50 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
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| L29 | 12     | L11 AND I3                                                                                                                                                                                                                                                                                                                                | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/18 08:50 PM |
| L30 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2023/11/18 08:50 PM |
| L31 | 2596   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/09/19 08:07 PM |
| L32 | 468828 | (348/135).ccls. OR                                                                                                                                                                                                                                                                                                                        | (US-PGPUB; USPAT;                                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/09/19          |

|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73 OR<br>G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56 OR<br>G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                 | USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                      |    |    |    | 08:07 PM               |
| L33 | 903    | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L34 | 208447 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L35 | 296319 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |         |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
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| L36 | 2914    | L4 AND I5                                                                                                  | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L37 | 18      | L6 AND I3                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L38 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3<br>OR distance) near60<br>beam near5 (light OR<br>profile) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L39 | 1447259 | filter near6 (image OR<br>median OR contrast OR                                                            | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L40 | 119738 | energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L41 | 92     | L8 AND I9                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L42 | 39     | L10 AND I6                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
|     |        | L11 AND I1                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |      |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L43 | 48   | L11 AND I2                                                     | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L44 | 18   | L11 AND I3                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L45 | 4    | "20180033146".pn.                                              | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L46 | 2596 | ( Friedrichr near3<br>Schoick) OR (Peter<br>near3 Schillen) OR | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
|     |        | ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           |    |    |    |                     |
| L47 | 468828 | (348/135).cls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                       | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L48 | 903    | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/09/19 08:07 PM |
| L49 | 208447 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                              | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/09/19 08:07 PM |

|     |         |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                        |
|-----|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L50 | 296319  | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L51 | 2914    | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L52 | 18      | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                      | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L53 | 1963130 | (reflect\$3 OR image OR<br>object OR evaluat\$3                                               | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,                                                                                                                                                                                                                                                                                                                                                                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |



|     |         |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|---------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |         | OR distance) near60<br>beam near5 (light OR<br>profile)                                                           | AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L54 | 1447259 | filter near6 (image OR<br>median OR contrast OR<br>energy OR<br>homogeneity OR<br>dissimilarity OR<br>smoothnwss) | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L55 | 119738  | L8 AND I9                                                                                                         | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L56 | 92      | L10 AND I6                                                                                                        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |

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| L57 | 39 | L11 AND I1        | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L58 | 48 | L11 AND I2        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L59 | 18 | L11 AND I3        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/09/19<br>08:07 PM |
| L60 | 4  | "20180033146".pn. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/09/19<br>08:07 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L61 | 2      | "11947013".pn.                                                                                                                                                                                                                                                                                                                            | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/09/19 09:42 PM |
| L62 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L63 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR                                                                                                                                                                                                 | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO,                                                                                                    | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L64 | 1005   | G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc.                                                                                                                                                                   | NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L65 | 219410 | (G01S17/46 OR<br>G06T7/521 OR<br>G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L66 | 318145 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                                                    | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L67 | 3134   | L4 AND I5                                                                                                                                                                                                                                       | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                          | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|---------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L68 | 18      | L6 AND I3                                                                                          | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L69 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L70 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |                        |
|-----|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L71 | 122964 | L8 AND I9  | SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L72 | 102    | L10 AND I6 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L73 | 41     | L11 AND I1 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L74 | 56     | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                                                                                 | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |      |                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L75 | 18   | L11 AND I3                                                                                                                                                                                                                    | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L76 | 4    | "20180033146".pn.                                                                                                                                                                                                             | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L77 | 2657 | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L78 | 496753 | (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv.<br>(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L79 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                   | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L80 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)                                                                                                                                                                                                                                        | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L81 | 318145 | longitudinal WITH coordinate AND                                                                                                                                                                                                                                                                                           | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |



|     |         |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |    |                     |
|-----|---------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L82 | 3134    | (material OR property OR evaluat\$3 OR distance)<br><br>L4 AND I5                              | AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br><br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L83 | 18      | L6 AND I3                                                                                      | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23 11:54 PM |
| L84 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE,                                                                                                                                                                                                                                                                                                                                                                      | OR | ON | ON | 2024/10/23 11:54 PM |

|     |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
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| L85 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L86 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L87 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L88 | 41      | L11 AND I1                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

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|-----|----|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
| L89 | 56 | L11 AND I2        | CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L90 | 18 | L11 AND I3        | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L91 | 4  | "20180033146".pn. | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,                                                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23<br>11:54 PM |

|     |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L92 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L93 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L94 | 1005   | (G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                                  | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L95 | 219410 | (detect\$3 OR sen\$3) near10 (identif\$ OR                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|     |        |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|-----|--------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|     |        | classify\$3) WITH<br>(material OR property)                                                   | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L96 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance) | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L97 | 3134   | L4 AND I5                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L98 | 18     | L6 AND I3                                                                                     | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
|------|---------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L99  | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L100 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L101 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L102 | 102     | L10 AND I6                                                                                         | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|      |    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |                        |
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| L103 | 41 | L11 AND I1 | CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)<br><br>(US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L104 | 56 | L11 AND I2 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L105 | 18 | L11 AND I3 | (US-PGPUB; USPAT;<br>USOCR; FIT (AP, AT,<br>AU, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, MC, MD, MY, NL,<br>NO, NZ, OA, PH, PL,<br>PT, RO, RS, RU, SE,<br>SG, SI, SK, SU, TH,<br>TN, TR, TW, UA, VN,<br>WO); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                                                                                                                                                                                                                                                                                       | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |        |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L106 | 4      | "20180033146".pn.                                                                                                                                                                                                                                                                                                                         | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L107 | 2657   | ( Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore) ).inv. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L108 | 496753 | (348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.                                                                                                                               | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L109 | 1005   | (G01S17/46 OR G06T7/521 OR                                                                                                                                                                                                                                                                                                                | (US-PGPUB; USPAT; USOCR; FIT (AU, AP,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |



|      |        |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |                        |
|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|------------------------|
|      |        | G06T7/73).cpc. AND<br>(G06V10/60 OR<br>G06V10/145 OR<br>G06V10/764 OR<br>H04N23/74 OR<br>H04N23/56).cpc. AND<br>(G01B11/22 OR<br>G01C21/16 OR<br>G01S17/66 OR<br>G06T5/002 OR<br>G06T5/20 OR<br>G06T7/20).cpc. | AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB)                                             |    |    |    |                        |
| L110 | 219410 | (detect\$3 OR sen\$3)<br>near10 (identif\$ OR<br>classify\$3) WITH<br>(material OR property)                                                                                                                   | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L111 | 318145 | longitudinal WITH<br>coordinate AND<br>(material OR property<br>OR evaluat\$3 OR<br>distance)                                                                                                                  | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,<br>RS, SE, SG, SI, SK,<br>TH, TN, TR, TW, UA,<br>VN); FPRS; EPO;<br>JPO; DERWENT;<br>IBM_TDB) | OR | ON | ON | 2024/10/23<br>11:54 PM |
| L112 | 3134   | L4 AND I5                                                                                                                                                                                                      | (US-PGPUB; USPAT;<br>USOCR; FIT (AU, AP,<br>AT, BE, BG, BR, BY,<br>CA, CH, CN, CS, CU,<br>CZ, DD, DE, DK, EA,<br>EE, EP, ES, FI, FR, GB,<br>HR, HU, ID, IE, IL, IS,<br>IT, JP, KR, LT, LU, LV,<br>MA, OA, RU, SU, WO,<br>MC, MD, MY, NL, NO,<br>NZ, PH, PL, PT, RO,                                                                                               | OR | ON | ON | 2024/10/23<br>11:54 PM |

|      |         |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |                     |
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| L113 | 18      | L6 AND I3                                                                                          | RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L114 | 2008062 | (reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)     | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L115 | 1497001 | filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                                                   | OR | ON | ON | 2024/10/23 11:54 PM |
| L116 | 122964  | L8 AND I9                                                                                          | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY,                                                                                                                                                                                                                                                                                                                                              | OR | ON | ON | 2024/10/23 11:54 PM |

|      |     |            |                                                                                                                                                                                                                                                                                                                      |    |    |    |                     |
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| L117 | 102 | L10 AND I6 | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                           | OR | ON | ON | 2024/10/23 11:54 PM |
| L118 | 41  | L11 AND I1 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L119 | 56  | L11 AND I2 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |

|      |    |                   |                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |                     |
|------|----|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---------------------|
| L120 | 18 | L11 AND I3        | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)<br>(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2024/10/23 11:54 PM |
| L121 | 4  | "20180033146".pn. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |
| L122 | 2  | "11947013".pn.    | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)                                                               | OR | ON | ON | 2024/10/23 11:54 PM |

## PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



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## NOTICE OF ALLOWANCE AND FEE(S) DUE

76656 7590 02/20/2025  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

EXAMINER

JEAN, FRANTZ B

ART UNIT

PAPER NUMBER

2454

DATE MAILED: 02/20/2025

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

| APPLN. TYPE    | ENTITY STATUS | ISSUE FEE DUE | PUBLICATION FEE DUE | PREV. PAID ISSUE FEE | TOTAL FEE(S) DUE | DATE DUE   |
|----------------|---------------|---------------|---------------------|----------------------|------------------|------------|
| nonprovisional | UNDISCOUNTED  | \$1290        | \$0.00              | \$0.00               | \$1290           | 05/20/2025 |

**THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.**

**THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.**

### HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

**IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at [www.uspto.gov/PatentMaintenanceFees](http://www.uspto.gov/PatentMaintenanceFees).**

# PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE  
Commissioner for Patents  
P.O. Box 1450  
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

**INSTRUCTIONS:** This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

76656 7590 02/20/2025  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

## Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

|                         |
|-------------------------|
| (Typed or printed name) |
| (Signature)             |
| (Date)                  |

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

| APPLN. TYPE    | ENTITY STATUS | ISSUE FEE DUE | PUBLICATION FEE DUE | PREV. PAID ISSUE FEE | TOTAL FEE(S) DUE | DATE DUE   |
|----------------|---------------|---------------|---------------------|----------------------|------------------|------------|
| nonprovisional | UNDISCOUNTED  | \$1290        | \$0.00              | \$0.00               | \$1290           | 05/20/2025 |

| EXAMINER       | ART UNIT | CLASS-SUBCLASS |
|----------------|----------|----------------|
| JEAN, FRANTZ B | 2454     | 348-135000     |

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 \_\_\_\_\_

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 \_\_\_\_\_

3 \_\_\_\_\_

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. \_\_\_\_\_

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature \_\_\_\_\_

Date \_\_\_\_\_

Typed or printed name \_\_\_\_\_

Registration No. \_\_\_\_\_



# UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE  
**United States Patent and Trademark Office**  
Address: COMMISSIONER FOR PATENTS  
P.O. Box 1450  
Alexandria, Virginia 22313-1450  
www.uspto.gov

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |

76656 7590 02/20/2025  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

|          |
|----------|
| EXAMINER |
|----------|

JEAN, FRANTZ B

|          |              |
|----------|--------------|
| ART UNIT | PAPER NUMBER |
|----------|--------------|

2454

DATE MAILED: 02/20/2025

## **Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)** (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

## OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

### Privacy Act Statement

**The Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.



|                               |                                      |                                      |                                 |
|-------------------------------|--------------------------------------|--------------------------------------|---------------------------------|
| <b>Notice of Allowability</b> | <b>Application No.</b><br>18/743,352 | <b>Applicant(s)</b><br>SCHICK et al. |                                 |
|                               | <b>Examiner</b><br>FRANTZ B JEAN     | <b>Art Unit</b><br>2454              | <b>AIA (FITF) Status</b><br>Yes |

**-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--**

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 05 February 2025.  
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on \_\_\_\_\_.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on \_\_\_\_\_; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-30. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see [http://www.uspto.gov/patents/init\\_events/pph/index.jsp](http://www.uspto.gov/patents/init_events/pph/index.jsp) or send an inquiry to [PPHfeedback@uspto.gov](mailto:PPHfeedback@uspto.gov).

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a) ☒ All      b) ☐ Some\*      c) ☐ None of the:

1. ☒ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. \_\_\_\_\_.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

\* Certified copies not received: \_\_\_\_\_.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.  
**THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.**

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.  
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date \_\_\_\_\_.

**Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

**Attachment(s)**

|                                                                                                                   |                                                                                      |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1. <input type="checkbox"/> Notice of References Cited (PTO-892)                                                  | 5. <input type="checkbox"/> Examiner's Amendment/Comment                             |
| 2. <input type="checkbox"/> Information Disclosure Statements (PTO/SB/08),<br>Paper No./Mail Date _____.          | 6. <input checked="" type="checkbox"/> Examiner's Statement of Reasons for Allowance |
| 3. <input type="checkbox"/> Examiner's Comment Regarding Requirement for Deposit<br>of Biological Material _____. | 7. <input type="checkbox"/> Other _____.                                             |
| 4. <input type="checkbox"/> Interview Summary (PTO-413),<br>Paper No./Mail Date. _____.                           |                                                                                      |

|                                                    |  |
|----------------------------------------------------|--|
| /FRANTZ B JEAN/<br>Primary Examiner, Art Unit 2454 |  |
|----------------------------------------------------|--|

***Notice of Pre-AIA or AIA Status***

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claims 1-30 are allowed over the prior art of record.

The following is an examiner's statement of reasons for allowance: the prior art of record fails to teach and suggest the combination of a detector for identifying at least one material property m comprising at least one light projector, wherein the light projector is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; a pseudo- random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines; at least one camera comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the camera is configured for recording at least one reflection image of a light beam originating from at least one object; and at least one processor configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the processor is configured for determining at least one material feature p2m by applying at least one material dependent image filter ... to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey- level-occurrence-based contrast filter; a grey-level-occurrence-based energy

Art Unit: 2454

filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; and a linear combination thereof; wherein the processor is configured for determining the material property  $m$  by evaluating the material feature  $p2m$ .

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANTZ B JEAN whose telephone number is (571)272-3937. The examiner can normally be reached 8-5 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Glenton B. Burgess can be reached on 5712723949. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like

Art Unit: 2454

assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/FRANTZ B JEAN/

Primary Examiner, Art Unit 2454



# UNITED STATES PATENT AND TRADEMARK OFFICE

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| APPLICATION NO.          | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|--------------------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352               | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |
| 76656                    | 7590        | 02/21/2025           | EXAMINER              |                  |
| Patent Docket Department |             |                      | JEAN, FRANTZ B        |                  |
| Armstrong Teasdale LLP   |             |                      | ART UNIT              |                  |
| 7700 Forsyth Boulevard   |             |                      | PAPER NUMBER          |                  |
| Suite 1800               |             |                      | 2454                  |                  |
| St. Louis, MO 63105      |             |                      | NOTIFICATION DATE     |                  |
|                          |             |                      | DELIVERY MODE         |                  |
|                          |             |                      | 02/21/2025            |                  |
|                          |             |                      | ELECTRONIC            |                  |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com

|                                                     |                                      |                                      |                                                   |                       |
|-----------------------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------------------|-----------------------|
| <b><i>Applicant-Initiated Interview Summary</i></b> | <b>Application No.</b><br>18/743,352 | <b>Applicant(s)</b><br>SCHICK et al. |                                                   |                       |
|                                                     | <b>Examiner</b><br>FRANTZ B JEAN     | <b>Art Unit</b><br>2454              | <b>AIA (First Inventor to File) Status</b><br>Yes | <b>Page</b><br>1 of 1 |

| <b>All Participants</b> (applicant, applicants representative, PTO personnel) | <b>Title</b>       | <b>Type</b> |
|-------------------------------------------------------------------------------|--------------------|-------------|
| FRANTZ B JEAN                                                                 | Primary Examiner   | Telephonic  |
| Christopher M Goff                                                            | Attorney of Record |             |

**Date of Interview:** 28 January 2025

**Issues Discussed:**

**35 U.S.C. 112**

Applicant's representative need some clarification in regard to the 112 rejection and feedback on the proposed amendment. Agreement was reached.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| /FRANTZ B JEAN/<br>Primary Examiner, Art Unit 2454                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
| <p><b>Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04</b></p> <p>Please further see:<br/>MPEP 713.04<br/>Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b)<br/>37 CFR § 1.2 Business to be transacted in writing</p> |  |

**Applicant recordation instructions:** The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

**Examiner recordation instructions:** Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.



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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

|                   |                                    |                              |
|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>02/24/2025 09:59:28 AM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Barbara McGregor |
| PATENT CENTER #           | 69303186                                                    | FILING DATE             | 06/14/2024       |
| CUSTOMER #                | 76656                                                       | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE<br>ADDRESS | -                                                           | AUTHORIZED BY           | Christopher Goff |

### Documents

**TOTAL DOCUMENTS: 1**

| DOCUMENT                                                    | PAGES | DESCRIPTION                                     | SIZE<br>(KB) |
|-------------------------------------------------------------|-------|-------------------------------------------------|--------------|
| 2025-02-24 Interview<br>Summary Response 27843-<br>2603.pdf | 1     | Applicant summary of interview<br>with examiner | 100 KB       |

### Digest

| DOCUMENT                                                | MESSAGE DIGEST(SHA-512)                                                                                                                  |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-02-24 Interview Summary<br>Response 27843-2603.pdf | 6E2ADD19190FF595310EF5291C1105C80C050BE73EEC2ABF4<br>8B00B22687240E7F78A5507E67D49805014DCA97516E312E9C<br>24EEF380F69C39406DEB82CB405AB |

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



**In the United States Patent & Trademark Office**

|                 |                                                         |                   |               |
|-----------------|---------------------------------------------------------|-------------------|---------------|
| First Inventor: | Friedrich Schick                                        | Art Unit:         | 2454          |
| Serial No.:     | 18/743,352                                              | Confirmation No.: | 2279          |
| Date Filed:     | June 14, 2024                                           | Examiner:         | Frantz B Jean |
| Title:          | DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY |                   |               |

February 24, 2025

**RESPONSE TO INTERVIEW SUMMARY**

TO THE COMMISSIONER FOR PATENTS:

In response to the Interview Summary dated February 21, 2025, please consider the following remarks.

Applicant thanks Examiner Jean for the Examiner Interview with Applicant's representative Zach Wiersma (Reg. No. 77,808) on January 28, 2025. During the interview, the clarity of the claims and proposed amendments were discussed. Agreement was reached.

Applicant authorizes the Commissioner to charge any fees due in connection with this response to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP. Please reference docket number 27843-2603.

Respectfully submitted,

/Christopher M. Goff/

Christopher M. Goff  
USPTO Registration 41,785  
ARMSTRONG TEASDALE LLP  
7700 Forsyth Blvd., Suite 1800  
St. Louis, Missouri 63105  
(314) 621-5070

CMG/EMF/ZSW

VIA EFS



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## ELECTRONIC ACKNOWLEDGEMENT RECEIPT

|                   |                                    |                              |
|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>05/13/2025 09:12:56 AM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Barbara McGregor |
| PATENT CENTER #           | 70402534                                                    | FILING DATE             | 06/14/2024       |
| CUSTOMER #                | 76656                                                       | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |
| CORRESPONDENCE<br>ADDRESS | -                                                           | AUTHORIZED BY           | Christopher Goff |

### Documents

**TOTAL DOCUMENTS: 1**

| DOCUMENT                                           | PAGES | DESCRIPTION                 | SIZE<br>(KB) |
|----------------------------------------------------|-------|-----------------------------|--------------|
| 2025-05-13 Issue Fee<br>Transmittal 27843-2603.pdf | 1     | Issue Fee Payment (PTO-85B) | 126 KB       |

### Digest

| DOCUMENT                                           | MESSAGE DIGEST(SHA-512)                                                                                                                  |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-05-13 Issue Fee<br>Transmittal 27843-2603.pdf | 2230B97414DFF58AC4E522778113006DC399D754B3F32227D6<br>A29D5A3460A51F793223418831FB5CCB3C39BD350144423752<br>94ECD103E1F8B8D0479BF36EF617 |

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

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## ELECTRONIC PAYMENT RECEIPT

|                   |                                    |                              |
|-------------------|------------------------------------|------------------------------|
| APPLICATION #     | RECEIPT DATE / TIME                | ATTORNEY DOCKET #            |
| <b>18/743,352</b> | <b>05/13/2025 09:12:56 AM Z ET</b> | <b>27843-2603 182037US03</b> |

### Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

### Application Information

|                           |                                                             |                         |                  |
|---------------------------|-------------------------------------------------------------|-------------------------|------------------|
| APPLICATION TYPE          | Utility - Nonprovisional Application<br>under 35 USC 111(a) | PATENT #                | -                |
| CONFIRMATION #            | 2279                                                        | FILED BY                | Barbara McGregor |
| PATENT CENTER #           | 70402534                                                    | AUTHORIZED BY           | Christopher Goff |
| CUSTOMER #                | 76656                                                       | FILING DATE             | 06/14/2024       |
| CORRESPONDENCE<br>ADDRESS | -                                                           | FIRST NAMED<br>INVENTOR | Friedrich SCHICK |

### Payment Information

|                |                        |                       |
|----------------|------------------------|-----------------------|
| PAYMENT METHOD | PAYMENT TRANSACTION ID | PAYMENT AUTHORIZED BY |
| DA / 012384    | E20255C013257491       | Barbara McGregor      |

| FEE CODE                 | DESCRIPTION       | ITEM PRICE(\$) | QUANTITY | ITEM TOTAL(\$)    |
|--------------------------|-------------------|----------------|----------|-------------------|
| 1501                     | UTILITY ISSUE FEE | 1290.00        | 1        | 1290.00           |
| <b>TOTAL<br/>AMOUNT:</b> |                   |                |          | <b>\$1,290.00</b> |

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#### New Applications Under 35 U.S.C. 111

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# PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE  
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By fax, send to: (571)-273-2885

**INSTRUCTIONS:** This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

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Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

76656 7590 02/20/2025  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

## Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

|                         |
|-------------------------|
| (Typed or printed name) |
| (Signature)             |
| (Date)                  |

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352      | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

| APPLN. TYPE    | ENTITY STATUS | ISSUE FEE DUE | PUBLICATION FEE DUE | PREV. PAID ISSUE FEE | TOTAL FEE(S) DUE | DATE DUE   |
|----------------|---------------|---------------|---------------------|----------------------|------------------|------------|
| nonprovisional | UNDISCOUNTED  | \$1290        | \$0.00              | \$0.00               | \$1290           | 05/20/2025 |

| EXAMINER       | ART UNIT | CLASS-SUBCLASS |
|----------------|----------|----------------|
| JEAN, FRANTZ B | 2454     | 348-135000     |

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Armstrong Teasdale LLP

2 \_\_\_\_\_

3 \_\_\_\_\_

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE  
TRINAMIX GMBH

(B) RESIDENCE: (CITY and STATE OR COUNTRY)  
LUDWIGSHAFEN AM RHEIN, GERMANY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 012384

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Christopher M. Goff/

Date May 13, 2025

Typed or printed name Christopher M. Goff

Registration No. 41,785



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| APPLICATION NO. | ISSUE DATE | PATENT NO. | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|-----------------|------------|------------|-----------------------|------------------|
| 18/743,352      | 06/17/2025 | 12332352   | 27843-2603 182037US03 | 2279             |

76656 7590 06/04/2025  
Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
Suite 1800  
St. Louis, MO 63105

## ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

### **Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)** (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

**INVENTOR(s)** (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Friedrich SCHICK, Ludwigshafen, GERMANY;  
Peter SCHILLEN, Ludwigshafen, GERMANY;  
Patrick SCHINDLER, Ludwigshafen, GERMANY;  
Andre SCHMIDT, Ludwigshafen, GERMANY;  
Michael EBERSPACH, Ludwigshafen, GERMANY;

**APPLICANT(s)** (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

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| APPLICATION NO.          | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO.   | CONFIRMATION NO. |
|--------------------------|-------------|----------------------|-----------------------|------------------|
| 18/743,352               | 06/14/2024  | Friedrich SCHICK     | 27843-2603 182037US03 | 2279             |
| 76656                    | 7590        | 06/17/2025           | EXAMINER              |                  |
| Patent Docket Department |             |                      | JEAN, FRANTZ B        |                  |
| Armstrong Teasdale LLP   |             |                      | ART UNIT              |                  |
| 7700 Forsyth Boulevard   |             |                      | PAPER NUMBER          |                  |
| Suite 1800               |             |                      | 2454                  |                  |
| St. Louis, MO 63105      |             |                      | NOTIFICATION DATE     |                  |
|                          |             |                      | DELIVERY MODE         |                  |
|                          |             |                      | 06/17/2025            |                  |
|                          |             |                      | ELECTRONIC            |                  |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com



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| APPLICATION NO. | ISSUE DATE  | PATENT NO. |
|-----------------|-------------|------------|
| 18/743,352      | 17-Jun-2025 | 12332352   |

Patent Docket Department  
Armstrong Teasdale LLP  
7700 Forsyth Boulevard  
St. Louis, MO 63105

### EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>